

Conditional Quotient-First Private PAC Learning Bounds

July 27, 2026

1 Theoretical Setup

1.1 Problem Setting

Let (X, Σ) be a nonempty measurable space and let $C \subseteq \{0, 1\}^X$ be a nonempty binary concept class. Define

$$x \equiv_C x' \iff c(x) = c(x') \text{ for every } c \in C,$$

let $Q_C = X/\equiv_C$, and write $\kappa : X \rightarrow Q_C$ for the quotient map. The quotient carries the discrete sigma-field 2^{Q_C} . For $c \in C$, write $\bar{c}(\kappa(x)) = c(x)$ and set $\bar{C} = \{\bar{c} : c \in C\}$. The released hypothesis space and decoder are

$$H_C = \{0, 1\}^{Q_C}, \quad \mathcal{H}_C = \bigotimes_{q \in Q_C} 2^{\{0, 1\}}, \quad \text{Dec}_C(\bar{h}) = \bar{h} \circ \kappa.$$

The released object is allowed to be improper: $\bar{h}$ need not lie in $\bar{C}$.

Write

$$v = \text{VC}(C), \quad d = \text{LD}(C),$$

and use natural logarithms. For $u \geq 0$, define $\log^* u = 0$ for $u \leq 1$ and, for $u > 1$, define $\log^* u = \min\{j \geq 1 : \log^{oj}(u) \leq 1\}$. For cardinality use

$$\log^+ |C| = \begin{cases} \max\{1, \log |C|\}, & |C| < \infty, \\ +\infty, & |C| = \infty. \end{cases}$$

Put $Z_X = X \times \{0, 1\}$ and $Z_Q = Q_C \times \{0, 1\}$, with the product sigma-fields. For $N \in \mathbb{N}_0$, define

$$T_N((x_r, y_r)_{r=1}^N) = ((\kappa(x_r), y_r))_{r=1}^N.$$

Raw labeled datasets use replace-one adjacency. If D is a probability measure on (X, Σ) and $c \in C$, let

$$\bar{D} = \kappa_{\#} D, \quad P_{D,c} = \mathcal{L}(x, c(x)), \quad P_{\bar{D}, \bar{c}} = \mathcal{L}(q, \bar{c}(q)),$$

where $x \sim D$ and $q \sim \bar{D}$. For $\bar{h} \in H_C$, define

$$\text{err}_D(\text{Dec}_C(\bar{h}), c) = D\{x : \bar{h}(\kappa(x)) \neq c(x)\}, \quad \text{err}_{\bar{D}}(\bar{h}, \bar{c}) = \bar{D}\{q : \bar{h}(q) \neq \bar{c}(q)\}.$$

A learner with sample size N is a Markov kernel $A_N : Z_X^N \rightsquigarrow H_C$. It is (ε, δ) -differentially private when, for every neighboring $s \sim s'$ and every $E \in \mathcal{H}_C$,

$$A_N(s, E) \leq e^\varepsilon A_N(s', E) + \delta.$$

The conditional-scope realizable private sample complexity is

$$m_C(\alpha, \beta; \varepsilon, \delta) := \inf \left\{ N \in \mathbb{N}_0 : \text{there exists a Markov kernel } A_N : Z_X^N \rightsquigarrow H_C \text{ that is } (\varepsilon, \delta)\text{-DP, and} \right. \\ \left. \sup_D \sup_{c \in C} \Pr_{S \sim P_{D,c}^N, \bar{H} \sim A_N(S, \cdot)} [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq \beta \right\}.$$

The first supremum ranges over all probability measures on (X, Σ) ; the DP predicate quantifies over every raw replace-one pair and every $E \in \mathcal{H}_C$ as above. Learners may be computationally unbounded and need not be proper.

1.2 Primitive Assumptions

Assumption 1 (Finite Littlestone dimension). *C is nonempty and $d = \text{LD}(C) < \infty$. No finite-cardinality, finite-domain, product, properness, efficiency, or computational assumption is imposed.*

Assumption 2 (Finite-or-countable measurable evaluation quotient). *$Q_C = X/\equiv_C$ is finite or countably infinite, and every cell $\kappa^{-1}(\{q\})$ belongs to Σ . Equivalently, $\kappa : (X, \Sigma) \rightarrow (Q_C, 2^{Q_C})$ is measurable. This is a static pre-sampling condition and does not assume measurability of restrictions, lists, selectors, events, or learner kernels.*

Assumption 3 (Realizable iid sampling). *For utility, D is arbitrary on (X, Σ) , c is arbitrary in C , and the labeled sample is iid from $P_{D,c}$. Privacy is required on every raw neighboring labeled input, including nonrealizable inputs.*

Assumption 4 (Learning, confidence, and approximate-DP parameters). *$0 < \alpha, \beta < 1/4$, $0 < \varepsilon \leq 1$, and $0 < \delta < 1$. For the positive-dimensional VC-sensitive comparison, the source convention $\delta \ll 1/N$ means that along the stated asymptotic sequences,*

$$\delta K \Lambda(d, v, \alpha, \beta, \varepsilon, \delta)^q R_{\text{VC}}(d, v, \alpha, \beta, \varepsilon, \delta) \longrightarrow 0$$

for the universal constants used below. No fixed positive δ is silently absorbed into that source-facing comparison.

1.3 Formalized Goal

This result is explicitly conditional on the four assumptions above. It constructs the quotient-first totalized law described in the next section, pulls it back through T_N , and proves measurability, all-input (ε, δ) -privacy, and realizable population error at most α with failure probability at most β . The finite-or-countable quotient is part of the stated scope; classes with uncountably many evaluation types remain outside the claim.

Define the three source-certified rate expressions

$$R_{\text{VC}} = \frac{d^4(v + \log(1/\beta)) \log(1/(\delta\beta))}{\varepsilon\alpha} + \frac{d + \log(1/\beta)}{\alpha}, \\ R_{\text{fin}} = \frac{\log^+ |C| + \log(1/\beta)}{\varepsilon\alpha} + \frac{\log(1/\beta)}{\alpha}, \quad R_{\text{old}} = \frac{d^5 \log(1/(\delta\beta))}{\varepsilon\alpha} + \frac{d + \log(1/\beta)}{\alpha}.$$

The logarithmic envelope used for hidden factors is

$$\Lambda = 1 + \log(e + d) + \log(e + v) + \log(e + \alpha^{-1}) + \log(e + \beta^{-1}) + \log(e + \varepsilon^{-1}) + \log(e + \log(e/\delta)).$$

The notation $N = \tilde{O}(R_{\text{VC}})$ means $N \leq K \Lambda^q R_{\text{VC}}$ for universal $K \geq 1$ and $q \in \mathbb{N}_0$; no positive power of d , v , or $\log^+ |C|$ may be hidden in the tilde.

2 Preliminaries

The quotient map κ and its recordwise extension T_N are the common input interface for all three learners. Every learner releases an element of $(H_C, \mathcal{H}_C)$ and applies Dec_C only after release. Fix once and for all a deterministic $\bar{c}_0 \in \bar{C}$ for every fallback path. When $d = 0$, $\bar{C} = \{\bar{c}_0\}$ and both Littlestone-based laws use $N = N_{\text{old}} = 0$ and release $\bar{c}_0$ deterministically.

We next give complete positive-dimensional definitions. Fix once and for all a universal constant $c_{\text{AT}} \geq 1$ large enough that the AboveThreshold privacy estimate in Lyu (Lyu, 2025, Lemma 3.2) holds, and fix the admissible universal block constants

$$C_{\text{blk}} := 144000, \quad C_o := 2^{20}.$$

For $d \geq 1$, put

$$\begin{aligned} \gamma &:= \alpha/16, & \rho &:= 1 - \frac{1}{2d}, \\ \beta_{\text{tr}} = \beta_{\text{AT}} = \beta_{\text{SS}} = \beta_{\text{gen}} &:= \beta/4, & \delta_{\text{AT}} = \delta_{\text{SS}} &:= \delta/2, & \varepsilon_{\text{SS}} &:= \varepsilon/8, \\ g_\delta &:= \log(4/\delta), & \eta &:= \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)}, & \tau_{\text{AT}} &:= \eta^{-1} \log \frac{d+1}{\beta_{\text{AT}}}. \end{aligned}$$

For any nonempty labeled quotient block $\bar{B} = ((q_j, y_j))_{j=1}^\ell$, write

$$\text{err}_{\bar{B}}(\bar{h}) := \frac{1}{\ell} \sum_{j=1}^\ell \mathbf{1}\{\bar{h}(q_j) \neq y_j\}.$$

For each integer $t \geq 2$, define the VC-sensitive candidate tuple

$$\begin{aligned} a(t) &:= v + \log(4t/\beta), & Q(t) &:= e + \frac{etd^2a(t)}{\alpha v}, \\ m(t) &:= \left\lceil C_{\text{blk}} \frac{d^2}{\alpha} a(t) \log Q(t) \right\rceil, & n(t) &:= tm(t), & p_r(t) &:= 2^r n(t)d \quad (0 \leq r \leq d), \\ L(t) &:= p_d(t)^d 2^{d^2}, & B(t) &:= \left\lceil \frac{10 \log(L(t)/\delta_{\text{SS}})}{\varepsilon_{\text{SS}}} \right\rceil, & \tau_{\text{SS}}(t) &:= \varepsilon_{\text{SS}}^{-1} \log \frac{tL(t) + 1}{\beta_{\text{SS}}}. \end{aligned}$$

The teacher count is the exact least feasible integer

$$k := \min \left\{ t \in \mathbb{Z} : t \geq 2, \frac{t}{2} - \tau_{\text{AT}} \geq B(t) + \tau_{\text{SS}}(t) + 2 \right\}.$$

The existence of this minimum is part of the theorem proved below. After the minimization, set

$$\begin{aligned} a &:= a(k), & Q_{\text{blk}} &:= Q(k), & m &:= m(k), & n_0 &:= km, & N &:= n_0, \\ p_r &:= 2^r n_0 d \quad (0 \leq r \leq d), & L &:= p_d^d 2^{d^2}, & B &:= B(k), & \tau_{\text{SS}} &:= \tau_{\text{SS}}(k). \end{aligned}$$

Thus, in particular,

$$a = v + \log(4k/\beta), \quad Q_{\text{blk}} = e + \frac{ekd^2a}{\alpha v}, \quad m = \left\lceil C_{\text{blk}} \frac{d^2}{\alpha} a \log Q_{\text{blk}} \right\rceil, \quad N = km.$$

These scalars define $K_C^{\text{VC-Lyu}}$ as follows. Draw one uniform labeled partition of the quotient master sample into fixed blocks $\bar{S}_1, \dots, \bar{S}_k$ of size m , and reuse it at every stage. For $i \in [k]$ and every endpoint-complete stage $r \in \{0, \dots, d\}$, set

$$H_i^r := \{\bar{h} \in \bar{C} : \text{err}_{\bar{S}_i}(\bar{h}) \leq \rho^{r+1}\gamma\}, \quad p_r = 2^r Nd.$$

Thus the corresponding source index is $s = r + 1 \in \{1, \dots, d + 1\}$, including its final endpoint. At stage r , use exactly the restriction H_i^r , a valid (p_r, d) -decomposition, and the essential-SOA list conventions of Lyu (Lyu, 2025, Definitions 4.2–4.3, Lemmas 4.1 and 4.3, Corollary 4.1, and the construction in Theorem 3). Empty restrictions receive the empty list; deterministic tie-breaking fixes every valid decomposition and list ordering before the sample is seen. Invalid, nonfinite, non- H_C , or oversized lists are replaced by the empty list.

For the resulting stage lists $\mathcal{L}_1^r, \dots, \mathcal{L}_k^r$, set

$$q_r := \max_{\bar{h} \in H_C} |\{i \in [k] : \bar{h} \in \mathcal{L}_i^r\}|,$$

with $q_r = 0$ when all lists are empty. Run one stopped AboveThreshold transcript, drawing independent $Z_r \sim \text{Lap}(1/\eta)$ and reporting Above exactly when $q_r + Z_r \geq k/2$. At the first Above stage, make one Sparse Sample call on the selected lists: each actual union item u has score $s(u) = |\{i : u \in \mathcal{L}_i^r\}|$, the distinct failure symbol $\perp$ has score B , and the output mass is proportional to $\exp(\varepsilon_{\text{SS}} s(u))$, exactly as in Lyu (Lyu, 2025, Algorithm 1 and Lemma 3.1). An actual union item is released. Every empty, invalid, failed, no-success, exhausted, or residual path releases $\bar{c}_0$. This fixes a pointwise law on every quotient input before sampling; its measurable-kernel realization is the law denoted $K_C^{\text{VC-Lyu}}$.

The old comparison arm is independently calibrated. Set

$$\begin{aligned} \gamma_o &:= \alpha/16, & \beta_{\text{tr},o} &= \beta_{\text{AT},o} = \beta_{\text{SS},o} = \beta_{\text{gen},o} := \beta/4, & \delta_{\text{AT},o} &= \delta_{\text{SS},o} := \delta/2, \\ \varepsilon_{\text{SS},o} &:= \varepsilon/8, & \eta_o &:= \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)}, & \tau_{\text{AT},o} &:= \eta_o^{-1} \log \frac{d+1}{\beta_{\text{AT},o}}. \end{aligned}$$

For every integer $t \geq 2$, define

$$\begin{aligned} a_o(t) &:= d + \log(4t/\beta), & Q_o(t) &:= e + \frac{etd^2 a_o(t)}{\alpha d}, \\ m_o(t) &:= \left\lceil C_o \frac{d^2}{\alpha} a_o(t) \log Q_o(t) \right\rceil, & n_o(t) &:= tm_o(t), & p_{o,r}(t) &:= 2^r n_o(t)d, \\ L_o(t) &:= p_{o,d}(t)d^{2d^2}, & B_o(t) &:= \left\lceil \frac{10 \log(L_o(t)/\delta_{\text{SS},o})}{\varepsilon_{\text{SS},o}} \right\rceil, & \tau_{\text{SS},o}(t) &:= \varepsilon_{\text{SS},o}^{-1} \log \frac{tL_o(t) + 1}{\beta_{\text{SS},o}}. \end{aligned}$$

Its least feasible teacher count and exact horizon are

$$\begin{aligned} k_o &:= \min \left\{ t \in \mathbb{Z} : t \geq 2, \frac{t}{2} - \tau_{\text{AT},o} \geq B_o(t) + \tau_{\text{SS},o}(t) + 2 \right\}, \\ a_o &:= a_o(k_o), & Q_o &:= Q_o(k_o), & m_o &:= m_o(k_o), & N_{\text{old}} &:= n_o := k_o m_o, \\ p_{o,r} &:= 2^r n_o d \quad (0 \leq r \leq d), & L_o &:= L_o(k_o), & B_o &:= B_o(k_o), & \tau_{\text{SS},o} &:= \tau_{\text{SS},o}(k_o). \end{aligned}$$

The law $K_C^{\text{old-Lyu}}$ is the same quotient-first, fixed-block procedure using this old tuple. Its quotient master sample is partitioned into fixed blocks $\bar{S}_{o,1}, \dots, \bar{S}_{o,k_o}$ of size m_o . For $i \in [k_o]$ and every endpoint-complete stage $r \in \{0, \dots, d\}$, its exact old-indexed restriction is

$$H_{o,i}^r := \{\bar{h} \in \bar{C} : \text{err}_{\bar{S}_{o,i}}(\bar{h}) \leq \rho^{r+1}\gamma_o\}, \quad p_{o,r} = 2^r N_{\text{old}} d.$$

The corresponding source index is again $s = r + 1 \in \{1, \dots, d + 1\}$. At each stage it uses the exact restriction $H_{o,i}^r$, a valid $(p_{o,r}, d)$ -decomposition, and the original d -based essential-SOA list construction of Lyu (Lyu, 2025, Definitions 4.2–4.3, Lemmas 4.1 and 4.3, Corollary 4.1, and Theorem 3). Empty restrictions receive the empty list, and every invalid, nonfinite, non- H_C , oversized, failure-symbol, no-success, exhausted, or residual path is totalized to $\bar{c}_0$. The law uses one stopped AboveThreshold transcript and one selected-stage Sparse Sample call. Its scores, Laplace scale, threshold, and categorical weights are the preceding VC-arm formulas with $(k, p_r, L, B, \eta, \varepsilon_{SS}, \tau_{SS})$ replaced by $(k_o, p_{o,r}, L_o, B_o, \eta_o, \varepsilon_{SS,o}, \tau_{SS,o})$. Hence both named laws are exact pointwise procedures rather than aliases for appendix conclusions.

When $M := |C| < \infty$, the finite comparison law is the everywhere-defined quotient exponential-weights kernel

$$K_{C,N}^{\text{fin}}(\bar{s}, E) = \frac{\sum_{\bar{h} \in \bar{C} \cap E} \exp \left[-\frac{\varepsilon}{2} \sum_{r=1}^N \mathbf{1}\{\bar{h}(q_r) \neq y_r\} \right]}{\sum_{\bar{g} \in \bar{C}} \exp \left[-\frac{\varepsilon}{2} \sum_{r=1}^N \mathbf{1}\{\bar{g}(q_r) \neq y_r\} \right]}.$$

Its certified horizon is

$$N_{\text{fin}} := \begin{cases} 0, & d = 0, \\ \left\lceil \frac{4(\log M + \log(1/\beta))}{\varepsilon \alpha} \right\rceil, & d \geq 1. \end{cases}$$

It is defined for every quotient input, including $N = 0$, and its raw pullback is its composition with T_N . If $|C| = \infty$, set $R_{\text{fin}} = +\infty$ and construct no finite surrogate. All three arms therefore share the output space, decoder, privacy quantifier, and realizable iid population-risk mode; the minimum theorem selects one whole law before observing the data.

3 Main Theorem

Theorem 3.1 (Conditional quotient-first private PAC bound). *Assume Assumptions 1, 2, 3, and 4. Let $K_C^{\text{VC-Lyu}}$ be the explicitly calibrated and totalized quotient-first law defined in the preliminaries, and let*

$$A_N^{\text{VC}}(s, E) := K_C^{\text{VC-Lyu}}(T_N(s), E).$$

If $d = 0$, then $N = 0$, the output is the unique member of $\bar{C}$, and A_0^{VC} is $(0, 0)$ -DP with zero decoded population error. If $d \geq 1$, the feasibility set defining k is nonempty, the exact horizon is $N = km$, and there is a universal constant $K_V \geq 1$ such that A_N^{VC} is a measurable Markov kernel, is (ε, δ) -DP for every raw replace-one pair (including nonrealizable labels), and satisfies

$$\sup_D \sup_{c \in C} \sup_{S \sim P_{D,c}^N} \Pr_{\bar{H} \sim K_C^{\text{VC-Lyu}}(T_N(S), \cdot)} [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq \beta,$$

with

$$N \leq K_V \Lambda^4 R_{\text{VC}}.$$

Here $(d, v, \alpha, \beta, \varepsilon, \delta)$ are the exposed variables; K_V is universal and may depend on none of (X, Σ, C, D, c) or on any decomposition, list, partition, event, or output. The probability mode is unconditional iid high probability over sampling, partition, and mechanism randomness; the horizon mode is fixed sample size N ; and the norm is the population binary zero-one error.

On the same quotient interface, the totalized old-Lyu (Lyu, 2025) law uses $N_{\text{old}} = 0$ on the singleton branch. If $d \geq 1$, its independently calibrated teacher-feasibility set is nonempty and its exact horizon is $N_{\text{old}} = k_{\text{o}}m_{\text{o}}$. In either branch it has the measurable raw pullback

$$A_{N_{\text{old}}}^{\text{old}}(s, E) := K_C^{\text{old-Lyu}}(T_{N_{\text{old}}}(s), E).$$

It is all-input (ε, δ) -DP and satisfies

$$\sup_D \sup_{c \in C} \sup_{S \sim P_{D,c}^{N_{\text{old}}}, \bar{H} \sim K_C^{\text{old-Lyu}}(T_{N_{\text{old}}}(S), \cdot)} \Pr [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq \beta,$$

with

$$N_{\text{old}} \leq K_{\text{O}} \Lambda^6 R_{\text{old}}$$

for a universal $K_{\text{O}} \geq 1$. When $|C| < \infty$, let

$$A_{N_{\text{fin}}}^{\text{fin}}(s, E) := K_{C, N_{\text{fin}}}^{\text{fin}}(T_{N_{\text{fin}}}(s), E).$$

This law has the same output and decoder, is pure ε -DP on all raw inputs, and satisfies

$$\sup_D \sup_{c \in C} \sup_{S \sim P_{D,c}^{N_{\text{fin}}}, \bar{H} \sim K_{C, N_{\text{fin}}}^{\text{fin}}(T_{N_{\text{fin}}}(S), \cdot)} \Pr [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq \beta,$$

with

$$N_{\text{fin}} \leq 8R_{\text{fin}};$$

when $|C| = \infty$, this arm is assigned cost $+\infty$ and no finite surrogate is substituted. Consequently, choosing the least certified arm gives a single learner (not a mixture of laws) satisfying

$$m_C(\alpha, \beta; \varepsilon, \delta) \leq K_* \Lambda^6 \min\{R_{\text{fin}}, R_{\text{old}}, R_{\text{VC}}\},$$

where $K_* := \max\{8, K_{\text{O}}, K_{\text{V}}\}$ is universal and the minimum uses the $+\infty$ convention above.

Corollary 3.1 (Rate specialization and baseline frontier). *Under Assumptions 1, 2, 3, and 4, fix $\alpha, \beta, \varepsilon$, specialize the auxiliary teacher and block parameters by the least feasible choices specified in the preliminaries, and retain all ceiling terms. For every fixed $0 < \delta < 1$, the complete finite-parameter comparison is valid. Along the selected-learner schedule*

$$\delta K_{\text{V}} \Lambda^4 R_{\text{VC}} \longrightarrow 0,$$

the least-threshold choice satisfies $N_ \delta \rightarrow 0$. If the standalone old arm is studied instead, its separate sufficient schedule is*

$$\delta K_{\text{O}} \Lambda^6 R_{\text{old}} \longrightarrow 0;$$

the selected-learner schedule is not asserted to imply this standalone condition. With $\alpha, \beta, \varepsilon$ fixed, the resulting class-complexity comparison is

$$m_C = \tilde{O}(\min\{\log^+ |C|, d^5, vd^4\}).$$

The auxiliary choices, verification of every technical condition, displayed term-absorption inequalities, probability conversion, and hidden-constant dependence are proved in Proposition A.71. The exact baseline reductions are: $N = 0$ when $d = 0$; $vd^4 = d^5$ when $v = d$ (and the same polynomial scale when $v = \Theta(d)$); the finite arm is available only for finite C . The unrestricted DP-PAC open problem, including uncountable evaluation quotients and a universal polynomial in v and $\log d$, remains unresolved.

4 Proof Sketch

The proof starts with the static evaluation quotient. The quotient factorization is Proposition A.4; preservation of VC and Littlestone dimensions is given by Lemmas A.4 and A.5; and the measurable record map, iid pushforward, and exact decoder-risk identity are Propositions A.6, A.7, and A.8, respectively. Because the quotient input is finite or countably discrete, the pointwise totalized procedure promotes to a Markov kernel; the corresponding event and output coordinates are measurable by Propositions A.9 and A.10.

On the positive-dimensional branch, the scalar dictionary and all fallback paths are fixed before sampling. The trace family is bounded by Sauer–Shelah (Sauer, 1972) and the fixed-trace without-replacement estimates produce the simultaneous source event. The ceiling-aware fixed point charges its failure to the allocated confidence share. The source/current stage map then gives exact factor-two restriction inclusions, finite essential lists, and irreducible leaf witnesses. A finite DDim potential drops by one at every accurately Below stage; at dimension zero the realizable target supplies a common list item, so a score- k stage is reached within the declared finite horizon.

The one AboveThreshold transcript and one Sparse Sample call use the teacher margin to produce an actual quotient function except on their two allocated mechanism events. The list-locality argument proves all-input privacy, independently of realizability. The selected function has an exact SOA identity and master empirical error at most $\alpha/8$. An analysis-only finite occurrence mark identifies a producer block; its projection is exact, and the independent complement of that block gives the one-sided holdout bound. The finite union over marks and candidate lists yields the quotient PAC guarantee, which transfers to raw risk by the quotient identity.

Finally, the VC-arm auxiliaries are eliminated with explicit inequalities, the old-Lyu (Lyu, 2025) arm is normalized independently, and the finite exponential-weight arm is proved on the same quotient interface. Only after all three arms have the same output space, decoder, privacy mode, and PAC mode is the deterministic minimum taken. The full derivations appear in the following appendix subsections; the final proof cites their theorem-style results and performs only this direct composition.

References

- LYU, X. (2025). Private learning of littlestone classes, revisited. [3](#), [4](#), [5](#), [6](#), [7](#), [9](#), [10](#), [11](#), [15](#), [24](#), [34](#), [35](#), [38](#), [39](#), [45](#), [47](#), [48](#), [49](#), [50](#), [53](#), [54](#), [55](#), [58](#), [59](#), [61](#), [65](#), [67](#), [68](#), [72](#), [73](#), [74](#), [76](#), [77](#), [78](#), [91](#), [92](#), [94](#), [96](#), [97](#), [98](#), [101](#), [102](#), [105](#), [114](#), [115](#)
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A Proof Details

A.1 Parameter Closure and Totalization

Proposition A.1 (Exact zero-dimensional law). *Under Assumption 1, if $d = 0$, then C and $\bar{C}$ are singletons. With $N = 0$, the law that releases their unique quotient concept is a Markov kernel into $(H_C, \mathcal{H}_C)$, is $(0, 0)$ -DP for replace-one raw adjacency, and has population binary error zero for every probability measure D and every $c \in C$.*

Proof. Suppose two concepts $c, c' \in C$ were distinct. There would be an $x \in X$ with $c(x) \neq c'(x)$. A depth-one Littlestone tree whose root is labeled by x would then have both binary branches realized by members of C , which would imply $\text{LD}(C) \geq 1$. Thus $d = 0$ forces every member of the nonempty set C to be the same function; hence $C = \{c_0\}$ and $\bar{C} = \{\bar{c}_0\}$.

The input space Z_X^0 consists of the single empty data set. Define $A_0(\emptyset, E) = \mathbf{1}\{\bar{c}_0 \in E\}$ for $E \in \mathcal{H}_C$. This is the Dirac probability measure at $\bar{c}_0$, and every coordinate of a kernel on a singleton input is measurable. The only replace-one comparison at sample size zero compares the empty input with itself, so the two output laws are identical and the kernel is $(0, 0)$ -DP, hence also (ε, δ) -DP for every allowed positive (ε, δ) . Finally,

$$\text{Dec}_C(\bar{c}_0)(x) = \bar{c}_0(\kappa(x)) = c_0(x) \quad \text{for every } x \in X,$$

so the error is identically zero and the event that it exceeds α has probability zero, uniformly in D . No positive-branch parameter, source mechanism, failure event, or small- δ schedule is activated. $\square$

Lemma A.1 (Quotient restrictions inherit the raw Littlestone bound). *Under Assumption 1, if $d \geq 1$, then every hypothesis class $\mathcal{H} \subseteq \bar{C}$, including the empty class, satisfies*

$$\text{LD}(\mathcal{H}) \leq d. \quad (\text{QuotientLDBound})$$

In particular, every source restriction $H_i^r \subseteq \bar{C}$ formed by the positive-dimensional pointwise quotient procedure satisfies the class- dimension premise in Lyu (Lyu, 2025) Definition 4.2.

Proof. The empty class shatters no positive-depth Littlestone tree, so under either standard empty-class dimension convention its dimension is strictly less than one and hence at most d . Now let $\mathcal{H} \neq \emptyset$, let $s \geq 1$, and suppose a finite depth- s Littlestone tree T_Q , whose nodes are labeled in Q_C , is shattered by $\mathcal{H}$. Index a node at depth $j - 1$ by the preceding edge string $b_{<j} \in \{0, 1\}^{j-1}$, and write its label as $q_{b_{<j}}$.

For each of the finitely many node occurrences, choose an arbitrary representative

$$x_{b_{<j}} \in \kappa^{-1}(\{q_{b_{<j}}\}).$$

Every such fiber is nonempty because κ is the quotient map. Replace the label of each node occurrence by its chosen representative, leaving the tree and all edge labels unchanged; call the resulting X -labeled tree T_X . This uses only finitely many existential choices and does not define a selector or a measurable object.

Fix an arbitrary path $b = (b_1, \dots, b_s) \in \{0, 1\}^s$. Since T_Q is shattered, there is $\bar{h}_b \in \mathcal{H}$ such that

$$\bar{h}_b(q_{b_{<j}}) = b_j \quad (1 \leq j \leq s). \quad (6.1)$$

Because $\mathcal{H} \subseteq \bar{C}$, choose $c_b \in C$ with $\bar{c}_b = \bar{h}_b$. The quotient factorization in the definition of $\bar{c}_b$ gives, at every node on the path,

$$c_b(x_{b_{<j}}) = \bar{c}_b(\kappa(x_{b_{<j}})) = \bar{h}_b(q_{b_{<j}}) = b_j. \quad (6.2)$$

Thus c_b realizes path b in T_X . Since b was arbitrary, C shatters T_X , so $s \leq \text{LD}(C) = d$.

Repeated quotient labels and representative collisions cause no loss in this pullback. If two node occurrences have the same quotient label, their chosen representatives may coincide or may be different points of the same fiber; by the definition of $\equiv_C$, every $c \in C$ is constant on that entire fiber, so (6.2) is unchanged. Moreover, if the same quotient label occurs twice on one path, a conflicting pair of edge labels could not be realized by any function on Q_C , and hence could not occur on a path of the shattered tree T_Q . Representatives of distinct quotient labels cannot collide because a point has a unique κ -image. Therefore the argument permits arbitrary fibers, repeated labels, and all possible representative coincidences. Since every finite shattered depth s is at most d , (QuotientLDBound) follows. $\square$

$\square$

Lemma A.2 (Legal positive-branch dictionary and source calibration). *Under Assumptions 1 and 4 and Lemma A.1, if $d \geq 1$, then $1 \leq v \leq d$. For every integer $t \geq 2$, all of the following quantities are finite and lie in their displayed domains:*

$$\gamma := \alpha/16, \quad \beta_{\text{tr}} = \beta_{\text{AT}} = \beta_{\text{SS}} = \beta_{\text{gen}} := \beta/4, \quad \delta_{\text{AT}} = \delta_{\text{SS}} := \delta/2, \quad (2.1)$$

$$g_\delta := \log(4/\delta), \quad \eta := \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)}, \quad \varepsilon_{\text{SS}} := \varepsilon/8, \quad (2.2)$$

$$a(t) := v + \log(4t/\beta), \quad Q(t) := e + \frac{etd^2a(t)}{\alpha v}, \quad (2.3)$$

$$m(t) := \left\lceil C_{\text{blk}} \frac{d^2}{\alpha} a(t) \log Q(t) \right\rceil, \quad n(t) := tm(t), \quad p_r(t) := 2^r n(t)d \quad (0 \leq r \leq d), \quad (2.4)$$

$$L(t) := p_d(t)d2^{d^2}, \quad B(t) := \left\lceil \frac{10 \log(L(t)/\delta_{\text{SS}})}{\varepsilon_{\text{SS}}} \right\rceil, \quad (2.5)$$

$$\tau_{\text{AT}} := \eta^{-1} \log((d+1)/\beta_{\text{AT}}), \quad \tau_{\text{SS}}(t) := \varepsilon_{\text{SS}}^{-1} \log((tL(t)+1)/\beta_{\text{SS}}). \quad (2.6)$$

For every $0 \leq r \leq d$ and $1 \leq i \leq t$, if $H_i^r \subseteq \bar{C}$ is nonempty, then $p_r(t), d \in \mathbb{N}$ and $\text{LD}(H_i^r) \leq d$. Consequently, if $\mathcal{L}_i^r$ is exactly an ordering without repetition of all Lyu (Lyu, 2025) $(p_r(t), d)$ -essential hypotheses of H_i^r , then

$$|\mathcal{L}_i^r| \leq p_r(t)d2^{d^2} \leq p_d(t)d2^{d^2} = L(t). \quad (\text{ListCap})$$

If $H_i^r = \emptyset$, define $\mathcal{L}_i^r := \emptyset$, and (ListCap) still holds. Conditional on each stage input list being one of these exact valid lists or a totalized empty-restriction list, the exact Sparse Sample threshold, the $(\varepsilon/4, \delta/2)$ Sparse Sample allocation, and the $(\varepsilon/4, \delta/2)$ conditional AboveThreshold allocation all hold with the current quotient objects and source conventions. The deterministic confidence calibrations satisfy

$$(d+1)e^{-\eta\tau_{\text{AT}}} = \beta_{\text{AT}}, \quad (tL(t)+1)e^{-\varepsilon_{\text{SS}}\tau_{\text{SS}}(t)} = \beta_{\text{SS}}. \quad (2.7)$$

Proof. A VC-shattered set of size s yields a complete depth- s Littlestone tree by querying those s points in one fixed order; every binary root-to-leaf labeling is realized because the set is VC-shattered. Thus $v \leq d$. Conversely, $d \geq 1$ supplies a shattered depth-one tree, so two concepts give opposite labels at its root and that singleton is VC-shattered. Hence $v \geq 1$.

The parameter ranges give $g_\delta > \log 4 > 1$, $\eta > 0$, $\varepsilon_{\text{SS}} > 0$, $a(t) > 0$, and $Q(t) > e$. Therefore $m(t), n(t), p_r(t), L(t)$ are positive integers, $B(t)$ is a nonnegative integer, and both tolerances in (2.6) are finite and positive. Equation (QuotientLDBound) gives $\text{LD}(H_i^r) \leq d$ for every nonempty

$H_i^r \subseteq \bar{C}$. Together with the positive integrality of $p_r(t)$ and d , this discharges all hypotheses of Lyu (Lyu, 2025) Definition 4.2 and Corollary 4.1. Thus an exact essential list obeys the first inequality in (ListCap); the second follows from $r \leq d$, hence $p_r(t) \leq p_d(t)$. A totalized empty-restriction list has cardinality zero and obeys the same bound without invoking the source corollary.

For an arbitrary domain $\mathcal{U}$, finite lists $\mathcal{L}_1, \dots, \mathcal{L}_t \subseteq \mathcal{U}$, $\varepsilon_s > 0$, and $B \geq 0$, Lyu (Lyu, 2025) Algorithm 1 assigns scores

$$s(u) = |\{i : u \in \mathcal{L}_i\}|, \quad s(\perp) = B,$$

and samples from $(\bigcup_i \mathcal{L}_i) \cup \{\perp\}$ according to

$$\Pr[w = u] = \frac{e^{\varepsilon_s s(u)}}{e^{\varepsilon_s B} + \sum_{z \in \bigcup_i \mathcal{L}_i} e^{\varepsilon_s s(z)}}. \quad (\text{SparseLaw})$$

The exponent has no factor $1/2$, and $\perp$ is a distinct support point. Its privacy lemma states that, if every list has size at most L and

$$B \geq \frac{10 \log(L/\delta_s)}{\varepsilon_s}, \quad (\text{SparsePrivacyThreshold})$$

then this law is $(2\varepsilon_s, \delta_s)$ -DP under addition, removal, or replacement of one entire list coordinate.

The definition of $B(t)$ is at least this exact threshold for every exact or totalized-empty list, including when its unrounded value is an integer, so the list-level privacy cost is $(2\varepsilon_{\text{SS}}, \delta_{\text{SS}}) = (\varepsilon/4, \delta/2)$.

For sensitivity-one adaptive queries, the AboveThreshold process with independent $\text{Lap}(1/\eta)$ query noises and at most K reported crossings has privacy cost

$$\left(\eta O\left(\sqrt{K \log(1/\delta_a)} + \log(1/\delta_a)\right), \delta_a \right). \quad (\text{ATPrivacy})$$

Fixing $c_{\text{AT}} \geq 1$ to dominate the implicit numerical constant for $K = 1$, substituting $\delta_a = \delta_{\text{AT}} = \delta/2$, and using $g_\delta \geq \log(2/\delta) = \log(1/\delta_{\text{AT}})$ with the value of η in (2.2) gives the stated conditional $(\varepsilon/4, \delta/2)$ allocation. This conclusion is conditional on the sensitivity-one interface proved later and makes no raw-input privacy claim. Finally, (2.7) follows by substituting (2.6); no probabilistic event is assumed in this algebra. $\square$

$\square$

Lemma A.3 (Ceiling-aware candidate envelope). *Under Assumptions 1 and 4, suppose $d \geq 1$, and use the dictionary of Lemma A.2. Define*

$$\ell := \log \frac{64}{\delta\beta}, \quad R_T := \frac{d^2 \ell \Lambda^2}{\varepsilon}, \quad A_{\log} := 80 + \log(1 + C_{\text{blk}}), \quad (3.1)$$

and, for a universal scalar $C \geq 1$,

$$u_C := 1 + \log C, \quad t_C := \lceil C R_T \rceil. \quad (3.2)$$

Then

$$\begin{aligned} \log t_C &\leq A_{\log} u_C \Lambda, & \log Q(t_C) &\leq A_{\log} u_C \Lambda, \\ \log m(t_C) &\leq A_{\log} u_C \Lambda, & \log n(t_C) &\leq A_{\log} u_C \Lambda, \\ \log L(t_C) &\leq A_{\log} u_C d^2 \Lambda. \end{aligned} \quad (3.3)$$

If

$$A_{\text{def}} := 256(A_{\log} + c_{\text{AT}} + 1), \quad (3.4)$$

then the entire defect opposing the teacher score obeys

$$\tau_{\text{AT}} + B(t_C) + \tau_{\text{SS}}(t_C) \leq A_{\text{def}} u_C R_T. \quad (3.5)$$

Proof. First, $\ell > \log 256 > 5$, $\Lambda \geq 1$, and $R_T > 5$. The definition of Λ gives

$$\log d, \log v, \log(1/\alpha), \log(1/\beta), \log(1/\varepsilon) \leq \Lambda. \quad (3.6)$$

It also gives $\log(e/\delta) \leq e^\Lambda$. Consequently

$$\ell = \log 64 + \log(1/\delta) + \log(1/\beta) \leq 7e^\Lambda, \quad \log \ell \leq 3\Lambda. \quad (3.7)$$

Here the last inequality uses $\Lambda \geq 1$. Therefore

$$\log R_T = 2 \log d + \log \ell + 2 \log \Lambda + \log(1/\varepsilon) \leq 8\Lambda. \quad (3.8)$$

Because $CR_T \geq 1$, the ceiling in (3.2) gives $CR_T \leq t_C \leq 2CR_T$, not an asymptotic replacement. Hence

$$\log t_C \leq \log 2 + \log C + \log R_T \leq 9u_C\Lambda. \quad (3.9)$$

Using $1 \leq v \leq d$, (3.6), and (3.9),

$$\log(4t_C/\beta) \leq 12u_C\Lambda, \quad \log a(t_C) \leq 6u_C\Lambda. \quad (3.10)$$

For the second inequality, use $\log(x+y) \leq \log 2 + \max\{\log x, \log y\}$, followed by $\log u_C \leq u_C$ and $\log \Lambda \leq \Lambda$.

Writing $x = t_C d^2 a(t_C)/(\alpha v) > 0$, $Q(t_C) = e(1+x)$ and $\log(1+x) \leq \log 2 + \max\{0, \log x\}$. Equations (3.6), (3.9), and (3.10) give

$$1 \leq \log Q(t_C) \leq 20u_C\Lambda. \quad (3.11)$$

The unrounded block expression $d^2 \alpha^{-1} a(t_C) \log Q(t_C)$ is greater than one. Therefore the ceiling is retained through

$$m(t_C) \leq (1 + C_{\text{blk}}) d^2 \alpha^{-1} a(t_C) \log Q(t_C).$$

Taking logarithms and using $\log \log Q(t_C) \leq \log Q(t_C)$ yields

$$\log m(t_C) \leq (29 + \log(1 + C_{\text{blk}})) u_C \Lambda. \quad (3.12)$$

Combining (3.9) and (3.12) bounds $\log n(t_C)$ by $(38 + \log(1 + C_{\text{blk}})) u_C \Lambda$. Finally, the exact list formula, with no ceiling or power suppressed, is

$$\begin{aligned} \log L(t_C) &= d \log(2^d n(t_C) d) + d^2 \log 2 \\ &= 2d^2 \log 2 + d \log n(t_C) + d \log d \\ &\leq (41 + \log(1 + C_{\text{blk}})) u_C d^2 \Lambda. \end{aligned} \quad (3.13)$$

The definition of $A_{\log}$ makes every inequality in (3.3) follow from (3.9), (3.11), (3.12), and (3.13).

It remains to bound the three teacher defects. Since $\varepsilon_{\text{SS}} = \varepsilon/8$ and $\delta_{\text{SS}} = \delta/2$, the retained ceiling gives

$$\begin{aligned} B(t_C) &\leq 1 + \frac{80}{\varepsilon} (\log L(t_C) + \log(2/\delta)) \\ &\leq 161 A_{\log} u_C R_T, \end{aligned} \quad (3.14)$$

where $\log(2/\delta) \leq \ell$ and each displayed summand, including the ceiling remainder 1, is bounded by its corresponding multiple of $R_T = d^2 \ell \Lambda^2 / \varepsilon$.

Because $t_C L(t_C) \geq 2$,

$$\begin{aligned} \tau_{\text{SS}}(t_C) &\leq \frac{8}{\varepsilon} (\log(8/\beta) + \log t_C + \log L(t_C)) \\ &\leq 24 A_{\log} u_C R_T. \end{aligned} \quad (3.15)$$

For AboveThreshold, $g_\delta > 1$, so $\sqrt{g_\delta} + g_\delta \leq 2g_\delta \leq 2\ell$, while

$$\log(4(d+1)/\beta) \leq \log 4 + \log(d+1) + \log(1/\beta) \leq 4\Lambda.$$

Thus

$$\tau_{\text{AT}} \leq \frac{32c_{\text{AT}}\ell\Lambda}{\varepsilon} \leq 32c_{\text{AT}}R_T. \quad (3.16)$$

Summing (3.14)-(3.16) and using (3.4) proves (3.5). Notice that the candidate tuple was fully defined before any feasibility statement was used; this is the noncircular part of the argument. $\square$

Proposition A.2 (Finite public teacher witness and least feasible count). *Under Assumptions 1 and 4, suppose $d \geq 1$. With $A_{\log}$ and A_{def} from Lemma A.3, fix the concrete finite universal constant*

$$C_{\text{teach}} := 2^{12}A_{\text{def}}^2 \quad (4.1)$$

and define

$$\bar{k} := \left\lceil C_{\text{teach}} \frac{d^2 \log(64/(\delta\beta))\Lambda^2}{\varepsilon} \right\rceil. \quad (4.2)$$

Then $\bar{k} \geq 2$ and

$$\frac{\bar{k}}{2} - \tau_{\text{AT}} \geq B(\bar{k}) + \tau_{\text{SS}}(\bar{k}) + 2. \quad (4.3)$$

Consequently the exact least feasible teacher count

$$k := \min \left\{ t \in \mathbb{Z} : t \geq 2, \quad \frac{t}{2} - \tau_{\text{AT}} \geq B(t) + \tau_{\text{SS}}(t) + 2 \right\} \quad (4.4)$$

exists and satisfies

$$2 \leq k \leq \bar{k} \leq 2C_{\text{teach}} \frac{d^2 \log(64/(\delta\beta))\Lambda^2}{\varepsilon}. \quad (4.5)$$

After this deterministic minimization, define exactly

$$m := m(k), \quad n_0 := km(k), \quad N := n_0, \quad p_r := 2^r n_0 d \quad (0 \leq r \leq d), \quad (4.6)$$

$$L := p_d^d 2^{d^2} = L(k), \quad B := B(k), \quad \tau_{\text{SS}} := \tau_{\text{SS}}(k). \quad (4.7)$$

These are finite public quantities, all integer ceilings from (2.4), (2.5), and (4.2) are retained, and

$$p_0 = n_0 d \geq \max\{n_0, d+1\}. \quad (4.8)$$

Proof. Put $C = C_{\text{teach}}$ in Lemma A.3. Since $A_{\text{def}} \geq 1$,

$$1 + \log C_{\text{teach}} = 1 + 12 \log 2 + 2 \log A_{\text{def}} \leq 15A_{\text{def}}.$$

Therefore

$$A_{\text{def}}(1 + \log C_{\text{teach}}) \leq 15A_{\text{def}}^2 \leq \frac{C_{\text{teach}}}{4}. \quad (4.9)$$

Equations (3.5) and (4.9) imply

$$\tau_{\text{AT}} + B(\bar{k}) + \tau_{\text{SS}}(\bar{k}) \leq \frac{C_{\text{teach}}R_T}{4}. \quad (4.10)$$

Also $C_{\text{teach}}R_T/4 > 2$, while $\bar{k}/2 \geq C_{\text{teach}}R_T/2$. Thus

$$\frac{\bar{k}}{2} \geq \tau_{\text{AT}} + B(\bar{k}) + \tau_{\text{SS}}(\bar{k}) + 2,$$

which is (4.3). The feasible set in (4.4) is a nonempty subset of the integers bounded below by two, so its least element exists and is no larger than $\bar{k}$. Since $C_{\text{teach}}R_T \geq 1$, the ceiling in (4.2) gives $\bar{k} \leq C_{\text{teach}}R_T + 1 \leq 2C_{\text{teach}}R_T$, proving (4.5).

All objects in (4.6)-(4.7) are now evaluated at a fixed public integer k ; there is no remaining equation in which k is defined using a quantity that itself awaits the choice of k . Finally, $n_0 \geq k \geq 2$ and $d \geq 1$, so $p_0 = n_0d \geq n_0$. Also $n_0d \geq 2d \geq d + 1$, proving (4.8). $\square$

Proposition A.3 (Pre-sampling totalization of the quotient procedure). *Under Assumptions 1 and 4, Lemma A.2, and Proposition A.2, the positive-dimensional pointwise quotient procedure can be specified before the master sample, partition, source events, or mechanism coins are drawn so that it returns an element of H_C on every labeled quotient input and every internal path. The completion assigns the empty list to every empty restriction; preserves every exact valid list and successful path; replaces every invalid, nonfinite, non- H_C , or oversized purported list by the empty list; gives every remaining routine-failure state a defined transcript continuation; and sends every terminal mechanism-failure, no-success, or residual fallback path to the setting-defined $\bar{c}_0$.*

Proof. Fix the tuple (2.1)-(2.6), (4.4), and (4.6)-(4.7), the default $\bar{c}_0$, and deterministic tie-breaking conventions once and for all. For every nonempty restriction on which the certified routine supplies valid objects, fix one optimal decomposition and an ordering without repetition of its exact essential set. By (ListCap), every such exact list is finite, lies in H_C , and has cardinality at most L ; hence the cap and finiteness checks below never change it, whether that exact list is empty or nonempty. Every empty restriction is assigned the empty list directly. Identical local states are assigned identical choices. This is a pointwise definition only. Its measurable-kernel realization on the countable quotient input is established independently after the pointwise law has been fixed; measurability is not assumed here.

For a totalized list tuple at stage r , define its occurrence score by

$$q_r := \begin{cases} \max_{\bar{h} \in \bigcup_{i=1}^k \mathcal{L}_i^r} |\{i : \bar{h} \in \mathcal{L}_i^r\}|, & \bigcup_{i=1}^k \mathcal{L}_i^r \neq \emptyset, \\ 0, & \bigcup_{i=1}^k \mathcal{L}_i^r = \emptyset. \end{cases} \quad (5.1)$$

Feed $q_0, \dots, q_d$, in order, to one stopped source AboveThreshold process with threshold $k/2$, noise scale $1/\eta$, and the fixed first-success rule. At the first legal Above stage r_* , invoke the exact law (SparseLaw) on $(\mathcal{L}_1^{r_*}, \dots, \mathcal{L}_k^{r_*})$ with parameters $(\varepsilon_{\text{SS}}, B)$. If it returns an actual union item, return that item; otherwise use the terminal convention below.

The pointwise procedure is totalized as follows. An empty stage restriction receives the empty essential list. If a source decomposition or list routine is undefined, returns no valid object, or fails a declared check, the local list is replaced by the empty list. A purported list that is invalid, nonfinite, outside H_C , or larger than L is treated the same way; valid lists are unchanged. If all lists at a stage are empty, its occurrence score is 0 and the fixed transcript continues. If AboveThreshold reports no legal stage, the transcript is invalid, or stage processing is exhausted, the output is $\bar{c}_0$. When a legal stage is selected and Sparse Sample returns an actual union item, that item in H_C is returned. A failure symbol, an out-of-support value, an invalid normalizer, or any remaining mechanism failure returns $\bar{c}_0$.

For finite capped lists, (SparseLaw) in fact always has a positive finite normalizer because the $\perp$ weight is positive; its listed failure completion is included to make the pointwise program total independently of that later path verification. On an exact valid path, every nonempty restriction's exact list satisfies (ListCap), every empty restriction already has the prescribed empty list, AboveThreshold names a legal stage, and Sparse Sample returns an actual union item. Thus neither the finite/cap sanitizer nor any fallback changes that path. Conversely, any invalid, nonfinite, non- H_C , or oversized purported list is totalized to the empty list, so it cannot leave the pointwise procedure undefined.

The partition rule is also fixed before data: after the complete quotient master sample is obtained, draw one data-independent uniform partition into the k indexed blocks of size m , and reuse that same partition at all $d + 1$ stages. The table itself, all tie breaking, all thresholds, and the fallback output are fixed before this draw. Thus no good event, realizability condition, mechanism-success event, privacy conclusion, PAC conclusion, or event-dependent choice is used to define the law.

Proposition A.1 proves the complete $d = 0$ branch: the exact sample size is zero, the output is the unique quotient concept, the kernel is deterministic, privacy is $(0, 0)$, and the uniform PAC failure probability is zero.

On the $d \geq 1$ branch, Lemma A.1 pulls every finite quotient Littlestone tree back to X and proves $\text{LD}(H_i^r) \leq d$ for every current restriction, including all repeated-label and arbitrary-fiber cases. With that exact class premise available, Lemma A.2 proves $1 \leq v \leq d$, defines every confidence, privacy, block, list, failure, and mechanism tolerance for each candidate integer $t \geq 2$, and checks the exact Lyu (Lyu, 2025) essential-list cap and Sparse Sample formulas in the current quotient conventions. Lemma A.3 retains both ceilings and proves a finite logarithmic envelope for the complete coupled tuple before assuming a feasible teacher. Proposition A.2 then gives the concrete finite public witness (4.2), proves its margin, takes the least feasible integer only after nonemptiness is known, fixes $(k, m, n_0, N, p_r, L, B)$, and exports the explicit bound (4.5), whose leading teacher dependence is d^2/ε . Finally, the totalization construction above fixes every empty, routine-failure, mechanism-failure, no-success, and fallback outcome before any random event while preserving all exact valid paths.

This establishes the complete parameter dictionary, finite teacher witness, least-feasible count, and all-input totalization used by the later measurable, privacy, and utility results. $\square$

A.2 Evaluation Quotient and Exact Transfers

The propositions below give the measurable quotient, product-law, and risk interfaces.

Proposition A.4 (Measurable evaluation factorization). *Under Assumption 2, for the nonempty class C in the formalized basic setup, the relation*

$$x \equiv_C x' \iff c(x) = c(x') \text{ for every } c \in C$$

is an equivalence relation. Its quotient Q_C is nonempty and finite or countably infinite, the canonical surjection $\kappa(x) = [x]_{\equiv_C}$ is measurable into $(Q_C, 2^{Q_C})$, and

$$\Phi : C \longrightarrow \bar{C}, \quad \Phi(c) = \bar{c}, \quad \bar{c}(\kappa(x)) = c(x),$$

is a bijection. Moreover, every $c \in C$ and every decoded hypothesis $\text{Dec}_C(\bar{h}) = \bar{h} \circ \kappa$, $\bar{h} \in H_C$, is $\Sigma/2^{\{0,1\}}$ -measurable.

Proof. Reflexivity, symmetry, and transitivity follow coordinatewise from equality of the evaluation vector $(c(x))_{c \in C}$, so the relation is an equivalence relation. Since $X \neq \emptyset$, its quotient is nonempty, and its finite-or-countable cardinality is exactly the first clause of Assumption 2. The quotient map is surjective by construction.

For every $A \subseteq Q_C$,

$$\kappa^{-1}(A) = \bigcup_{q \in A} \kappa^{-1}(\{q\}). \quad (1)$$

The union is finite or countable, and each cell belongs to Σ by the assumption. Hence κ is measurable. This is a direct cell argument; it makes no choice of a representative of any cell.

If $\kappa(x) = \kappa(x')$, then $c(x) = c(x')$ for every $c \in C$, so $\bar{c}(\kappa(x)) := c(x)$ is well-defined. Surjectivity of Φ is the definition of $\bar{C}$. If $\Phi(c) = \Phi(c')$, then for every $x \in X$,

$$c(x) = \bar{c}(\kappa(x)) = \bar{c}'(\kappa(x)) = c'(x),$$

and therefore $c = c'$ as functions. Thus Φ is injective and hence bijective.

Every map from the discrete space Q_C to $\{0, 1\}$ is measurable. More explicitly, for $b \in \{0, 1\}$,

$$c^{-1}(\{b\}) = \bigcup_{q: \bar{c}(q)=b} \kappa^{-1}(\{q\}), \quad (\bar{h} \circ \kappa)^{-1}(\{b\}) = \bigcup_{q: \bar{h}(q)=b} \kappa^{-1}(\{q\}). \quad (2)$$

Both are countable unions of measurable cells. Thus the originally given concepts and every possibly improper decoded output are measurable. The argument uses the values of a quotient function on cells, never a selector from the cells. $\square$

$\square$

Proposition A.5 (Countable quotient and output Borel structure). *Under Assumption 2 and Proposition A.4, Q_C , $Z_Q = Q_C \times \{0, 1\}$, and Z_Q^N for every $N \in \mathbb{N}_0$ are countable discrete standard-Borel spaces (with Z_Q^0 the one-point empty-tuple space). The output measurable space*

$$(H_C, \mathcal{H}_C) = \left(\{0, 1\}^{Q_C}, \bigotimes_{q \in Q_C} 2^{\{0, 1\}} \right)$$

is standard Borel, and $\mathcal{H}_C$ is generated by the finite-coordinate cylinders. In particular, each coordinate evaluation $e_q(\bar{h}) = \bar{h}(q)$ is measurable.

Proof. A finite or countably infinite discrete space is Polish under the discrete metric: the metric is complete, and the whole countable space is a countable dense set. Hence Q_C , Z_Q , and every finite power Z_Q^N are standard Borel. Their Borel sigma-fields are their full power sets. For $N = 0$, the only point is the empty tuple.

If Q_C is finite, then H_C is finite discrete and the conclusion is immediate. If Q_C is countably infinite, fix an enumeration $Q_C = \{q_1, q_2, \dots\}$ and put

$$\rho(\bar{h}, \bar{g}) := \sum_{j=1}^{\infty} 2^{-j} |\bar{h}(q_j) - \bar{g}(q_j)|. \quad (3)$$

A ρ -Cauchy sequence is eventually constant in each coordinate; the coordinatewise limit belongs to H_C , and the tail of the series in (3) shows convergence to it. Thus ρ is complete. Functions that vanish outside a finite initial segment form a countable dense subset. The topology of ρ is exactly the product topology: fixing finitely many coordinates gives a clopen cylinder, and sufficiently long initial-coordinate agreement makes the tail in (3) arbitrarily small. Consequently this topology is

Polish and its Borel sigma-field is precisely the sigma-field generated by finite-coordinate cylinders, namely $\mathcal{H}_C$. Each e_q is a coordinate map and therefore measurable. The enumeration is only an enumeration of the already formed quotient; no raw representative is chosen. $\square$

Lemma A.4 (VC dimension is invariant under the evaluation quotient). *Under Assumptions 1 and 2 and Proposition A.4, a finite subset of X is shattered by C only if its quotient image has the same cardinality and is shattered by $\bar{C}$; conversely, every finite subset of Q_C shattered by $\bar{C}$ has a same-cardinality raw lift shattered by C . Therefore*

$$\text{VC}(\bar{C}) = \text{VC}(C) = v. \quad (4)$$

Proof. Let $S = \{x_1, \dots, x_s\} \subseteq X$ be shattered by C . If two distinct points x_i, x_j had $\kappa(x_i) = \kappa(x_j)$, shattering would provide a concept taking value 0 at one and 1 at the other, contrary to the definition of $\equiv_C$. Hence κ is injective on S , so $|\kappa(S)| = s$. Given any labeling $\ell : \kappa(S) \rightarrow \{0, 1\}$, shattering of S supplies a $c \in C$ with $c(x) = \ell(\kappa(x))$ on S . Its image $\bar{c} = \Phi(c)$ realizes ℓ on $\kappa(S)$. Thus $\kappa(S)$ is shattered by $\bar{C}$.

Conversely, let $A = \{q_1, \dots, q_s\} \subseteq Q_C$ be shattered by $\bar{C}$. For these finitely many cells, choose one $x_i \in \kappa^{-1}(\{q_i\})$. The points are distinct because the cells are disjoint. For any labeling of $\{x_1, \dots, x_s\}$, use the same labels on A ; quotient shattering supplies $\bar{c} \in \bar{C}$, and the unique $c = \Phi^{-1}(\bar{c})$ realizes the raw labels. Thus the raw lift is shattered. Only finitely many representatives are chosen for this one combinatorial witness; no representative map on Q_C , measurable or otherwise, is constructed. The attainable finite shattering cardinalities are the same in both classes, proving (4). $\square$

Lemma A.5 (Littlestone dimension is invariant under the evaluation quotient). *Under Assumptions 1 and 2 and Proposition A.4, a finite complete binary tree is shattered by C if and only if a tree of the same depth obtained by quotienting or finitely lifting its node labels is shattered by $\bar{C}$. Consequently,*

$$\text{LD}(\bar{C}) = \text{LD}(C) = d. \quad (5)$$

If $d = 0$, then C and $\bar{C}$ are singletons and $v = 0$.

Proof. A complete binary X -labeled tree of depth t is a family $(x_\sigma)_{\sigma \in \{0,1\}^{<t}}$. It is shattered by C when, for each $b = (b_1, \dots, b_t) \in \{0, 1\}^t$, some $c_b \in C$ satisfies

$$c_b(x_{b_{<r}}) = b_r, \quad r = 1, \dots, t, \quad (6)$$

where $b_{<r} = (b_1, \dots, b_{r-1})$.

Suppose the raw tree is shattered and set $q_\sigma := \kappa(x_\sigma)$. For each path b , the quotient concept $\Phi(c_b)$ satisfies

$$\Phi(c_b)(q_{b_{<r}}) = c_b(x_{b_{<r}}) = b_r,$$

so the quotient-labeled tree is shattered. If one's tree convention requires distinct instances along a path, that condition is automatic. Indeed, if a quotient label repeated at two nodes on a root-to-node path, extend the path at the later node using the bit opposite to the bit taken at the earlier node. Equation (6) would then require one concept to assign two values to the same quotient cell, a contradiction. Repetitions across incomparable branches are harmless and are allowed by the usual tree definition.

Conversely, let $(q_\sigma)_{\sigma \in \{0,1\}^{<t}}$ be a quotient tree shattered by $\bar{C}$. It has only $2^t - 1$ nodes. Choose, separately for these finitely many nodes, an $x_\sigma \in \kappa^{-1}(\{q_\sigma\})$. For each path, choose the witnessing $\bar{c}_b \in \bar{C}$ and let $c_b = \Phi^{-1}(\bar{c}_b)$. Then

$$c_b(x_{b_{<r}}) = \bar{c}_b(q_{b_{<r}}) = b_r,$$

so the lifted raw tree is shattered at the same depth. Again, this finite witness lift is not a global selector. The same finite depths are attainable on both sides, proving (5).

Finally suppose $d = 0$. If two members $c_0, c_1 \in C$ were distinct, they would disagree at some x . Because the labels are binary, the one-node tree labeled by x would have both branch labels realized and hence would be shattered, contradicting $d = 0$. Nonemptiness therefore makes C a singleton, and the bijection Φ makes $\bar{C}$ a singleton. A singleton binary class shatters no one-point set, so Lemma A.4 gives $v = 0$. Notice that Q_C itself need not be a singleton: the unique concept may take both binary values, in which case it has two evaluation cells. $\square$

Proposition A.6 (Measurable record transport and replacement adjacency). *Under Assumption 2 and Propositions A.4–A.5, the one-record map*

$$t : Z_X \rightarrow Z_Q, \quad t(x, y) = (\kappa(x), y), \quad (7)$$

and every data map $T_N = t^N$, $N \in \mathbb{N}_0$, are measurable. If $s, s' \in Z_X^N$ differ in at most one coordinate, then their images are equal or differ in exactly one coordinate. This conclusion holds on every raw labeled input, including arbitrary nonrealizable labels and same-cell replacements.

Proof. For each atom $(q, b) \in Z_Q$,

$$t^{-1}(\{(q, b)\}) = \kappa^{-1}(\{q\}) \times \{b\} \in \Sigma \otimes 2^{\{0,1\}}. \quad (8)$$

The codomain Z_Q is countable discrete, so the preimage of an arbitrary subset is a countable union of the sets in (8). Thus t is measurable. For $N \geq 1$, the preimage of an atom of Z_Q^N is the measurable rectangle

$$T_N^{-1}(\{((q_r, b_r))_{r=1}^N\}) = \prod_{r=1}^N (\kappa^{-1}(\{q_r\}) \times \{b_r\}). \quad (9)$$

Since Z_Q^N is countable discrete, another countable-union argument proves measurability of T_N . For $N = 0$, T_0 is the unique map between one-point empty-tuple spaces and is measurable.

If $s = s'$, their images are equal. Otherwise, if s and s' differ only at coordinate j , all image coordinates other than j agree. At coordinate j , replacing (x, y) by (x', y') has the following exhaustive behavior:

if $\kappa(x) = \kappa(x')$ and $y = y'$, the quotient record is unchanged, so the full image datasets are equal; if $\kappa(x) = \kappa(x')$ and $y \neq y'$, only the quotient label at coordinate j changes; if $\kappa(x) \neq \kappa(x')$, only coordinate j changes, regardless of the two labels.

Thus the images are equal or strict replace-one neighbors and, under the setting's "differ in at most one coordinate" convention, are always replace-one neighbors. No realizability condition was used. $\square$

Proposition A.7 (Exact realizable iid pushforward). *Under Assumptions 2 and 3, Proposition A.4, and Proposition A.6, for every probability measure D on (X, Σ) , every $c \in C$, and every $N \in \mathbb{N}_0$,*

$$t_{\#} P_{D,c} = P_{\bar{D}, \bar{c}}, \quad (T_N)_{\#} P_{D,c}^N = P_{\bar{D}, \bar{c}}^N, \quad \bar{D} = \kappa_{\#} D, \quad (10)$$

where $\bar{c} = \Phi(c)$.

Proof. Proposition A.4 shows that c is measurable. Hence $x \mapsto (x, c(x))$ is measurable: the preimage of a generating rectangle $A \times \{b\}$ is $A \cap c^{-1}(\{b\})$. Thus $P_{D,c}$ is well-defined. Likewise, $q \mapsto (q, \bar{c}(q))$ is measurable on the discrete quotient, so $P_{\bar{D},\bar{c}}$ is well-defined. For each atom $(q, b) \in Z_Q$,

$$\begin{aligned} (t_{\#}P_{D,c})(\{(q, b)\}) &= D\{x : \kappa(x) = q, c(x) = b\} \\ &= \mathbf{1}\{b = \bar{c}(q)\} D(\kappa^{-1}(\{q\})) \\ &= \mathbf{1}\{b = \bar{c}(q)\} \bar{D}(\{q\}) = P_{\bar{D},\bar{c}}(\{(q, b)\}). \end{aligned} \quad (11)$$

The second equality uses that c is exactly constant with value $\bar{c}(q)$ on the whole fiber $\kappa^{-1}(\{q\})$. Equality on atoms implies equality on all subsets of the countable discrete space Z_Q .

For $N \geq 1$, evaluate both sides of the second identity in (10) on an atom $((q_r, b_r))_{r=1}^N$. By (9), the product-measure formula, and (11), its probability under the left side is

$$\prod_{r=1}^N P_{D,c}(t^{-1}(\{(q_r, b_r)\})) = \prod_{r=1}^N P_{\bar{D},\bar{c}}(\{(q_r, b_r)\}), \quad (12)$$

which is its probability under $P_{\bar{D},\bar{c}}^N$. Since Z_Q^N is countable discrete, atomwise equality proves equality of the measures. Equation (12) also shows directly that deterministic quotienting preserves iid sampling even when several raw draws lie in the same fiber or produce repeated quotient records. For $N = 0$, both sides are the point mass at the empty tuple. $\square$

Proposition A.8 (Exact risk transfer for every quotient output). *Under Assumption 2 and Propositions A.4–A.5, for every probability measure D on (X, Σ) , every $c \in C$, and every $\bar{h} \in H_C$, with no requirement that $\bar{h} \in \bar{C}$, the disagreement sets are measurable and*

$$\text{err}_D(\text{Dec}_C(\bar{h}), c) = \text{err}_{\bar{D}}(\bar{h}, \bar{c}), \quad \bar{D} = \kappa_{\#}D, \quad \bar{c} = \Phi(c). \quad (13)$$

For fixed D, c , the common value in (13), viewed as a function of $\bar{h} \in (H_C, \mathcal{H}_C)$, is measurable.

Proof. Define the quotient disagreement set

$$A_{\bar{h},\bar{c}} := \{q \in Q_C : \bar{h}(q) \neq \bar{c}(q)\}. \quad (14)$$

It is measurable because Q_C is discrete. Proposition A.4 gives $c = \bar{c} \circ \kappa$ and a measurable decoder, and therefore

$$\{x : \text{Dec}_C(\bar{h})(x) \neq c(x)\} = \kappa^{-1}(A_{\bar{h},\bar{c}}). \quad (15)$$

Taking D -measure and using the definition of pushforward gives

$$\begin{aligned} \text{err}_D(\text{Dec}_C(\bar{h}), c) &= D(\kappa^{-1}(A_{\bar{h},\bar{c}})) \\ &= (\kappa_{\#}D)(A_{\bar{h},\bar{c}}) = \bar{D}\{\bar{h}(q) \neq \bar{c}(q)\} \\ &= \text{err}_{\bar{D}}(\bar{h}, \bar{c}), \end{aligned}$$

which proves (13) with no fiber-size, injectivity, propriety, or support condition.

For measurability in $\bar{h}$, the finite- Q_C case is a finite sum of coordinate-cylinder indicators. In the countably infinite case, use the enumeration from Proposition A.5. Then the right side of (13) is the pointwise increasing limit

$$\lim_{J \rightarrow \infty} \sum_{j=1}^J \bar{D}(\{q_j\}) \mathbf{1}\{\bar{h}(q_j) \neq \bar{c}(q_j)\}. \quad (16)$$

Each partial sum is $\mathcal{H}_C$ -measurable because it depends on finitely many coordinate evaluations. Hence the limit is measurable. Equation (16) uses quotient atoms and cylinders only; it does not select a raw point from any fiber. Also, for every fixed $x \in X$, the output-coordinate map $\bar{h} \mapsto \text{Dec}_C(\bar{h})(x) = e_{\kappa(x)}(\bar{h})$ is $\mathcal{H}_C$ -measurable by Proposition A.5. Proposition A.4 constructs the canonical evaluation quotient and the bijection $\Phi : C \leftrightarrow \bar{C}$, while also deriving measurability of all raw concepts and every decoded quotient output. Proposition A.5 establishes the exact countable-discrete input spaces and standard-Borel product output required by the setting. Lemma A.4 and Lemma A.5 transfer, in both directions, every finite VC witness and every finite Littlestone-tree witness, proving $\text{VC}(\bar{C}) = v$ and $\text{LD}(\bar{C}) = d$, with the singleton-class conclusion at $d = 0$.

Proposition A.6 proves measurability of the record map and every T_N by atom/cylinder preimages, and proves exact equal-or-replace-one transport for same-cell changes, label changes, and all other arbitrary raw replacements. Proposition A.7 then proves the realizable iid product-law identity for every finite sample size, including zero. Finally, the disagreement-set calculation above identifies the raw disagreement set as the preimage of the quotient disagreement set and proves the exact population-risk equality for every $\bar{h} \in H_C$, including improper outputs and arbitrary fibers. These propositions give the common quotient/raw data, adjacency, sampling, and risk interfaces. $\square$

A.3 Countable-Domain Kernel Construction

No paper result is used in this argument. The preceding kernel construction and the standard measurable-space facts supply the interfaces used below.

Lemma A.6 (Countable-atom promotion of pointwise coordinates and laws). *Under Assumption 2 and Proposition A.5, fix any $N \in \mathbb{N}_0$ and put $S_N := Z_Q^N$. Then:*

1. for every measurable space $(Y, \mathcal{Y})$, every function $f : S_N \rightarrow Y$ is $2^{S_N}/\mathcal{Y}$ -measurable; 2. for every family $(\mu_{\bar{s}})_{\bar{s} \in S_N}$ of probability measures on $(Y, \mathcal{Y})$,

$$K(\bar{s}, B) := \mu_{\bar{s}}(B), \quad \bar{s} \in S_N, \quad B \in \mathcal{Y}, \quad (1)$$

is a Markov kernel from S_N to Y ; and 3. the same conclusions hold after adjoining any finite or countable discrete collection of partition, stage, or history indices to S_N .

No compatibility, enumeration, or selector condition is required of the pointwise family in (1).

Proof. Proposition A.5 gives $\sigma(S_N) = 2^{S_N}$, including when $N = 0$, where S_0 is a singleton. For $B \in \mathcal{Y}$, the preimage $f^{-1}(B)$ is simply a subset of S_N , hence belongs to 2^{S_N} . This proves the first claim.

For the second claim, $B \mapsto K(\bar{s}, B)$ is the probability measure $\mu_{\bar{s}}$ for each fixed $\bar{s}$. For fixed $B \in \mathcal{Y}$, $\bar{s} \mapsto \mu_{\bar{s}}(B)$ is an arbitrary function from S_N to $[0, 1]$, so the first claim makes it measurable. These are exactly the two Markov-kernel axioms. A finite or countable product of S_N with a discrete index set is again countable discrete, so the same proof gives the third claim. At no point is a choice made from a quotient cell or from an uncountable family of source classes. $\square$

Proposition A.9 (Measurable coding of restrictions, finite lists, and transcripts). *Under Assumption 2, Propositions A.3 and A.5, and Lemma A.6, fix $N \in \mathbb{N}_0$. Every restriction, decomposition, finite-list, validity, event-status, finite-stage transcript, and terminal-output coordinate of the totalized quotient procedure has a measurable interface. More precisely:*

1. every realized restriction or decomposition may be coded in its countable realized range with the discrete sigma-field; 2. every sanitized finite list is a measurable coordinate with values in the

standard-Borel space

$$\text{List}(H_C) := \bigsqcup_{\ell=0}^{\infty} H_C^\ell, \quad (2)$$

and its empty-list, length, item, and membership coordinates are measurable; 3. the finite transcript may be coded in a standard-Borel space built from finite/countable discrete flags, finitely many copies of (2), finitely many real or uniform randomization coordinates, and

$$H_C^\dagger := H_C \sqcup \{\perp, \dagger\}, \quad (3)$$

where $\perp$ is the Sparse Sample failure symbol and $\dagger$ is an optional invalid-state token; and 4. every family of pointwise transcript laws indexed by the quotient input is a Markov kernel, and every measurable transcript coordinate has a Markov-kernel marginal.

The construction includes empty input, empty restrictions and lists, invalid source paths, no-success exhaustion, and terminal fallback. It does not put a sigma-field on the full hyperspace 2^{H_C} and does not choose a representative from any raw quotient cell.

Proof. Let $\mathfrak{P}_N$ denote the finite set of data-independent indexed partitions used by the procedure at its declared sample size; when no partition is used, including the $N = 0$ branch, let $\mathfrak{P}_N$ be a singleton. Let $\mathfrak{A}_N$ collect the finitely many stage indices and finite-alphabet structural-history flags that can precede a list or restriction coordinate. The structural state space

$$\mathbf{U}_N := S_N \times \mathfrak{P}_N \times \mathfrak{A}_N \quad (4)$$

is countable discrete. Duplicate records, arbitrary binary labels, and nonrealizable samples remain ordinary atoms of S_N ; they do not alter this conclusion.

Fix one restriction coordinate $R : \mathbf{U}_N \rightarrow 2^{H_C}$ generated by the pointwise procedure. Its realized range

$$\mathcal{R}_R := \{R(u) : u \in \mathbf{U}_N\} \quad (5)$$

is countable. Equip only this token set with its full power-set sigma-field. Then $R : \mathbf{U}_N \rightarrow \mathcal{R}_R$ is measurable by Lemma A.6. The same construction codes every realized decomposition or leaf-class object. This is the precise measurable interface required by the algorithm: emptiness, the fixed chosen decomposition, the sanitized list, and later finite coordinates are functions of the token and the countable state. We neither assert nor need that an arbitrary member of 2^{H_C} , or the possibly non-Borel class $\bar{C}$, is itself a Borel subset of H_C .

We make the standard-Borel interfaces used below explicit. By Proposition A.5, there are a Polish space P , a Borel set $B \subseteq P$, and a Borel isomorphism $\phi : H_C \rightarrow B$. For every $\ell \geq 1$, the product map ϕ^ℓ realizes H_C^ℓ as the Borel subset B^ℓ of the Polish space P^ℓ ; the zeroth power is a singleton. Thus every finite power H_C^ℓ is standard Borel.

For completeness, the product and tagged-union closure used here follows directly from the same realization. If finitely many standard-Borel spaces are realized as Borel sets B_i in Polish spaces P_i , their product is realized as the Borel set $\prod_i B_i$ in the Polish space $\prod_i P_i$. For a finite or countable family, choose bounded complete compatible metrics of diameter less than one on the P_i 's and give $\bigsqcup_i P_i$ the metric that equals the chosen metric within a component and equals one between distinct components. This space is Polish: a Cauchy sequence eventually stays in one complete component, and the union of countable dense sets is countable and dense. Every component is clopen, so $\bigsqcup_i B_i$ is Borel in $\bigsqcup_i P_i$. Hence finite products and finite or countable tagged disjoint unions of the spaces used here are standard Borel.

Applying this construction to the powers above proves that (2) is standard Borel. Its component inclusions are measurable, the component $\ell = 0$ is the unique empty list, and length is

the measurable disjoint-union tag. The same finite tagged-union construction makes (3) standard Borel: it is the disjoint union of H_C and two singleton spaces, with $\perp$ and $\dagger$ isolated. On every component of length at least j , the j -th item map is a measurable coordinate projection; countable pasting over those components is measurable. Sending lists of length below j to $\dagger$ therefore gives a total measurable coordinate $\text{List}(H_C) \rightarrow H_C^\dagger$.

The diagonal needed for dynamic membership is also Borel directly from the realization. The diagonal Δ_P is closed in $P \times P$, so

$$\Delta_B = (B \times B) \cap \Delta_P$$

is Borel, and the Borel isomorphism $\phi \times \phi$ transports it back to $\Delta_{H_C} = \{(\bar{h}, \bar{h}) : \bar{h} \in H_C\}$. The list-membership relation

$$\mathbf{M} := \{(\lambda, \bar{h}) \in \text{List}(H_C) \times H_C : \bar{h} \text{ occurs in } \lambda\} \quad (6)$$

is Borel: on the component $H_C^\ell \times H_C$, it is

$$\bigcup_{j=1}^{\ell} \{((\bar{h}_1, \dots, \bar{h}_\ell), \bar{h}) : \bar{h}_j = \bar{h}\}, \quad (7)$$

a finite union of inverse images of the Borel diagonal of H_C . Taking the countable union over ℓ proves (6). Proposition A.3 ensures that every list reaching the measured interface is either a genuine finite H_C -list or the empty list. Thus each list coordinate $\lambda : \mathbf{U}_N \rightarrow \text{List}(H_C)$ is measurable by Lemma A.6, regardless of how the fixed pointwise source choice varies among atoms.

Invalidity, oversize, nonfiniteness, non- H_C output, source failure, empty union, and fallback are predicates on the countable state (4). Hence each corresponding flag set is measurable before sanitization. After sanitization, only the empty list, valid finite H_C -lists, actual H_C -items, and the isolated tokens in (3) enter the transcript. This avoids any need to impose a measurable structure on an untrusted source object outside H_C .

There are finitely many stages and, on the positive branch, finitely many teacher/list coordinates. Consequently a complete transcript fits in a finite product, or in a finite or countable tagged disjoint union of finite products, of spaces of the following types:

$$\mathfrak{P}_N, \quad \mathfrak{A}_N, \quad \text{List}(H_C), \quad \mathbb{R}, \quad [0, 1], \quad H_C^\dagger. \quad (8)$$

The spaces $\mathfrak{P}_N$ and $\mathfrak{A}_N$, as well as any additional finite or countable structural flag range, are countable discrete and hence Polish under the discrete metric; $\mathbb{R}$ and $[0, 1]$ are Polish, and the remaining two factors were handled above. For each fixed structural and stopping-time tag, the active transcript coordinates form a finite product of these factors and are therefore standard Borel by the explicit product realization above. The declared procedure has only finitely many stopping-time tags, and adjoining any countable structural tags is covered by the explicit tagged-union realization. Thus the resulting container, denoted by Tr_N , is standard Borel. Its tag map and its coordinate projections on every tagged component are measurable by the construction. Variable stopping times are represented by these measurable tags, not by an infinite transcript.

For a fixed input atom $\bar{s}$, the totalized procedure gives a pointwise probability law $\nu_{\bar{s}}$ on Tr_N . Concretely, the partition has a finite law, AboveThreshold uses finitely many real noise coordinates, and a Sparse Sample call has a finite categorical law because its sanitized input lists are finite and its failure-symbol weight is positive. Concatenate the already fixed ordered lists and retain each candidate at its first occurrence; this gives a deterministic finite support order on every countable structural state and hence introduces no global enumeration of H_C . Comparisons with

fixed thresholds are Borel subsets of the real-noise coordinates; the categorical draw in that finite order can be realized by consecutive intervals in one $[0, 1]$ coordinate. Invalid or exhausted paths use the constant terminal convention. Thus every fixed-input transcript is a Borel random element and $\nu_{\bar{s}}$ is a Borel probability measure.

Lemma A.6 now makes

$$\Gamma_N(\bar{s}, B) := \nu_{\bar{s}}(B), \quad B \in \mathcal{B}(\text{Tr}_N), \quad (9)$$

a Markov kernel. If $(Y, \mathcal{Y})$ is a measurable space and $\pi : \text{Tr}_N \rightarrow Y$ is any measurable list, flag, event-status, stage, or terminal-output coordinate, then, for $A \in \mathcal{Y}$,

$$\Gamma_N^\pi(\bar{s}, A) := \Gamma_N(\bar{s}, \pi^{-1}(A)) \quad (10)$$

is a kernel: it is a probability measure in A for fixed $\bar{s}$, and is measurable in $\bar{s}$ for fixed A by (9). This proves all four claims. For $N = 0$, (4), (8), and (9) reduce to singleton structural and transcript spaces, so no positive-branch coordinate is silently activated. $\square$

$\square$

Proposition A.10 (Measurable list, event, prediction, and error interfaces). *Under Assumption 2, Propositions A.5 and A.8, and Proposition A.9, all of the following interfaces are measurable for every $N \in \mathbb{N}_0$:*

1. *quotient evaluation and joint decoded evaluation; 2. empirical-error and fixed-target population-error events, including arbitrary labels, duplicate records, and improper outputs; 3. dynamic finite-list membership, empty-list, list-validity, transcript-status, invalid-path, and fallback events; and 4. on the declared block horizon $N = n_0 = km$, the exact source event $E_{\text{good}}(\bar{S}, \mathcal{P})$ and every one of its finite stage/block sections.*

More generally, any family of Borel output-event sections indexed by a countable quotient state has a measurable joint graph.

Proof. The quotient evaluation map is jointly measurable. Indeed,

$$\{(\bar{h}, q) \in H_C \times Q_C : \bar{h}(q) = 1\} = \bigcup_{q \in Q_C} e_q^{-1}(\{1\}) \times \{q\}, \quad (11)$$

a countable union of measurable rectangles. Using the measurable quotient cells, the joint decoded evaluation is also measurable because

$$\{(\bar{h}, x) \in H_C \times X : \bar{h}(\kappa(x)) = 1\} = \bigcup_{q \in Q_C} e_q^{-1}(\{1\}) \times \kappa^{-1}(\{q\}). \quad (12)$$

This proves the output-coordinate interface for every $\bar{h} \in H_C$; no membership in $\bar{C}$ is required.

For $\bar{s} = ((q_r, y_r))_{r=1}^N$, define the total empirical-error map

$$\widehat{R}_N(\bar{s}, \bar{h}) := \begin{cases} 0, & N = 0, \\ \frac{1}{N} \sum_{r=1}^N \mathbf{1}\{\bar{h}(q_r) \neq y_r\}, & N \geq 1. \end{cases} \quad (13)$$

For each fixed $\bar{s}$, (13) is a finite sum of measurable evaluation indicators. Since S_N is countable, for every Borel $B \subseteq [0, 1]$,

$$\{(\bar{s}, \bar{h}) : \widehat{R}_N(\bar{s}, \bar{h}) \in B\} = \bigcup_{\bar{s} \in S_N} \{\bar{s}\} \times \{\bar{h} : \widehat{R}_N(\bar{s}, \bar{h}) \in B\} \quad (14)$$

is a countable union of measurable rectangles. Thus (13) is jointly measurable. Repeated q_r 's and arbitrary y_r 's merely repeat or change finitely many summands and do not affect (14). For fixed $(\bar{D}, \bar{c})$, Proposition A.8 makes $\bar{h} \mapsto \text{err}_{\bar{D}}(\bar{h}, \bar{c})$ measurable, so every threshold event for population error is in $\mathcal{H}_C$, also for improper $\bar{h}$.

If $\lambda : \mathcal{U}_N \rightarrow \text{List}(H_C)$ is any generated list coordinate, then

$$\{(u, \bar{h}) : \bar{h} \text{ occurs in } \lambda(u)\} = \{(u, \bar{h}) : (\lambda(u), \bar{h}) \in \mathbf{M}\} \quad (15)$$

is measurable by Proposition A.9. The empty-list and length/cap events are inverse images under measurable list coordinates. List validity, source validity, Above/Below status, first legal stage, no-success exhaustion, invalid transcript, Sparse Sample failure, and fallback are measurable discrete flags in the same proposition. In particular, the terminal output map sends every fallback flag to the constant $\bar{c}_0$, whose singleton is Borel in the standard-Borel space H_C .

At the actual block horizon, $S_N \times \mathfrak{P}_N$ is countable discrete. The exact event

$$E_{\text{good}} := \{(\bar{s}, \mathcal{P}) : E_{\text{good}}(\bar{s}, \mathcal{P}) \text{ holds}\} \quad (16)$$

is therefore measurable as a subset of that space, as are its finitely many stage, block, high-error, low-error, and complement-error sections. This argument does not take a supremum over a non-measurable function class and does not select a representative trace: the exact pointwise predicate in the setting is evaluated on each countable input/partition atom.

Finally, let $(B_u)_{u \in \mathcal{U}}$ be any family with $B_u \in \mathcal{H}_C$, where $\mathcal{U}$ is countable discrete. Then

$$\{(u, \bar{h}) : \bar{h} \in B_u\} = \bigcup_{u \in \mathcal{U}} \{u\} \times B_u \quad (17)$$

is measurable. Formula (17) covers every remaining pointwise Borel output event and proves the general assertion. $\square$

$\square$

Proposition A.11 (Quotient kernel for every totalized pointwise output law). *Under Assumption 2, Propositions A.3 and A.1, Lemma A.6, and Proposition A.9, fix $N \in \mathbb{N}_0$. Let a totalized pointwise quotient law assign to every $\bar{s} \in Z_Q^N$ a Borel probability measure $\mu_{\bar{s}}$ on the exact output space $(H_C, \mathcal{H}_C)$. Then*

$$K_N(\bar{s}, E) := \mu_{\bar{s}}(E), \quad E \in \mathcal{H}_C, \quad (18)$$

is a Markov kernel

$$K_N : (Z_Q^N, 2^{Z_Q^N}) \rightsquigarrow (H_C, \mathcal{H}_C). \quad (19)$$

This applies to every deterministic or randomized totalization, without a support-cardinality or propriety restriction. In particular it applies to the exact totalized quotient-first Lyu (Lyu, 2025) law, its totalized old-Lyu (Lyu, 2025) analogue, and, when defined, the totalized finite-class pointwise law. If $d = 0$, $N = 0$, and the supplied law is one of these setting-defined arm laws, Proposition A.1 gives $\mu_{\emptyset} = \delta_{\bar{c}_0}$, so (18) is exactly that Dirac law. For an arbitrary supplied pointwise law, the universal kernel assertion places no Dirac restriction on $\mu_{\emptyset}$. At the public sample size of any setting-defined arm, its notation K_C means the corresponding kernel K_N in (18).

Proof. A pointwise output law is, by definition, a probability measure $\mu_{\bar{s}}$ on $(H_C, \mathcal{H}_C)$ for each fixed input. For the procedure exported by Proposition A.3, this law exists directly as the terminal-output marginal of the pointwise transcript law constructed in Proposition A.9. All exact valid paths output their actual H_C -item, while invalid, empty, exhausted, or failure paths output the

fixed $\bar{c}_0$. Hence no mass is assigned outside H_C , and the output marginal has total mass one. More explicitly, on the positive branch its support for fixed $\bar{s}$ is contained in

$$\{\bar{c}_0\} \cup \bigcup_{\mathcal{P} \in \mathfrak{P}_N} \bigcup_{r=0}^d \bigcup_{i=1}^k \mathcal{L}_i^r(\bar{s}, \mathcal{P}), \quad (18a)$$

which is finite by totalization; continuous mechanism coins change only the masses on this finite Borel set. A separately supplied totalized pointwise law needs no transcript representation or finite-support property; its Borel probability measures $\mu_{\bar{s}}$ are already the complete pointwise input.

Applying Lemma A.6 with $Y = H_C$ proves (19): for fixed $\bar{s}$, $E \mapsto K_N(\bar{s}, E)$ is the probability measure $\mu_{\bar{s}}$, and for fixed $E \in \mathcal{H}_C$, $\bar{s} \mapsto K_N(\bar{s}, E)$ is measurable on the countable discrete input. This remains true if the pointwise supports vary arbitrarily with $\bar{s}$, are not uniformly finite, or contain improper hypotheses. Thus the result does not obtain measurability by choosing a global list enumeration or a selector.

When $N = 0$, $Z_Q^0 = \{\emptyset\}$, so the universal conclusion says only that (18) is the constant-input kernel with output measure $\mu_{\emptyset}$. Even when $d = 0$, an arbitrary Borel probability measure on the improper output space H_C need not be Dirac. If the supplied law is the null law of one of the three setting-defined learner arms, Proposition A.1 instead gives $\mu_{\emptyset} = \delta_{\bar{c}_0}$, yielding the exact deterministic baseline. When $d \geq 1$, all quotient datasets, including duplicate records and arbitrary/nonrealizable labels, are atoms of Z_Q^N on which the procedure is total. Therefore none is removed from the kernel domain. $\square$

$\square$

Proposition A.12 (Measurable raw pullback of the quotient kernel). *Under Assumption 2, Proposition A.6, and Proposition A.11, for every $N \in \mathbb{N}_0$ and every quotient kernel K_N produced there, define*

$$A_N(s, E) := K_N(T_N(s), E), \quad s \in Z_X^N, \quad E \in \mathcal{H}_C. \quad (20)$$

Then

$$A_N : (Z_X^N, \mathcal{Z}_X^{\otimes N}) \rightsquigarrow (H_C, \mathcal{H}_C) \quad (21)$$

is a Markov kernel. Its output is the same quotient hypothesis in H_C ; decoding occurs only after release and no propriety condition is introduced. For $N = 0$, universally $A_0(\emptyset, \cdot) = K_0(\emptyset, \cdot)$; this common law is Dirac only under the arm-specific premise $K_0(\emptyset, \cdot) = \delta_{\bar{c}_0}$.

Proof. For fixed $s \in Z_X^N$, $E \mapsto A_N(s, E)$ is the probability measure $K_N(T_N(s), \cdot)$. For fixed $E \in \mathcal{H}_C$, the kernel property of K_N makes $\bar{s} \mapsto K_N(\bar{s}, E)$ measurable on Z_Q^N , while Proposition A.6 makes T_N measurable on the raw product space. Their composition in (20) is therefore $\mathcal{Z}_X^{\otimes N}$ -measurable. This proves both kernel axioms.

For $N = 0$, both input spaces are singletons and T_0 is the unique empty-tuple map, so for every $E \in \mathcal{H}_C$,

$$A_0(\emptyset, E) = K_0(T_0(\emptyset), E) = K_0(\emptyset, E). \quad (22)$$

Thus the universal empty-input conclusion is exactly $A_0 = K_0$. If the supplied kernel is a setting-defined learner-arm null law, Proposition A.1 additionally gives $K_0 = A_0 = \delta_{\bar{c}_0}$; no Dirac conclusion follows for a generic supplied law. For $N \geq 1$, raw duplicates, same-cell records, and arbitrary labels are all in the domain of the measurable T_N ; no realizability condition is used. Equation (20) is an exact pullback into H_C , not a kernel on a trace, list, representative, or raw-selector surrogate. The neighbor property of T_N and all pointwise privacy inequalities are deliberately unused here.

Proposition A.3 supplies the complete positive-dimensional pointwise quotient procedure and ensures that every valid, empty, invalid, failure, exhausted, and fallback path terminates in H_C .

Proposition A.1 separately supplies the exact $d = 0, N = 0$ Dirac law at the unique $\bar{c}_0$. Proposition A.5 supplies the exact countable-discrete input and standard-Borel output spaces.

Lemma A.6 proves the central promotion principle: on every Z_Q^N , arbitrary pointwise coordinates are measurable and arbitrary pointwise Borel probability laws are kernels. Proposition A.9 applies that principle to the actual source state space, coding realized restrictions and decompositions only in their countable ranges, coding finite lists in $\text{List}(H_C)$, and coding the finite random transcript in a standard-Borel space. This proves list/transcript measurability without a hyperspace sigma-field or a selector. Proposition A.10 then proves the exact list-membership, source-event, transcript-status, fallback, prediction, empirical-error, and population-error interfaces, including E_{good} and potentially improper outputs.

Proposition A.11 turns every totalized pointwise H_C -law into the required quotient Markov kernel K_N , with every arbitrary-label input included. Only for the three setting-defined learner-arm null laws does Proposition A.1 identify the $d = 0, N = 0$ kernel as $\delta_{\bar{c}_0}$; an arbitrary supplied law remains arbitrary. Finally, Proposition A.6 makes T_N measurable, so composition with the quotient kernel proves that $A_N = K_N \circ T_N$ is a raw-input learner kernel for every $N \in \mathbb{N}_0$, with $A_0 = K_0$ universally and the exact Dirac pullback only under the arm-specific premise. Together, these propositions establish exactly the stated conclusion. The occurrence-mark, privacy, utility, and rate properties are established in the subsequent named results. $\square$

A.4 Marked Kernel and Released Projection

Proposition A.11 realizes the exact released law as the terminal-output marginal of the complete transcript kernel: for every quotient input $\bar{s} \in Z_Q^N$ and every $E \in \mathcal{H}_C$,

$$K_C(\bar{s}, E) = \int_{\text{Tr}_N} \mathbf{1}_E(\bar{H}(\tau)) \Gamma_N(\bar{s}, d\tau). \quad (1)$$

This identity uses the same transcript coordinate that is marked below; no new coupling or output law is introduced.

Lemma A.7 (Measurable finite producer-block occurrence set). *Under Propositions A.9, A.10, and A.11, fix the declared branch $d \geq 1$, its finite teacher count $k \geq 2$, and its sample size $N = n_0$. There are measurable transcript coordinates*

$$\lambda_{i,r} : \text{Tr}_N \longrightarrow \text{List}(H_C), \quad i \in [k], \quad r \in \{0, \dots, d\}, \quad (2)$$

the terminal output $\bar{H} : \text{Tr}_N \rightarrow H_C$, and a measurable actual-output event $\text{Act} \subseteq \text{Tr}_N$. Define the all-stage producer list by concatenation,

$$\lambda_i(\tau) := \lambda_{i,0}(\tau) \parallel \dots \parallel \lambda_{i,d}(\tau), \quad (3)$$

and define

$$a_i(\tau) := \mathbf{1}\{\bar{H}(\tau) \text{ occurs in } \lambda_i(\tau)\}, \quad I(\tau) := \{i \in [k] : a_i(\tau) = 1\}, \quad R(\tau) := \sum_{i=1}^k a_i(\tau). \quad (4)$$

Then every a_i and R is measurable, $I(\tau)$ is finite for every transcript, and

$$\tau \in \text{Act} \implies 1 \leq R(\tau) = |I(\tau)| \leq k. \quad (5)$$

The set $I(\tau)$ is exactly the setting-defined set of producer-block coordinates whose all-stage candidate union contains the actual output.

Proof. Proposition A.9 supplies every coordinate in (2), including a sanitized empty list when a stage/block list is absent or invalid. Concatenation in (3) is Borel. To see this directly, restrict $\text{List}(H_C)^{d+1}$ to a component with length vector $(\ell_0, \dots, \ell_d)$. On that component concatenation is merely the coordinate map

$$H_C^{\ell_0} \times \dots \times H_C^{\ell_d} \longrightarrow H_C^{\ell_0 + \dots + \ell_d}$$

that preserves the displayed order. It is measurable. Taking the countable disjoint union over all finite length vectors proves measurability on the whole list space. This derivation keeps every list position, including repeated positions; it makes no first-occurrence selection.

Proposition A.10 makes dynamic membership of the measurable terminal output in a measurable finite-list coordinate a Borel event. Therefore each a_i is a measurable $\{0, 1\}$ -valued function. The finite sum R is measurable with range $\{0, 1, \dots, k\}$, and (4) makes $I(\tau)$ a subset of the finite set $[k]$.

It remains to prove nonemptiness on actual paths rather than assume it. By the exact transcript construction exported by Proposition A.9, a transcript in Act records a selected stage $r_*(\tau)$ and an actual Sparse Sample categorical outcome. That categorical outcome is one of the items in the sanitized tuple $(\lambda_{1,r_*}(\tau), \dots, \lambda_{k,r_*}(\tau))$. Hence some $i \in [k]$ satisfies $\bar{H}(\tau) \in \lambda_{i,r_*}(\tau)$. The selected-stage list is a sublist of the concatenation (3), so $a_i(\tau) = 1$, proving the lower bound in (5). The upper bound follows from $I(\tau) \subseteq [k]$.

This reasoning is pathwise and retains adaptive selection: the realized r_* , lists, and output are read from the same transcript. If the output occurs in several producer blocks, every corresponding block index remains in $I(\tau)$. Repetitions across stages or positions inside one block are also retained in (3), but they intentionally map to that one block mark i , because the setting's mark records producer blocks rather than list positions. If all lists are empty, (4) gives $R = 0$, and the contrapositive of (5) shows that the transcript is nonactual. A failure or fallback output may happen to equal $\bar{c}_0$ appearing in a list; its status remains nonactual and is handled by mark 0 below. $\square$

Lemma A.8 (Uniform pathwise occurrence-mark kernel). *Under Lemma A.7, put $\mathbf{M}_k := \{0, 1, \dots, k\}$ with its power-set sigma-field and define, with the convention that the second line is zero off Act ,*

$$w_0(\tau) := \mathbf{1}_{\text{Act}^c}(\tau), \quad w_i(\tau) := \mathbf{1}_{\text{Act}}(\tau) \frac{a_i(\tau)}{R(\tau)}, \quad i \in [k]. \quad (6)$$

Then

$$W(\tau, B) := \sum_{i \in B} w_i(\tau), \quad B \subseteq \mathbf{M}_k, \quad (7)$$

is a Markov kernel $W : \text{Tr}_N \rightsquigarrow \mathbf{M}_k$. On an actual path it is uniform on $I(\tau)$, assigns no mass to 0, and retains every distinct producer-block occurrence. On every fallback or otherwise nonactual path it is δ_0 , even when the terminal fallback value also occurs in a candidate list.

Proof. Lemma A.7 makes Act , every a_i , and R measurable, and (5) guarantees that division in (6) is only performed when $R \in \{1, \dots, k\}$. Equivalently, define the displayed quotient to be zero at $R = 0$; this is a measurable finite-valued function. Thus every w_i is measurable.

If $\tau \notin \text{Act}$, then $w_0 = 1$ and all positive weights are zero. If $\tau \in \text{Act}$, then $w_0 = 0$, and

$$\sum_{i=1}^k w_i(\tau) = \frac{\sum_{i=1}^k a_i(\tau)}{R(\tau)} = 1. \quad (8)$$

Hence the weights are nonnegative and sum to one on every path. For fixed τ , (7) is consequently a probability measure on the finite space $\mathbf{M}_k$; for fixed B , it is a finite sum of measurable functions. These are the two kernel axioms.

When $I(\tau) = \{i\}$, (6) assigns mass one to i . When $I(\tau) = [k]$, it assigns mass $1/k$ to every producer. In particular, at the boundary $k = 2$, the only actual-path denominators are 1 and 2. If there is no occurrence, Lemma A.7 rules out an actual path, so (7) assigns mark 0. Empty-list, invalid, no-success, and Sparse Sample failure paths are all nonactual and obey the same rule. No representative producer is selected; the kernel splits mass over all producer coordinates. $\square$

Proposition A.13 (Marked lift of the exact quotient output law). *Under Lemma A.6, Proposition A.9, and Lemma A.8, for every $\bar{s} \in Z_Q^N$ and every $B \in \mathcal{H}_C \otimes 2^{\mathbf{M}_k}$, let*

$$B_i := \{\bar{h} \in H_C : (\bar{h}, i) \in B\}, \quad i \in \mathbf{M}_k, \quad (9)$$

and define

$$\tilde{K}_C(\bar{s}, B) := \sum_{i=0}^k \int_{\text{Tr}_N} \mathbf{1}_{B_i}(\bar{H}(\tau)) w_i(\tau) \Gamma_N(\bar{s}, d\tau). \quad (10)$$

Then

$$\tilde{K}_C : (Z_Q^N, 2^{Z_Q^N}) \rightsquigarrow (H_C \times \mathbf{M}_k, \mathcal{H}_C \otimes 2^{\mathbf{M}_k}) \quad (11)$$

is a Markov kernel. It is the pathwise lift of the exact output law, not a new released mechanism.

Proof. Since $\mathbf{M}_k$ is finite and discrete, every $B \in \mathcal{H}_C \otimes 2^{\mathbf{M}_k}$ decomposes uniquely as

$$B = \bigcup_{i=0}^k B_i \times \{i\}, \quad (12)$$

with each $B_i \in \mathcal{H}_C$. The terminal map $\bar{H}$ and every w_i are measurable, so each integrand in (10) is bounded and measurable.

Fix $\bar{s}$. Nonnegativity is immediate. Equation (8) gives

$$\tilde{K}_C(\bar{s}, H_C \times \mathbf{M}_k) = \int \sum_{i=0}^k w_i(\tau) \Gamma_N(\bar{s}, d\tau) = 1. \quad (13)$$

If $(B^{(n)})_{n \geq 1}$ are pairwise disjoint marked events, their sections $(B_i^{(n)})_{n \geq 1}$ are pairwise disjoint for every fixed i . Applying monotone convergence inside each of the finitely many nonnegative integrals gives

$$\tilde{K}_C\left(\bar{s}, \bigcup_{n \geq 1} B^{(n)}\right) = \sum_{n \geq 1} \tilde{K}_C(\bar{s}, B^{(n)}). \quad (14)$$

Thus $B \mapsto \tilde{K}_C(\bar{s}, B)$ is a probability measure.

For fixed B , (10) assigns one real number in $[0, 1]$ to every $\bar{s} \in Z_Q^N$. Lemma A.6 makes this coordinate measurable on the countable-discrete quotient input. Together with the fixed-input probability property, this proves both Markov-kernel axioms in (11). In particular, for every $E \in \mathcal{H}_C$,

$$\tilde{K}_C(\bar{s}, E \times \{i\}) = \int \mathbf{1}_E(\bar{H}(\tau)) w_i(\tau) \Gamma_N(\bar{s}, d\tau). \quad (15)$$

Formula (10) first marks each full internal path and only then integrates it out. Therefore two paths with the same terminal $\bar{h}$ but different partitions, selected stages, repeated occurrences, or occurrence sets keep their own weights; no conditional law given only $\bar{h}$ and no global selector is introduced. $\square$

Proposition A.14 (Exact released projection and zero privacy residual). *Under Proposition A.11 and Proposition A.13, let $\pi_H : H_C \times \mathbf{M}_k \rightarrow H_C$ be the first-coordinate projection. Then, for every quotient input $\bar{s}$ and every $E \in \mathcal{H}_C$,*

$$(\pi_{H\#}\tilde{K}_C)(\bar{s}, E) = \sum_{i=0}^k \tilde{K}_C(\bar{s}, E \times \{i\}) = K_C(\bar{s}, E). \quad (16)$$

Consequently the analysis-only mark changes neither the released output law nor any privacy parameter of that released law. No privacy assertion is made for the unreleased pair $(\bar{H}, J)$.

Proof. Sum (15) over the finite mark space and apply (8):

$$\begin{aligned} \sum_{i=0}^k \tilde{K}_C(\bar{s}, E \times \{i\}) &= \int \mathbf{1}_E(\bar{H}(\tau)) \left(\sum_{i=0}^k w_i(\tau) \right) \Gamma_N(\bar{s}, d\tau) \\ &= \int \mathbf{1}_E(\bar{H}(\tau)) \Gamma_N(\bar{s}, d\tau) \\ &= K_C(\bar{s}, E), \end{aligned} \quad (17)$$

where the last equality is the exact terminal-marginal identity (1). This proves (16), with no approximation, exceptional event, or residual mass.

For completeness, for any two quotient inputs $\bar{s}, \bar{s}'$, any $\varepsilon, \delta \geq 0$, and any released event E , (16) gives the identity

$$\begin{aligned} (\pi_{H\#}\tilde{K}_C)(\bar{s}, E) - e^\varepsilon (\pi_{H\#}\tilde{K}_C)(\bar{s}', E) - \delta \\ = K_C(\bar{s}, E) - e^\varepsilon K_C(\bar{s}', E) - \delta. \end{aligned} \quad (18)$$

Thus every future privacy inequality for the actually released kernel is literally unchanged: the mark consumes zero privacy slack. Equation (18) does not claim that privacy has already been proved, and it does not claim privacy if J were released.

The same zero-residual statement holds on the raw interface by direct substitution: for every raw input $s \in Z_X^N$,

$$\sum_{i=0}^k \tilde{K}_C(T_N(s), E \times \{i\}) = K_C(T_N(s), E). \quad (19)$$

Thus a later DP comparison on raw neighboring inputs sees exactly the same two released-event probabilities before and after this analysis-only lift; no marked raw kernel or additional privacy theorem is being asserted here.

The status split is essential. On an actual path, $w_0 = 0$, including when there is exactly one occurrence, all k occurrences, or $k = 2$. On a nonactual path, $w_0 = 1$, including when the fallback value $\bar{c}_0$ coincides with an item in one or more lists. Hence mark 0 is used only for fallback/nonactual paths, and no fallback mass leaks into a positive producer mark. Proposition A.9 supplies the exact standard-Borel transcript kernel, every stage/block list, the adaptive selected-stage status, the actual/fallback split, and the terminal output. Proposition A.10 supplies measurable dynamic membership and status events. Lemma A.7 combines those interfaces into the exact pathwise occurrence set from the setting and proves, rather than assumes, that every actual output has a nonempty finite set of producer coordinates.

Lemma A.8 gives the unique required support rule: uniform mass on all positive occurrence coordinates for an actual path and unit mass at 0 for every nonactual path. It explicitly covers repeated occurrences, adaptively selected stages, empty lists, one/all/no occurrence, failure/fallback, and $k = 2$, without choosing one producer. Proposition A.13 integrates those pathwise weights

against the transcript kernel and verifies countable additivity, unit mass, and input-event measurability on the exact codomain $H_C \times \{0, \dots, k\}$.

Finally, Proposition A.11 identifies the unmarked terminal marginal with the exact released K_C , and summing the marked weights above proves the finite-sum identity

$$\sum_{i=0}^k \tilde{K}_C(\bar{s}, E \times \{i\}) = K_C(\bar{s}, E)$$

for every input and every measurable released event. The same proposition shows that the unreleased mark introduces exactly zero output-law and privacy residual. Thus the mark is an analysis variable only; it adds no selector, algorithmic branch, or released output coordinate.

The setting defines k and the occurrence analysis only on $d \geq 1$. On the $d = 0, N = 0$ branch, Proposition A.1 gives the exact Dirac release at $\bar{c}_0$; the canonical inactive convention $\mathbf{M}_0 = \{0\}$ and $\tilde{K}_C(\emptyset, E \times \{0\}) = K_C(\emptyset, E)$ satisfies the same projection identity. It activates no positive-dimensional list, stage, or teacher coordinate and preserves the no-data baseline. $\square$

A.5 VC Trace Counting

Lemma A.9 (Null and positive quotient-VC branches). *Under Assumptions 1 and 2, Proposition A.4, Lemma A.4, and Lemma A.5, for every indexed labeled quotient sample $\bar{S} = ((q_r, y_r))_{r=1}^{n_0}$, define*

$$\mathcal{E}_{\bar{C}}(\bar{S}) = \{(\mathbf{1}\{\bar{c}(q_r) \neq y_r\})_{r=1}^{n_0} : \bar{c} \in \bar{C}\},$$

and define the indexed-sample growth function by

$$\Pi_{\bar{C}}(n) := \sup_{(q_1, \dots, q_n) \in Q_C^n} |\{(\bar{c}(q_r))_{r=1}^n : \bar{c} \in \bar{C}\}|, \quad n \in \mathbb{N}_0,$$

with the unique empty-vector convention when $n = 0$. Then

$$0 \leq v \leq d.$$

Moreover, the following are equivalent:

$$v = 0, \quad d = 0, \quad |\bar{C}| = 1. \tag{1}$$

In this case, for every $n_0 \in \mathbb{N}_0$ and every indexed labeled sample $\bar{S} \in Z_Q^{n_0}$,

$$|\mathcal{E}_{\bar{C}}(\bar{S})| = 1. \tag{2}$$

If $d \geq 1$, then $1 \leq v \leq d$.

Proof. If $A = \{q_1, \dots, q_t\} \subseteq Q_C$ is shattered by $\bar{C}$, label every node at level r of a complete binary tree of depth t by q_r . Every root-to-leaf bit string is a labeling of A , so shattering of A supplies a quotient concept realizing that path. Thus this tree is shattered by $\bar{C}$. Consequently every finite VC witness of size t gives a Littlestone witness of depth t , and the identities imply

$$v = \text{VC}(\bar{C}) \leq \text{LD}(\bar{C}) = d < \infty. \tag{3}$$

The class $\bar{C}$ is nonempty by the factorization proposition. If it contained distinct $\bar{c}_0, \bar{c}_1$, they would disagree at some $q \in Q_C$. Because the labels are binary, these two functions would realize both labels on $\{q\}$, so $\{q\}$ would be shattered. Hence $v = 0$ forces $|\bar{C}| = 1$. Conversely, a singleton

binary class cannot shatter a point or a depth-one tree, so it has $v = d = 0$. The Littlestone quotient lemma supplies the remaining implication $d = 0 \Rightarrow |\bar{C}| = 1$, proving (1).

When $\bar{C}$ is a singleton, the definition of $\mathcal{E}_{\bar{C}}(\bar{S})$ contains exactly the one vector generated by its unique concept. This remains true for $n_0 = 0$, when that vector is the empty vector, and proves (2). Finally, (1) and (3) show that $d \geq 1$ implies $1 \leq v \leq d$. $\square$

Proposition A.15 (Indexed xor identity and duplicate-support collapse). *Under Proposition A.4, fix any $n_0 \in \mathbb{N}_0$ and any indexed labeled quotient sample*

$$\bar{S} = ((q_r, y_r))_{r=1}^{n_0}.$$

Define its prediction-trace family and distinct quotient support by

$$\mathcal{P}_{\bar{C}}(\bar{S}) := \{(\bar{c}(q_r))_{r=1}^{n_0} : \bar{c} \in \bar{C}\}, \quad A_{\bar{S}} := \{q_r : 1 \leq r \leq n_0\}. \quad (4)$$

Then coordinatewise xor with the fixed label vector is a bijection

$$\chi_{\mathbf{y}} : \mathcal{P}_{\bar{C}}(\bar{S}) \longrightarrow \mathcal{E}_{\bar{C}}(\bar{S}), \quad \chi_{\mathbf{y}}(z)_r = z_r \oplus y_r, \quad (5)$$

and replication along the indexed sample is a bijection

$$\rho_{\bar{S}} : \bar{C}|_{A_{\bar{S}}} \longrightarrow \mathcal{P}_{\bar{C}}(\bar{S}), \quad \rho_{\bar{S}}(f) = (f(q_r))_{r=1}^{n_0}. \quad (6)$$

Consequently,

$$|\mathcal{E}_{\bar{C}}(\bar{S})| = |\mathcal{P}_{\bar{C}}(\bar{S})| = |\bar{C}|_{A_{\bar{S}}}. \quad (7)$$

These identities hold with arbitrary repeated quotient points, exact duplicate labeled records, or different labels attached to repeated points.

Proof. For bits z_r, y_r , $z_r \oplus y_r = \mathbf{1}\{z_r \neq y_r\}$. Therefore (5) sends the prediction vector of each $\bar{c}$ to exactly its error vector. On the whole cube $\{0, 1\}^{n_0}$, xor with $\mathbf{y} = (y_r)_{r=1}^{n_0}$ is an involution:

$$\chi_{\mathbf{y}}(\chi_{\mathbf{y}}(z)) = z.$$

Its restriction in (5) is thus bijective onto the error-trace family.

The map in (6) is surjective by the definition of the prediction-trace family. If $f, g \in \bar{C}|_{A_{\bar{S}}}$ are distinct, then they differ at some $q \in A_{\bar{S}}$. By definition of the support, $q = q_r$ for at least one index r , and the corresponding replicated coordinates differ. Thus $\rho_{\bar{S}}$ is injective, proving (7).

Repeated points only repeat coordinates under $\rho_{\bar{S}}$ and cannot create another trace. The label vector is fixed before xor is applied, so even if $q_r = q_t$ but $y_r \neq y_t$, the global involution argument still preserves cardinality exactly. When $n_0 = 0$, the support is empty and all three sets in (7) contain the unique empty function or empty vector. $\square$

Lemma A.10 (Sauer–Shelah (Sauer, 1972) counting on a finite quotient restriction). *Under Lemma A.9, Lemma A.4, and Proposition A.15, let $\bar{S}$ be any fixed indexed labeled quotient sample of length n_0 , and put $s := |A_{\bar{S}}| \leq n_0$. Then*

$$|\mathcal{E}_{\bar{C}}(\bar{S})| \leq \sum_{j=0}^{\min\{v, s\}} \binom{s}{j} \leq \sum_{j=0}^{\min\{v, n_0\}} \binom{n_0}{j}. \quad (8)$$

Equivalently, for the setting's growth function (defined on indexed samples),

$$|\mathcal{E}_{\bar{C}}(\bar{S})| \leq \Pi_{\bar{C}}(n_0) \leq \sum_{j=0}^{\min\{v, n_0\}} \binom{n_0}{j}. \quad (9)$$

The conclusion does not require $\bar{C}$ to be finite.

Proof. We first prove the finite-cube form of Sauer–Shelah (Sauer, 1972). For an integer $t \geq 0$, a class $\mathcal{F} \subseteq \{0, 1\}^B$ on a t -point set B , and an integer $r \geq 0$, suppose $\text{VC}(\mathcal{F}) \leq r$. We claim

$$|\mathcal{F}| \leq S(t, r) := \sum_{j=0}^r \binom{t}{j}, \quad (10)$$

where $\binom{t}{j} = 0$ for $j > t$, and set $S(t, -1) := 0$.

The proof is by induction on t . At $t = 0$, the Boolean cube contains only the empty function, so (10) holds. For $t \geq 1$, fix $x \in B$, write $B' = B \setminus \{x\}$, and define

$$\begin{aligned} \mathcal{F}_b &:= \{f|_{B'} : f \in \mathcal{F}, f(x) = b\}, \quad b \in \{0, 1\}, \\ \mathcal{U} &:= \mathcal{F}_0 \cup \mathcal{F}_1, \quad \mathcal{I} := \mathcal{F}_0 \cap \mathcal{F}_1. \end{aligned} \quad (11)$$

Partitioning $\mathcal{F}$ by its value at x , followed by the finite set identity $|F_0| + |F_1| = |F_0 \cup F_1| + |F_0 \cap F_1|$, gives

$$|\mathcal{F}| = |\mathcal{U}| + |\mathcal{I}|. \quad (12)$$

Any set shattered by $\mathcal{U}$ is shattered by $\mathcal{F}$, so $\text{VC}(\mathcal{U}) \leq r$. If a set $D \subseteq B'$ is shattered by $\mathcal{I}$, then for every labeling of D , membership in both $\mathcal{F}_0$ and $\mathcal{F}_1$ supplies both extensions at x . Hence $D \cup \{x\}$ is shattered by $\mathcal{F}$, and $\text{VC}(\mathcal{I}) \leq r - 1$. For $r = 0$, this says $\mathcal{I} = \emptyset$: otherwise $\{x\}$ would be shattered. The induction hypothesis and Pascal's identity now yield

$$|\mathcal{F}| \leq S(t - 1, r) + S(t - 1, r - 1) = S(t, r), \quad (13)$$

which proves (10).

Apply (10) to the restriction family

$$\mathcal{F}_{\bar{S}} := \bar{C}|_{A_{\bar{S}}} \subseteq \{0, 1\}^{A_{\bar{S}}}.$$

This family is finite because the ambient cube has 2^s elements, even when $\bar{C}$ is infinite. If a subset of $A_{\bar{S}}$ is shattered by $\mathcal{F}_{\bar{S}}$, it is shattered by $\bar{C}$; the VC identity therefore gives $\text{VC}(\mathcal{F}_{\bar{S}}) \leq v$. Equations (7) and (10) give the first inequality in (8). Since $s \leq n_0$, for every $0 \leq j \leq s$ one has $\binom{s}{j} \leq \binom{n_0}{j}$; adding the remaining nonnegative terms proves the second inequality.

For any indexed n_0 -sample, Proposition A.15 shows that xor preserves the prediction-trace count. Taking the supremum over the quotient-point sequences gives (9). No concept representative is selected for a trace, and no cardinality of $\bar{C}$ enters the proof. $\square$

Lemma A.11 (Positive-dimensional Sauer-sum compression). *Under Lemma A.10, if $1 \leq v \leq n_0$, then every fixed indexed labeled quotient sample $\bar{S}$ of length n_0 satisfies*

$$|\mathcal{E}_{\bar{C}}(\bar{S})| \leq \sum_{j=0}^v \binom{n_0}{j} \leq \left(\frac{en_0}{v}\right)^v. \quad (14)$$

This statement includes $v = 1$, remains unchanged when $v = d$, and uses a nonstrict premise $v \leq n_0$, so it includes $n_0 = v$.

Proof. Put $x := v/n_0 \in (0, 1]$. Since $x^j \geq x^v$ for $0 \leq j \leq v$, the binomial theorem gives

$$(1+x)^{n_0} = \sum_{j=0}^{n_0} \binom{n_0}{j} x^j \geq x^v \sum_{j=0}^v \binom{n_0}{j}. \quad (15)$$

Using $1+x \leq e^x$ and $n_0 x = v$,

$$\sum_{j=0}^v \binom{n_0}{j} \leq x^{-v} (1+x)^{n_0} \leq \left(\frac{n_0}{v}\right)^v e^{n_0 x} = \left(\frac{en_0}{v}\right)^v. \quad (16)$$

Together with (8), this proves (14).

At $v = 1$, (14) reads $|\mathcal{E}_{\bar{C}}(\bar{S})| \leq 1 + n_0 \leq en_0$ for $n_0 \geq 1$. At $n_0 = v$, the exact Sauer sum is $\sum_{j=0}^v \binom{v}{j} = 2^v \leq e^v$, which is precisely the right side of (14). If $v = d$, substitution in (14) gives $(en_0/d)^d$ with no extra factor or exponent; no class-cardinality term appears in any case. $\square$

Proposition A.16 (The positive branch lies in the Sauer range). *Under Assumptions 1 and 4, Lemma A.2, Proposition A.2, and Lemma A.9, suppose $d \geq 1$ and fix the setting's stipulated sufficiently large universal block constant so that, in particular, $C_{\text{blk}} \geq 1$. Then the preceding calibration fixes*

$$\begin{aligned} a(k) &= v + \log(4k/\beta), & Q(k) &= e + \frac{ekd^2 a(k)}{\alpha v}, \\ m &= m(k) = \left\lceil C_{\text{blk}} \frac{d^2}{\alpha} a(k) \log Q(k) \right\rceil, & n_0 &= km(k), \end{aligned}$$

with $k \geq 2$, and this realized tuple satisfies $1 \leq v \leq n_0$.

Proof. Lemma A.2 gives $1 \leq v \leq d$ and the displayed candidate dictionary for every integer $t \geq 2$. Proposition A.2 fixes the least feasible $k \geq 2$ and the realized identities $m = m(k)$ and $n_0 = km(k)$. These conclusions agree with the positive branch isolated by Lemma A.9. Since $0 < \beta < 1/4$ and $0 < \alpha < 1/4$, substitution of $t = k$ in the dictionary gives

$$a(k) > v, \quad Q(k) > e, \quad \log Q(k) > 1, \quad \alpha^{-1} > 4.$$

Since $d \geq 1$ and $C_{\text{blk}} \geq 1$,

$$m(k) \geq C_{\text{blk}} \frac{d^2}{\alpha} a(k) \log Q(k) > 4v. \quad (17)$$

Thus $n_0 = km(k) \geq m(k) \geq v$. Requiring $C_{\text{blk}} \geq 1$ does not add a new parameter assumption: it is one of the finite lower bounds included in the setting's choice of a "sufficiently large universal" block constant. The zero-dimensional case is separate. By Lemma A.9, $v = 0$ is exactly the singleton, $d = 0$ case, and $|\mathcal{E}_{\bar{C}}(\bar{S})| = 1$, including for the empty sample. The setting's no-data learner therefore bypasses the positive-dimensional trace union. The undefined expression $(en_0/v)^v$ is never invoked at $v = 0$.

On the positive branch, Lemma A.2 supplies the exact legal dictionary, Proposition A.2 fixes the realized k, m, n_0 , and the positive-range calculation above derives $1 \leq v \leq n_0$ from those outputs. Proposition A.15 then gives the exact chain

$$|\mathcal{E}_{\bar{C}}(\bar{S})| = |\mathcal{P}_{\bar{C}}(\bar{S})| = |\bar{C}|_{A_{\bar{S}}}. \quad (18)$$

Thus neither xor nor repeated quotient records cost a multiplicative factor. Lemma A.10 bounds this finite restriction family, and Lemma A.11 compresses the resulting sum. Hence

$$|\mathcal{E}_{\bar{C}}(\bar{S})| \leq \Pi_{\bar{C}}(n_0) \leq \sum_{j=0}^v \binom{n_0}{j} \leq \left(\frac{en_0}{v}\right)^v, \quad (19)$$

which is the exact VC-trace bound.

The proof of Lemma A.10 operates only on the finite set of restrictions inside $\{0,1\}^{A_{\bar{S}}}$. It therefore applies when $\bar{C}$ is infinite and does not select or count concepts. The boundary calculations in Lemma A.11 explicitly retain $v = 1$, $v = d$, and $n_0 = v$, while Proposition A.15 covers every duplicate-record pattern. All downstream source lists may continue to contain actual quotient functions; (19) counts only the exact error vectors used in the finite trace union. $\square$

A.6 Fixed-Trace Concentration Event

The source event uses its internal accuracy parameter at thresholds one third and one half of that parameter. Here the parameter is $\gamma = \alpha/16$, so the thresholds and relative tolerance below are the exact source thresholds in the present notation.

For every fixed positive-branch quotient master sample $\bar{S}$, Lemma A.11 and Proposition A.16 give the exact finite trace interface

$$|\mathcal{E}_{\bar{C}}(\bar{S})| \leq \Pi_{\bar{C}}(n_0) \leq (en_0/v)^v, \quad 1 \leq v \leq n_0. \quad (1)$$

The only external tail used in the high-error branch is Lyu's without-replacement proposition (Lyu, 2025): for fixed bits $x_1, \dots, x_N \in \{0,1\}$, their mean $p = N^{-1} \sum_r x_r$, a uniformly random subset $A \subseteq [N]$ of size t , and $0 < \zeta < 1$,

$$\Pr \left[\left| t^{-1} \sum_{r \in A} x_r - p \right| > \zeta p \right] \leq 2 \exp(-\zeta^2 t p / 3). \quad (2)$$

Here $N = n_0$, $t = m$, $x_r = z_r$, $p = \mu(z)$, and $\zeta = 1/(5d)$. The marginal block law proved next discharges uniform sampling; $n_0 = km$ and $k \geq 2$ give $t \leq N$, and $d \geq 1$ gives $0 < \zeta < 1$. Equation (2) is not used for the near-zero branch.

Proposition A.17 (Marginal law of a uniform labeled block partition). *Under Assumption 4 and the formalized setting's positive-branch construction, let $d \geq 1$, $k \geq 2$, let m be the exact integer produced by the displayed ceiling, and let $n_0 = km$. Fix any indexed master sample $\bar{S} = ((q_r, y_r))_{r=1}^{n_0}$. If $\mathcal{P} = (B_1, \dots, B_k)$ is uniform over all labeled partitions of $[n_0]$ with $|B_i| = m$, then, conditional on $\bar{S}$, for every $i \in [k]$ and every $A \subseteq [n_0]$ of size m ,*

$$\Pr_{\mathcal{P}}[B_i = A \mid \bar{S}] = \binom{n_0}{m}^{-1}. \quad (3)$$

This is a marginal statement only. The blocks are pairwise disjoint and jointly cover $[n_0]$, and hence are generally dependent. In particular, when $k = 2$, $B_2 = [n_0] \setminus B_1$.

Proof. The partition is drawn independently of the record values, so conditioning on $\bar{S}$ does not change its law. The number of labeled equal-size partitions is

$$\frac{n_0!}{(m!)^k}.$$

After fixing $B_i = A$, the remaining $n_0 - m$ indices can be assigned to the remaining labeled blocks in

$$\frac{(n_0 - m)!}{(m!)^{k-1}}$$

ways. Their ratio is

$$\frac{(n_0 - m)!}{(m!)^{k-1}} \frac{(m!)^k}{n_0!} = \frac{m!(n_0 - m)!}{n_0!} = \binom{n_0}{m}^{-1},$$

which proves (3). Disjointness and coverage are part of the definition of a partition and give the dependence statements, including the complement identity for $k = 2$. No later argument replaces (3) by a joint product law. $\square$

Lemma A.12 (High-error relative fixed-trace tail). *Under Assumption 4, the positive-branch definitions $d \geq 1$, $\gamma = \alpha/16$, and $a_d = 1/(5d)$, Proposition A.17, and Lyu (Lyu, 2025) v1 Proposition 2, instantiated in (2), fix an arbitrary trace $z = (z_r)_{r=1}^{n_0} \in \{0, 1\}^{n_0}$ and define*

$$\mu(z) := \frac{1}{n_0} \sum_{r=1}^{n_0} z_r, \quad \hat{\mu}_i(z) := \frac{1}{m} \sum_{r \in B_i} z_r. \quad (4)$$

If $\mu(z) > \gamma/3$, then for every $i \in [k]$,

$$\begin{aligned} \Pr_{\mathcal{P}} [|\hat{\mu}_i(z) - \mu(z)| > a_d \mu(z) \mid \bar{S}] \\ \leq 2 \exp\left(-\frac{m\mu(z)}{75d^2}\right) \leq 2 \exp\left(-\frac{m\alpha}{3600d^2}\right). \end{aligned} \quad (5)$$

The good relative interval is closed, so equality at either relative endpoint is not a failure event in (5).

Proof. Conditional on $\bar{S}$, the vector z is fixed, and Proposition A.17 makes B_i a uniformly random size- m subset of the n_0 indexed coordinates. In (2), instantiate

$$N = n_0, \quad t = m, \quad p = \mu(z), \quad \zeta = a_d = \frac{1}{5d}.$$

The conditions $t \leq N$ and $0 < \zeta < 1$ hold because $n_0 = km$, $k \geq 2$, and $d \geq 1$. Thus (2) gives

$$2 \exp\left(-\frac{a_d^2 m \mu(z)}{3}\right) = 2 \exp\left(-\frac{m\mu(z)}{75d^2}\right).$$

The branch premise and $\gamma = \alpha/16$ give $\mu(z) > \gamma/3 = \alpha/48$. Hence

$$\frac{m\mu(z)}{75d^2} > \frac{m\alpha}{75 \cdot 48 d^2} = \frac{m\alpha}{3600d^2},$$

which proves the second inequality in (5). Lyu (Lyu, 2025)'s proposition controls the strict event **greater than the relative tolerance**; its complement is therefore exactly the closed interval used in the event definition. $\square$

$\square$

Lemma A.13 (Finite-population exponential-moment domination). *Under Proposition A.17, fix an arbitrary binary trace z , define $\mu(z)$ and $\hat{\mu}_i(z)$ by (4), and fix a block $i \in [k]$. Then for every $\lambda \geq 0$, conditional on the fixed master sample,*

$$\mathbb{E}_{\mathcal{P}} \left[e^{\lambda m \hat{\mu}_i(z)} \mid \bar{S} \right] \leq \left(1 - \mu(z) + \mu(z) e^{\lambda} \right)^m. \quad (6)$$

No independence between selected coordinates or between different blocks is asserted.

Proof. Write $N = n_0$, $a_r = e^{\lambda z_r} \geq 0$, and let $e_m(a_1, \dots, a_N)$ be the degree- m elementary symmetric sum. By the exact marginal law (3),

$$\begin{aligned} \mathbb{E}_{\mathcal{P}} \left[e^{\lambda \sum_{r \in B_i} z_r} \mid \bar{S} \right] &= \binom{N}{m}^{-1} \sum_{A \subseteq [N]: |A|=m} \prod_{r \in A} a_r \\ &= \binom{N}{m}^{-1} e_m(a_1, \dots, a_N). \end{aligned} \quad (7)$$

We prove the symmetric-mean inequality needed to bound (7). For two coordinates a, b and the vector $\mathbf{c}$ of all remaining coordinates,

$$e_m(a, b, \mathbf{c}) = e_m(\mathbf{c}) + (a + b)e_{m-1}(\mathbf{c}) + ab e_{m-2}(\mathbf{c}), \quad (8)$$

with the usual conventions $e_0 = 1$ and $e_j = 0$ outside the available degrees. Replacing a, b by their average preserves their sum and changes their product by

$$\left(\frac{a+b}{2} \right)^2 - ab = \frac{(a-b)^2}{4} \geq 0.$$

Since every elementary symmetric sum of the nonnegative coordinates $\mathbf{c}$ is nonnegative, (8) shows that pairwise averaging cannot decrease e_m . To make the limiting step explicit, repeatedly replace a current maximum coordinate a and a current minimum coordinate b by their average. The coordinate sum is preserved, while the sum of squares decreases by

$$a^2 + b^2 - 2 \left(\frac{a+b}{2} \right)^2 = \frac{(a-b)^2}{2}.$$

The nonnegative sum of squares cannot decrease by a fixed positive amount infinitely often. Hence the maximum-minus-minimum range tends to zero, and, because the sum stays fixed, the vector converges to its constant mean vector. Continuity of e_m now gives

$$e_m(a_1, \dots, a_N) \leq \binom{N}{m} \left(\frac{1}{N} \sum_{r=1}^N a_r \right)^m. \quad (9)$$

For the current binary population,

$$\frac{1}{N} \sum_{r=1}^N a_r = \frac{1}{N} \sum_{r=1}^N e^{\lambda z_r} = 1 - \mu(z) + \mu(z) e^{\lambda}.$$

Substitution in (7) proves (6). This proof uses only the single-block marginal law. $\square$

$\square$

Lemma A.14 (Low-error one-sided fixed-trace tail). *Under Assumption 4, the positive-branch definitions $d \geq 1$ and $\gamma = \alpha/16$, and Lemma A.13, fix an arbitrary trace z with $0 \leq \mu(z) \leq \gamma/3$. Then every $i \in [k]$ satisfies*

$$\begin{aligned} \Pr_{\mathcal{P}}[\hat{\mu}_i(z) > \gamma/2 \mid \bar{S}] &\leq \exp(-m\gamma/30) \\ &= \exp(-m\alpha/480) \\ &\leq \exp\left(-\frac{m\alpha}{3600d^2}\right). \end{aligned} \tag{10}$$

If $\mu(z) = 0$, the failure event is empty. The conclusion includes the branch boundary $\mu(z) = \gamma/3$.

Proof. Put $p = \mu(z)$ and $q = \gamma/2$. Because $0 < \alpha < 1/4$, one has $0 < q < 1$. If $p = 0$, every coordinate of the binary trace is zero. Therefore $\hat{\mu}_i(z) = 0$ for every partition, and the claimed failure probability is zero.

Now suppose $0 < p \leq \gamma/3 < q$. For every $\lambda > 0$, exponential Markov and Lemma A.13 give

$$\begin{aligned} \Pr_{\mathcal{P}}[\hat{\mu}_i(z) > q \mid \bar{S}] &\leq e^{-\lambda m q} \mathbb{E}_{\mathcal{P}}[e^{\lambda m \hat{\mu}_i(z)} \mid \bar{S}] \\ &\leq \exp\{m[-\lambda q + \log(1 - p + pe^{\lambda})]\}. \end{aligned} \tag{11}$$

The minimizing value is

$$\lambda_* := \log \frac{q(1-p)}{p(1-q)} > 0.$$

Direct substitution in (11) yields

$$\Pr_{\mathcal{P}}[\hat{\mu}_i(z) > q \mid \bar{S}] \leq e^{-mD(q\|p)}, \tag{12}$$

where

$$D(q\|p) := q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p} \tag{13}$$

is the binary relative entropy. For fixed q and $0 < p < q$,

$$\frac{\partial}{\partial p} D(q\|p) = \frac{p-q}{p(1-p)} < 0. \tag{14}$$

Consequently (13) is minimized over $0 < p \leq p_0 := \gamma/3$ at $p = p_0$. At this endpoint, $q = (3/2)p_0$. For any $0 < p_0 < q < 1$, set $u = (q - p_0)/(1 - p_0)$. The elementary inequality

$$\log(1-u) \geq -\frac{u}{1-u}, \quad 0 \leq u < 1, \tag{15}$$

follows because the difference between the two sides is zero at $u = 0$ and has derivative $u/(1-u)^2 \geq 0$. Since $1-q = (1-p_0)(1-u)$, (15) gives

$$(1-q) \log \frac{1-q}{1-p_0} \geq -(q-p_0). \tag{16}$$

It follows that

$$\begin{aligned} D(q\|p_0) &\geq q \log(q/p_0) - (q-p_0) \\ &= p_0 \left(\frac{3}{2} \log \frac{3}{2} - \frac{1}{2} \right). \end{aligned} \tag{17}$$

For $x \geq 0$,

$$\log(1+x) \geq \frac{2x}{2+x}, \quad (18)$$

because the difference is zero at zero and its derivative is $x^2/((1+x)(2+x)^2) \geq 0$. Taking $x = 1/2$ in (18) gives $\log(3/2) \geq 2/5$. Thus (17) implies

$$D(q\|p_0) \geq p_0 \left(\frac{3}{2} \frac{2}{5} - \frac{1}{2} \right) = \frac{p_0}{10} = \frac{\gamma}{30}. \quad (19)$$

Equations (12), (14), and (19) prove the first line of (10), including $p = p_0 = \gamma/3$. Finally, $\gamma/30 = \alpha/480$, and $d \geq 1$ gives

$$\frac{1}{480} \geq \frac{1}{3600d^2}.$$

This proves the remaining lines of (10). $\square$

$\square$

Proposition A.18 (Exact simultaneous source trace event). *Under Assumption 4, the setting-defined positive branch, Lemma A.11, Proposition A.16, Lemma A.12, and Lemma A.14, fix an arbitrary labeled quotient master sample $\bar{S}$ and let $\mathcal{P} = (B_1, \dots, B_k)$ be the uniform labeled partition. For every $z \in \mathcal{E}_{\bar{C}}(\bar{S})$, use the means in (4), and define*

$$\begin{aligned} E_{\text{good}}(\bar{S}, \mathcal{P}) := & \bigcap_{z \in \mathcal{E}_{\bar{C}}(\bar{S})} \bigcap_{i=1}^k \left(\{\mu(z) > \gamma/3\} \cap \{|\hat{\mu}_i(z) - \mu(z)| \leq a_d \mu(z)\} \right. \\ & \left. \cup \{\mu(z) \leq \gamma/3\} \cap \{0 \leq \hat{\mu}_i(z) \leq \gamma/2\} \right). \end{aligned} \quad (20)$$

Then, with $c_{\text{tr}} := 1/3600$,

$$\begin{aligned} \Pr_{\mathcal{P}}[E_{\text{good}}(\bar{S}, \mathcal{P})^c \mid \bar{S}] & \leq 4k |\mathcal{E}_{\bar{C}}(\bar{S})| \exp\left(-c_{\text{tr}} \frac{m\alpha}{d^2}\right) \\ & \leq 4k \Pi_{\bar{C}}(n_0) \exp\left(-c_{\text{tr}} \frac{m\alpha}{d^2}\right) \\ & \leq 4k \left(\frac{en_0}{v}\right)^v \exp\left(-c_{\text{tr}} \frac{m\alpha}{d^2}\right). \end{aligned} \quad (21)$$

The event (20) is exactly Lyu (Lyu, 2025)'s source event after mapping the source's internal accuracy to γ : for every $\bar{c} \in \bar{C}$, its error trace gives its master and block empirical errors, and concepts with the same trace impose the same inequalities.

Proof. For every trace-block pair, define the two branch-qualified failure events

$$\begin{aligned} F_{z,i}^{\text{H}} & := \{\mu(z) > \gamma/3\} \cap \{|\hat{\mu}_i(z) - \mu(z)| > a_d \mu(z)\}, \\ F_{z,i}^{\text{L}} & := \{\mu(z) \leq \gamma/3\} \cap \{\hat{\mu}_i(z) > \gamma/2\}. \end{aligned} \quad (22)$$

The lower endpoint in the low branch needs no failure event because every block mean of a binary trace is nonnegative. The predicates in (22) split all possible master means and assign the equality $\mu(z) = \gamma/3$ to the low branch. Hence

$$E_{\text{good}}(\bar{S}, \mathcal{P})^c \subseteq \bigcup_{z \in \mathcal{E}_{\bar{C}}(\bar{S})} \bigcup_{i=1}^k (F_{z,i}^{\text{H}} \cup F_{z,i}^{\text{L}}). \quad (23)$$

Lemma A.12 gives

$$\Pr(F_{z,i}^H \mid \bar{S}) \leq 2 \exp\left(-\frac{m\alpha}{3600d^2}\right), \quad (24)$$

where the probability is zero if its branch predicate is false. Likewise, Lemma A.14 gives

$$\Pr(F_{z,i}^L \mid \bar{S}) \leq \exp\left(-\frac{m\alpha}{3600d^2}\right) \leq 2 \exp\left(-\frac{m\alpha}{3600d^2}\right). \quad (25)$$

Apply the union bound to (23), retain the four-term envelope per trace-block pair from (24)–(25), and use the trace estimate supplied by Lemma A.11 and Proposition A.16 in (1). This proves all three lines of (21). Only the one-block marginal law (3) entered (24) and (25). The proof neither factors a joint probability nor multiplies block success probabilities, so the dependence of the single partition’s blocks is irrelevant.

For exact object matching, if $z_r = \mathbf{1}\{\bar{c}(q_r) \neq y_r\}$, then $\mu(z)$ is precisely $\text{err}_{\bar{S}}(\bar{c})$, and $\hat{\mu}_i(z)$ is precisely $\text{err}_{\bar{S}_i}(\bar{c})$. Thus (20) is neither a trace representative event nor a changed empirical metric. The positive branch fixes the actual integer m through its ceiling and sets $n_0 = km$. Proposition A.17 therefore makes every block, conditional on an arbitrary fixed master sample, a uniform size- m sample without replacement. This remains true at $k = 2$, even though then the second block is completely determined by the first.

For a fixed exact error trace above $\gamma/3$, Lemma A.12 instantiates the Lyu (Lyu, 2025) proposition with the unchanged relative tolerance $1/(5d)$ and gives exponent $m\alpha/(3600d^2)$. For a trace at or below $\gamma/3$, Lemma A.13 derives the without-replacement exponential moment directly, and Lemma A.14 turns it into the one-sided threshold $\gamma/2$. That lemma treats mean zero deterministically and includes the equality boundary $\mu = \gamma/3$.

Definition (20) and the union-bound calculation (23)–(25) give the exact two-clause source event simultaneously over the finite trace family produced above and all k blocks. Its finite union retains exactly the required $4k$ envelope and proves

$$\Pr_{\mathcal{P}}[E_{\text{good}}(\bar{S}, \mathcal{P})^c \mid \bar{S}] \leq 4k\Pi_{\bar{C}}(n_0) \exp\left(-\frac{m\alpha}{3600d^2}\right). \quad (26)$$

No factor $d + 1$ is present: the single event (20) is defined once from master and block errors and is reused at every fixed stage. No independence among blocks is used. This is the conditional trace interface used by the fixed-point calculation below.

□

A.7 Trace Fixed Point and Confidence Budget

The deterministic positive-branch dictionary furnished by Lemma A.2 and Proposition A.2 is

$$\beta_{\text{tr}} := \beta/4, \quad a(t) := v + \log(4t/\beta), \quad Q(t) := e + \frac{etd^2a(t)}{\alpha v}, \quad (1)$$

$$m(t) := \left\lceil C_{\text{blk}} \frac{d^2}{\alpha} a(t) \log Q(t) \right\rceil, \quad n(t) := tm(t). \quad (2)$$

For the least feasible public teacher count $k \geq 2$, set

$$a := a(k), \quad Q := Q(k) = Q_{\text{blk}}, \quad m := m(k), \quad n_0 := n(k) = km. \quad (3)$$

No sample, partition, trace, event, stage, or mechanism output enters (1)–(3). Fix the universal block constant once and for all so that

$$C_{\text{blk}} > 0, \quad \frac{C_{\text{blk}}}{3600} \geq 4 + \log(1 + C_{\text{blk}}). \quad (4)$$

For example, $C_{\text{blk}} = 144000$ is admissible: the left side is 40, whereas $1 + 144000 < 2^{18} < e^{18}$ gives $4 + \log(1 + 144000) < 22$. This is a universal specialization of the setting's sufficiently large block constant, not a parameter-dependent condition.

Proposition A.18 supplies, for every fixed master sample, the exact conditional estimate

$$\Pr_{\mathcal{P}}[E_{\text{good}}(\bar{S}, \mathcal{P})^c \mid \bar{S}] \leq 4k \left(\frac{en_0}{v} \right)^v \exp\left(-\frac{m\alpha}{3600d^2}\right). \quad (5)$$

This is the only probabilistic input needed for the fixed-point calculation below; all remaining steps are algebraic.

Lemma A.15 (Exact ceiling-aware trace fixed point). *Under Assumption 4, Lemma A.2, and Proposition A.2, suppose $d \geq 1$, and use the exact positive-branch dictionary (1)-(3). Define the appendix-local universal constant*

$$C_{\text{fp}} := 2 + \log(1 + C_{\text{blk}}). \quad (6)$$

Then

$$\frac{en_0}{v} \leq (1 + C_{\text{blk}})Q \log Q, \quad \log \frac{en_0}{v} \leq C_{\text{fp}} \log Q, \quad (7)$$

and the exact block size also obeys

$$m \leq (1 + C_{\text{blk}}) \frac{d^2}{\alpha} a \log Q. \quad (8)$$

Proof. Lemma A.2 gives $1 \leq v \leq d$, while the parameter regime gives $0 < \alpha < 1/4$. Moreover $a = v + \log(4k/\beta) > v \geq 1$, and $Q > e$, hence $\log Q > 1$. Put

$$x := C_{\text{blk}} \frac{d^2}{\alpha} a \log Q. \quad (9)$$

The ceiling is represented exactly as

$$m = \lceil x \rceil = x + \theta, \quad 0 \leq \theta < 1. \quad (10)$$

This includes the integer case $\theta = 0$; no ceiling remainder is dropped. Since

$$\frac{d^2}{\alpha} a \log Q > 4, \quad (11)$$

we have $1 \leq d^2 a \log Q / \alpha$. Therefore (10) gives

$$m \leq C_{\text{blk}} \frac{d^2}{\alpha} a \log Q + 1 \leq (1 + C_{\text{blk}}) \frac{d^2}{\alpha} a \log Q,$$

which proves (8), including the additive ceiling contribution.

For the fixed point itself, substitute $n_0 = k(x + \theta)$ and separate the ceiling term:

$$\begin{aligned} \frac{en_0}{v} &= C_{\text{blk}} \log Q \left(\frac{ekd^2a}{\alpha v} \right) + \frac{ek\theta}{v} \\ &\leq C_{\text{blk}} Q \log Q + Q. \end{aligned} \quad (12)$$

Indeed, the first term uses $ekd^2a/(\alpha v) \leq Q$. For the second, $d^2a/\alpha \geq 1$ implies

$$\frac{ek\theta}{v} \leq \frac{ek}{v} \leq \frac{ekd^2a}{\alpha v} \leq Q. \quad (13)$$

Because $\log Q > 1$, the last Q in (12) is at most $Q \log Q$, which proves the first inequality in (7). Taking logarithms and using $\log \log Q \leq \log Q$ gives

$$\begin{aligned} \log \frac{en_0}{v} &\leq \log(1 + C_{\text{blk}}) + \log Q + \log \log Q \\ &\leq (2 + \log(1 + C_{\text{blk}})) \log Q = C_{\text{fp}} \log Q. \end{aligned} \quad (14)$$

The last step also uses $\log Q > 1$ to bound the fixed positive constant $\log(1 + C_{\text{blk}})$ by $\log(1 + C_{\text{blk}}) \log Q$. Equations (12)-(14) close the fixed point using the same $n_0 = km$ that appears in Q, m , and the trace count. $\square$

$\square$

Lemma A.16 (Public-witness sample envelope). *Under Assumption 4, Lemma A.3, Proposition A.2, and Lemma A.15, suppose $d \geq 1$. Define the appendix quantities*

$$\ell := \log \frac{64}{\delta\beta}, \quad R_T := \frac{d^2 \ell \Lambda^2}{\varepsilon}, \quad A_{\log} := 80 + \log(1 + C_{\text{blk}}), \quad (15)$$

and the appendix-local universal constant

$$H := A_{\log}(1 + \log C_{\text{teach}}). \quad (16)$$

Then the exact sample size obeys

$$n_0 \leq (1 + C_{\text{blk}}) \bar{k} \frac{d^2}{\alpha} a(\bar{k}) \log Q(\bar{k}), \quad (17)$$

and, more explicitly,

$$n_0 \leq 2(1 + C_{\text{blk}}) C_{\text{teach}} H \frac{d^4 \ell \Lambda^3}{\varepsilon \alpha} [v + (H + 3) \Lambda]. \quad (18)$$

In particular, with the explicit universal constant

$$K_{\text{fp}} := 2(1 + C_{\text{blk}}) C_{\text{teach}} H(H + 3), \quad (19)$$

one has the fully parameter-exposed intermediate envelope

$$n_0 \leq K_{\text{fp}} \frac{d^4 \log(64/(\delta\beta)) \Lambda^3}{\varepsilon \alpha} (v + \Lambda). \quad (20)$$

No hidden constant in (18)-(20) depends on $d, v, \alpha, \beta, \varepsilon, \delta$.

Proof. For integers $2 \leq s \leq t$, the definition (1) gives $a(s) \leq a(t)$. Because all factors are positive, $sa(s) \leq ta(t)$, and hence $Q(s) \leq Q(t)$ and $\log Q(s) \leq \log Q(t)$. Thus the right side of (8), multiplied by k , and the inequality $k \leq \bar{k}$ give

$$\begin{aligned} n_0 = km(k) &\leq (1 + C_{\text{blk}}) k \frac{d^2}{\alpha} a(k) \log Q(k) \\ &\leq (1 + C_{\text{blk}}) \bar{k} \frac{d^2}{\alpha} a(\bar{k}) \log Q(\bar{k}), \end{aligned} \quad (21)$$

which is (17). This comparison uses monotonicity only after the least feasible k and the independently verified public witness $\bar{k}$ have both been produced; it does not assume that the teacher feasible set is monotone.

The witness is exactly $\bar{k} = \lceil C_{\text{teach}} R_T \rceil$. Lemma A.3, instantiated with $C = C_{\text{teach}}$, and Proposition A.2 give

$$\bar{k} \leq 2C_{\text{teach}}R_T, \quad \log \bar{k} \leq H\Lambda, \quad \log Q(\bar{k}) \leq H\Lambda. \quad (22)$$

Since $\Lambda \geq 1$, $\log(1/\beta) \leq \Lambda$, and $\log 4 < 2$,

$$\begin{aligned} a(\bar{k}) &= v + \log 4 + \log \bar{k} + \log(1/\beta) \\ &\leq v + (H + 3)\Lambda. \end{aligned} \quad (23)$$

Insert (22)-(23) and $R_T = d^2\ell\Lambda^2/\varepsilon$ into (17) to obtain (18). Finally, $H + 3 \geq 1$ implies

$$v + (H + 3)\Lambda \leq (H + 3)(v + \Lambda),$$

so (19) turns (18) into (20). Every use of a ceiling is upstream in the bound $\bar{k} \leq 2C_{\text{teach}}R_T$ or explicitly retained in (8); no additive integer term has been hidden. $\square$

$\square$

Proposition A.19 (Exact trace-exponent domination and conditional charge). *Under Assumption 4, Proposition A.18, Lemma A.15, and the universal block calibration (4), suppose $d \geq 1$. Then*

$$\boxed{\frac{m\alpha}{3600d^2} \geq v \log \frac{en_0}{v} + \log \frac{4k}{\beta_{\text{tr}}}} \quad (24)$$

and, for every fixed indexed quotient master sample $\bar{S}$,

$$\Pr_{\mathcal{P}}[E_{\text{good}}(\bar{S}, \mathcal{P})^c \mid \bar{S}] \leq \beta_{\text{tr}} = \beta/4. \quad (25)$$

Proof. The lower side of the exact ceiling (2) is

$$m \geq C_{\text{blk}} \frac{d^2}{\alpha} a \log Q,$$

so

$$\frac{m\alpha}{3600d^2} \geq \frac{C_{\text{blk}}}{3600} a \log Q. \quad (26)$$

The two positive terms that must be paid are controlled separately. First, Lemma A.15 and $v \leq a$ give

$$v \log \frac{en_0}{v} \leq C_{\text{fp}} v \log Q \leq C_{\text{fp}} a \log Q. \quad (27)$$

Second, put $b := \log(4k/\beta) > 0$, so $a = v + b$. Since $\beta_{\text{tr}} = \beta/4$, $\log 4 < 2 \leq 2v$, and $\log Q > 1$,

$$\begin{aligned} \log \frac{4k}{\beta_{\text{tr}}} &= \log \frac{16k}{\beta} = b + \log 4 \\ &\leq b + 2v \leq 2(v + b) = 2a \leq 2a \log Q. \end{aligned} \quad (28)$$

Combining (27)-(28), and recalling $C_{\text{fp}} = 2 + \log(1 + C_{\text{blk}})$, yields the exact opposing-term bound

$$\begin{aligned} v \log \frac{en_0}{v} + \log \frac{4k}{\beta_{\text{tr}}} &\leq (C_{\text{fp}} + 2)a \log Q \\ &= (4 + \log(1 + C_{\text{blk}}))a \log Q \\ &\leq \frac{C_{\text{blk}}}{3600} a \log Q \leq \frac{m\alpha}{3600d^2}, \end{aligned} \quad (29)$$

where the penultimate inequality is precisely (4), and the last is (26). This proves (24) with every multiplicity and confidence term visible.

Now apply the conditional finite-union estimate (5). Taking the logarithm of its deterministic upper bound and using (24),

$$\begin{aligned} \log(4k) + v \log \frac{en_0}{v} - \frac{m\alpha}{3600d^2} \\ \leq \log(4k) - \log \frac{4k}{\beta_{\text{tr}}} = \log \beta_{\text{tr}}. \end{aligned} \quad (30)$$

Exponentiating (30) proves (25). The finite trace/block union is already the one in Proposition A.18; no new factor, stage union, or independence assertion is inserted. $\square$

Proposition A.20 (Tower conversion for the trace ledger). *Under Assumption 4, Proposition A.18, and Proposition A.19, suppose $d \geq 1$. Let ν be any probability law on the setting-defined discrete quotient master-sample space and, after drawing the sample, draw the data-independent uniform labeled partition $\mathcal{P}$ prescribed by the setting. Then*

$$\Pr_{\bar{S} \sim \nu, \mathcal{P}} [E_{\text{good}}(\bar{S}, \mathcal{P})^c] \leq \beta_{\text{tr}}. \quad (31)$$

Proof. The partition space is finite. For each fixed sample, Proposition A.18 forms the bad section by a finite union over its finite exact trace set and the k labeled blocks. Thus its conditional section probability is the function

$$r(\bar{s}) := \Pr_{\mathcal{P}} [E_{\text{good}}(\bar{s}, \mathcal{P})^c \mid \bar{S} = \bar{s}].$$

On the setting's quotient sample interface this section is measurable; in particular, the quotient sample space is discrete in the declared branch, and the finite-partition sum has measurable sections. Proposition A.19 proves the pointwise inequality $0 \leq r(\bar{s}) \leq \beta_{\text{tr}}$ for every $\bar{s}$, without a distributional or realizability hypothesis. Therefore the tower identity is simply

$$\begin{aligned} \Pr[E_{\text{good}}^c] &= \int r(\bar{s}) \nu(d\bar{s}) \\ &\leq \int \beta_{\text{tr}} \nu(d\bar{s}) = \beta_{\text{tr}}. \end{aligned} \quad (32)$$

This is a mixture of one-partition failure probabilities. It neither conditions on a successful generated event nor multiplies block success probabilities. The same event is reused across the fixed source stages, so there is no additional stage union in (32). $\square$

Proposition A.21 (Boundary integrity and null-branch bypass). *Under Assumption 4, Proposition A.1, Proposition A.18, Lemma A.15, Lemma A.16, and Proposition A.19, the following piecewise conclusions hold: Item 7 applies when $d = 0$, and Items 1-6 apply when $d \geq 1$.*

1. At $v = 1$, (7), (18), (24), and the charge (25) remain valid with no division by zero and with trace logarithm $\log(en_0)$. 2. At $v = d \geq 1$, (7), (18), and (24) remain valid, the trace term is $d \log(en_0/d)$, and the factor $d^4(v + \Lambda)$ in (20) becomes $d^4(d + \Lambda)$, whose polynomial v -term is d^5 . This is only the fixed-point baseline specialization, not a final sample-complexity theorem. 3. At $k = 2$, all scalar inequalities remain valid. No proof line requires block independence; (5) already treats the two complementary blocks through their marginal laws and a finite union. 4. At $d = v = 1$, every denominator in (1)-(3), (7), and (24) is positive, $Q = e + eka/\alpha > e$, and the same constants

apply. 5. The ceiling is retained in both useful directions: $m \geq C_{\text{blk}} d^2 a \log Q / \alpha$ in (26), while (10)-(13) pay its entire additive remainder in the logarithmic fixed point and (8), (17) pay it in the sample envelope. 6. For every $0 < \alpha < 1/4$, the proof is uniform and uses no positive lower cutoff on α . Thus it remains valid along any sequence $\alpha \downarrow 0$; the explicit $1/\alpha$, Q , and logarithmic growth remain visible. The excluded value $\alpha = 0$ is not claimed. 7. If $d = 0$, Proposition A.1 gives the exact $N = 0$ singleton law. The positive-branch fact $v \geq 1$, the partition, m, k, n_0, Q , and E_{good} are not constructed, so no expression containing v^{-1} or d^{-2} is evaluated and the trace ledger contributes zero failure.

Proof. For Item 1, the positive-branch range is $1 \leq v \leq d$, and substitution $v = 1$ in (1), (7), (18), and (24) leaves every quantity finite. For Item 2, substitution $v = d$ turns $d^4(v + \Lambda)$ in (20) into $d^4(d + \Lambda)$, while the exact trace power in (5) becomes d ; no inequality changes direction. Item 3 follows because $k \geq 2$ is used only through positive logarithms and (5), whose proof did not factor the block law. Item 4 is the common specialization of Items 1-3 and the strict parameter ranges.

Item 5 is exactly the pair of ceiling inequalities (10), (12)-(13), and (26). For Item 6, every use of α is through its strict positive range, (11), or an explicitly displayed $1/\alpha$ term; the constants in (4), (6), (16), and (19) are independent of α . Item 7 is precisely the branch conclusion of Proposition A.1; it is logically prior to every positive-branch definition. Lemma A.2 and Proposition A.2 first produce the exact public dictionary (1)-(3), including the least feasible k , its independent witness $\bar{k}$, the ceiled block size m , and $n_0 = km$, before a sample or event exists. Lemma A.15 then separates the ceiling as $m = x + \theta$ and proves

$$\log(en_0/v) \leq (2 + \log(1 + C_{\text{blk}})) \log Q. \quad (33)$$

Thus (33) closes the fixed-point relation noncircularly at the exact public quantity $Q = Q(k)$ and the actual sample size. Lemma A.16 next uses $k \leq \bar{k}$ and the ceiling-aware witness bounds to prove both the exact auxiliary envelope (17) and the fully exposed intermediate envelope (20).

Proposition A.18 supplies the conditional finite-union bound (5) with the actual m . Proposition A.19 proves the exact bridge inequality

$$\frac{m\alpha}{3600d^2} \geq v \log(en_0/v) + \log(4k/\beta_{\text{tr}}),$$

so that (5) is at most β_{tr} for every fixed master sample. Proposition A.20 integrates this pointwise bound over the master sample and proves the same unconditional trace charge. Finally, the boundary calculations above verify $v = 1$, $v = d$, $k = 2$, $d = v = 1$, the complete ceiling remainder, arbitrary positive α tending to zero, and the exact $d = 0$ bypass.

Consequently, the fixed point, trace confidence charge, and explicit intermediate sample envelope all hold with the stated parameters. □

A.8 Source-Current Restrictions and Essential Lists

On the positive branch $d \geq 1$, put

$$\gamma := \alpha/16, \quad a_d := \frac{1}{5d}, \quad \rho := 1 - \frac{1}{2d}. \quad (1)$$

For an indexed quotient master sample $\bar{S}$, its block $\bar{S}_i$, and $\bar{h} \in \bar{C}$, write

$$e(\bar{h}) := \text{err}_{\bar{S}}(\bar{h}), \quad e_i(\bar{h}) := \text{err}_{\bar{S}_i}(\bar{h}). \quad (2)$$

The endpoint-complete current restrictions and structural parameters are

$$H_i^r := \{\bar{h} \in \bar{C} : e_i(\bar{h}) \leq \rho^{r+1}\gamma\}, \quad p_r := 2^r n_0 d, \quad 0 \leq r \leq d. \quad (3)$$

On the event produced by Proposition A.18, simultaneously for every $\bar{h} \in \bar{C}$ and $i \in [k]$,

$$\begin{cases} (1 - a_d)e(\bar{h}) \leq e_i(\bar{h}) \leq (1 + a_d)e(\bar{h}), & e(\bar{h}) > \gamma/3, \\ 0 \leq e_i(\bar{h}) \leq \gamma/2, & e(\bar{h}) \leq \gamma/3. \end{cases} \quad (4)$$

Equality at $e(\bar{h}) = \gamma/3$ belongs to the second branch.

We also record the exact structural interfaces used below from Lyu (Lyu, 2025). A nonempty binary class $\mathcal{A}$ on Q_C is q -irreducible when, for every q -tuple $x_1, \dots, x_q \in Q_C$,

$$\text{LD}(\mathcal{A}|_{(x_1, \text{SOA}_{\mathcal{A}}(x_1)), \dots, (x_q, \text{SOA}_{\mathcal{A}}(x_q))}) = \text{LD}(\mathcal{A}). \quad (5)$$

For positive integers p, d and nonempty $\mathcal{A} \subseteq \{0, 1\}^{Q_C}$ with $\text{LD}(\mathcal{A}) \leq d$, a valid (p, d) -decomposition is a finite binary restriction tree whose node classes $\mathcal{A}_u$ and leaf classes $\mathcal{A}_\ell$ obey

$$\text{depth}(u) \leq p(2^{d - \text{LD}(\mathcal{A}_u) + 1} - 1), \quad (6)$$

$$\text{depth}(\ell) \leq p(2^{d - \text{LD}(\mathcal{A}_\ell)} - 1), \quad \mathcal{A}_\ell \text{ is } p2^{d - \text{LD}(\mathcal{A}_\ell)}\text{-irreducible}. \quad (7)$$

Its degree is the largest leaf Littlestone dimension, and

$$\text{DDim}_{p,d}(\mathcal{A}) := \min\{\text{degree}(T) : T \text{ is valid}\}. \quad (8)$$

Such a valid tree exists, the minimum is attained, and every valid tree has at most

$$p^d 2^{d^2} \quad (9)$$

leaves. If nonempty $\mathcal{G} \subseteq \mathcal{H}$ have Littlestone dimension at most d , then arbitrary optimal $(2p, d)$ - and (p, d) -decompositions satisfy

$$\text{DDim}_{2p,d}(\mathcal{G}) \leq \text{DDim}_{p,d}(\mathcal{H}). \quad (10)$$

Only when

$$\text{DDim}_{2p,d}(\mathcal{G}) = \text{DDim}_{p,d}(\mathcal{H}) = t \quad (11)$$

does every dimension- t leaf $\mathcal{G}_v$ of the first tree have a dimension- t leaf $\mathcal{H}_u$ of the second tree with

$$\text{SOA}_{\mathcal{H}_u} = \text{SOA}_{\mathcal{G}_v} \quad \text{as functions } Q_C \rightarrow \{0, 1\}. \quad (12)$$

Finally, with $t = \text{DDim}_{p,d}(\mathcal{A})$, an actual function $f : Q_C \rightarrow \{0, 1\}$ is (p, d) -essential to $\mathcal{A}$ when every optimal decomposition has a leaf satisfying

$$\text{LD}(\mathcal{A}_\ell) = t, \quad \text{SOA}_{\mathcal{A}_\ell} = f \quad \text{pointwise on } Q_C. \quad (13)$$

The corresponding corollary gives at most $p^d 2^{d^2}$ essential functions, inheritance from a subclass with the same (p, d) -DDim, existence when the $(2p, d)$ - and (p, d) -DDim values agree, and, at DDim zero, equality of the essential set with the finite nonempty class.

Lemma A.17 (Endpoint-complete source/current stage and dimension map). *Under Assumption 1 and the setting's positive branch $d \geq 1$, use the endpoint-complete current restrictions and structural parameters in (1)–(3). Resolve the source endpoint by defining, for every $i \in [k]$,*

$$H_{i,\text{src}}^s := \{\bar{h} \in \bar{C} : e_i(\bar{h}) \leq \rho^s \gamma\}, \quad p_{s,\text{src}} := 2^s n_0 d, \quad 1 \leq s \leq d+1. \quad (14)$$

Then, for every $0 \leq r \leq d$,

$$H_i^r = H_{i,\text{src}}^{r+1}, \quad p_r = 2^r n_0 d = \frac{p_{r+1,\text{src}}}{2}, \quad (15)$$

and, for every $0 \leq r < d$,

$$p_{r+1} = 2p_r. \quad (16)$$

Every nonempty H_i^r is a binary class on Q_C with $\text{LD}(H_i^r) \leq d$, and every p_r is a positive integer. Moreover, if an actual $\bar{c}^ \in \bar{C}$ has $e_i(\bar{c}^*) = 0$ for every i , then $\bar{c}^* \in H_i^r$ for all i, r ; hence all current restrictions are nonempty under that explicit condition.*

Proof. The construction uses the range $j \leq d+1$, and in the proof prints the restriction display for $j \in [d]$, but its algorithm immediately operates in $d+1$ stages and consumes H_i^{d+1} . Thus the source's operational endpoint convention is the same formula at $s = d+1$. Equation (14) makes that necessary endpoint explicit; it does not introduce an additional stage or a new threshold.

Substituting $s = r+1$ in (14) and comparing with (3) proves both identities in (15). Equation (16) is the exact computation

$$p_{r+1} = 2^{r+1} n_0 d = 2(2^r n_0 d) = 2p_r.$$

In particular, the current (p_r, d) -essential list on $H_i^r = H_{i,\text{src}}^{r+1}$ is not mislabeled as the source algorithm's literal stage- $r+1$ list: its parameter is exactly one half of $p_{r+1,\text{src}}$. The source structural statements are uniform in the positive integer p , and (16) is the exact $(2p, p)$ pairing used below.

It remains to discharge the dimension premise on the quotient without assuming class finiteness. Suppose $\bar{C}$ Littlestone-shatters a finite binary tree of depth s whose nodes are labeled by elements of Q_C . For each of the finitely many node labels q , choose an $x_q \in X$ with $\kappa(x_q) = q$. Replacing every node label q by x_q gives a tree shattered by C , because each witnessing $\bar{c}$ is induced by the corresponding c and $\bar{c}(q) = c(x_q)$. Hence $s \leq \text{LD}(C) = d$, so $\text{LD}(\bar{C}) \leq d$. Monotonicity under subclasses gives $\text{LD}(H_i^r) \leq d$ whenever the outer restriction is nonempty. Finally, $n_0, d \in \mathbb{N}$ on this branch, so $p_r \in \mathbb{N}$. If $e_i(\bar{c}^*) = 0$, positivity of $\rho^{r+1}\gamma$ makes the last membership assertion immediate. This last statement is a proved conditional implication, not a realizability assumption. $\square$

$\square$

Proposition A.22 (Exact eventwise interleaving inclusion). *Under Assumption 1, Proposition A.18, and Lemma A.17, suppose $d \geq 1$ and the exact derived event E_{good} occurs. Then, for every $r \in \{0, \dots, d-1\}$ and every $i_* \in [k]$,*

$$H_{i_*}^{r+1} \subseteq \bigcap_{i=1}^k H_i^r. \quad (17)$$

This conclusion holds whether or not either side is empty.

Proof. Fix $r < d$, i_* , and $\bar{h} \in H_{i_*}^{r+1}$. By (3),

$$e_{i_*}(\bar{h}) \leq \rho^{r+2}\gamma. \quad (18)$$

First suppose $e(\bar{h}) > \gamma/3$. The high-master-error clause of the exact event gives, for every i ,

$$e_i(\bar{h}) \leq (1 + a_d)e(\bar{h}), \quad e_{i_*}(\bar{h}) \geq (1 - a_d)e(\bar{h}). \quad (19)$$

Since $1 - a_d > 0$, (18)-(19) imply

$$e_i(\bar{h}) \leq \frac{1 + a_d}{1 - a_d} e_{i_*}(\bar{h}) \leq \frac{1 + a_d}{1 - a_d} \rho^{r+2} \gamma. \quad (20)$$

The scalar comparison that turns (20) into the required inclusion is

$$\frac{1 + a_d}{1 - a_d} \rho \leq 1. \quad (21)$$

Indeed,

$$\begin{aligned} (1 - a_d) - \rho(1 + a_d) &= \frac{1}{2d} - \frac{1}{5d} \left(2 - \frac{1}{2d} \right) \\ &= \frac{d + 1}{10d^2} > 0. \end{aligned} \quad (22)$$

Both denominators in (21) are positive, so (22) proves (21), and (20) becomes $e_i(\bar{h}) \leq \rho^{r+1} \gamma$.

Now suppose $e(\bar{h}) \leq \gamma/3$. The low-master-error clause of the same event gives $e_i(\bar{h}) \leq \gamma/2$ for every i . For $x \in [0, 1]$ and an integer $s \geq 1$, the elementary inequality $(1 - x)^s \geq 1 - sx$ follows by induction: after the base case, multiplication of the induction hypothesis by $1 - x \geq 0$ gives

$$(1 - x)^{s+1} \geq (1 - sx)(1 - x) = 1 - (s + 1)x + sx^2 \geq 1 - (s + 1)x.$$

Taking $x = 1/(2d)$ and $s = d$ gives

$$\rho^d = \left(1 - \frac{1}{2d} \right)^d \geq 1 - \frac{d}{2d} = \frac{1}{2}. \quad (23)$$

Because $r + 1 \leq d$ and $0 < \rho < 1$, $\rho^{r+1} \geq \rho^d \geq 1/2$. Therefore

$$e_i(\bar{h}) \leq \gamma/2 \leq \rho^{r+1} \gamma. \quad (24)$$

In both branches $\bar{h} \in H_i^r$ for every i , proving (17). If $H_{i_*}^{r+1}$ is empty, (17) is vacuous, so the proof never assumes nonemptiness or a generated success event. $\square$

$\square$

Proposition A.23 (Finite optimal decompositions for current restrictions). *Under Assumption 1, Lemma A.17, the current-notation versions of Lyu (Lyu, 2025) Definition 4.2, its valid-decomposition existence claim, and Lyu (Lyu, 2025) Lemma 4.1, let $\mathcal{A}$ be any nonempty current restriction H_i^r , and let $p \in \mathbb{N}$. Then:*

1. *at least one finite valid (p, d) -decomposition of $\mathcal{A}$ exists;*
2. *an optimal (p, d) -decomposition exists, although it need not be unique;*
3. *every valid or optimal tree has at most $p^d 2^{d^2}$ leaves; and*
4. *if a nonempty leaf $\mathcal{A}_\ell$ has $t = \text{LD}(\mathcal{A}_\ell)$, then it is $p 2^{d-t}$ -irreducible.*

None of these conclusions assumes that $Q_C, \bar{C}$, or $\mathcal{A}$ is finite.

Proof. Lemma A.17 gives $\text{LD}(\mathcal{A}) \leq d$. Together with $p, d \in \mathbb{N}$ and nonemptiness, this discharges every premise of the source existence claim, which supplies a valid tree. In its greedy construction, only nonempty reducible leaves are expanded. Every such node has Littlestone dimension at least zero. The node-depth clause of Lyu (Lyu, 2025) Definition 4.2 therefore bounds its depth by

$$B := p(2^{d+1} - 1).$$

An empty child is never expanded. Hence the constructed binary tree has depth at most $B + 1$, and therefore at most $2^{B+2} - 1$ nodes. In particular, the produced valid tree is finite; no cardinality of $\mathcal{A}$ enters this conclusion.

Let

$$\mathcal{D} := \{\text{degree}(T) : T \text{ is a valid } (p, d)\text{-decomposition of } \mathcal{A}\}.$$

The existence claim makes $\mathcal{D}$ nonempty. Its values are integers between 0 and d : the tree covers the nonempty root class, so some leaf is nonempty, and every nonempty leaf is a subclass of $\mathcal{A}$. Hence the finite set of possible integer degrees has a least attained value t , and a tree whose degree equals t is optimal. This proves optimal attainment without compactness, enumeration, or a cardinality assumption on $\mathcal{A}$. It also leaves arbitrary nonuniqueness intact.

Lyu (Lyu, 2025) Lemma 4.1 gives at most $p^d 2^{d^2}$ leaves for every valid tree. Finally, the valid-leaf clause of Definition 4.2 says directly that a leaf of dimension t is $p2^{d-t}$ -irreducible. $\square$

$\square$

Proposition A.24 (Exact factor-two transition and actual SOA inheritance). *Under Assumption 1, Lemma A.17, Propositions A.22 and A.23, and Lyu (Lyu, 2025) Lemma 4.3, suppose E_{good} occurs. Fix $0 \leq r < d$ and $i_*, i \in [k]$. If $H_{i_*}^{r+1} \neq \emptyset$, then $H_i^r \neq \emptyset$, and the only Lemma 4.3 instantiation made here is*

$$\mathcal{G} = H_{i_*}^{r+1}, \quad \mathcal{H} = H_i^r, \quad (2p, p) = (p_{r+1}, p_r). \quad (25)$$

It yields

$$\text{DDim}_{p_{r+1}, d}(H_{i_*}^{r+1}) \leq \text{DDim}_{p_r, d}(H_i^r). \quad (26)$$

If the two sides in (26) are both equal to t , then, for every arbitrarily chosen optimal (p_{r+1}, d) -decomposition of $H_{i_*}^{r+1}$, every dimension- t leaf $\mathcal{G}_v$, and every arbitrarily chosen optimal (p_r, d) -decomposition of H_i^r , there is a dimension- t leaf $\mathcal{H}_u$ satisfying

$$\text{SOA}_{\mathcal{H}_u} = \text{SOA}_{\mathcal{G}_v} \quad \text{pointwise on } Q_C. \quad (27)$$

Consequently, under that same equal-DDim premise only,

$$\text{Ess}_{p_{r+1}, d}(H_{i_*}^{r+1}) \subseteq \text{Ess}_{p_r, d}(H_i^r), \quad (28)$$

where $\text{Ess}_{p, d}$ denotes the actual essential-function set of Definition 4.3.

Proof. Proposition A.22 gives $H_{i_*}^{r+1} \subseteq H_i^r$; thus nonemptiness of the former proves nonemptiness of the latter. Proposition A.23 supplies both arbitrary optimal trees, and Lemma A.17 supplies their $\text{LD} \leq d$ premises. Equation (16) gives

$$p_{r+1} = 2p_r,$$

so (25) is literally the source pair $(2p, p)$, with no same-scale substitution. Lyu (Lyu, 2025) Lemma 4.3 now gives (26) in the displayed direction. Its second conclusion is conditional on equality, and under precisely that premise it gives (27) as equality of actual SOA functions, not merely equality on $\bar{S}$ or a block.

For (28), let f be essential to $H_{i_*}^{r+1}$, and fix an arbitrary optimal (p_r, d) -decomposition of H_i^r . Choose any optimal (p_{r+1}, d) -decomposition of $H_{i_*}^{r+1}$. By essentiality, some dimension- t leaf $\mathcal{G}_v$ has $\text{SOA}_{\mathcal{G}_v} = f$. Equation (27) produces a dimension- t leaf in the arbitrary H_i^r tree with the same actual function f . Since the H_i^r tree was arbitrary, Definition 4.3 says that f is essential to H_i^r . This proves (28).

No conclusion stronger than (26) is drawn when the DDim values are unequal. In particular, this proposition does not turn (26) into a strict decrease and does not prove the later finite-potential descent. $\square$

$\square$

Proposition A.25 (Finite actual essential lists and the full source corollary). *Under Assumption 1, Proposition A.23, Lyu (Lyu, 2025) Definition 4.3, and Lyu (Lyu, 2025) Corollary 4.1, define, for $0 \leq r \leq d$,*

$$\mathcal{L}_i^r := \begin{cases} \text{Ess}_{p_r, d}(H_i^r), & H_i^r \neq \emptyset, \\ \emptyset, & H_i^r = \emptyset. \end{cases} \quad (29)$$

The setting's pre-fixed deterministic ordering of this finite set is the algorithmic list ordering. The mathematical set and every function identity below are independent of that ordering.

For a nonempty H_i^r , this is a finite set of actual functions $Q_C \rightarrow \{0, 1\}$, and

$$|\mathcal{L}_i^r| \leq p_r^d 2^{d^2}. \quad (30)$$

More generally, for all nonempty quotient classes to which the source corollary applies, its four conclusions remain exactly:

1. the bound (30); 2. same- (p, d) , same-DDim inheritance from a subclass to a superclass; 3. nonemptiness of the (p, d) -essential set when $\text{DDim}_{2p, d} = \text{DDim}_{p, d}$; and 4. if $\text{DDim}_{p, d}(\mathcal{A}) = 0$, then $\mathcal{A}$ is finite and $\text{Ess}_{p, d}(\mathcal{A}) = \mathcal{A}$.

The last clause is never applied to an empty class, and (29) does not identify an empty restriction with DDim zero.

Proof. For nonempty H_i^r , Proposition A.23 supplies an optimal decomposition. Definition 4.3 makes every essential item an actual SOA function of a maximal-dimensional leaf in every optimal decomposition. Fix just one such tree. Every essential function must occur among its maximal-leaf SOA functions. The tree has at most $p_r^d 2^{d^2}$ leaves, so this one finite set contains every essential function and proves (30). Duplicate SOA functions at different leaves only reduce the set cardinality.

This argument also explains why no cardinality assumption on H_i^r or $\bar{C}$ is present: a possibly infinite intersection over optimal trees is contained in the finite SOA set of any one fixed optimal tree. The remaining three conclusions are Corollary 4.1 of Lyu (Lyu, 2025) after replacing its domain by Q_C . All its hypotheses are discharged by nonemptiness, Proposition A.23, and the explicit DDim premises in the respective clauses.

If $H_i^r = \emptyset$, (29) is instead the setting's predetermined arbitrary-input totalization. No source SOA or decomposition is evaluated. If $H_i^r \neq \emptyset$, its essential set may nevertheless be empty unless a source existence clause applies; the algorithmic empty-list path is totalized in that case as well. Thus neither class nonemptiness nor list nonemptiness is smuggled into the proof. $\square$

$\square$

Proposition A.26 (Authoritative stage-list envelope). *Under Assumption 1 and Proposition A.25, use the positive-branch dictionary*

$$p_d = p_d(k) = 2^d n_0 d, \quad L(k) := p_d(k)^d 2^{d^2}. \quad (31)$$

Then, for every $i \in [k]$ and $0 \leq r \leq d$,

$$|\mathcal{L}_i^r| \leq p_r^d 2^{d^2} \leq p_d^d 2^{d^2} = L(k). \quad (32)$$

The logarithm is exactly

$$\log L(k) = d \log n_0 + d \log d + 2d^2 \log 2. \quad (33)$$

For the actual all-stage set $\mathcal{G}_i := \bigcup_{r=0}^d \mathcal{L}_i^r$,

$$|\mathcal{G}_i| \leq (d+1)L(k). \quad (34)$$

Proof. Equation (30) proves the first inequality in (32), including the totalized-empty case. For $0 \leq r \leq d$,

$$p_r = 2^r n_0 d \leq 2^d n_0 d = p_d.$$

Since $d \geq 1$, raising both positive sides to the d -th power and multiplying by 2^{d^2} proves the second inequality and the equality with the authoritative definition (31). There is no additional factor, power, asymptotic comparison, or absorbed logarithm.

Expanding (31) gives

$$\begin{aligned} \log L(k) &= d \log(2^d n_0 d) + d^2 \log 2 \\ &= d \log n_0 + d \log d + 2d^2 \log 2, \end{aligned}$$

which is (33). Finally, subadditivity of cardinality for a union of the exactly $d+1$ stage sets and (32) gives (34). Duplicates across stages or within different leaf witnesses can only decrease the union size. $\square$

Proposition A.27 (Exact essential-leaf identity and irreducibility export). *Under Assumption 1 and Propositions A.23 and A.25, fix i, r and $\bar{f} \in \mathcal{L}_i^r$. Then $H_i^r \neq \emptyset$, and in every optimal (p_r, d) -decomposition of H_i^r there is a leaf $\mathcal{A}_\ell$ such that, with*

$$t := \text{DDim}_{p_r, d}(H_i^r) = \text{LD}(\mathcal{A}_\ell), \quad (35)$$

one has the exact actual-function identity

$$\bar{f} = \text{SOA}_{\mathcal{A}_\ell} \quad \text{on all of } Q_C, \quad (36)$$

and $\mathcal{A}_\ell$ is $p_r 2^{d-t}$ -irreducible. Its scale satisfies

$$p_r 2^{d-t} \geq p_0 = n_0 d \geq \max\{n_0, d+1\}. \quad (37)$$

Consequently, the same leaf is both n_0 -irreducible and $(d+1)$ -irreducible.

Proof. Membership in the first case of (29) and Definition 4.3 give (35)-(36) in every optimal decomposition. The SOA is a function in H_C ; it need not belong to $\bar{C}$, so the identity preserves the source's potentially improper actual output rather than silently asserting properness. The valid-leaf clause of Lyu (Lyu, 2025) Definition 4.2 gives $p_r 2^{d-t}$ -irreducibility.

Because $0 \leq t \leq d$, $2^{d-t} \geq 1$. Because $r \geq 0$, $p_r = 2^r p_0 \geq p_0$. Hence the first inequality in (37) follows. On the positive branch $d \geq 1$, so $n_0 d \geq n_0$. Also $n_0 = km \geq 2$, whence

$$n_0 d \geq 2d \geq d+1.$$

This proves all of (37).

For completeness, K -irreducibility implies q -irreducibility for every $1 \leq q \leq K$. Given a q -tuple, extend it to length K by repeating any point of the nonempty domain Q_C . If restriction by the first q SOA-labeled points lowered Littlestone dimension, further restrictions could not increase it, contradicting K -irreducibility. Apply this observation with $K = p_r 2^{d-t}$ and $q = n_0, d+1$.

The two exported scales have distinct later roles. The n_0 scale is the input reserved for the later full-master-sample empirical contradiction, and the $d + 1$ scale places the exact SOA function in the later fixed irreducible-SOA family. This proposition supplies those interfaces only; it does not prove either later empirical or fixed-family conclusion. $\square$

$\square$

Proposition A.28 (Boundary, totalization, and object integrity). *Under Assumption 1, Lemma A.17, and Propositions A.22, A.23, A.24, A.25, A.26, and A.27, the endpoint, empty-restriction/list, DDim-zero, scalar/endpoint/list/scale parts of $d = 1$, infinite countable-quotient class, $v = d$, duplicate-trace, nonunique-decomposition, and actual-function-versus-trace conclusions below hold unconditionally. If E_{good} occurs, the first-transition inclusion in Item 1 and the $d = 1$ transition inclusion in Item 5 also hold.*

1. The first transition index is $r = 0$, and the exact source pair is $(p_1, p_0) = (2n_0d, n_0d)$ unconditionally. If E_{good} occurs, then $H_{i_*}^1 \subseteq \cap_i H_i^0$. 2. The last current restriction is $H_i^d = H_{i, \text{src}}^{d+1}$. It is used as the endpoint stage; no undefined H_i^{d+1} or transition beyond $r = d - 1$ is claimed. 3. If an arbitrary-input restriction is empty, its list is empty and no DDim, SOA, decomposition, or source lemma is evaluated for it. If a zero-error actual concept exists, Lemma A.17 proves all restrictions nonempty rather than assuming that fact. 4. DDim zero is considered only for a nonempty class and gives exactly the finite-class/list identity in Corollary 4.1; it is not a synonym for an empty restriction. 5. At $d = 1$, the map has current stages $r = 0, 1$, the transition index $r = 0$ corresponds to the source endpoint $s = 2$, $\rho = 1/2$, and every scalar inequality, endpoint, leaf scale, and list bound above remains valid unconditionally. If E_{good} occurs, the only transition inclusion is $H_{i_*}^1 \subseteq \cap_i H_i^0$. 6. The proof permits finite or countably infinite Q_C and arbitrary possibly infinite $\bar{C}$. Its finite lists come from finite trees, not from enumerating or bounding $|\bar{C}|$. 7. The specialization $v = d$ changes none of the structural statements: v does not appear in (14)-(37), except indirectly in already fixed public parameters. 8. Duplicate empirical traces do not identify actual concepts or SOA functions. They only deduplicate the finite union defining E_{good} . 9. Optimal decompositions need not be unique. Essentiality quantifies over every optimal tree, while Lemma 4.3 applies to arbitrary chosen optimal trees and preserves the exact actual function.

Proof. For Item 1, (16) at $r = 0$ gives the exact source pair unconditionally, while (17) at $r = 0$ gives the stated inclusion on E_{good} . Item 2 is (15) at $r = d$ together with the range $r < d$ in Propositions A.22 and A.24. Item 3 is the piecewise definition (29) and the proved zero-error implication in Lemma A.17; no realized event or mechanism success is used. Item 4 is Proposition A.25.

For Item 5, when $d = 1$, (22) remains strict, (23) is equality, $p_0 = n_0$, $p_1 = 2n_0$, and $p_r 2^{1-t} \geq n_0 \geq 2 = d + 1$, all unconditionally. Equations (15)-(16) identify the two current stages and the unique source-endpoint pair, while (32)-(34) preserve the list bounds. On E_{good} , (17) at $r = 0$ gives the only transition inclusion. Thus the smallest positive dimension loses no endpoint, scalar, list, or scale interface, and its cross-block inclusion retains the event condition. Items 6 and 9 follow from the cardinality-free source statements and the finite-tree arguments in Propositions A.23 and A.25. Item 7 is immediate from inspecting the exact displayed interfaces. Item 8 follows because (4) holds for each actual function sharing a trace, while the SOA identities in Propositions A.24 and A.27 are pointwise equalities on all of Q_C .

These checks establish no strict DDim drop, common list item, mechanism event, selected output, privacy inequality, empirical error, or population error. Lemma A.17 resolves the source's printed endpoint omission using its explicit $d + 1$ -stage algorithm and proves, for every $0 \leq r \leq d$,

$$H_i^r = H_{i, \text{src}}^{r+1}, \quad p_r = 2^r n_0 d = p_{r+1, \text{src}} / 2.$$

It also proves the source-required $\text{LD} \leq d$ condition directly on the quotient. Proposition A.22 then uses both exact branches of the generated event. The high branch is closed by the displayed inequality

$$\frac{1 + a_d}{1 - a_d} \rho \leq 1,$$

and the low branch by $\gamma/2 \leq \rho^d \gamma \leq \rho^{r+1} \gamma$. Thus it proves exactly

$$H_{i_*}^{r+1} \subseteq \bigcap_i H_i^r \quad (0 \leq r < d).$$

Proposition A.23 discharges Definition 4.2, the valid-decomposition existence claim, optimal attainment, Lemma 4.1, and leaf irreducibility pointwise for every nonempty current restriction. Proposition A.24 then invokes Lemma 4.3 only with

$$(\mathcal{G}, \mathcal{H}, 2p, p) = (H_{i_*}^{r+1}, H_i^r, p_{r+1}, p_r).$$

It preserves the direction of the DDim inequality and uses the leaf SOA identity only under exact DDim equality. The resulting equality is of actual functions on Q_C , and under the same premise it yields the correct cross-stage essential-list inheritance.

Proposition A.25 discharges Definition 4.3 and all four clauses of Corollary 4.1, defines the empty outer-restriction path without assigning it a DDim, and proves finite actual lists without assuming $\bar{C}$ finite. Proposition A.26 applies the authoritative dictionary without hiding a power:

$$|\mathcal{L}_i^r| \leq p_r^d 2^{d^2} \leq p_d^d 2^{d^2} = L(k), \quad |\mathcal{G}_i| \leq (d+1)L(k),$$

with the exact logarithm (33).

Finally, Proposition A.27 gives every essential actual function an exact maximal-leaf SOA witness and exports

$$p_r 2^{d-t} \geq p_0 = n_0 d \geq \max\{n_0, d+1\}.$$

The boundary calculations above cover the endpoint, empty, zero-dimensional, infinite-class, $v = d$, duplicate-trace, nonuniqueness, actual-function, and scalar/list/scale parts of the smallest- d boundary unconditionally; on E_{good} , it also verifies the first-transition and $d = 1$ transition inclusions.

Thus the cross-block inclusion, exact lists and SOA witnesses, finite list multiplicity, factor-two DDim comparison, and both irreducibility scales hold simultaneously. $\square$

A.9 DDim Descent and Common Stage

Use the restrictions, scales, and actual essential-function lists

$$H_i^r = \{\bar{h} \in \bar{C} : \text{err}_{\bar{S}_i}(\bar{h}) \leq \rho^{r+1} \gamma\}, \quad p_r = 2^r n_0 d, \quad \mathcal{L}_i^r = \text{Ess}_{p_r, d}(H_i^r), \quad (1)$$

for $i \in [k]$ and $r \in \{0, \dots, d\}$, with the empty-restriction list convention of Proposition A.25. The structural conclusion proved in this subsection is that, on E_{good} , some $r_* \in \{0, \dots, d\}$ and actual $\bar{f}_* \in H_C$ satisfy

$$\bar{f}_* \in \bigcap_{i=1}^k \mathcal{L}_i^{r_*}. \quad (2)$$

Proposition A.22 gives, on the same event,

$$H_{i_*}^{r+1} \subseteq \bigcap_{i=1}^k H_i^r \quad (0 \leq r < d, \ i_* \in [k]), \quad (3)$$

and Proposition A.24 uses only the exact pairing

$$(\mathcal{G}, \mathcal{H}, 2p, p) = (H_{i_*}^{r+1}, H_i^r, p_{r+1}, p_r). \quad (4)$$

In particular, Lemma A.17 states

$$H_i^r = \{\bar{h} \in \bar{C} : \text{err}_{\bar{S}_i}(\bar{h}) \leq \rho^{r+1}\gamma\}, \quad p_r = 2^r n_0 d, \quad 0 \leq r \leq d, \quad (5)$$

and every nonempty restriction has an attained integer value

$$D_{i,r} := \text{DDim}_{p_r,d}(H_i^r) \in \{0, 1, \dots, d\}. \quad (6)$$

For reference, the factor-two comparison from Lyu (Lyu, 2025) is

$$\text{DDim}_{2p,d}(\mathcal{G}) \leq \text{DDim}_{p,d}(\mathcal{H}), \quad (7)$$

and equality at a common value t gives maximum-leaf SOA identity

$$\text{SOA}_{\mathcal{H}_u} = \text{SOA}_{\mathcal{G}_v} \quad \text{as functions on } Q_C. \quad (8)$$

The current-notation substitution is exactly

$$\mathcal{G} = H_{i_*}^{r+1}, \quad \mathcal{H} = H_i^r, \quad (2p, p) = (p_{r+1}, p_r). \quad (9)$$

For a nonempty restriction, Definition 4.3 in Lyu (Lyu, 2025) is represented by

$$\mathcal{L}_i^r = \text{Ess}_{p_r,d}(H_i^r). \quad (10)$$

Its Corollary 4.1 preserves essential functions under equal DDim and says that at DDim zero the essential set is the entire finite nonempty class. These are the two support interfaces used in the equality and zero-potential arguments below.

Lemma A.18 (Legal realizable stage state, potential, and actual-function score). *Under Assumptions 1 and 3 and Lemma A.17 and Propositions A.23 and A.25, suppose $d \geq 1$, fix arbitrary D and $c \in C$, and let the raw master sample be generated as*

$$S = ((x_j, c(x_j)))_{j=1}^{n_0}, \quad x_1, \dots, x_{n_0} \stackrel{\text{iid}}{\sim} D. \quad (11)$$

Let $\bar{S} = T_{n_0}(S)$, let $\bar{c} \in \bar{C}$ be induced by c , and use the fixed blocks of $\bar{S}$. Then

$$\text{err}_{\bar{S}_i}(\bar{c}) = 0, \quad \bar{c} \in H_i^r \quad (i \in [k], 0 \leq r \leq d). \quad (12)$$

In particular, every H_i^r is nonempty. It is therefore legal to define, only on this realizable path,

$$D_{i,r} := \text{DDim}_{p_r,d}(H_i^r), \quad M_r := \max_{i \in [k]} D_{i,r}. \quad (13)$$

These maxima are attained and

$$D_{i,r}, M_r \in \{0, 1, \dots, d\}. \quad (14)$$

For every actual quotient function $\bar{f} \in H_C$, define its stage- r teacher-occurrence score and the maximum score by

$$s_r(\bar{f}) := |\{i \in [k] : \bar{f} \in \mathcal{L}_i^r\}|, \quad q_r := \max_{\bar{f} \in H_C} s_r(\bar{f}). \quad (15)$$

The maximum in (15) is attained, $q_r \in \{0, 1, \dots, k\}$, and

$$q_r = k \iff \text{there is an actual } \bar{f} \in H_C \text{ with } \bar{f} \in \bigcap_{i=1}^k \mathcal{L}_i^r. \quad (16)$$

Proof. For every generated record in (11), put $q_j = \kappa(x_j)$. By the definition of the induced quotient concept,

$$c(x_j) = \bar{c}(\kappa(x_j)) = \bar{c}(q_j).$$

Thus $\bar{c}$ makes no error on any record and hence no error on any block, proving the first assertion in (12). The thresholds $\rho^{r+1}\gamma$ in Lemma A.17 are positive on the positive branch, so the second assertion in (12) follows. This derives nonemptiness from the realizable labels; it does not treat nonemptiness as a primitive or generated-event assumption. Independence in (11) is not used in this pointwise implication.

Proposition A.23 now applies to every nonempty restriction. Its integer-valued degree conclusion makes (13) well-defined on the finite nonempty index set $[k]$, its maximum is attained, and (14) follows.

Each list supplied by Proposition A.25 consists of actual elements of H_C . For any $\bar{f} \in H_C$, (15) counts block indices rather than leaf duplicates, so $s_r(\bar{f}) \in \{0, \dots, k\}$. The space H_C is nonempty. Hence the nonempty set of attained integer scores is a subset of the finite set $\{0, \dots, k\}$ and has a largest attained value. This remains true if every list is empty: then every score is zero. Finally, a score equals k exactly when the same actual function is present in every one of the k set-valued lists, proving (16). $\square$

$\square$

Proposition A.29 (Equality of consecutive maxima forces one common actual SOA). *Under Assumptions 1 and 3, Propositions A.22, A.23, and A.24, Lyu (Lyu, 2025) Lemma 4.3, and Lemma A.18, suppose E_{good} occurs, $0 \leq r < d$, and*

$$M_{r+1} = M_r =: t. \quad (17)$$

Then there is one actual function $\bar{f} \in H_C$ that is (p_r, d) -essential to every H_i^r . Consequently,

$$\bar{f} \in \bigcap_{i=1}^k \mathcal{L}_i^r \quad \text{and} \quad q_r = k. \quad (18)$$

Proof. Because $[k]$ is finite and every next-stage DDim is defined by Lemma A.18, choose

$$i_* \in \arg \max_{j \in [k]} D_{j,r+1}, \quad D_{i_*,r+1} = M_{r+1} = t. \quad (19)$$

The same lemma gives $H_{i_*}^{r+1} \neq \emptyset$, and the eventwise inclusion gives

$$H_{i_*}^{r+1} \subseteq H_i^r \quad \text{for every } i \in [k]. \quad (20)$$

For every current block i , apply Proposition A.24 to $(H_{i_*}^{r+1}, H_i^r, p_{r+1}, p_r)$. Its factor-two inequality, (17), and (19), together with the definition of M_r , give the two-sided comparison

$$t = M_{r+1} = D_{i_*,r+1} \leq D_{i,r} \leq M_r = t \quad \text{for every } i \in [k]. \quad (21)$$

Thus every quantity in (21) equals t . Notice that the factor-two comparison was made against every current block, not only a current maximizer.

Now fix one arbitrary optimal (p_{r+1}, d) -decomposition T_* of $H_{i_*}^{r+1}$, and in that tree fix one arbitrary leaf v_* whose Littlestone dimension equals its degree t . Such a leaf exists because the preceding proposition supplies a finite optimal tree whose degree is the maximum leaf dimension. Define the actual pointwise function

$$\bar{f} := \text{SOA}_{(H_{i_*}^{r+1})_{v_*}} \in H_C. \quad (22)$$

The function in (22) may be improper, meaning $\bar{f} \notin \bar{C}$; no properness assertion is needed.

Fix a block $i \in [k]$, and then fix an arbitrary optimal (p_r, d) -decomposition T_i of H_i^r . Pair this arbitrary current tree with the fixed next tree T_* . The source Lemma 4.3 explicitly allows arbitrarily chosen optimal trees. Its equality premise holds by (21), so its second conclusion, applied to the fixed arbitrary maximum leaf v_* , produces a dimension- t leaf u_i of T_i with

$$\text{SOA}_{(H_i^r)_{u_i}} = \bar{f} \quad \text{pointwise on all of } Q_C. \quad (23)$$

Because T_i was arbitrary, Definition 4.3 says that the one function $\bar{f}$ in (22) is (p_r, d) -essential to H_i^r . The next tree T_* , the next maximum leaf v_* , and hence $\bar{f}$ were held fixed while i and T_i varied. Therefore the same actual function is essential to every current block. By Proposition A.25, it lies in every $\mathcal{L}_i^r$, and (16) gives $q_r = k$.

This proof does not assert that $\bar{f}$ is essential to the next-stage class: one maximum leaf of one fixed next optimal tree is sufficient. That distinction avoids assuming the extra same-class, two-scale equality required by Corollary 4.1 Item 3. $\square$

Proposition A.30 (Below-common-score stages cause one-unit DDim descent). *Under Assumptions 1 and 3, Propositions A.22 and A.24, Lemma A.18, and Proposition A.29, suppose E_{good} occurs, $0 \leq r < d$, and*

$$q_r < k. \quad (24)$$

Then

$$M_{r+1} \leq M_r - 1. \quad (25)$$

Proof. Choose a next-stage maximizer i_* exactly as in (19). For every current block i , realizable nonemptiness, Proposition A.22, and the exact pair $(p_{r+1}, p_r) = (2p_r, p_r)$ discharge Proposition A.24. Hence

$$M_{r+1} = D_{i_*, r+1} \leq D_{i, r} \leq M_r \quad \text{for every } i \in [k]. \quad (26)$$

In particular, $M_{r+1} \leq M_r$. If equality held, Proposition A.29 would give $q_r = k$, contradicting (24). Therefore

$$M_{r+1} < M_r. \quad (27)$$

By (14), both sides are integers. Thus strict inequality (27), rather than mere nonincrease, is exactly the signed one-unit bound (25). There is no additive defect, stochastic forcing, or approximation term in this recurrence. $\square$

Proposition A.31 (DDim-zero support contains the realizable target in every list). *Under Assumptions 1 and 3, Proposition A.25, Lyu (Lyu, 2025) Corollary 4.1 Item 4, and Lemma A.18, fix any stage $r \in \{0, \dots, d\}$. If*

$$M_r = 0, \quad (28)$$

then

$$\bar{c} \in \bigcap_{i=1}^k \mathcal{L}_i^r, \quad q_r = k. \quad (29)$$

Proof. Every $D_{i,r}$ is a nonnegative integer and is at most M_r . Thus (28) gives

$$\text{DDim}_{p_r,d}(H_i^r) = D_{i,r} = 0 \quad \text{for every } i \in [k]. \quad (30)$$

Lemma A.18 proved that every H_i^r is nonempty and that $\bar{c} \in H_i^r$. Therefore the nonempty-class premise of Corollary 4.1 Item 4 is discharged separately for every block. That item and the list definition give

$$\mathcal{L}_i^r = \text{Ess}_{p_r,d}(H_i^r) = H_i^r \quad \text{for every } i \in [k]. \quad (31)$$

Combining target membership with (31) proves (29), and (16) gives the score statement. In particular, the common item is supplied by the realizable target and Item 4; it is not inferred from potential nonnegativity alone. $\square$

Proposition A.32 (Finite DDim budget produces a score- k stage by stage d). *Under Assumptions 1 and 3, Lemma A.18, and Propositions A.30 and A.31, suppose $d \geq 1$ and E_{good} occurs. Then there is a stage*

$$r_* \in \{0, 1, \dots, M_0\} \subseteq \{0, 1, \dots, d\} \quad (32)$$

and an actual quotient function $\bar{f}_* \in H_C$ such that

$$q_{r_*} = k, \quad \bar{f}_* \in \bigcap_{i=1}^k \mathcal{L}_i^{r_*}. \quad (33)$$

The function $\bar{f}_*$ is an actual member of the stage lists; it may be an improper SOA function unless the zero-potential branch selects the target $\bar{c}$.

Proof. By (14), let

$$\tau := M_0 \in \{0, 1, \dots, d\}. \quad (34)$$

If $\tau = 0$, Proposition A.31 applied at stage (0) gives $q_0 = k$, so take $r_* = 0$.

Now suppose $\tau \geq 1$. If some $r < \tau$ already satisfies $q_r = k$, take the least such r . Otherwise,

$$q_r < k \quad \text{for every } r = 0, 1, \dots, \tau - 1. \quad (35)$$

Because $\tau \leq d$, every index in (35) is a legal transition index: $r \leq \tau - 1 \leq d - 1$. Proposition A.30 therefore applies successively and gives, by induction,

$$M_j \leq M_0 - j = \tau - j \quad (j = 0, 1, \dots, \tau). \quad (36)$$

At $j = \tau$, (36) gives $M_\tau \leq 0$. Nonnegativity in (14) gives $M_\tau = 0$, so Proposition A.31 yields

$$q_\tau = k, \quad \bar{c} \in \bigcap_i \mathcal{L}_i^\tau. \quad (37)$$

Take $r_* = \tau$ and $\bar{f}_* = \bar{c}$. Together with the earlier-success case and (16), this proves (32)-(33). The boundary traces are exact:

1. If $M_0 = 0$, no transition is taken and stage (0) succeeds by (31). 2. If $d = 1$, then $M_0 \in \{0, 1\}$. The first case succeeds at stage (0); in the second case, either stage (0) already has score k , or the sole legal transition $r = 0$ gives $M_1 = 0$ and stage (1) succeeds. 3. If $M_0 = d$ and every earlier stage has score below k , (36) uses exactly the d transitions $r = 0, \dots, d - 1$ and reaches $M_d = 0$. Corollary 4.1 Item 4 is then applied at the last defined stage d . No H_i^{d+1} , M_{d+1} , or transition out of stage d is introduced. 4. On nonrealizable arbitrary inputs, some restrictions may be empty; the totalization assigns them empty lists without evaluating DDim or SOA. This proposition makes no common-stage utility claim on those paths. On the realizable path, (12) proves nonemptiness at all $d + 1$ stages before the recurrence begins. 5. In the equality case used inside the descent proof, nonunique optimal trees are handled by fixing an arbitrary next tree/maximum leaf and quantifying over every arbitrary current tree. The resulting common object is the same actual function on Q_C , even when it is improper.

Finally, the induction is deterministic conditional on the one simultaneous event E_{good} . It draws no new data, introduces no union over stages, invokes no mechanism report or noise variable, and multiplies no sample or confidence bound by $d + 1$. Lemma A.18 first derives the missing entry fact: for every sample actually generated from c , the quotient target $\bar{c}$ has zero error on every teacher block and therefore belongs to every H_i^r . This makes all quantities

$$D_{i,r} = \text{DDim}_{p_r,d}(H_i^r), \quad M_r = \max_i D_{i,r}$$

legal on their nonempty domains and integer-valued in $\{0, \dots, d\}$. The same lemma defines the exact actual-function occurrence query q_r and proves that $q_r = k$ is equivalent to one actual function lying in all k lists.

For each transition $r < d$, choose a next-stage maximizer i_* . The interleaving inclusion and the exact factor-two source pair give, against every current block,

$$M_{r+1} = D_{i_*,r+1} \leq D_{i,r} \leq M_r.$$

Proposition A.29 proves that equality of the outer terms forces a common current-stage actual SOA. It does so with the full source quantifiers: one arbitrary optimal next tree and one arbitrary maximum leaf define a single function, and Lemma 4.3 maps that same function into every arbitrary optimal current tree of every block. Hence it is essential to every current class and has score k . Therefore, when $q_r < k$, Proposition A.30 gives the strict integer recurrence

$$M_{r+1} \leq M_r - 1.$$

Proposition A.31 separately supplies the terminal mechanism source: at $M_r = 0$, Corollary 4.1 Item 4 makes each list equal its nonempty restriction, and the already proved common target $\bar{c}$ belongs to all lists. Thus zero potential implies score k for a concrete actual function.

Finally, the finite-potential argument above starts at the integer $M_0 \leq d$, charges one unit for every score-below- k stage, and after at most M_0 transitions reaches either an earlier common stage or a zero-potential stage. It produces $r_* \leq M_0 \leq d$ and $\bar{f}_* \in \cap_i \mathcal{L}_i^{r_*}$. Its boundary trace covers $M_0 = 0$, $d = 1$, and first success at stage d , while preserving the empty arbitrary-input totalization and making no utility claim on that path.

Therefore, on realizable E_{good} paths with the stated list interface, finite DDim descent produces a common actual quotient function of score k by stage d .

□

A.10 Mechanism Accuracy and Actual Output

Proposition A.33 (Calibrated quotient mechanism interfaces). *Under Assumption 4, Lemma A.2, Propositions A.2 and A.3, Lemma A.18, Proposition A.32, Lemma A.7, and Propositions A.13 and A.14, the positive-dimensional quotient construction has the following two mechanism interfaces.*

The fixed allocations and thresholds are

$$\begin{aligned}\beta_{\text{AT}} &= \beta_{\text{SS}} = \beta/4, & \varepsilon_{\text{SS}} &= \varepsilon/8, & \tau_{\text{AT}} &= \eta^{-1} \log((d+1)/\beta_{\text{AT}}), \\ \tau_{\text{SS}} &= \varepsilon_{\text{SS}}^{-1} \log((kL+1)/\beta_{\text{SS}}).\end{aligned}\tag{1}$$

The least feasible teacher count satisfies the exact margin

$$\frac{k}{2} - \tau_{\text{AT}} \geq B + \tau_{\text{SS}} + 2.\tag{2}$$

On every realizable path, the current lists determine attained occurrence scores

$$s_r(\bar{h}) := |\{i \in [k] : \bar{h} \in \mathcal{L}_i^r\}|, \quad q_r := \max_{\bar{h} \in H_C} s_r(\bar{h}) \in \{0, \dots, k\}.\tag{3}$$

On E_{good} , Proposition A.32 produces a stage $r^\circ \in \{0, \dots, d\}$ and an actual $\bar{f}^\circ \in H_C$ with

$$\bar{f}^\circ \in \bigcap_{i=1}^k \mathcal{L}_i^{r^\circ}, \quad q_{r^\circ} = k.\tag{4}$$

For an arbitrary domain $\mathcal{U}$, lists $\mathcal{L}_1, \dots, \mathcal{L}_k \subseteq \mathcal{U}$, $\varepsilon_s > 0$, and $B \geq 0$, define

$$s(u) = |\{i : u \in \mathcal{L}_i\}| \quad (u \in \bigcup_i \mathcal{L}_i), \quad s(\perp) = B,$$

and sample from $\bigcup_i \mathcal{L}_i \cup \{\perp\}$ with weights

$$\exp(\varepsilon_s s(u)).\tag{5}$$

The exponent has no factor $1/2$, and $\perp$ is a separate support point of score B . If every list has size at most L and

$$B \geq 10 \log(L/\delta_s)/\varepsilon_s,$$

the Lyu (Lyu, 2025) Sparse Sample interface is $(2\varepsilon_s, \delta_s)$ -DP for addition, removal, or replacement of one entire list coordinate. In the present construction $\mathcal{U} = H_C$, $\varepsilon_s = \varepsilon_{\text{SS}}$, $\delta_s = \delta_{\text{SS}}$, and the selected tuple is $(\hat{\mathcal{L}}_1, \dots, \hat{\mathcal{L}}_k)$.

*The AboveThreshold interface draws independent $Z_r \sim \text{Lap}(1/\eta)$ and reports **Above** exactly when*

$$q_r + Z_r \geq k/2;\tag{6}$$

*otherwise it reports **Below** and stops at the first **Above**. For sensitivity-one adaptive queries and at most K counted crossings, its privacy cost is*

$$\left(\eta O(\sqrt{K \log(1/\delta_a)} + \log(1/\delta_a)), \delta_a \right),$$

with the same Above/Below symmetry and concurrent composition as in Lyu (Lyu, 2025). Here it is one first-crossing process with $K = 1$, threshold $k/2$, and at most $d+1$ stage queries.

The calibration identities and margin are

$$(d+1)e^{-\eta\tau_{\text{AT}}} = \beta_{\text{AT}}, \quad (kL+1)e^{-\varepsilon_{\text{SS}}\tau_{\text{SS}}} = \beta_{\text{SS}}, \quad \frac{k}{2} - \tau_{\text{AT}} \geq B + \tau_{\text{SS}} + 2.\tag{7}$$

On a realizable source-good path, the stage scores are

$$s_r(\bar{h}) = |\{i \in [k] : \bar{h} \in \mathcal{L}_i^r\}|, \quad q_r = \max_{\bar{h} \in H_C} s_r(\bar{h}),$$

and the DDim descent result supplies a stage with an actual common function of score k . An actual Sparse Sample output has a nonempty finite producer-block occurrence set; the marked lift splits this set and the projection sums the mark back to the released quotient law K_C .

Proof. The two displayed mechanism laws and their list-coordinate privacy bounds are Lyu (Lyu, 2025)'s Algorithm 1/Lemma 3.1 Sparse Sample interface and Algorithm 2/Lemma 3.2 AboveThreshold interface, with the current domain, threshold, and privacy parameters substituted. Lemma A.2 supplies the list cap, the exact ceiling-defined failure score, and the first two identities in (7), while Proposition A.2 supplies the last inequality with all ceilings retained. Proposition A.3 defines the stopped transcript and every deterministic fallback, so these interfaces are defined on arbitrary quotient inputs.

Lemma A.2 bounds every current exact list by L , and Proposition A.3 replaces every invalid list by the empty list. Hence the selected union has at most kL elements and the law in (5) has a positive finite normalizer. The AboveThreshold statement here is conditional on sensitivity-one adaptive queries; that list-interface condition is proved before the mechanism accuracy result consumes it. Lemma A.18 identifies the scores and exact list membership on the realizable path, and Proposition A.32 supplies the score- k stage without an additional confidence or sample charge. Finally, Lemma A.7 gives the finite producer set and Propositions A.13 and A.14 give its marked kernel and exact released projection. Raw-record locality and raw privacy are established separately by the later record-locality propositions; the present result is the quotient list/mechanism interface used for utility. $\square$

Lemma A.19 (Sensitivity-one occurrence scores and finite mechanism support). *Under Lemma A.2, Proposition A.3, and Lemma A.18, fix any stage $r \in \{0, \dots, d\}$. For a tuple $\mathbf{L} = (L_1, \dots, L_k)$ of current sanitized lists, define*

$$s_{\mathbf{L}}(\bar{h}) := \sum_{i=1}^k \mathbf{1}\{\bar{h} \in L_i\}, \quad q(\mathbf{L}) := \max_{\bar{h} \in H_C} s_{\mathbf{L}}(\bar{h}). \quad (8)$$

Then:

1. if $\mathbf{L}$ and $\mathbf{L}'$ differ in at most one entire list coordinate, then

$$|q(\mathbf{L}) - q(\mathbf{L}')| \leq 1; \quad (9)$$

2. every current list has size at most L , so

$$U(\mathbf{L}) := \bigcup_{i=1}^k L_i, \quad |U(\mathbf{L})| \leq kL; \quad (10)$$

3. the Sparse Sample support $U(\mathbf{L}) \cup \{\perp\}$ has size at most $kL + 1$, is finite even when H_C is infinite, and has a positive finite normalizer under (5); and 4. if all lists are empty, then $q(\mathbf{L}) = 0$.

These are list-interface statements only. In particular, (9) does not claim that one raw record replacement changes only one current list.

Proof. If $\mathbf{L}$ and $\mathbf{L}'$ differ only at coordinate i_0 , then for every $\bar{h} \in H_C$,

$$|s_{\mathbf{L}}(\bar{h}) - s_{\mathbf{L}'}(\bar{h})| = |\mathbf{1}\{\bar{h} \in L_{i_0}\} - \mathbf{1}\{\bar{h} \in L'_{i_0}\}| \leq 1. \quad (11)$$

Consequently,

$$q(\mathbf{L}) = \max_{\bar{h}} s_{\mathbf{L}}(\bar{h}) \leq \max_{\bar{h}} (s_{\mathbf{L}'}(\bar{h}) + 1) = q(\mathbf{L}') + 1.$$

Interchanging the tuples proves (9). This proof remains valid for ordered list encodings and repeated positions because the score uses set membership, not positional multiplicity.

At an adaptive transcript prefix consisting of r preceding **Below** reports, both compared executions request the same stage- r query. Thus (9) is the required sensitivity-one statement for every next query conditioned on a common transcript prefix. It is still only a statement at the list-tuple interface and makes no raw-record locality claim.

Lemma A.2 bounds every exact current list by L , while Proposition A.3 replaces every invalid, nonfinite, non- H_C , or oversized purported list by the empty list. Thus the same cap holds on every totalized path. The finite union bound gives (10), and adjoining one distinguished symbol gives support size at most $kL + 1$. Every score is finite and belongs to $[0, k]$, so every exponential weight in (5) is finite. The $\perp$ weight $e^{\varepsilon_{\text{SS}} B}$ is strictly positive; hence the finite sum of weights is positive and finite, proving the normalizer claim. If every list is empty, each summand in (8) is zero for every $\bar{h}$, so the maximum is zero. $\square$

$\square$

Proposition A.34 (One AboveThreshold transcript selects a score-separated stage). *Under Assumption 4, Proposition A.2, Proposition A.32, and Lemma A.19, fix an arbitrary realizable source-good path on which (4) holds. Realize in advance independent*

$$Z_0, \dots, Z_d \sim \text{Lap}(1/\eta)$$

and run the one first-crossing transcript (6). Define

$$E_{\text{AT}} := \bigcap_{r=0}^d \{|Z_r| \leq \tau_{\text{AT}}\}. \quad (12)$$

Then

$$\Pr(E_{\text{AT}}^c) \leq \beta_{\text{AT}}. \quad (13)$$

On E_{AT} :

1. every reported Below stage satisfies $q_r < k$; 2. every stage with $q_r = k$ is reported Above if reached; 3. the stopped transcript has a first legal selected stage $\hat{r} \in \{0, \dots, d\}$; and 4. its true selected score obeys

$$q_{\hat{r}} \geq \frac{k}{2} - \tau_{\text{AT}} \geq B + \tau_{\text{SS}} + 2. \quad (14)$$

Thus an early Above at a stage with score below k is a safe false positive rather than a mechanism failure: (14) is still sufficient for the Sparse Sample call. An all-empty stage cannot be selected on E_{AT} . Immediate score- k success is selected at stage 0, and a first score- k stage at d is either reached and selected or is preceded by an already safe false positive.

Proof. A random variable $Z \sim \text{Lap}(1/\eta)$ has density $(\eta/2)e^{-\eta|z|}$. Direct integration gives, for $t \geq 0$,

$$\Pr(|Z| > t) = 2 \int_t^\infty \frac{\eta}{2} e^{-\eta z} dz = e^{-\eta t}. \quad (15)$$

There are at most $d + 1$ queries. Sampling all possible noises in advance does not alter the adaptive stopped transcript, and a union bound with (7) gives

$$\Pr(E_{\text{AT}}^c) \leq (d+1)e^{-\eta\tau_{\text{AT}}} = \beta_{\text{AT}},$$

which proves (13). No independence among stages is needed for this union bound, although the source tests are independent.

The teacher margin (2) has a strictly positive right-hand side, so

$$\tau_{\text{AT}} < k/2. \quad (16)$$

If the process reports **Below** at stage r , the exact threshold rule (6) gives $q_r + Z_r < k/2$. On E_{AT} ,

$$q_r < k/2 - Z_r \leq k/2 + \tau_{\text{AT}} < k, \quad (17)$$

proving the first assertion. If $q_r = k$, then on the same event

$$q_r + Z_r \geq k - \tau_{\text{AT}} > k/2, \quad (18)$$

so that stage is reported **Above**. Proposition A.32 supplies at least one such stage by stage d . Therefore the stopped transcript has a first legal **Above** no later than a score- k stage, proving existence of $\hat{r}$ and excluding no-success exhaustion.

At the selected stage, the **Above** inequality and $Z_{\hat{r}} \leq \tau_{\text{AT}}$ give

$$q_{\hat{r}} \geq k/2 - Z_{\hat{r}} \geq k/2 - \tau_{\text{AT}}.$$

Combining this with (2) proves (14). If all lists at a stage are empty, Lemma A.19 gives $q_r = 0$, and (16) yields

$$q_r + Z_r \leq \tau_{\text{AT}} < k/2;$$

hence that stage must report **Below** on E_{AT} . This also proves the empty-stage boundary.

For endpoint handling, let $r_{\min}$ be the least stage with score k , which exists by (4). If $r_{\min} = 0$, (18) selects the first query. If $r_{\min} = d$, then either a previous query reports **Above**, in which case (14) makes it a safe selected stage, or all previous reports are **Below** and (18) selects stage d . Thus the transcript uses one sequence of at most $d+1$ queries and no per-stage success event. $\square$

$\square$

Lemma A.20 (Sparse Sample score tail and exclusion of the failure symbol). *Under Lyu (Lyu, 2025) Algorithm 1 in the form (5), Lemma A.2, and Proposition A.34, condition on an arbitrary source-good path, a transcript in E_{AT} , its selected stage $\hat{r}$, and the selected current list tuple. Put*

$$U_{\hat{r}} := \bigcup_{i=1}^k \mathcal{L}_i^{\hat{r}}, \quad q := q_{\hat{r}},$$

and let $W \in U_{\hat{r}} \cup \{\perp\}$ be the Sparse Sample outcome, with $s(\perp) = B$. Define

$$E_{\text{SS}} := \{s(W) \geq q - \tau_{\text{SS}}\}. \quad (19)$$

Then, uniformly over every such selected tuple,

$$\Pr(E_{\text{SS}}^c \mid \text{selected lists and AboveThreshold transcript}) \leq \beta_{\text{SS}}. \quad (20)$$

Moreover,

$$\Pr(W = \perp \mid \text{selected lists and AboveThreshold transcript}) \leq e^{-2\varepsilon_{\text{SS}}} \frac{\beta_{\text{SS}}}{kL + 1} \leq \beta_{\text{SS}}, \quad (21)$$

and on E_{SS} ,

$$W \in U_{\hat{r}}, \quad W \in \mathcal{L}_i^{\hat{r}} \text{ for at least one } i \in [k]. \quad (22)$$

Thus W is an actual current list member. It is an element of H_C but need not be in $\bar{C}$.

Proof. Equation (14) gives

$$q \geq B + \tau_{\text{SS}} + 2 > 0. \quad (23)$$

Therefore $U_{\hat{r}}$ is nonempty and contains an attained maximum-score item $\bar{h}_*$ with $s(\bar{h}_*) = q$. By Lemma A.19, the complete support has at most $kL + 1$ points. Let

$$\mathcal{B} := \{z \in U_{\hat{r}} \cup \{\perp\} : s(z) < q - \tau_{\text{SS}}\}.$$

Under the exact law (5), the denominator contains the term $e^{\varepsilon_{\text{SS}}q}$ of $\bar{h}_*$, while every point in $\mathcal{B}$ has weight at most $e^{\varepsilon_{\text{SS}}(q - \tau_{\text{SS}})}$. Hence

$$\begin{aligned} \Pr(E_{\text{SS}}^c \mid \text{selected state}) &= \frac{\sum_{z \in \mathcal{B}} e^{\varepsilon_{\text{SS}}s(z)}}{e^{\varepsilon_{\text{SS}}B} + \sum_{u \in U_{\hat{r}}} e^{\varepsilon_{\text{SS}}s(u)}} \\ &\leq |\mathcal{B}| e^{-\varepsilon_{\text{SS}}\tau_{\text{SS}}} \\ &\leq (kL + 1) e^{-\varepsilon_{\text{SS}}\tau_{\text{SS}}} \\ &= \beta_{\text{SS}}, \end{aligned} \quad (24)$$

where the last equality is (7). This is the full finite-effective-domain tail; it charges low-score actual items as well as $\perp$ and does not invoke the source's commented utility statement.

For the exact failure-symbol mass, the same maximum-score denominator term gives

$$\begin{aligned} \Pr(W = \perp \mid \text{selected state}) &= \frac{e^{\varepsilon_{\text{SS}}B}}{e^{\varepsilon_{\text{SS}}B} + \sum_{u \in U_{\hat{r}}} e^{\varepsilon_{\text{SS}}s(u)}} \\ &\leq e^{-\varepsilon_{\text{SS}}(q - B)} \\ &\leq e^{-\varepsilon_{\text{SS}}(\tau_{\text{SS}} + 2)} \\ &= e^{-2\varepsilon_{\text{SS}}} \frac{\beta_{\text{SS}}}{kL + 1}, \end{aligned} \quad (25)$$

which proves (21) without dropping the extra teacher slack.

Finally, (23) gives

$$q - \tau_{\text{SS}} \geq B + 2 > B = s(\perp). \quad (26)$$

Thus E_{SS} is incompatible with $W = \perp$. By the exact support of (5), $W \neq \perp$ implies $W \in U_{\hat{r}}$, proving (22). The current lists contain actual quotient functions in H_C ; an essential SOA may be improper, so no conclusion $W \in \bar{C}$ is made. $\square$

$\square$

Proposition A.35 (Mechanism-good event, actual selected item, and exact fallback ledger). *Under Assumption 4, Propositions A.3, A.13, A.14, A.18, and A.34, and Lemmas A.7 and A.20, define on a realizable source-good path*

$$E_{\text{mech}} := E_{\text{AT}} \cap E_{\text{SS}}, \quad (27)$$

where E_{SS} is declared false if no legal selected Sparse Sample call occurs. Together with the source event produced by Proposition A.18, define the core event

$$E_{\text{core}} := E_{\text{good}} \cap E_{\text{mech}}. \quad (27a)$$

Then, conditional on any fixed realizable sample and partition for which the E_{good} antecedent holds,

$$\Pr(E_{\text{mech}}^c \mid \bar{S}, \mathcal{P}, E_{\text{good}}) \leq \beta_{\text{AT}} + \beta_{\text{SS}} = \frac{\beta}{2}. \quad (28)$$

On E_{mech} , there are a selected stage $\hat{r} \in \{0, \dots, d\}$, a producer $i \in [k]$, and a terminal quotient output $\bar{H} \in H_C$ such that

$$\bar{H} \in \mathcal{L}_i^{\hat{r}} \subseteq \mathcal{G}_i := \bigcup_{r=0}^d \mathcal{L}_i^r. \quad (29)$$

The transcript is in the actual-output status event, not a fallback status. Under the marked lift its occurrence set is nonempty, $J \in [k]$, and $\bar{H} \in \mathcal{G}_J$; summing out J gives exactly the same released K_C law. These statements remain status-correct if the actual selected value happens to equal the default $\bar{c}_0$.

Proof. Proposition A.34 gives (13) and, on E_{AT} , a legal selected tuple satisfying (14). Conditional on every possible selected tuple and preceding AboveThreshold transcript, Lemma A.20 gives (20). Therefore the tower property and a union bound yield

$$\begin{aligned} \Pr(E_{\text{mech}}^c \mid \bar{S}, \mathcal{P}, E_{\text{good}}) &\leq \Pr(E_{\text{AT}}^c) + \Pr(E_{\text{AT}} \cap E_{\text{SS}}^c \mid \bar{S}, \mathcal{P}, E_{\text{good}}) \\ &\leq \beta_{\text{AT}} + \beta_{\text{SS}}. \end{aligned} \quad (30)$$

This argument is uniform over the adaptively selected stage; it does not multiply β_{SS} by $d + 1$. Equation (1) and the allocation give the final equality in (28).

On E_{mech} , equation (22) supplies a producer i and proves the first inclusion in (29); the second is the definition of the producer's all-stage union. The valid selected tuple is finite, its normalizer is positive and finite by Lemma A.19, and the outcome lies in the exact categorical support. Consequently none of the following totalized terminal branches occurs on E_{mech} :

1. an all-empty selected-stage tuple (excluded already by (14));
2. no **Above** or stage exhaustion (excluded by (18));
- an invalid report or a named stage outside $\{0, \dots, d\}$ has probability zero under the exact finite threshold transcript (6);
3. a nonfinite, non- H_C , oversized, or otherwise invalid selected list (the exact current lists pass the sanitizer and obey the cap);
4. an invalid or zero Sparse Sample normalizer (excluded by finiteness and the positive $\perp$ weight);
5. the failure symbol $\perp$ (excluded by (26));
6. an out-of-support outcome or residual mechanism failure (the exact finite categorical law has support only $U_{\hat{r}} \cup \{\perp\}$);
- or 7. the no-success or default terminal output as a status path.

Thus Proposition A.3 returns the same actual $\bar{H} = W$ rather than invoking $\bar{c}_0$ as fallback. If $W = \bar{c}_0$ as a function, its categorical actual status, not its value, still distinguishes it from fallback.

Lemma A.7 now applies to this actual transcript and gives a nonempty finite producer occurrence set. The marked kernel chooses J only from that set, so $J \in [k]$ and $\bar{H} \in \mathcal{G}_J$. Proposition A.14 then removes the analysis-only mark with exact equality of the released marginal. No privacy statement for $(\bar{H}, J)$ is made.

The boundary cases do not create a hidden branch or factor:

If the first score- k stage is 0, it is selected immediately on E_{AT} . If it first occurs at d , the process either safely selects an earlier false positive or selects stage d . If a queried stage has all lists empty, it reports **Below** on E_{AT} . Empty lists in only some blocks are harmless; the selected score in (14) still supplies an actual maximum-score item. The candidate dictionary is defined at $t = 2$, and all sensitivity, support, and marking formulas remain well-formed there. For the realized least feasible teacher, however, $k = 2$ is impossible: (2) would require $1 - \tau_{\text{AT}} \geq B + \tau_{\text{SS}} + 2$, whose left side is less than (1) and whose right side is greater than (2). Thus no proof line silently divides by $k - 1$ or assumes a nonexistent feasible $k = 2$ mechanism path. The output may be improper. Actual membership means membership in a current H_C -valued essential list, not membership in $\bar{C}$.

There is one master sample, one fixed block partition reused at all stages, one AboveThreshold transcript, and one Sparse Sample call. The only stage multiplicity in this argument is the $d + 1$ Laplace union already present inside τ_{AT} ; no sample factor, privacy composition, or additional confidence share is introduced. Lemma A.2 and Proposition A.2 provide the exact source parameters, list cap, confidence identities, and teacher margin. Proposition A.3 provides the one-transcript, one-call procedure and every fallback branch. Lemma A.19 proves directly that the maximum occurrence query is sensitivity one at the list interface and that every Sparse Sample effective support is finite of size at most $kL + 1$, even though H_C can be infinite.

Proposition A.32 supplies a literal score- k actual function at some current stage, conditional on its realizable source-good antecedent. Proposition A.34 realizes all $d + 1$ possible Laplace noises within one stopped process. Its simultaneous event costs exactly one β_{AT} share, makes every **Below** report structurally accurate enough ($q_r < k$), detects a score- k stage, and gives the selected-stage lower bound

$$q_{\hat{r}} \geq k/2 - \tau_{\text{AT}} \geq B + \tau_{\text{SS}} + 2.$$

Thus empty stages cannot be selected on the good-noise event, while an early false positive remains safe.

Lemma A.20 then applies the exact source exponential weights to the one selected finite support. Its maximum-score denominator term and support size give

$$\Pr[s(W) < q_{\hat{r}} - \tau_{\text{SS}}] \leq (kL + 1)e^{-\varepsilon_{\text{SS}}\tau_{\text{SS}}} = \beta_{\text{SS}},$$

and the teacher margin puts the failure symbol's score strictly below the output-score threshold. Hence the good Sparse Sample outcome is literally in one selected current list, not merely equal to a trace or a fallback value.

The conditioning argument above composes these two finite tails on the single selected list tuple and obtains the mechanism ledger

$$\Pr(E_{\text{mech}}^c \mid \bar{S}, \mathcal{P}, E_{\text{good}}) \leq \beta_{\text{AT}} + \beta_{\text{SS}} = \beta/2.$$

The totalized branches are all covered by (29). Lemma A.7 turns actual selected-stage membership into a nonempty all-stage producer occurrence set, while Proposition A.14 confirms that the analysis-only mark changes no released mass.

Hence one adaptive AboveThreshold transcript and one Sparse Sample call produce, on E_{mech} , an actual current quotient-list member rather than fallback.

□

A.11 All-Input Privacy and Raw Transfer

The two mechanism charges fixed before sampling are

$$(\varepsilon_{\text{AT}}, \delta_{\text{AT}}) = (\varepsilon/4, \delta/2), \quad (\varepsilon_{\text{sp}}, \delta_{\text{SS}}) = (\varepsilon/4, \delta/2), \quad (1)$$

where $\varepsilon_{\text{sp}} = 2\varepsilon_{\text{SS}}$ and $\varepsilon_{\text{SS}} = \varepsilon/8$. The raw learner kernel supplied by Proposition A.12 is

$$A_N(s, E) = K_C(T_N(s), E), \quad E \in \mathcal{H}_C. \quad (2)$$

Every totalized current list has the all-input cap

$$|\mathcal{L}_i^r| \leq L \quad (i \in [k], 0 \leq r \leq d), \quad (3)$$

and is determined blockwise from

$$H_i^r = \{\bar{h} \in \bar{C} : \text{err}_{\bar{S}_i}(\bar{h}) \leq \rho^{r+1}\gamma\}, \quad 0 \leq r \leq d. \quad (4)$$

For finite lists $L_1, \dots, L_k \subseteq H_C$, define

$$q(L_1, \dots, L_k) := \max_{\bar{h} \in H_C} \sum_{i=1}^k \mathbf{1}\{\bar{h} \in L_i\}. \quad (5)$$

Lemma A.19 shows that this query changes by at most one under replacement of one list coordinate and that the Sparse Sample support has size at most $kL + 1$, including $\perp$.

The Sparse Sample privacy interface of Lyu (Lyu, 2025) is

$$B \geq \frac{10 \log(L/\delta_s)}{\varepsilon_s} \quad (6)$$

for $(2\varepsilon_s, \delta_s)$ -DP under addition, removal, or replacement of one list coordinate. Here

$$\varepsilon_s = \varepsilon_{\text{SS}} := \varepsilon/8, \quad \delta_s = \delta_{\text{SS}} := \delta/2, \quad B = \left\lceil \frac{10 \log(L/\delta_{\text{SS}})}{\varepsilon_{\text{SS}}} \right\rceil, \quad (7)$$

so the conditional selected-stage law has charge

$$(2\varepsilon_{\text{SS}}, \delta_{\text{SS}}) = (\varepsilon/4, \delta/2). \quad (8)$$

For sensitivity-one adaptive queries, the stopped AboveThreshold interface in the same source has privacy parameters

$$\left(\eta O\left(\sqrt{K \log(1/\delta_a)} + \log(1/\delta_a) \right), \delta_a \right). \quad (9)$$

With $K = 1$, $\delta_a = \delta_{\text{AT}} := \delta/2$, and the fixed universal c_{AT} , its first coordinate is at most

$$c_{\text{AT}} \eta \left(\sqrt{\log(1/\delta_{\text{AT}})} + \log(1/\delta_{\text{AT}}) \right). \quad (10)$$

Finally,

$$g_\delta := \log(4/\delta), \quad \eta := \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)}, \quad (11)$$

and $\log(1/\delta_{\text{AT}}) = \log(2/\delta) \leq g_\delta$, so (10) is at most $\varepsilon/4$. The results below establish the missing record-to-one-list premise and then compose these two interfaces without using any utility event.

Lemma A.21 (Recordwise quotient adjacency and fixed-block localization). *Under Assumption 2 and Proposition A.6, let $N \in \mathbb{N}_0$. If $s, s' \in Z_X^N$ are replace-one neighbors, then $T_N(s)$ and $T_N(s')$ are equal or replace-one neighbors in Z_Q^N . If $\bar{s}, \bar{s}' \in Z_Q^N$ are equal or replace-one neighbors and a fixed partition π assigns the N coordinate indices to k ordered blocks, then equal quotient tuples give equal ordered block tuples; otherwise the block tuples are equal except possibly at the unique block containing the changed coordinate. For $N = 0$, all these tuples are the same empty tuple.*

Proof. Write

$$s = ((x_j, y_j))_{j=1}^N, \quad s' = ((x'_j, y'_j))_{j=1}^N.$$

If the raw tuples are equal, their quotient images are equal. Otherwise, there is one index j_* such that $(x_j, y_j) = (x'_j, y'_j)$ for every $j \neq j_*$. Since T_N acts coordinatewise,

$$T_N(s)_j = (\kappa(x_j), y_j) = (\kappa(x'_j), y'_j) = T_N(s')_j \quad (j \neq j_*). \quad (12)$$

In this latter case, at j_* the two quotient records may also coincide, for example when the raw points lie in the same evaluation cell and the labels agree. Thus the quotient tuples are either equal or differ at precisely that one coordinate. No realizability relation between x_j and y_j was used.

Now fix quotient tuples $\bar{s}, \bar{s}'$ and an index partition π . If $\bar{s} = \bar{s}'$, every ordered block projection is identical, so no changed block need be named. Otherwise, let j_* be their unique differing coordinate. Every coordinate belongs to exactly one block; call the block containing j_* by $i_* = i_*(\pi, j_*)$. The ordered block projection for every $i \neq i_*$ uses only unchanged coordinates and is therefore identical under $\bar{s}$ and $\bar{s}'$. Only block i_* can differ, and within it only one ordered record differs. This conclusion is deterministic conditional on π , holds for every possible changed label, and is unaffected by duplicate records. If $N = 0$, there is only the empty raw tuple, its empty quotient image, and the unique empty block state; the assertions are identities. $\square$

$\square$

Proposition A.36 (All-stage one-list locality and sensitivity at every common prefix). *Under Proposition A.9, Lemma A.17, Propositions A.25 and A.26, Lemma A.19, and Lemma A.21, suppose $d \geq 1$. Fix a partition π and quotient neighbors $\bar{s} \sim \bar{s}'$. There is a block $i_* \in [k]$ such that, simultaneously for every $0 \leq r \leq d$,*

$$\mathcal{L}_i^r(\bar{s}, \pi) = \mathcal{L}_i^r(\bar{s}', \pi) \quad (i \neq i_*), \quad (13)$$

while both possibly different i_ -lists are finite subsets of H_C of size at most L . Consequently the two stage- r list tuples differ by replacement of at most one entire list coordinate, and*

$$|q_r(\bar{s}, \pi) - q_r(\bar{s}', \pi)| \leq 1, \quad 0 \leq r \leq d, \quad (14)$$

where

$$q_r(\bar{z}, \pi) := \max_{\bar{h} \in H_C} \sum_{i=1}^k \mathbf{1}\{\bar{h} \in \mathcal{L}_i^r(\bar{z}, \pi)\}. \quad (15)$$

At every common prefix of the stopped AboveThreshold transcript, both executions request the same next stage r , so (14) is the exact sensitivity-one adaptive-query condition. The statement includes arbitrary nonrealizable labels, empty restrictions and lists, pointwise sanitation, fixed list orderings, repeated encodings, and duplicate functions across different block lists.

Proof. If $\bar{s} = \bar{s}'$, all block data, sanitized lists, and scores are equal; since the positive branch has $k \geq 2$, choose any $i_* \in [k]$, and (13)-(14) follow. Hence assume that the quotient inputs are strict replace-one neighbors. By Lemma A.21, only the ordered data of their unique changed block i_* can differ. The restriction definition in Lemma A.17 shows that H_i^r is a function only of the ordered data in block i , the fixed public stage, and the fixed public thresholds. The binding construction fixes a decomposition and a list ordering by a lookup on this local quotient-block state. Therefore identical local states use identical restrictions, decomposition choices, essential-function sets, and orderings. This proves (13) for every $i \neq i_*$ and every stage simultaneously.

If a local restriction is empty, Proposition A.25 assigns that coordinate the empty list. If a purported local list is nonfinite, outside H_C , oversized, or otherwise invalid, the totalization sanitizes that one coordinate to the empty list before either mechanism reads it. This sanitation is pointwise in the same local block state and therefore cannot change a second coordinate. For a nonempty valid restriction, the cited source construction supplies its finite actual-function set; Proposition A.26 gives $|\mathcal{L}_i^r| \leq L$. Hence both possible i_* -lists are legal and capped even though they may be completely different. No stability of individual candidates is required.

For completeness, fix r and abbreviate the two list tuples by $\mathbf{L} = (L_1, \dots, L_k)$ and $\mathbf{L}' = (L'_1, \dots, L'_k)$. For every $\bar{h} \in H_C$, (13) gives

$$\left| \sum_i \mathbf{1}\{\bar{h} \in L_i\} - \sum_i \mathbf{1}\{\bar{h} \in L'_i\} \right| = |\mathbf{1}\{\bar{h} \in L_{i_*}\} - \mathbf{1}\{\bar{h} \in L'_{i_*}\}| \leq 1. \quad (16)$$

Taking maxima and applying (16) in each direction proves (14). The maximum is well-defined even if H_C is infinite: if the union is nonempty it is attained on the finite union of lists, and if all lists are empty its value is zero. An algorithmic ordering is only an encoding of each finite set. Repeated positions are setified by the list interface, and the score counts membership in a block list once. The same function appearing in several different block lists is intentionally counted once per block and does not affect (16).

Before the first **Above**, a common transcript prefix consists of the same number r of preceding **Below** reports, so both executions request the same setting-defined query q_r . If a prefix is exhausted or invalid, no further data query is made. Thus every next query at every common prefix is either the same sensitivity-one function in (14) or a constant no-query continuation. This argument is pointwise for all quotient inputs and never uses E_{good} , realizability, or a mechanism-accuracy event. $\square$

Proposition A.37 (All-input privacy of the stopped AboveThreshold prefix). *Under Assumption 4, Lemma A.2, Propositions A.33 and A.36, and Lyu (Lyu, 2025) Lemma 3.2, fix a data-independent partition π . Let M_{AT}^π be the finite transcript kernel that queries $q_0, \dots, q_d$ in order, stops at its first **Above**, and otherwise records exhaustion after stage d . Include the selected stage or the no-selection tag in the transcript. Then, on all quotient inputs and for every measurable transcript event F ,*

$$M_{\text{AT}}^\pi(\bar{s}, F) \leq e^{\varepsilon/4} M_{\text{AT}}^\pi(\bar{s}', F) + \delta/2 \quad (\bar{s} \sim \bar{s}'). \quad (17)$$

The same inequality holds with $\bar{s}, \bar{s}'$ interchanged.

Proof. At every common transcript prefix, Proposition A.36 gives exactly the sensitivity-one premise of Lyu (Lyu, 2025) Lemma 3.2. The process counts at most one **Above**. Its termination after at most $d + 1$ queries and its retention of only the first-crossing prefix are deterministic postprocessings of the source transcript; neither operation increases privacy loss.

Put $\delta_{\text{AT}} = \delta/2$. Lemma A.2 sets

$$g_\delta = \log(4/\delta), \quad \eta = \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)}.$$

Moreover,

$$\log(1/\delta_{\text{AT}}) = \log(2/\delta) \leq \log(4/\delta) = g_\delta$$

Proposition A.33 gives the $K = 1$ AboveThreshold cost bound, so

$$\begin{aligned} \varepsilon_{\text{AT}} &\leq c_{\text{AT}}\eta \left(\sqrt{\log(1/\delta_{\text{AT}})} + \log(1/\delta_{\text{AT}}) \right) \\ &\leq c_{\text{AT}} \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)} (\sqrt{g_\delta} + g_\delta) = \frac{\varepsilon}{4}. \end{aligned} \tag{18}$$

Lyu (Lyu, 2025) Lemma 3.2 therefore gives (17). Its DP statement is for every ordered neighbor pair and every transcript event. Because replacement adjacency is symmetric, applying the same statement to the ordered pair $(\bar{s}', \bar{s})$ gives the reverse inequality; it is not obtained by algebraically reversing (17). Empty-list scores, arbitrary labels, and an eventual no-selection transcript are ordinary inputs and outputs of this same source process. No accuracy condition on its Laplace noises was conditioned upon. $\square$

Proposition A.38 (Uniform selected-stage Sparse Sample privacy and totalized no-call continuations). *Under Assumption 4, Lemma A.2, Propositions A.33 and A.36, and Lyu (Lyu, 2025) Lemma 3.1, fix a partition π and any possible AboveThreshold history t . Define a total conditional H_C -valued kernel $M_{2,t}^\pi$ as follows:*

1. *if t selects a stage $r \in \{0, \dots, d\}$, apply Sparse Sample to the sanitized tuple $(\mathcal{L}_1^r, \dots, \mathcal{L}_k^r)$ with $\varepsilon_{\text{SS}} = \varepsilon/8$, $\delta_{\text{SS}} = \delta/2$, and*

$$B = \left\lceil \frac{10 \log(L/\delta_{\text{SS}})}{\varepsilon_{\text{SS}}} \right\rceil,$$

then map $\perp$ to $\bar{c}_0$ and retain every actual H_C -item; and 2. if t records no selection, exhaustion, an out-of-range stage, or an invalid/no-call tag, return $\bar{c}_0$ deterministically.

Then for every history t , every quotient neighboring pair, and every $E \in \mathcal{H}_C$,

$$M_{2,t}^\pi(\bar{s}, E) \leq e^{\varepsilon/4} M_{2,t}^\pi(\bar{s}', E) + \delta/2. \tag{19}$$

This uniform statement does not require the selected stages in two actual neighboring executions to agree.

Proof. First suppose that t selects the common history-indexed stage r . Proposition A.36 says that the two stage- r input tuples differ in at most one entire list coordinate, and Proposition A.26 bounds every list by L . Lemma A.2 defines the displayed ceiling, which implies the exact Sparse Sample threshold in Proposition A.33. Lyu (Lyu, 2025) Lemma 3.1 therefore makes the Sparse Sample law on the fixed global space $H_C \cup \{\perp\}$

$$(2\varepsilon_{\text{SS}}, \delta_{\text{SS}}) = (\varepsilon/4, \delta/2)\text{-DP}. \tag{20}$$

The deterministic map

$$\phi(z) := \begin{cases} z, & z \in H_C, \\ \bar{c}_0, & z = \perp \end{cases} \tag{21}$$

is measurable. For any $E \in \mathcal{H}_C$, the probability after totalization is the pre-totalization probability of $\phi^{-1}(E)$. Applying (20) to this measurable preimage proves (19). This remains valid when an actual categorical item equals $\bar{c}_0$: the preimage then contains both that actual item and $\perp$, and postprocessing still applies to the union.

All stage lists have already been sanitized before the occurrence query and before Sparse Sample. Thus a private-data-dependent invalid list does not choose between an unaccounted mechanism and a constant branch: its one coordinate is replaced by the empty list, and the same source mechanism is run on the resulting legal tuple. If all selected lists are empty, the source support is only $\{\perp\}$, its normalizer is the positive weight $e^{\varepsilon_{\text{ss}} B}$, and (21) makes the continuation constant. Partially empty tuples require no special case. A zero normalizer or an out-of-support categorical outcome has probability zero under the exact source law; its predeclared fallback is still the same measurable postprocessing.

If t is a no-call, exhausted, out-of-range, or invalid history, the conditional law is $\delta_{\bar{c}_0}$ for every input. It is $(0, 0)$ -DP and hence satisfies (19). We define this constant continuation even for histories having probability zero so that the adaptive family is total on the whole finite transcript space.

The proof is indexed by a *common value* t of the first transcript. If one actual execution selects stage r and the other selects stage r' , we never compare $\mathcal{L}^r(\bar{s})$ directly with $\mathcal{L}^{r'}(\bar{s}')$. The private first transcript accounts for the different-history probabilities; for each possible shared history, the conditional second kernel satisfies (19). This is precisely the interface used by adaptive composition. No actual-output or good-noise event has been used. $\square$

$\square$

Lemma A.22 (Exact finite-transcript adaptive composition and postprocessing). *Let M_1 be a kernel from a database space to a finite discrete transcript space $(\mathbb{T}, 2^{\mathbb{T}})$. For every $t \in \mathbb{T}$, let $M_{2,t}$ be a kernel from the same database space to a measurable space $(\mathbb{Y}, \mathcal{Y})$. Let $\varepsilon_1, \varepsilon_2 \geq 0$ and $0 \leq \delta_1, \delta_2 \leq 1$. Suppose, for every ordered neighboring pair, M_1 is $(\varepsilon_1, \delta_1)$ -DP and every $M_{2,t}$ is $(\varepsilon_2, \delta_2)$ -DP, uniformly in t . Then the adaptive joint law*

$$J_x(dt, dy) := M_1(x, dt)M_{2,t}(x, dy) \quad (22)$$

is $(\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2)$ -DP. For every measurable output space $(\mathbb{O}, \mathcal{O})$ and measurable map $g : \mathbb{T} \times \mathbb{Y} \rightarrow \mathbb{O}$, the postprocessed law $g_{\#}J_x$ has the same guarantee.

Proof. Fix an ordered neighbor pair $x \sim x'$. Write P, Q for the two first-transcript laws. The one-direction DP inequality is

$$P(F) \leq e^{\varepsilon_1} Q(F) + \delta_1 \quad (F \subseteq \mathbb{T}). \quad (23)$$

We record an exact submeasure form of (23). With $\lambda = P + Q$, let $p = dP/d\lambda$, $q = dQ/d\lambda$, and define

$$dP^0 := \min\{p, e^{\varepsilon_1} q\} d\lambda. \quad (24)$$

Then $P^0 \leq P$, $P^0 \leq e^{\varepsilon_1} Q$, and

$$\begin{aligned} P(\mathbb{T}) - P^0(\mathbb{T}) &= \int (p - e^{\varepsilon_1} q)_+ d\lambda \\ &= \sup_{F \subseteq \mathbb{T}} \{P(F) - e^{\varepsilon_1} Q(F)\} \leq \delta_1. \end{aligned} \quad (25)$$

For each $t \in \mathbb{T}$, apply the same construction to the two conditional second-output laws

$$P_t := M_{2,t}(x, \cdot), \quad Q_t := M_{2,t}(x', \cdot).$$

It yields a subprobability $P_t^0 \leq P_t$ satisfying

$$P_t^0 \leq e^{\varepsilon_2} Q_t, \quad P_t^0(\mathbf{Y}) \geq 1 - \delta_2. \quad (26)$$

Because $\mathbf{T}$ is finite, these finitely many choices form a measurable history-indexed family without a selection issue. Define the joint submeasure

$$J_x^0(dt, dy) := P^0(dt)P_t^0(dy). \quad (27)$$

It obeys $J_x^0 \leq J_x$, and (25)-(26) give

$$J_x^0(\mathbf{T} \times \mathbf{Y}) \geq (1 - \delta_1)(1 - \delta_2) \geq 1 - \delta_1 - \delta_2. \quad (28)$$

For any measurable $G \subseteq \mathbf{T} \times \mathbf{Y}$, with section $G_t := \{y : (t, y) \in G\}$, (24) and (26) imply

$$\begin{aligned} J_x^0(G) &= \sum_{t \in \mathbf{T}} P^0(\{t\}) P_t^0(G_t) \\ &\leq e^{\varepsilon_1 + \varepsilon_2} \sum_{t \in \mathbf{T}} Q(\{t\}) Q_t(G_t) \\ &= e^{\varepsilon_1 + \varepsilon_2} J_{x'}(G). \end{aligned} \quad (29)$$

Since the missing mass of the submeasure in (27) is at most $\delta_1 + \delta_2$,

$$\begin{aligned} J_x(G) &\leq J_x^0(G) + (J_x - J_x^0)(\mathbf{T} \times \mathbf{Y}) \\ &\leq e^{\varepsilon_1 + \varepsilon_2} J_{x'}(G) + \delta_1 + \delta_2. \end{aligned} \quad (30)$$

This derivation explains why the additive term is $\delta_1 + \delta_2$, with no erroneous exponential multiplier. Repeating it for the reversed ordered pair supplies the reverse DP inequality.

Finally, for a measurable output event $E \in \mathcal{O}$,

$$(g_{\#} J_x)(E) = J_x(g^{-1}(E)).$$

Using $G = g^{-1}(E)$ in (30) proves the postprocessing statement for every measurable event. $\square$

$\square$

Proposition A.39 (All-input quotient privacy after adaptive composition and partition mixing). *Under Assumption 4, Propositions A.9 and A.11, Propositions A.37 and A.38, and Lemma A.22, the positive-branch released quotient kernel satisfies, for all quotient replace-one neighbors and all $E \in \mathcal{H}_C$,*

$$K_C(\bar{s}, E) \leq e^{\varepsilon/2} K_C(\bar{s}', E) + \delta. \quad (31)$$

This holds on every input, without realizability or conditioning on any generated event.

Proof. Fix a partition π . Use the finite first transcript

$$\mathbf{T}_{\text{AT}} = \{t_r : 0 \leq r \leq d\} \cup \{t_{\text{none}}\} \quad (32)$$

where t_r records r preceding **Below** reports followed by **Above** at stage r , and t_{none} records $d+1$ **Below** reports. Any predeclared zero-probability invalid transcript tags can be adjoined to this finite space with the constant continuation of Proposition A.38.

Proposition A.37 gives

$$(\varepsilon_1, \delta_1) = (\varepsilon/4, \delta/2)$$

for the first kernel. Proposition A.38 gives, uniformly for every t ,

$$(\varepsilon_2, \delta_2) = (\varepsilon/4, \delta/2)$$

for the totalized second kernel; on no-call histories this is a conservative upper bound on its exact $(0, 0)$ cost. Lemma A.22 therefore makes the augmented history/output law

$$(\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2) = (\varepsilon/2, \delta)\text{-DP}. \quad (33)$$

The released quotient output is a measurable postprocessing: retain the Sparse Sample H_C -item on a selected actual outcome and otherwise return $\bar{c}_0$. Proposition A.9 supplies the measurable status and output coordinates. Lemma A.22 therefore proves (31) for the conditional released kernel K_C^π .

The terminal routes before removing π are as follows:

1. An empty restriction is represented by an empty list before either mechanism. An all-empty stage still has the legal query value zero.
2. An **Above** on an all-empty or partially empty tuple invokes the same private Sparse Sample kernel; an all-empty union returns only $\perp$, which postprocesses to $\bar{c}_0$.
3. If no **Above** occurs by stage d , the private first transcript is followed by the constant no-call output $\bar{c}_0$.
4. A selected Sparse Sample $\perp$ outcome maps to $\bar{c}_0$. An actual selected function equal to $\bar{c}_0$ may collide in released value with this fallback, but collision under a deterministic map cannot increase privacy loss.
5. A nonfinite, non- H_C , oversized, or otherwise invalid local list is sanitized coordinatewise before querying. Proposition A.26 guarantees that valid nonempty restrictions in fact produce finite capped lists on every input, so the remaining structural-invalid tags are unreachable; their predeclared constant continuation is harmless.
6. An out-of-range transcript, zero normalizer, or out-of-support mechanism outcome has probability zero under the exact finite mechanisms. Its totalized value is nevertheless defined and is included in the same measurable postprocessing.

Thus no private-data-dependent branch outside the two composed mechanisms is released. In particular, none of these cases assumes that the output is an actual list item, that the first crossing is accurate, or that a source-good path occurred.

It remains to average over the uniform data-independent partition. Let Π_N be its finite set of index partitions and let ν be its law, which is the same for $\bar{s}$ and $\bar{s}'$. For every fixed $\pi \in \Pi_N$, the preceding argument gives

$$K_C^\pi(\bar{s}, E) \leq e^{\varepsilon/2} K_C^\pi(\bar{s}', E) + \delta. \quad (34)$$

Consequently,

$$\begin{aligned} K_C(\bar{s}, E) &= \sum_{\pi \in \Pi_N} \nu(\pi) K_C^\pi(\bar{s}, E) \\ &\leq e^{\varepsilon/2} \sum_{\pi \in \Pi_N} \nu(\pi) K_C^\pi(\bar{s}', E) + \delta \sum_{\pi \in \Pi_N} \nu(\pi) \\ &= e^{\varepsilon/2} K_C(\bar{s}', E) + \delta. \end{aligned} \quad (35)$$

The additive term is paid once, not once per partition. Mechanism coins are already integrated inside each conditional kernel. The proof applies to every quotient input atom, including arbitrary labels and duplicate records. Since E was arbitrary in $\mathcal{H}_C$, (31) holds for all measurable outputs, whether proper, improper, default-valued, or outside all current lists. $\square$

$\square$

Proposition A.40 (Exact raw pullback, privacy monotonicity, and null boundary). *Under Assumption 4, Proposition A.12, Lemma A.21, and Proposition A.39, the setting-defined raw learner kernel A_N is $(\varepsilon/2, \delta)$ -DP and hence (ε, δ) -DP for every raw replace-one pair and every $E \in \mathcal{H}_C$. If $d = 0$, the setting uses $N = 0$ and the law is exactly $(0, 0)$ -DP.*

Proof. Proposition A.12 gives the exact kernel identity (2). Fix raw neighbors $s \sim s'$ and an arbitrary $E \in \mathcal{H}_C$. Lemma A.21 leaves two cases.

If $T_N(s) = T_N(s')$, then exact pullback gives

$$A_N(s, E) = K_C(T_N(s), E) = K_C(T_N(s'), E) = A_N(s', E). \quad (36)$$

This covers a raw point replacement within one quotient cell when the label is unchanged. If the labels differ, the quotient records differ and the next case applies.

Otherwise the quotient images are distinct, and Lemma A.21 makes them strict replace-one neighbors. Proposition A.39 gives

$$\begin{aligned} A_N(s, E) &= K_C(T_N(s), E) \\ &\leq e^{\varepsilon/2} K_C(T_N(s'), E) + \delta \\ &= e^{\varepsilon/2} A_N(s', E) + \delta. \end{aligned} \quad (37)$$

Equations (36)-(37) are pointwise for arbitrary labels; neither case uses a data distribution or target concept. Adjacency is symmetric, so applying the same two-case proof to the ordered pair (s', s) gives the reverse inequality. Again, the reverse is not inferred by rearranging an approximate-DP inequality.

Because $0 < \varepsilon$, $e^{\varepsilon/2} \leq e^\varepsilon$. Therefore (37) implies

$$A_N(s, E) \leq e^\varepsilon A_N(s', E) + \delta, \quad (38)$$

which is exactly the setting's raw replacement- (ε, δ) -DP definition for the released H_C -valued output. No decoder is needed for (38); applying the fixed decoder after release would itself be postprocessing.

For $d = 0$, Proposition A.11 supplies $N = 0$ and the exact law $\delta_{\bar{e}_0}$ on the unique empty quotient input. Proposition A.12 gives the same Dirac law on the unique empty raw input. Thus all event probabilities are identical and the kernel is $(0, 0)$ -DP; no partition, AboveThreshold query, or Sparse Sample call exists.

For the formal $k = 2$ positive-branch boundary, the privacy proof remains well-formed: Lyu (Lyu, 2025) Sparse Sample permits every $k \geq 1$, the list cap and one-coordinate relation remain valid, AboveThreshold counts one crossing, and no line divides by $k - 1$ or uses the teacher utility margin. Whether the least-feasible utility calibration actually selects $k = 2$ is irrelevant to privacy. First-stage selection, stage- d selection, different selected stages, no selection, and every fallback have already been included in the finite transcript composition. This closes every claimed boundary without changing the mechanism. Lemma A.21 first proves the exact pointwise transport: one raw replacement becomes either no change or one quotient replacement, and, conditional on a fixed data-independent index partition, only one ordered block can change. Proposition A.36 combines that fact with the block-local restriction lists, fixed choices, empty-list totalization, and cap. It proves simultaneously over all $d + 1$ stages that only one entire list coordinate can change. The direct membership calculation then gives sensitivity one for every next maximum-occurrence query at every common transcript prefix. This derivation holds for arbitrary labels and does not consume eventwise inclusion from the restriction-list proposition.

Proposition A.37 applies Lyu (Lyu, 2025) Lemma 3.2 with $K = 1$, $\delta_{AT} = \delta/2$, and the exact η . The displayed calculation (18) gives the first allocation $(\varepsilon/4, \delta/2)$ without multiplying privacy

by $d + 1$. Proposition A.38 applies Lyu (Lyu, 2025) Lemma 3.1 at every possible common selected-stage history with $\varepsilon_{\text{SS}} = \varepsilon/8$, $\delta_{\text{SS}} = \delta/2$, the exact ceiling B , and one-list replacement. It gives the second allocation $(\varepsilon/4, \delta/2)$. No-call and invalid-history continuations are constant, while $\perp$ and all other fallbacks are postprocessed to $\bar{c}_0$.

Lemma A.22 proves, with an explicit submeasure argument, that these history-dependent kernels compose to

$$(\varepsilon/4 + \varepsilon/4, \delta/2 + \delta/2) = (\varepsilon/2, \delta)\text{-DP}.$$

This proof handles differing stopping times and selected stages because it compares the second kernel only at a common transcript value and charges the transcript mismatch to the first private kernel. Proposition A.39 then applies measurable postprocessing and averages over the common data-independent partition, paying neither a new privacy term nor a partition multiplier. It yields the all-input quotient inequality (31) for every $E \in \mathcal{H}_C$.

Finally, Proposition A.12 and the quotient privacy inequality just proved transfer (31) exactly through T_N . Equal quotient images give equality of laws; neighboring images use (31). Thus the raw learner is in fact $(\varepsilon/2, \delta)$ -DP and therefore is (ε, δ) -DP. The exact $d = 0, N = 0$ branch is $(0, 0)$ -DP. This completes the all-input privacy and raw-transfer argument. $\square$

A.12 Exact SOA Identity and Empirical Utility

Fix the positive branch $d \geq 1$, an indexed quotient master sample, and its partition:

$$\bar{S} = ((q_j, y_j))_{j=1}^{n_0}, \quad \mathcal{P} = (B_1, \dots, B_k), \quad |B_i| = m, \quad n_0 = km. \quad (1)$$

For $f \in H_C$, define the master and block empirical errors

$$e_{\bar{S}}(f) := \frac{1}{n_0} \sum_{j=1}^{n_0} \mathbf{1}\{f(q_j) \neq y_j\}, \quad e_i(f) := \frac{1}{m} \sum_{j \in B_i} \mathbf{1}\{f(q_j) \neq y_j\}. \quad (2)$$

Retain

$$\gamma := \frac{\alpha}{16}, \quad a_d := \frac{1}{5d}, \quad \rho := 1 - \frac{1}{2d}, \quad (3)$$

and

$$H_i^r := \{\bar{c} \in \bar{C} : e_i(\bar{c}) \leq \rho^{r+1}\gamma\}, \quad p_r := 2^r n_0 d, \quad 0 \leq r \leq d. \quad (4)$$

Write $E_{\text{core}} := E_{\text{good}} \cap E_{\text{mech}}$ only for the joint path condition produced by the trace and mechanism results. The statements proved below are that an actual selected output has the exact form

$$\bar{H} = \text{SOA}_{\mathcal{G}} \quad \text{for a nonempty } \mathcal{G} \subseteq \bar{C} \text{ that is both } n_0\text{- and } (d+1)\text{-irreducible}, \quad (5)$$

that

$$\bar{H} \in \hat{C}_{d+1} := \{\text{SOA}_{\mathcal{A}} : \emptyset \neq \mathcal{A} \subseteq \bar{C} \text{ is } (d+1)\text{-irreducible}\}, \quad \text{LD}(\hat{C}_{d+1}) \leq d, \quad (6)$$

and that

$$e_{\bar{S}}(\bar{H}) \leq 2\gamma = \frac{\alpha}{8}. \quad (7)$$

The family in (6) is fixed by $\bar{C}$ and may be improper.

On E_{good} , Proposition A.18 gives, for every actual $\bar{c} \in \bar{C}$ and every $i \in [k]$,

$$\begin{cases} (1 - a_d)e_{\bar{S}}(\bar{c}) \leq e_i(\bar{c}) \leq (1 + a_d)e_{\bar{S}}(\bar{c}), & e_{\bar{S}}(\bar{c}) > \gamma/3, \\ 0 \leq e_i(\bar{c}) \leq \gamma/2, & e_{\bar{S}}(\bar{c}) \leq \gamma/3. \end{cases} \quad (8)$$

For every $\bar{f} \in \mathcal{L}_i^r$, Proposition A.27 supplies a maximal leaf $\mathcal{G}$ with

$$\bar{f} = \text{SOA}_{\mathcal{G}}, \quad \mathcal{G} \subseteq H_i^r \subseteq \bar{C}, \quad \mathcal{G} \text{ is } p_r 2^{d-t}\text{-irreducible}, \quad (9)$$

where $t = \text{DDim}_{p_r, d}(H_i^r) = \text{LD}(\mathcal{G})$, and

$$p_r 2^{d-t} \geq p_0 = n_0 d \geq \max\{n_0, d+1\}. \quad (10)$$

On E_{core} , Proposition A.35 gives a selected stage and producer with

$$\bar{H} \in \mathcal{L}_i^{\hat{r}}. \quad (11)$$

The exact source/current identities from Lemma A.17 are

$$H_i^r = H_{i, \text{src}}^{r+1}, \quad p_r = 2^r n_0 d = \frac{p_{r+1, \text{src}}}{2}, \quad (12)$$

and, only when $r < d$,

$$p_{r+1} = 2p_r. \quad (13)$$

For a nonempty class $\mathcal{A}$ with $t = \text{DDim}_{p, d}(\mathcal{A})$, Lyu's essential-function definition (Lyu, 2025) gives, in every optimal decomposition, a leaf $\mathcal{A}_\ell$ satisfying

$$\text{LD}(\mathcal{A}_\ell) = t, \quad \text{SOA}_{\mathcal{A}_\ell} = f \quad \text{pointwise on all of } Q_C. \quad (14)$$

The empirical restriction argument in the proof of Lyu's Theorem 3 is instantiated here with

$$\mathcal{X} = Q_C, \quad S = \bar{S}, \quad n = n_0, \quad h = \bar{H}, \quad u = \gamma. \quad (15)$$

It says that if a nonempty $\mathcal{G}$ is n -irreducible, $h = \text{SOA}_{\mathcal{G}}$, and every $g \in \mathcal{G}$ has empirical error at most $2u$, then h has empirical error at most $2u$; the contradiction is reproduced below before the interface is used. Finally, Lyu (Lyu, 2025, Lemma 4.2) defines

$$\hat{\mathcal{H}}_{d+1} := \{\text{SOA}_{\mathcal{A}} : \emptyset \neq \mathcal{A} \subseteq \mathcal{H} \text{ is } (d+1)\text{-irreducible}\} \quad (16)$$

and concludes

$$\text{LD}(\hat{\mathcal{H}}_{d+1}) \leq d \quad (17)$$

whenever $\text{LD}(\mathcal{H}) \leq d$. We apply it only with $\mathcal{H} = \bar{C}$.

Proposition A.41 (Actual selected item has an exact current maximal-leaf SOA). *Under Assumption 1, Lemma A.17, Propositions A.18, A.25, A.27, and A.35, suppose $d \geq 1$ and the fixed path lies in $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$. Then there are $\hat{r} \in \{0, \dots, d\}$, $i \in [k]$, and, in every optimal $(p_{\hat{r}}, d)$ -decomposition of $H_i^{\hat{r}}$, a nonempty leaf $\mathcal{G}$ such that, with*

$$t := \text{DDim}_{p_{\hat{r}}, d}(H_i^{\hat{r}}) = \text{LD}(\mathcal{G}), \quad (18)$$

one has

$$\bar{H} \in \mathcal{L}_i^{\hat{r}}, \quad \mathcal{G} \subseteq H_i^{\hat{r}} \subseteq \bar{C}, \quad \bar{H} = \text{SOA}_{\mathcal{G}} \quad \text{pointwise on } Q_C. \quad (19)$$

The leaf $\mathcal{G}$ is $p_{\hat{r}} 2^{d-t}$ -irreducible, and

$$p_{\hat{r}} 2^{d-t} \geq p_0 = n_0 d \geq \max\{n_0, d+1\}. \quad (20)$$

Thus $\mathcal{G}$ is both n_0 -irreducible and $(d+1)$ -irreducible. The exact source/current identities are

$$H_i^{\hat{r}} = H_{i,\text{src}}^{\hat{r}+1}, \quad p_{\hat{r}} = 2^{\hat{r}} n_0 d = \frac{p_{\hat{r}+1,\text{src}}}{2}. \quad (21)$$

If $\hat{r} < d$, the next current scale satisfies $p_{\hat{r}+1} = 2p_{\hat{r}}$. If $\hat{r} = d$, there is no current transition beyond the selected endpoint. The conclusion records actual selected status even if $\bar{H} = \bar{c}_0$ as a function.

Proof. Proposition A.35 gives the selected stage, producer, actual-status output, and first relation in (19). Because the output is a literal member of $\mathcal{L}_i^{\hat{r}}$, this list is nonempty. By the piecewise definition in Proposition A.25, the corresponding outer restriction $H_i^{\hat{r}}$ is therefore nonempty and the selected function is genuinely $(p_{\hat{r}}, d)$ -essential to it. No DDim or SOA is evaluated on an empty class.

Apply Proposition A.27 to this exact list membership. It gives, in every optimal current decomposition, a maximal leaf $\mathcal{G}$, (18), the subset relation in (19), the pointwise identity in (19), and the full scale (20). That proposition also proves that irreducibility at the scale in (20) implies irreducibility at each smaller positive integer scale, giving n_0 and $d + 1$.

Finally, Lemma A.17 gives (21). Its factor-two statement gives $p_{\hat{r}+1} = 2p_{\hat{r}}$ only when $\hat{r} < d$. The selected list throughout this argument remains the current half-source-scale list at $p_{\hat{r}}$; it is not relabeled as a literal source-stage list. At $\hat{r} = d$, (21) uses the source endpoint $d + 1$ and does not invent a current stage $d + 1$.

Actual versus fallback status comes from the transcript, not from the function value. Hence equality $\bar{H} = \bar{c}_0$ does not change any of the preceding memberships or identities. $\square$

Lemma A.23 (Every member of the selected leaf has small master error). *Under Assumption 1, Propositions A.18, A.35, and A.41, suppose $d \geq 1$ and $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$ holds. Then every $g \in \mathcal{G}$ satisfies*

$$e_{\bar{S}}(g) \leq 2\gamma. \quad (22)$$

This is a statement about actual members $g \in \bar{C}$; it makes no application of E_{good} to the potentially improper $\bar{H} = \text{SOA}_{\mathcal{G}}$.

Proof. Fix $g \in \mathcal{G}$. By (19) and the selected current restriction defined in Lemma A.17,

$$e_i(g) \leq \rho^{\hat{r}+1} \gamma \leq \gamma, \quad (23)$$

because $0 < \rho < 1$ and $\hat{r} + 1 \geq 1$. Also $g \in \mathcal{G} \subseteq H_i^{\hat{r}} \subseteq \bar{C}$, so the exact event produced by Proposition A.18 applies to this same actual function.

If $e_{\bar{S}}(g) \leq \gamma/3$, then immediately

$$e_{\bar{S}}(g) \leq \gamma/3 \leq 2\gamma. \quad (24)$$

This branch includes equality at $\gamma/3$. If instead $e_{\bar{S}}(g) > \gamma/3$, the lower half of the high-error relative clause in Proposition A.18, together with (23), gives

$$(1 - a_d)e_{\bar{S}}(g) \leq e_i(g) \leq \rho^{\hat{r}+1} \gamma \leq \gamma. \quad (25)$$

Since $d \geq 1$,

$$a_d = \frac{1}{5d} \leq \frac{1}{5}, \quad 1 - a_d \geq \frac{4}{5}, \quad \frac{1}{1 - a_d} \leq \frac{5}{4} \leq 2. \quad (26)$$

Dividing (25) by the positive $1 - a_d$ and using (26) yields

$$e_{\bar{S}}(g) \leq \frac{\gamma}{1 - a_d} \leq \frac{5}{4}\gamma \leq 2\gamma. \quad (27)$$

The two exhaustive event branches prove (22). The argument is uniform in the selected stage, including 0 and d , and in every leaf member; no union, independence, or new confidence factor is used. $\square$

$\square$

Lemma A.24 (Indexed irreducible-SOA empirical contradiction). *Under Assumption 1, Propositions A.18, A.35, and A.41, Lemma A.23, and the empirical argument in Lyu (Lyu, 2025)’s proof of Theorem 3, suppose $d \geq 1$ and $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$ holds. Then*

$$e_{\bar{S}}(\bar{H}) \leq 2\gamma = \frac{\alpha}{8}. \quad (28)$$

The empirical error in (28) uses all n_0 indexed master records with normalization $1/n_0$, including repetitions.

Proof. Suppose, for contradiction, that

$$e_{\bar{S}}(\bar{H}) > 2\gamma. \quad (29)$$

Restrict the nonempty leaf $\mathcal{G}$ along the complete ordered sequence of master inputs, labeled by the exact SOA function:

$$\mathcal{R} := \mathcal{G}|_{(q_1, \bar{H}(q_1)), \dots, (q_{n_0}, \bar{H}(q_{n_0}))}. \quad (30)$$

If $g \in \mathcal{R}$, then $g(q_j) = \bar{H}(q_j)$ for every indexed coordinate j . Consequently, for every j , including every repeated occurrence of a quotient point,

$$\mathbf{1}\{g(q_j) \neq y_j\} = \mathbf{1}\{\bar{H}(q_j) \neq y_j\}. \quad (31)$$

Summing (31) and dividing by the same n_0 gives

$$e_{\bar{S}}(g) = e_{\bar{S}}(\bar{H}) > 2\gamma, \quad (32)$$

contradicting Lemma A.23. Hence

$$\mathcal{R} = \emptyset. \quad (33)$$

On the other hand, Proposition A.41 gives $\bar{H} = \text{SOA}_{\mathcal{G}}$ pointwise and says that $\mathcal{G}$ is n_0 -irreducible. Applying the definition of n_0 -irreducibility to the exact point sequence $q_1, \dots, q_{n_0}$ gives

$$\text{LD}(\mathcal{R}) = \text{LD}(\mathcal{G}) = t \geq 0. \quad (34)$$

Thus $\mathcal{R}$ must be nonempty: an empty binary class has no depth-zero Littlestone witness and cannot have the nonnegative dimension of the nonempty leaf $\mathcal{G}$. This contradicts (33). Repeated sample points cause no difficulty, because irreducibility quantifies over sequences of points and permits repetitions; moreover the labels in (30) are the single function values $\bar{H}(q_j)$.

Therefore (29) is false and $e_{\bar{S}}(\bar{H}) \leq 2\gamma$. Finally,

$$2\gamma = 2 \left(\frac{\alpha}{16} \right) = \frac{\alpha}{8}, \quad (35)$$

which proves (28). The contradiction began with a strict $>$, so the exported conclusion correctly permits exact equality at $\alpha/8$. $\square$

$\square$

Proposition A.42 (Membership in the fixed improper irreducible-SOA family). *Under Assumption 1, Lemma A.17, Propositions A.18, A.35, and A.41, and Lyu (Lyu, 2025) Lemma 4.2, suppose $d \geq 1$ and $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$ holds. Define the static family*

$$\hat{C}_{d+1} := \{\text{SOA}_{\mathcal{A}} : \emptyset \neq \mathcal{A} \subseteq \bar{C} \text{ is } (d+1)\text{-irreducible}\}. \quad (36)$$

Then

$$\bar{H} \in \hat{C}_{d+1}, \quad \text{LD}(\hat{C}_{d+1}) \leq d. \quad (37)$$

The first membership in (37) is literal equality of functions on Q_C . It does not assert $\bar{H} \in \bar{C}$, and the second conclusion does not assert any measurability, empirical, population, or privacy property.

Proof. Proposition A.41 gives a nonempty $\mathcal{G} \subseteq H_i^{\hat{r}} \subseteq \bar{C}$, proves that it is $(d+1)$ -irreducible, and gives the pointwise identity $\bar{H} = \text{SOA}_{\mathcal{G}}$. Thus the same function $\bar{H}$ is one of the elements in the sample-independent definition (36). This proves the first part of (37) without selecting a trace representative or a proper concept.

Lemma A.17 proves $\text{LD}(\bar{C}) \leq d$. Every hypothesis of Lyu (Lyu, 2025) Lemma 4.2, applied to the static family in (36), is now discharged with $\mathcal{H} = \bar{C}$. Its conclusion is precisely the second part of (37). $\square$

$\square$

Proposition A.43 (Endpoint, normalization, status, and improperness boundary cases). *Under Assumption 1, Lemma A.24, and Propositions A.18, A.35, A.41, and A.42, suppose $d \geq 1$ and $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$ holds. Then the exact SOA, fixed-family, and empirical-error conclusions remain valid in all of the following boundary and identity regimes.*

1. If $\hat{r} = 0$, the selected current class is source stage 1, $H_i^0 = H_{i,\text{src}}^1$, and the selected-list parameter is $p_0 = n_0 d = p_{1,\text{src}}/2$. The proof uses this exact current list, not a same-scale or literal source-stage substitute.
2. If $\hat{r} = d$, the selected current class is the explicit source endpoint $H_i^d = H_{i,\text{src}}^{d+1}$, with $p_d = p_{d+1,\text{src}}/2$. No current H_i^{d+1} , no next-stage essential list, and no transition beyond d is used.
3. If $d = 1$, the only possible selected stages are 0 and 1, $p_0 = n_0$, $p_1 = 2n_0$, and $p_{\hat{r}} 2^{1-t} \geq n_0 \geq 2 = d+1$. Thus both the full-sample and fixed-family irreducibility uses remain legal, including $t = 0$.
4. The proof is unchanged at $v = 1$ and at $v = d$. The parameter v does not enter (18)-(37); it only affects already fixed upstream sample quantities. No additional v - or d -power is introduced here.
5. A singleton current list, one function repeated in several lists, or duplicate encodings of the same list item still supplies one actual function $\bar{H}$ and one producer i . Essentiality and SOA equality concern that function, not its multiplicity. If the selected function equals $\bar{c}_0$, actual transcript status keeps it actual; equality of values never turns it into fallback.
6. An empty selected tuple, an empty selected list, $\perp$, and every totalized fallback path are absent on the joint source/mechanism-good path. Specifically, Proposition A.35 supplies literal selected-list membership, which implies that the chosen list and current restriction are nonempty. The source event E_{good} supplies the error transfer, but is not by itself misrepresented as a mechanism-status event.
7. If the selected maximal leaf has $t = 0$, it is still nonempty and the same scale (20), indexed restriction (30), and fixed-family argument apply. DDim zero is never assigned to an empty outer restriction.
8. Repeated quotient points or repeated labeled records remain separate indices in both sums in (2). Equality of functions on all Q_C gives coordinatewise equality (31), so no deduplication changes the $1/n_0$ normalization or shortens the n_0 -point irreducibility sequence.
9. The threshold identity is exactly $2\gamma = \alpha/8$. The proof excludes only $e_{\bar{S}}(\bar{H}) > \alpha/8$ and therefore makes no unjustified strict-error claim at equality.
10. The output is guaranteed to lie in H_C and in the fixed family $\hat{C}_{d+1}$. It may lie outside $\bar{C}$; no properness of the quotient output or decoded raw hypothesis is asserted.

Proof. Items 1 and 2 are the endpoint cases of (21), with the transition range recorded in Proposition A.41. Item 3 substitutes $d = 1$ into (18)-(21); the scale inequality is the exact bound (20), not an asymptotic estimate. Item 4 follows because no derivation after the fixed upstream dictionary contains v .

For Item 5, mechanism-good status uses set membership rather than positional multiplicity, while Definition 4.3 and (19) identify actual functions pointwise. Item 6 follows from the selected-list status in Proposition A.35; the separate role of Proposition A.18 remains visible. Item 7 follows from the nonempty leaf in Proposition A.41 and does not require a positive t .

Items 8 and 9 are the explicit indexed calculation (31)-(35). Item 10 is exactly Proposition A.42 together with the actual-output boundary in (19). None of these specializations adds a privacy, probability, population-risk, raw-risk, properness, or sample-size conclusion. On the joint source-good plus mechanism-good path, Proposition A.35 first supplies a literal actual current-list member and its selected stage. Proposition A.41 combines that status with Definition 4.3 and leaf-scale interfaces to obtain one nonempty $\mathcal{G} \subseteq H_i^{\hat{r}} \subseteq \bar{C}$ satisfying the exact pointwise identity

$$\bar{H} = \text{SOA}_{\mathcal{G}}$$

and the exact scale

$$\mathcal{G} \text{ is } p_{\hat{r}} 2^{d-t}\text{-irreducible,} \quad p_{\hat{r}} 2^{d-t} \geq n_0 d \geq \max\{n_0, d + 1\}.$$

The same proposition preserves the selected current/source stage identity and the half-scale convention, with a factor-two next-current pairing only when $\hat{r} < d$.

Lemma A.23 next uses the exact two-clause E_{good} interface on each actual $g \in \mathcal{G}$. Membership in the selected restriction gives $e_i(g) \leq \rho^{\hat{r}+1} \gamma \leq \gamma$; the low event branch is already below 2γ , while the high branch gives

$$e_{\bar{S}}(g) \leq \frac{\gamma}{1 - a_d} \leq \frac{5}{4} \gamma \leq 2\gamma.$$

Lemma A.24 then applies Lyu (Lyu, 2025) Theorem 3 contradiction to the same actual function and the same indexed master sample. If $e_{\bar{S}}(\bar{H}) > 2\gamma$, every member of $\mathcal{G}$ is excluded from the restriction along the complete $\bar{H} = \text{SOA}_{\mathcal{G}}$ -labeled n_0 -tuple. The restriction is empty, contradicting n_0 -irreducibility. Hence

$$e_{\bar{S}}(\bar{H}) \leq 2\gamma = \alpha/8.$$

Finally, Proposition A.42 uses the same leaf, not a surrogate class or trace, to prove

$$\bar{H} \in \hat{C}_{d+1}, \quad \text{LD}(\hat{C}_{d+1}) \leq d.$$

The endpoint and object-integrity calculations above verify that the two exported conclusions retain literal identity, normalization, actual status, endpoint scope, equality allowance, and improperness in every boundary regime. Together, these conclusions give the exact SOA utility guarantee and the empirical-error/actual-SOA interface used in the PAC conversion. $\square$

A.13 Marked Holdout and PAC Conversion

Proposition A.44 (Core-event and marked-holdout interfaces). *Under Assumptions 1, 3, and 4, Lemmas A.7, A.15, and A.24, and Propositions A.13, A.14, A.21, A.20, A.26, and A.35, suppose $d \geq 1$. Then the analysis-only mark is assigned pathwise to the nonempty producer set of an actual*

terminal output, while nonactual paths receive mark 0. Its marked kernel has the exact released marginal

$$\sum_{j=0}^k \tilde{K}_C(\bar{s}, B \times \{j\}) = K_C(\bar{s}, B), \quad B \in \mathcal{H}_C. \quad (1)$$

This identity makes no privacy assertion for the unreleased pair $(\bar{H}, J)$.

Write

$$b := \log(4k/\beta), \quad a := v + b, \quad Q := e + \frac{ekd^2a}{\alpha v}, \quad n_0 := km, \quad (2)$$

and

$$C_{\text{fp}} := 2 + \log(1 + C_{\text{blk}}). \quad (3)$$

Then the fixed-point and exact-ceiling bounds are

$$\log \frac{en_0}{v} \leq C_{\text{fp}} \log Q, \quad m \geq C_{\text{blk}} \frac{d^2}{\alpha} a \log Q, \quad (4)$$

where the second inequality is the lower side of the exact ceiling, and the universal block calibration is

$$\frac{C_{\text{blk}}}{3600} \geq 4 + \log(1 + C_{\text{blk}}) = C_{\text{fp}} + 2. \quad (5)$$

$$\Pr(E_{\text{good}}^c) \leq \beta_{\text{tr}} = \beta/4. \quad (6)$$

The complementary $d = 0$ branch is the exact $N = 0$ singleton law of Proposition A.21.

For the actual current essential lists, including the totalized empty list for an empty outer restriction, define

$$\mathcal{G}_i := \bigcup_{r=0}^d \mathcal{L}_i^r, \quad |\mathcal{G}_i| \leq (d+1)L, \quad (7)$$

with

$$L = (2^d n_0 d)^d 2^{d^2}, \quad \log L = d \log n_0 + d \log d + 2d^2 \log 2. \quad (8)$$

The elements in (7) are actual functions in H_C , not empirical-trace representatives. For every fixed realizable sample and partition satisfying E_{good} , the measurable event E_{mech} satisfies

$$\Pr(E_{\text{mech}}^c \mid \bar{S}, \mathcal{P}, E_{\text{good}}) \leq \beta_{\text{AT}} + \beta_{\text{SS}} = \beta/2. \quad (9)$$

On E_{mech} , the terminal output is an actual selected-stage list item, not fallback; its marked occurrence is positive and obeys

$$J = i \implies \bar{H} \in \mathcal{G}_i(\bar{S}_i). \quad (10)$$

Equation (10) is pathwise; it asserts no independence conditional on $J = i$. On the previously produced event $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$, the same literal actual output satisfies

$$\widehat{\text{err}}_{\bar{S}}(\bar{H}) := \frac{1}{n_0} \sum_{(q,y) \in \bar{S}} \mathbf{1}\{\bar{H}(q) \neq y\} \leq \frac{\alpha}{8}. \quad (12)$$

This is a deterministic full-master empirical certificate on the core path.

For the downstream holdout argument, fix arbitrary D and $c \in C$, write $\bar{D} = \kappa_{\#} D$ and $\bar{c}$ for the quotient target, and let

$$\bar{S} = (Z_1, \dots, Z_{n_0}) \sim P_{\bar{D}, \bar{c}}^{n_0}, \quad Z_j = (Q_j, \bar{c}(Q_j)). \quad (13)$$

After drawing this iid quotient master sample, draw the setting's data-independent uniform indexed partition $\mathcal{P}$, the learner transcript, and the analysis-only occurrence mark. Denote the resulting full marked law by $\tilde{\mathbb{P}}_{\bar{D}, \bar{c}}$. For a realized partition π , let $\bar{S}_i^\pi$ be the ordered m -record producer block and $\bar{S}_{-i}^\pi$ its ordered $M = (k-1)m$ -record complement, and define

$$\widehat{\text{err}}_{-i}^\pi(h) := \frac{1}{M} \sum_{(q,y) \in \bar{S}_{-i}^\pi} \mathbf{1}\{h(q) \neq y\}, \quad r(h) := \text{err}_{\bar{D}}(h, \bar{c}). \quad (14)$$

The superscript π is suppressed when the realized partition is clear. Finally, let

$$F_\alpha := \{r(\bar{H}) > \alpha\}. \quad (15)$$

These definitions name the common probability space and assert no independence beyond Assumption 3.

Proof. Lemma A.7 and Propositions A.13 and A.14 give the mark and (1). Lemma A.15 and Proposition A.21 give (2)–(5), while Proposition A.20 gives (6). Proposition A.26 gives (7)–(8), and Proposition A.35 gives (9)–(10) and defines $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$, while Lemma A.24 gives (12). Assumption 3 gives the iid quotient law (13); the data-independent partition and Proposition A.13 then define the full marked experiment, and (14)–(15) are the stated complement loss, population risk, and failure-event definitions. They add no independence assertion. Together, the cited results prove all displayed interfaces without requiring an additional event or tail bound. $\square$

Lemma A.25 (Measurable block-local all-stage candidate family). *Under Lemma A.7 and Propositions A.26 and A.44, suppose $d \geq 1$. For every producer $i \in [k]$ and local labeled block $s_i \in Z_Q^m$, let $\mathcal{L}_i^r(s_i)$ be the actual ordered current essential list at stage r , with its empty-list totalization, and set*

$$\mathcal{G}_i(s_i) := \bigcup_{r=0}^d \mathcal{L}_i^r(s_i). \quad (16)$$

Then:

1. $\mathcal{G}_i(s_i)$ depends only on the m labeled records in s_i and the public parameter tuple, not on s_{-i} , the selected stage, mechanism coins, output, or mark; 2. its concatenated finite-list encoding and membership graph

$$\{(s_i, h) \in Z_Q^m \times H_C : h \in \mathcal{G}_i(s_i)\} \quad (17)$$

are measurable; 3. for every local block state,

$$|\mathcal{G}_i(s_i)| \leq (d+1)L; \quad (18)$$

4. for every partition π , the event

$$A_i^\pi := \bigcup_{h \in \mathcal{G}_i(\bar{S}_i^\pi) : r(h) > \alpha} \{\widehat{\text{err}}_{-i}^\pi(h) \leq \alpha/4\} \quad (19)$$

is measurable. This is a finite random union, not an uncountable supremum event.

Proof. The restriction list at stage r is computed from the block empirical errors of s_i alone. Its decomposition/list convention was fixed before sampling as a deterministic lookup on the local quotient state. Consequently each actual list $\mathcal{L}_i^r(s_i)$, and therefore their all-stage union (16), is a function only of s_i and public parameters. Taking the union over all stages rather than only the selected stage is what removes any dependence of the candidate family on the adaptive mechanism transcript.

The local input space Z_Q^m is countable discrete. Thus the pre-fixed finite-list coordinate $s_i \mapsto \mathcal{L}_i^r(s_i)$ is measurable for every r . Concatenating the $d+1$ ordered list encodings is measurable, and dynamic membership in the concatenation is the finite-list membership relation already proved measurable by Lemma A.7. This proves (17). Removing repeated entries across stages is unnecessary for measurability; it only converts the encoding to the mathematical set in (16).

Proposition A.26 gives $|\mathcal{L}_i^r(s_i)| \leq L$ at every stage, including an empty totalized list. Subadditivity over exactly $d+1$ stages proves (18).

It remains to check the section in (19). Because Q_C is finite or countable on the quotient interface, enumerate it as (q_ℓ) . For $h \in H_C$,

$$r(h) = \sum_{\ell} \bar{D}(q_\ell) \mathbf{1}\{h(q_\ell) \neq \bar{c}(q_\ell)\} \quad (20)$$

is the monotone limit of measurable finite coordinate sums. Hence $h \mapsto r(h)$ is $\mathcal{H}_C$ -measurable. The complement empirical error in (14) is a finite measurable sum in $(h, \bar{S}_{-i})$. Combining these two facts with the measurable finite-list graph (17), (19) is a finite measurable disjunction over the entries of the concatenated list. If all lists are empty, that disjunction is empty. If a function occurs at several stages, retaining several encoding positions can only repeat the same measurable event and never enlarges the cardinality bound beyond (18). $\square$

$\square$

Proposition A.45 (Full-path marked inclusion and Holdout). *Under Propositions A.13, A.35, and A.44, Lemma A.24, and Lemma A.25, suppose $d \geq 1$. For every complete realization of the sample, partition, learner transcript, mechanism coins, and analysis mark, and every $i \in [k]$,*

$$E_{\text{core}} \cap \{J = i\} \implies \bar{H} \in \mathcal{G}_i(\bar{S}_i) \quad \text{and} \quad \widehat{\text{err}}_{-i}(\bar{H}) \leq \frac{k\alpha}{8(k-1)} \leq \frac{\alpha}{4}. \quad (21)$$

Consequently the following inclusion holds pathwise:

$$\boxed{E_{\text{core}} \cap \{J = i\} \cap F_\alpha \subseteq \bigcup_{h \in \mathcal{G}_i(\bar{S}_i): r(h) > \alpha} \{\widehat{\text{err}}_{-i}(h) \leq \alpha/4\}.} \quad (22)$$

No probability conditioning, independence statement, or distributional claim about J is used in (21)–(22).

Proof. On E_{core} , Proposition A.35 gives actual rather than fallback status. The occurrence mark is therefore positive, and if its realized value is $J = i$, its support rule (10) gives the literal membership

$$\bar{H} \in \mathcal{G}_i(\bar{S}_i). \quad (23)$$

Lemma A.24 applies to this same function and the same full indexed master sample, giving

$$\sum_{(q,y) \in \bar{S}} \mathbf{1}\{\bar{H}(q) \neq y\} \leq n_0 \frac{\alpha}{8}. \quad (24)$$

Every summand is nonnegative, so deleting the m producer-block terms can only reduce the numerator. Since $n_0 = km$ and $n_0 - m = (k-1)m$,

$$\begin{aligned}\widehat{\text{err}}_{-i}(\bar{H}) &\leq \frac{n_0}{n_0 - m} \frac{\alpha}{8} \\ &= \frac{k\alpha}{8(k-1)} \leq \frac{\alpha}{4},\end{aligned}\tag{25}$$

where the last inequality is exactly $k/(k-1) \leq 2$ for $k \geq 2$. This is (Holdout). It compares the same literal $\bar{H}$, the same target labels $y = \bar{c}(q)$, and the same zero-one loss in the full and complement samples.

If F_α also occurs, then the function $h = \bar{H}$ supplied by (23) satisfies $r(h) > \alpha$, and (25) supplies its right-hand event in (22). Thus that literal function witnesses the finite union, proving the inclusion.

The proof has not conditioned on $J = i$; $\{J = i\}$ is simply one measurable event on the already realized full path. In particular, it does not claim that $\bar{S}_{-i}$ is independent of J , the selected stage, the transcript, or $\bar{H}$. The independent fixed-candidate integration is applied only after (22). $\square$

Lemma A.26 (Direct one-sided iid Bernoulli lower tail). *Under Assumption 3, let $h \in H_C$ be fixed, put $p = r(h) > \alpha$, and let $M = (k-1)m \geq 1$ fresh quotient examples be iid from $P_{\bar{D}, \bar{c}}$. If Y is the number of errors of h on those examples, then*

$$\begin{aligned}\Pr\left[\frac{Y}{M} \leq \frac{\alpha}{4}\right] &\leq \Pr[Y \leq Mp/4] \\ &\leq e^{-9Mp/32} \\ &\leq e^{-9\alpha(k-1)m/32}.\end{aligned}\tag{26}$$

Proof. The error indicators are iid Bernoulli(p), so $Y \sim \text{Bin}(M, p)$. Because $p > \alpha$, the first event in (26) is contained in $\{Y \leq Mp/4\}$.

For $\lambda = \log 4 > 0$, exponential Markov applied to $e^{-\lambda Y}$ gives

$$\begin{aligned}\Pr[Y \leq Mp/4] &\leq e^{\lambda Mp/4} \mathbb{E}[e^{-\lambda Y}] \\ &= e^{\lambda Mp/4} (1 - p + pe^{-\lambda})^M \\ &= e^{\lambda Mp/4} (1 - 3p/4)^M \\ &\leq \exp\left\{Mp\left(\frac{\lambda}{4} - \frac{3}{4}\right)\right\},\end{aligned}\tag{27}$$

where $1 - x \leq e^{-x}$ was used in the last line. The elementary series bound

$$e^{3/2} > 1 + \frac{3}{2} + \frac{(3/2)^2}{2} + \frac{(3/2)^3}{6} > 4$$

implies $\log 4 < 3/2$. Hence the exponent in (27) is at most $-3Mp/8$, which is no larger than $-9Mp/32$. This proves the second line of (26). The last line follows from $p > \alpha$ and $M = (k-1)m$. Every constant is explicit, and no cited Chernoff statement or asymptotic estimate is used. $\square$

Proposition A.46 (Finite producer integration after the pathwise inclusion). *Under Assumption 3, Lemmas A.25 and A.26, and Propositions A.44 and A.45, suppose $d \geq 1$. Then*

$$\tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{core}} \cap F_\alpha) \leq k(d+1)L \exp\left(-\frac{9\alpha(k-1)m}{32}\right). \quad (28)$$

Proof. Proposition A.45 has already proved (22) on the full sample/partition/mechanism/mark path. We now, and only now, condition on the data-independent partition and producer block.

Fix a deterministic indexed partition π and $i \in [k]$. Because the master coordinates in (13) are iid and π is fixed independently of them, the random vectors $\bar{S}_i^\pi$ and $\bar{S}_{-i}^\pi$ are independent; the complement consists of exactly $M = (k-1)m$ iid quotient examples. This is conditioning on the partition, not conditioning on the complete master sample. No independence among blocks conditional on the complete sample is asserted.

Next condition on $\bar{S}_i^\pi = s_i$. By Lemma A.25, the set $\mathcal{G}_i(s_i)$ is now a fixed finite set of at most $(d+1)L$ literal functions and does not depend on the complement. For every member h of this set with $r(h) > \alpha$, Lemma A.26 applies to the independent complement and gives

$$\Pr[\widehat{\text{err}}_{-i}^\pi(h) \leq \alpha/4 \mid \mathcal{P} = \pi, \bar{S}_i^\pi = s_i] \leq e^{-9\alpha(k-1)m/32}. \quad (29)$$

Here the product regular conditional law is chosen to be the iid complement law also on producer states of probability zero; this is legitimate on the countable discrete block space and makes (29) pointwise in s_i . The function in (29) is fixed by s_i . It is not obtained by conditioning on $J = i$, the selected stage, $\bar{H}$, or any mechanism event. Applying the finite union bound to (19) therefore yields

$$\Pr[A_i^\pi \mid \mathcal{P} = \pi, \bar{S}_i^\pi = s_i] \leq (d+1)L e^{-9\alpha(k-1)m/32}. \quad (30)$$

Empty lists give an empty union, and repeated candidates only reduce the number of distinct terms. The bound in (30) is uniform in s_i and π . Measurability from Lemma A.25 permits the tower integral, first over $\bar{S}_i^\pi$ and then over the finite partition law. Writing $A_i := A_i^\mathcal{P}$, this gives

$$\tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(A_i) \leq (d+1)L e^{-9\alpha(k-1)m/32}. \quad (31)$$

The event A_i depends only on the sample and partition, so integrating the learner and mark kernels contributes unit mass. On E_{core} , actual status excludes mark 0. The positive mark events are disjoint, and (22) gives

$$\begin{aligned} \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{core}} \cap F_\alpha) &= \sum_{i=1}^k \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{core}} \cap \{J = i\} \cap F_\alpha) \\ &\leq \sum_{i=1}^k \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(A_i) \\ &\leq k(d+1)L e^{-9\alpha(k-1)m/32}. \end{aligned} \quad (32)$$

This proves (28). The mark is used only to obtain the finite pathwise decomposition in the first line of (32); it is never used as a source of conditional independence. $\square$

$\square$

Lemma A.27 (Exact finite-multiplicity domination by the block calibration). *Under Lemma A.15 and Propositions A.21, A.26, and A.44, suppose $d \geq 1$. Then*

$$k(d+1)L \exp\left(-\frac{9\alpha(k-1)m}{32}\right) \leq \beta_{\text{gen}} = \frac{\beta}{4}. \quad (33)$$

Proof. Retain a, b, Q, C_{fp} from (2)–(3). The positive branch gives $1 \leq v \leq d, k \geq 2$, and $a = v + b \geq 1$. Since $a \geq v$ and $0 < \alpha < 1/4$,

$$Q = e + \frac{ekd^2a}{\alpha v} > e + 8ed^2 > d, \quad (34)$$

so $\log Q > 1$, $\log d \leq \log Q$, and $\log 2 \leq \log Q$. From the fixed point (4),

$$\begin{aligned} \log n_0 &= \log(en_0/v) + \log v - 1 \\ &\leq C_{\text{fp}} \log Q + \log d \\ &\leq (C_{\text{fp}} + 1) \log Q. \end{aligned} \quad (35)$$

Substituting (35) into the exact list formula (8) gives

$$\begin{aligned} \log L &= d \log n_0 + d \log d + 2d^2 \log 2 \\ &\leq (C_{\text{fp}} + 1)d \log Q + d \log Q + 2d^2 \log Q \\ &\leq (C_{\text{fp}} + 4)d^2 \log Q. \end{aligned} \quad (36)$$

The exact confidence multiplicity is

$$\begin{aligned} \log \frac{k(d+1)L}{\beta_{\text{gen}}} &= \log \frac{4k}{\beta} + \log(d+1) + \log L \\ &= b + \log(d+1) + \log L. \end{aligned} \quad (37)$$

Here $b \leq a$, $d+1 \leq 2d$, $d^2a \geq 1$, and $\log Q > 1$. Consequently

$$b \leq d^2a \log Q, \quad \log(d+1) \leq 2 \log Q \leq 2d^2a \log Q, \quad (38)$$

and (36)–(38) yield the fully displayed multiplicity bound

$$\log \frac{k(d+1)L}{\beta_{\text{gen}}} \leq (C_{\text{fp}} + 7)d^2a \log Q. \quad (39)$$

The same confidence-budget result block calibration pays this term. Indeed, (5) implies

$$\frac{9C_{\text{blk}}}{32} = \frac{2025}{2} \frac{C_{\text{blk}}}{3600} \geq \frac{2025}{2} (C_{\text{fp}} + 2) \geq C_{\text{fp}} + 7, \quad (40)$$

where the last difference is positive because $C_{\text{fp}} > 0$. Using the lower ceiling inequality from Proposition A.21, displayed in (4), and $k-1 \geq 1$,

$$\begin{aligned} \frac{9\alpha(k-1)m}{32} &\geq \frac{9C_{\text{blk}}}{32} (k-1)d^2a \log Q \\ &\geq (C_{\text{fp}} + 7)d^2a \log Q \\ &\geq \log \frac{k(d+1)L}{\beta_{\text{gen}}}. \end{aligned} \quad (41)$$

Exponentiating (41) proves (33). Thus the exact trace fixed point and the already fixed universal block constant pay the producer, stage, list, and confidence multiplicities. No new parameter-dependent constant is chosen and no term is absorbed by prose. $\square$

$\square$

Proposition A.47 (Unconditional released quotient PAC ledger and boundary integrity). *Under Assumption 3, Propositions A.14, A.20, A.21, A.35, A.44, Lemma A.24, Proposition A.46, and Lemma A.27, the released quotient law obeys, for every allowed D and $c \in C$,*

$$\Pr_{\bar{S} \sim P_{D,\bar{c}}^{n_0}, \bar{H} \sim K_C(\bar{S}, \cdot)} [\text{err}_{\bar{D}}(\bar{H}, \bar{c}) > \alpha] \leq \beta \quad (42)$$

on the positive branch. Equivalently, the released quotient output has population error at most α with probability at least $1 - \beta$. The $d = 0$ branch satisfies the same conclusion with zero failure probability and no data.

Proof. First suppose $d \geq 1$. Since E_{good} is determined by the sample and partition, integrating the uniform conditional bound (9) over its successful sections gives

$$\begin{aligned} \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{good}} \cap E_{\text{mech}}^c) &= \mathbb{E} [\mathbf{1}_{E_{\text{good}}} \Pr(E_{\text{mech}}^c \mid \bar{S}, \mathcal{P}, E_{\text{good}})] \\ &\leq \beta_{\text{AT}} + \beta_{\text{SS}}. \end{aligned} \quad (43)$$

There is no additional empirical-certificate failure on $E_{\text{core}} = E_{\text{good}} \cap E_{\text{mech}}$: Lemma A.24 is deterministic on that path. Proposition A.20, (43), Proposition A.46, and Lemma A.27 therefore give

$$\begin{aligned} \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(F_\alpha) &\leq \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{good}}^c) + \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{good}} \cap E_{\text{mech}}^c) + \tilde{\mathbb{P}}_{\bar{D}, \bar{c}}(E_{\text{core}} \cap F_\alpha) \\ &\leq \beta_{\text{tr}} + \beta_{\text{AT}} + \beta_{\text{SS}} + \beta_{\text{gen}} \\ &= \frac{\beta}{4} + \frac{\beta}{4} + \frac{\beta}{4} + \frac{\beta}{4} = \beta. \end{aligned} \quad (44)$$

Let

$$B_\alpha := \{h \in H_C : r(h) > \alpha\}.$$

It is measurable by (20), and F_α is its pullback by the terminal output coordinate. The exact projection (1), integrated over the iid sample, says

$$\tilde{\mathbb{P}}_{\bar{D}, \bar{c}}[\bar{H} \in B_\alpha] = \Pr_{\bar{S}, \bar{H} \sim K_C(\bar{S}, \cdot)} [\bar{H} \in B_\alpha]. \quad (45)$$

Combining (44)–(45) proves (42). This projects out the mark; it neither releases J nor assigns it a privacy property.

The remaining boundary cases preserve the same argument:

1. Proposition A.21 gives the setting's $d = 0$ branch: $\bar{C}$ has the unique released member, $N = 0$, and its quotient population error against the unique target is zero. The teacher, blocks, $\mathcal{G}_i$, positive marks, and every denominator containing $k - 1$ are bypassed.
2. At $k = 2$, the complement size is $M = m$, and (25) is exactly $2(\alpha/8) = \alpha/4$. No block-independence claim conditional on the complete sample is introduced.
3. At $d = 1$, $\mathcal{G}_i$ is the union of the two actual endpoint lists $r = 0, 1$, and (18) is $|\mathcal{G}_i| \leq 2L$. Selection at stage 0 or stage d is included because (16) uses all stages.
4. At $v = 1$ and $v = d$, the fixed point and inequalities (34)–(41) remain valid; the argument introduces no extra structural power or rate claim.
5. Empty stage lists give empty finite sections. Repeated functions within the all-stage encoding or across producer blocks only repeat union terms and cannot increase the bound. Repeated sample values remain independent indexed iid draws before realization and retain their literal multiplicities in every empirical sum.
6. A fallback path lies in the complement of E_{mech} whenever E_{good} holds and is paid in (43); outside E_{good} it is paid by β_{tr} . Its mark is 0, and it is never inserted into a positive producer union. An actual selected

function equal in value to $\bar{c}_0$ remains actual by status and is treated normally. 7. The space H_C may be infinite or uncountable as a set. The proof uses only the finite families in (16), measurable coordinate risks, and finite unions; it never forms an uncountable supremum over H_C .

Thus all fallback/core, multiplicity, empty-list, selected-endpoint, smallest-parameter, repeated-record, and infinite-output-space regimes obey the same unconditional quotient PAC conclusion. The restriction-list propositions first supply the actual block-local lists. Lemma A.25 takes their union over all $d + 1$ stages, not over a selected stage, and proves that this family depends only on the producer block, is measurably encoded, and has size at most $(d + 1)L$.

Proposition A.45 then performs the high-risk step in the required order. On every complete path, actual marking gives $\bar{H} \in \mathcal{G}_i(\bar{S}_i)$ when $J = i$, and Lemma A.24 gives the full-master error of that same function. Deleting the producer block yields

$$\widehat{\text{err}}_{-i}(\bar{H}) \leq \frac{k}{k-1} \frac{\alpha}{8} \leq \frac{\alpha}{4}.$$

Together with population failure, this proves the full-path inclusion (22) before any conditional independence is invoked.

Lemma A.26 proves directly that a fixed bad candidate has complement lower-tail probability at most $e^{-9\alpha(k-1)m/32}$. Only after (22) is available does Proposition A.46 fix the independent partition, condition on $\bar{S}_i$, apply that fixed-candidate bound to the now deterministic finite family, and integrate. It obtains exactly

$$\tilde{\mathbb{P}}(E_{\text{core}} \cap F_\alpha) \leq k(d+1)L e^{-9\alpha(k-1)m/32}.$$

Lemma A.27 uses the confidence-budget fixed point and lower ceiling inequality, the exact restriction-list formula for $\log L$, and the same universal block calibration to prove that this term is at most β_{gen} , with every logarithmic domination displayed in (34)–(41).

Finally, Proposition A.46 and Lemma A.27 integrate the generalization charge, and the mechanism ledger assembles

$$\beta_{\text{tr}} + \beta_{\text{AT}} + \beta_{\text{SS}} + \beta_{\text{gen}} = \beta.$$

Proposition A.14 transfers the marked bound exactly to the released quotient law. This proves the exact PAC conversion without conditioning on J for independence, releasing the mark, using an uncountable union, or recharging privacy. □

A.14 VC-Arm Rate Specialization

Proposition A.48 (Exact quotient-to-raw private PAC interface). *Under Assumptions 1, 2, 3, and 4, Propositions A.7, A.8, A.40, and A.47, suppose $d \geq 1$ and use the exact sample size $N = n_0$. Then $K_C = K_C^{\text{VC-Ly}}$ is the released quotient Markov kernel and*

$$A_N(s, E) := K_C(T_N(s), E), \quad s \in Z_X^N, \quad E \in \mathcal{H}_C, \quad (1)$$

is its raw-input Markov-kernel pullback. For every raw replace-one pair $s \sim s'$, including nonrealizable inputs and arbitrary labels,

$$A_N(s, E) \leq e^\varepsilon A_N(s', E) + \delta \quad (E \in \mathcal{H}_C). \quad (2)$$

Moreover, for every probability measure D on (X, Σ) and every $c \in C$,

$$\begin{aligned} & \Pr_{S \sim P_{D,c}^N, \bar{H} \sim A_N(S, \cdot)} [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \\ &= \Pr_{\bar{S} \sim P_{D,\bar{c}}^N, \bar{H} \sim K_C(\bar{S}, \cdot)} [\text{err}_{\bar{D}}(\bar{H}, \bar{c}) > \alpha] \leq \beta. \end{aligned} \quad (3)$$

The equality in (3) is exact for every possibly improper output.

Proof. Proposition A.40, together with the kernel inputs discharged in its proof, gives both Markov kernels in (1), the exact pullback identity, and (2). Its conclusion is deterministic and pointwise over every raw labeled input; it does not use realizability or any utility event.

Fix D, c . Proposition A.8 makes

$$B_{\alpha,D,c} := \{\bar{h} \in H_C : \text{err}_{\bar{D}}(\bar{h}, \bar{c}) > \alpha\} \quad (4)$$

measurable and gives, pointwise for every $\bar{h} \in H_C$,

$$\mathbf{1}\{\text{err}_D(\text{Dec}_C(\bar{h}), c) > \alpha\} = \mathbf{1}\{\bar{h} \in B_{\alpha,D,c}\}. \quad (5)$$

Using (1), the raw failure probability is therefore

$$\int_{Z_X^N} K_C(T_N(s), B_{\alpha,D,c}) P_{D,c}^N(ds). \quad (6)$$

The integrand is measurable because K_C is a kernel and T_N is measurable. By the defining change-of-variables identity for the pushforward measure and Proposition A.7,

$$\begin{aligned} (6) &= \int_{Z_Q^N} K_C(\bar{s}, B_{\alpha,D,c}) (T_N)_\# P_{D,c}^N(d\bar{s}) \\ &= \int_{Z_Q^N} K_C(\bar{s}, B_{\alpha,D,c}) P_{D,\bar{c}}^N(d\bar{s}). \end{aligned} \quad (7)$$

This is exactly the quotient failure probability in (3), including all partition and mechanism randomness already integrated into K_C . Proposition A.47 bounds it by β after the analysis-only mark has been projected out. Thus (3) preserves the released law, target function, and binary population-risk metric with zero residual. $\square$

$\square$

Lemma A.28 (Ceiling-aware elimination of every VC-arm auxiliary). *Under Assumptions 1 and 4, Lemma A.3, Proposition A.2, and Lemmas A.15 and A.16, suppose $d \geq 1$. Put*

$$\ell := \log \frac{64}{\delta\beta}, \quad A_{\log} := 80 + \log(1 + C_{\text{blk}}), \quad A_{\text{def}} := 256(A_{\log} + c_{\text{AT}} + 1), \quad (8)$$

$$C_{\text{teach}} := 2^{12} A_{\text{def}}^2, \quad H := A_{\log}(1 + \log C_{\text{teach}}), \quad (9)$$

and

$$K_{\text{fp}} := 2(1 + C_{\text{blk}})C_{\text{teach}}H(H + 3). \quad (10)$$

All constants in (8)-(10) are universal. For the exact tuple

$$a(t) = v + \log(4t/\beta), \quad Q(t) = e + \frac{etd^2a(t)}{\alpha v}, \quad (11)$$

$$m(t) = \left\lceil C_{\text{blk}} \frac{d^2}{\alpha} a(t) \log Q(t) \right\rceil, \quad N = n_0 = km(k), \quad (12)$$

one has, with every ceiling retained,

$$N \leq K_{\text{fp}} \frac{d^4 \ell \Lambda^3}{\varepsilon \alpha} (v + \Lambda). \quad (13)$$

Thus $k, a(k), Q(k), m$, and $N = km$ have all been replaced by displayed parameters and universal constants.

Proof. Proposition A.2 defines

$$\bar{k} = \left\lceil C_{\text{teach}} \frac{d^2 \ell \Lambda^2}{\varepsilon} \right\rceil \quad (14)$$

before sampling and proves

$$2 \leq k \leq \bar{k} \leq 2C_{\text{teach}} \frac{d^2 \ell \Lambda^2}{\varepsilon}. \quad (15)$$

The final inequality is the exact ceiling bound $\lceil y \rceil \leq y + 1 \leq 2y$, applied only after the proof establishes $y = C_{\text{teach}} d^2 \ell \Lambda^2 / \varepsilon \geq 1$.

We next retain the independent ceiling in m . With

$$x := C_{\text{blk}} \frac{d^2}{\alpha} a(k) \log Q(k),$$

write exactly $m = x + \theta$, where $0 \leq \theta < 1$. Lemma A.15 proves $d^2 a(k) \log Q(k) / \alpha > 4$, hence

$$\begin{aligned} m &\leq C_{\text{blk}} \frac{d^2}{\alpha} a(k) \log Q(k) + 1 \\ &\leq (1 + C_{\text{blk}}) \frac{d^2}{\alpha} a(k) \log Q(k). \end{aligned} \quad (16)$$

The additive integer remainder has therefore been paid explicitly, rather than hidden in asymptotic notation.

For integers $2 \leq s \leq t$, (11) gives $a(s) \leq a(t)$, then $sa(s) \leq ta(t)$, and therefore $Q(s) \leq Q(t)$. Applying this deterministic monotonicity only after $k \leq \bar{k}$ is known, (15)-(16) give

$$N = km \leq (1 + C_{\text{blk}}) \bar{k} \frac{d^2}{\alpha} a(\bar{k}) \log Q(\bar{k}). \quad (17)$$

No monotonicity of the teacher feasible set is asserted.

Lemma A.3, at the fixed universal $C = C_{\text{teach}}$, gives

$$\log \bar{k} \leq H\Lambda, \quad \log Q(\bar{k}) \leq H\Lambda. \quad (18)$$

Writing $b = \log(1/\beta)$, the parameter ranges give $b \leq \Lambda$, $\Lambda \geq 1$, and $\log 4 < 2$, so

$$\begin{aligned} a(\bar{k}) &= v + \log 4 + \log \bar{k} + b \\ &\leq v + (H + 3)\Lambda. \end{aligned} \quad (19)$$

Substituting (15), (18), and (19) into (17) yields the fully visible chain

$$N \leq 2(1 + C_{\text{blk}}) C_{\text{teach}} H \frac{d^4 \ell \Lambda^3}{\varepsilon \alpha} [v + (H + 3)\Lambda]. \quad (20)$$

Finally,

$$v + (H + 3)\Lambda \leq (H + 3)(v + \Lambda), \quad (21)$$

because $H + 3 \geq 1$. Equations (10), (20), and (21) prove (13). This line-by-line chain eliminates k through (15), $a(k)$ through (19), $Q(k)$ through its logarithm in (18), m through the exact ceiling calculation (16), and $N = km$ through (17)-(20). $\square$

$\square$

Proposition A.49 (Exact normalized VC rate and structural boundary profile). *Under Assumption 4 and Lemma A.28, suppose $d \geq 1$. Define*

$$K_V := \max\{1, 4K_{\text{fp}}\}, \quad q := 4. \quad (22)$$

Then $K_V \geq 1$ is universal, $q \in \mathbb{N}_0$, and

$$N \leq K_V \Lambda(d, v, \alpha, \beta, \varepsilon, \delta)^q R_{\text{VC}}(d, v, \alpha, \beta, \varepsilon, \delta). \quad (23)$$

The bound is pointwise valid for every fixed $0 < \delta < 1$, has only displayed-parameter logarithms inside Λ , has the d^4 profile at $v = 1$, and at $v = d$ has the exact polynomial numerator $d^5 + d^4 \log(1/\beta)$.

Proof. For clarity only inside this proof, set

$$b := \log(1/\beta), \quad s := \log(1/(\delta\beta)), \quad \ell := \log(64/(\delta\beta)). \quad (24)$$

Since $0 < \beta < 1/4$ and $0 < \delta < 1$,

$$b > \log 4 > 1, \quad s > \log 4 > 1. \quad (25)$$

Moreover $\log 64 = 3 \log 4$, so

$$\ell = s + \log 64 \leq 4s. \quad (26)$$

Because $v \geq 1$ and $\Lambda \geq 1$,

$$v + \Lambda \leq \Lambda(v + 1) \leq \Lambda(v + b). \quad (27)$$

Applying (26)-(27) to (13) gives

$$N \leq 4K_{\text{fp}} \Lambda^4 \frac{d^4(v + b)s}{\varepsilon\alpha}. \quad (28)$$

This is the only logarithmic absorption in the rate bridge, and both inequalities proving it are displayed.

The setting's exact rate is

$$R_{\text{VC}} = \frac{d^4(v + b)s}{\varepsilon\alpha} + \frac{d + b}{\alpha}. \quad (29)$$

The second term has not been silently discarded. Indeed $d \geq 1$, $v \geq 1$, $s > 1$, and $\varepsilon \leq 1$ give

$$\frac{d^4(v + b)s}{\varepsilon\alpha} \geq \frac{d^4v + d^4b}{\alpha} \geq \frac{d + b}{\alpha}. \quad (30)$$

Thus the two terms in (29) are explicitly comparable, while in the direction needed for (23),

$$\frac{d^4(v + b)s}{\varepsilon\alpha} \leq R_{\text{VC}}. \quad (31)$$

Equations (22), (28), and (31) prove (23).

At $v = 1$, (28) reads

$$N \leq K_V \Lambda^4 \frac{d^4(1+b)s}{\varepsilon \alpha}, \quad (32)$$

so no additional positive power of v or d is hidden. At $v = d$, the first numerator in (29) is exactly

$$d^4(d+b) = d^5 + d^4b, \quad (33)$$

which is the claimed d^5 structural scale, with its confidence term still explicit. These are specializations of the same VC arm and do not invoke or compare another arm.

Finally, (25)-(31) require only the fixed allowed range $0 < \delta < 1$. They remain finite and valid for an arbitrary such δ , whether moderate or tending to zero. No limiting claim has been used to prove the fixed-sample theorem. $\square$

Proposition A.50 (Exact zero-dimensional no-data baseline). *Under Assumptions 1, 2, 3, and 4, and Propositions A.1, A.8, A.40, and A.47, if $d = 0$, then $v = 0$, $N = 0$, the quotient law and its raw pullback are the exact Dirac law at the unique $\bar{c}_0 \in \bar{C}$, the raw law is $(0,0)$ -DP, and for every D and $c \in C$,*

$$\text{err}_D(\text{Dec}_C(\bar{c}_0), c) = 0 \quad (34)$$

deterministically. Consequently the PAC failure probability is exactly zero.

Proof. Proposition A.1 proves that two distinct concepts would shatter a depth-one Littlestone tree. Hence $d = 0$ makes the nonempty class C , and therefore $\bar{C}$, a singleton; it also implies $v = 0$. The construction bypasses the teacher, blocks, stages, lists, partition, mechanisms, and every expression containing v^{-1} , and sets $N = 0$.

Proposition A.40 identifies both quotient and raw empty-input laws as the same Dirac kernel and proves exact $(0,0)$ -DP. Since the unique released quotient concept is the quotient factor of the unique target, Proposition A.8 gives (34). Proposition A.47 records the same released-law conclusion with zero failure probability. This is an exact baseline, not a limit of the positive-dimensional formula. $\square$

Proposition A.51 (Conditional normalized VC-arm theorem and exact small- δ schedule). *Under Assumptions 1, 2, 3, and 4, and Propositions A.48, A.49, and A.50, the quotient-first totalized VC-sensitive law has the following exact common interface.*

If $d = 0$, it uses $N = 0$, is a quotient/raw Markov-kernel Dirac law, is $(0,0)$ -DP, has identically zero decoded population risk, and therefore $m_C(\alpha, \beta; \varepsilon, \delta) = 0$. If $d \geq 1$, it uses the exact $N = n_0 = km$, its raw pullback is (ε, δ) -DP on every raw neighboring labeled input, and

$$\sup_D \sup_{c \in C} \Pr_{S \sim P_{D,c}^N, \bar{H} \sim A_N(S, \cdot)} [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq \beta, \quad (35)$$

while the universal constants in (22) and the definition of m_C give

$$m_C(\alpha, \beta; \varepsilon, \delta) \leq N \leq K_V \Lambda^q R_{VC}. \quad (36)$$

Along any parameter sequence satisfying exactly

$$\delta K_V \Lambda^q R_{VC} \rightarrow 0, \quad (37)$$

the positive-dimensional sample sizes satisfy $N\delta \rightarrow 0$. The fixed-parameter theorem, including privacy and PAC utility, remains valid for every $0 < \delta < 1$; no $N\delta$ limit is claimed for fixed positive δ absent (37).

This is an explicitly conditional theorem for finite-or-countable measurable evaluation quotients. It makes no claim for uncountable evaluation quotients, proves no universal polynomial in $(v, \log d)$ or $(v, \log^ d)$, and makes no comparison with the old-Lyu (Lyu, 2025) or finite-class arms.*

Proof. On the positive branch, Proposition A.48 supplies the exact quotient/raw kernel identity, all-input privacy, exact decoder-risk transfer, and (35). Proposition A.49 supplies (36) with the universal $K_V = \max\{1, 4K_{\text{fp}}\}$ and integer $q = 4$, including every ceiling and structural boundary checks; the first inequality in (36) follows directly because Proposition A.48 supplies an admissible learner at sample size N . On the null branch, Proposition A.50 supplies the exact no-data conclusion.

For a sequence satisfying (37), nonnegativity and (36) give the displayed squeeze

$$0 \leq N\delta \leq \delta K_V \Lambda^q R_{VC} \longrightarrow 0. \quad (38)$$

This is the entire small- δ argument. It does not infer (37) from $\delta \rightarrow 0$, and it does not infer $N\delta \rightarrow 0$ for a fixed positive δ . The scope statements merely record the unchanged primitive restriction and the arms outside this result. Proposition A.48 first composes the quotient factorization, privacy, and holdout results; its exact integral identity keeps the released quotient law, raw pullback, decoder, target, and binary population risk aligned. Lemma A.28 then composes the parameter closure and trace fixed point; equations (14)–(21) pay both ceilings and eliminate $k, a(k), Q(k), m, N = km$ with the explicit universal constant K_{fp} . Proposition A.49 proves each logarithmic domination, retains the exact two-term R_{VC} , fixes $K_V = \max\{1, 4K_{\text{fp}}\}$ and $q = 4$, and checks $v = 1, v = d$, and every allowed fixed δ . Proposition A.50 separately preserves the exact $d = 0$ baseline.

Together these named results establish the normalized VC-arm theorem. Its final squeeze proves $N\delta \rightarrow 0$ only under the displayed schedule, while fixed positive δ remains covered by the finite-parameter statement. □

A.15 Old-Lyu Comparison Arm

Proposition A.52 (Old-arm quotient and source interfaces). *Under Assumptions 1, 2, and 4, Lemmas A.6, A.7, and Propositions A.6, A.7, A.8, A.11, A.12, A.13, and A.14, suppose $d \geq 1$. The following transport and source interfaces are available for the old-arm construction on Q_C .*

The map T_N is measurable and sends raw replace-one neighbors to equal or replace-one quotient samples; realizable iid samples satisfy $T_N(S) \sim P_{\bar{D}, \bar{c}}^N$; and, for every $\bar{h} \in H_C$,

$$\text{err}_D(\text{Dec}_C(\bar{h}), c) = \text{err}_{\bar{D}}(\bar{h}, \bar{c}). \quad (\text{D1})$$

The countable quotient promotes every pointwise totalized law to a Markov kernel. Whenever a totalized finite-list transcript has a nonempty producer occurrence set, Lemma A.7 and Propositions A.13 and A.14 supply an analysis-only mark whose marginal is exactly the released quotient law.

For a fixed binary population of size N , mean μ , and a uniformly sampled subset of size m , the without-replacement bound from Lyu (Lyu, 2025) is, for $0 < \xi < 1$,

$$\Pr(|\hat{\mu} - \mu| > \xi\mu) \leq 2 \exp(-\xi^2 m\mu/3). \quad (\text{C1})$$

If $\text{LD}(\mathcal{H}) \leq d$, valid (p, d) -decompositions exist, a valid tree has at most $p^d 2^{d^2}$ leaves, and a maximum-dimensional leaf of dimension t is $p 2^{d-t}$ -irreducible. For $\mathcal{G} \subseteq \mathcal{H}$,

$$\text{DDim}_{2p, d}(\mathcal{G}) \leq \text{DDim}_{p, d}(\mathcal{H}). \quad (\text{C2})$$

When both sides equal t , every dimension- t SOA from an optimal $(2p, d)$ -decomposition of $\mathcal{G}$ occurs in every optimal (p, d) -decomposition of $\mathcal{H}$, and there are at most $p^d 2^{d^2}$ essential hypotheses; at DDim zero the essential set is the whole nonempty class.

The fixed family

$$\widehat{C}_{d+1} := \{\text{SOA}_{\mathcal{G}} : \mathcal{G} \subseteq \bar{C} \text{ is } (d+1)\text{-irreducible}\} \quad (\text{C3})$$

has Littlestone dimension at most d . For finite lists $\mathcal{L}_1, \dots, \mathcal{L}_k$ of size at most L , the Sparse Sample law uses score $s(h) = |\{i : h \in \mathcal{L}_i\}|$, a distinct failure symbol of score B , and satisfies

$$B \geq \frac{10 \log(L/\delta_s)}{\varepsilon_s} \implies (2\varepsilon_s, \delta_s)\text{-DP} \quad (\text{C4})$$

under one-list replacement. The stopped one-crossing AboveThreshold interface for sensitivity-one queries has

$$(c_{\text{AT}}\eta[\sqrt{\log(1/\delta_a)} + \log(1/\delta_a)], \delta_a)\text{-DP}, \quad (\text{C5})$$

for a suitable universal $c_{\text{AT}} \geq 1$. The accuracy event for the Laplace noises of the at most $d+1$ stage queries is derived directly from their elementary tails when it is first used; it is not part of this source-privacy interface.

The empirical implication used from Lyu (Lyu, 2025), Theorem 3 is the following precise restriction argument: if a nonempty class $\mathcal{G}$ is irreducible for at least the indexed sample length, $h = \text{SOA}_{\mathcal{G}}$, and every $g \in \mathcal{G}$ has empirical error at most u , then h also has empirical error at most u . Indeed, a larger error for h would make the restriction of $\mathcal{G}$ along the full SOA-labeled sample empty, contradicting irreducibility. No privacy or probability-one conclusion is part of this interface.

Finally, Sauer–Shelah (Sauer, 1972) gives at most $\sum_{j=0}^d \binom{n}{j} \leq (en/d)^d$ traces when $n \geq d \geq 1$. The finite-population and Bernoulli lower-tail inequalities needed later are proved locally at their first use, rather than asserted by this source wrapper or imported from an external authority. For adaptive composition, let M_1 be a kernel from a database space to a finite transcript space $\mathbb{T}$, and, for each $t \in \mathbb{T}$, let $M_{2,t}$ be a kernel from the same database space to a measurable space $(\mathcal{Y}, \mathcal{Y})$. If M_1 is $(\varepsilon_1, \delta_1)$ -DP and the history-indexed kernels $M_{2,t}$ are uniformly $(\varepsilon_2, \delta_2)$ -DP, then their adaptive joint law is $(\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2)$ -DP. Measurable postprocessing and mixing over common data-independent randomness preserve these parameters.

Proof. Propositions A.6, A.7, and A.8 give the transport and risk identities in (D1). Lemma A.6 and Propositions A.11 and A.12 give the kernel promotions, while Lemma A.7 and Propositions A.13 and A.14 give the stated analysis-only mark and exact-projection interface.

Equation (C1) is Lyu (Lyu, 2025), Proposition 2. The decomposition, leaf, irreducibility, DDim, and essential-set assertions in (C2) are the current-notation forms of Lyu (Lyu, 2025), Definitions 4.2–4.3, Lemmas 4.1 and 4.3, and Corollary 4.1. Equation (C3) is Lyu (Lyu, 2025), Lemma 4.2, applied with $\text{LD}(\bar{C}) = d$ from Lemma A.5. Equations (C4)–(C5) are the privacy interfaces of Lyu (Lyu, 2025), Algorithms 1–2 and Lemmas 3.1–3.2. The displayed trace bound is Sauer–Shelah (Sauer, 1972). The finite-population and Bernoulli lower-tail inequalities needed later are proved locally at their first use; this source wrapper does not depend on those later results.

It remains to verify the stated adaptive-composition interface. Fix an ordered neighboring pair x, x' , and write

$$P = M_1(x), \quad Q = M_1(x'), \quad P_t = M_{2,t}(x), \quad Q_t = M_{2,t}(x').$$

Because $\mathbb{T}$ is finite, the subprobability measure

$$P^0(\{t\}) := \min\{P(\{t\}), e^{\varepsilon_1} Q(\{t\})\}$$

satisfies $P^0 \leq P$, $P^0 \leq e^{\varepsilon_1} Q$, and $P^0(\mathsf{T}) \geq 1 - \delta_1$. Indeed, its missing mass is the positive-part sum $\sup_{A \subseteq \mathsf{T}} \{P(A) - e^{\varepsilon_1} Q(A)\}$, which is at most δ_1 by privacy of M_1 .

For each t , take densities p_t, q_t with respect to $P_t + Q_t$ and define

$$dP_t^0 := \min\{p_t, e^{\varepsilon_2} q_t\} d(P_t + Q_t).$$

The same positive-part identity and the uniform privacy of $M_{2,t}$ give

$$P_t^0 \leq P_t, \quad P_t^0 \leq e^{\varepsilon_2} Q_t, \quad P_t^0(\mathsf{Y}) \geq 1 - \delta_2 \quad \text{for every } t.$$

If J_x is the adaptive joint law, define, for every measurable $A \subseteq \mathsf{T} \times \mathsf{Y}$ with sections $A_t = \{y : (t, y) \in A\}$,

$$J_x^0(A) := \sum_{t \in \mathsf{T}} P^0(\{t\}) P_t^0(A_t).$$

Then $J_x^0 \leq J_x$, and

$$J_x^0(\mathsf{T} \times \mathsf{Y}) \geq (1 - \delta_2) P^0(\mathsf{T}) \geq 1 - \delta_1 - \delta_2.$$

Moreover,

$$\begin{aligned} J_x^0(A) &\leq e^{\varepsilon_2} \sum_{t \in \mathsf{T}} P^0(\{t\}) Q_t(A_t) \\ &\leq e^{\varepsilon_1 + \varepsilon_2} \sum_{t \in \mathsf{T}} Q(\{t\}) Q_t(A_t) = e^{\varepsilon_1 + \varepsilon_2} J_{x'}(A). \end{aligned}$$

Consequently

$$J_x(A) \leq e^{\varepsilon_1 + \varepsilon_2} J_{x'}(A) + \delta_1 + \delta_2.$$

Repeating the construction with x, x' reversed proves the reverse neighbor inequality. Postprocessing follows by applying this bound to inverse images, and integration over the same data-independent mixing law preserves it. Combining these interfaces gives the stated old-arm package without changing any source claim. $\square$

Proposition A.53 (Exact old-arm singleton branch). *Under Assumptions 1 and 2, if $d = 0$, then $\bar{C}$ has a unique element $\bar{c}_0$. The old arm with $N_{\text{old}} = 0$, quotient law*

$$K_o(\emptyset, E) = \mathbf{1}_E(\bar{c}_0), \tag{1.1}$$

and raw pullback along T_0 is a measurable $(0, 0)$ -DP Markov kernel. For every D and $c \in C$, its decoded population error is zero.

Proof. Lemma A.5 gives $\text{LD}(\bar{C}) = 0$. If two binary functions in $\bar{C}$ differed at a quotient point, that point would form a depth-one Littlestone tree; hence $\bar{C}$ is a singleton. Equation (1.1) is a Dirac probability law on the standard-Borel output space and is constant on the unique empty input. It is therefore a Markov kernel and is $(0, 0)$ -DP. Proposition A.4 identifies the unique $\bar{c}_0$ with every raw target, and (D1) gives zero decoded error. No data, partition, list, mechanism, or asymptotic condition is present. $\square$

Lemma A.29 (Independent old-arm scalar dictionary). *Under Assumptions 1 and 4 and Proposition A.52, suppose $d \geq 1$. Fix a universal $c_{\text{AT}} \geq 1$ for which the bound (C5) holds, and fix the universal old block constant*

$$C_o := 2^{20}. \tag{2.1}$$

Define

$$\gamma_o := \frac{\alpha}{16}, \quad \rho := 1 - \frac{1}{2d}, \quad \xi_d := \frac{1}{5d}, \quad (2.2)$$

$$\beta_{\text{tr},o} = \beta_{\text{AT},o} = \beta_{\text{SS},o} = \beta_{\text{gen},o} := \frac{\beta}{4}, \quad \delta_{\text{AT},o} = \delta_{\text{SS},o} := \frac{\delta}{2}, \quad (2.3)$$

$$g_\delta := \log(4/\delta), \quad \eta_o := \frac{\varepsilon}{4c_{\text{AT}}(\sqrt{g_\delta} + g_\delta)}, \quad \varepsilon_{\text{SS},o} := \frac{\varepsilon}{8}. \quad (2.4)$$

For every integer $t \geq 2$, define

$$a_o(t) := d + \log(4t/\beta), \quad Q_o(t) := e + \frac{etd^2a_o(t)}{\alpha d}, \quad (2.5)$$

$$m_o(t) := \left\lceil C_o \frac{d^2}{\alpha} a_o(t) \log Q_o(t) \right\rceil, \quad n_o(t) := t m_o(t), \quad (2.6)$$

$$p_{o,r}(t) := 2^r n_o(t) d \quad (0 \leq r \leq d), \quad L_o(t) := p_{o,d}(t) d 2^{d^2}, \quad (2.7)$$

$$B_o(t) := \left\lceil \frac{10 \log(L_o(t)/\delta_{\text{SS},o})}{\varepsilon_{\text{SS},o}} \right\rceil, \quad (2.8)$$

$$\tau_{\text{AT},o} := \eta_o^{-1} \log((d+1)/\beta_{\text{AT},o}), \quad \tau_{\text{SS},o}(t) := \varepsilon_{\text{SS},o}^{-1} \log((tL_o(t) + 1)/\beta_{\text{SS},o}). \quad (2.9)$$

Every displayed quantity is finite and positive; $m_o(t)$, $n_o(t)$, $p_{o,r}(t)$, $L_o(t)$, and $B_o(t)$ are integers. Moreover

$$B_o(t) \geq \frac{10 \log(L_o(t)/\delta_{\text{SS},o})}{\varepsilon_{\text{SS},o}}, \quad |\mathcal{L}| \leq p_{o,r}(t) d 2^{d^2} \leq L_o(t) \quad (2.10)$$

for every valid $(p_{o,r}(t), d)$ -essential list $\mathcal{L}$.

Proof. The parameter ranges give $g_\delta > \log 4 > 0$, $a_o(t) > 0$, and $Q_o(t) > e$. Hence all logarithms and reciprocals in (2.2)-(2.9) are legal and finite. The stated integrality follows from the ceilings and integer products and powers. The first inequality in (2.10) is exactly (2.8). For the second, Lyu (Lyu, 2025, Lemma 4.1 and Corollary 4.1) gives $|\mathcal{L}| \leq p_{o,r}(t) d 2^{d^2}$; monotonicity $p_{o,r}(t) \leq p_{o,d}(t)$ gives the final cap. Thus the privacy threshold (C4) and every finite list cap are fixed before any input is observed. $\square$

$\square$

Lemma A.30 (Ceiling-aware old teacher envelope). *Under Lemma A.29, put*

$$\ell := \log(64/(\delta\beta)), \quad R_{\text{T},o} := \frac{d^2 \ell \Lambda^2}{\varepsilon}. \quad (3.1)$$

There is a universal constant $A_o \geq 1$, depending only on the two fixed universal constants c_{AT} and C_o , such that, for every universal $C \geq 1$ and

$$t_C := \lceil C R_{\text{T},o} \rceil, \quad (3.2)$$

one has

$$\begin{aligned} \log t_C + \log m_o(t_C) + \log Q_o(t_C) &\leq A_o(1 + \log C) \Lambda^2, \\ \log L_o(t_C) &\leq A_o(1 + \log C) d^2 \Lambda^2, \end{aligned} \quad (3.3)$$

and

$$B_o(t_C) + \tau_{\text{SS},o}(t_C) + \tau_{\text{AT},o} \leq A_o(1 + \log C) R_{\text{T},o}. \quad (3.4)$$

Proof. First note that $\ell > 1$, $R_{T,o} > 1$, and

$$t_C \leq 2CR_{T,o}. \quad (3.5)$$

The definition of Λ gives

$$\log d, \log(1/\alpha), \log(1/\beta), \log(1/\varepsilon), \log \Lambda \leq \Lambda. \quad (3.6)$$

Also, with $u = \log(e/\delta)$,

$$\ell \leq \log 64 + u + \log(1/\beta), \quad \log \ell \leq 3\Lambda. \quad (3.7)$$

Indeed $\log u \leq \log(e + \log(e/\delta)) \leq \Lambda$, and the logarithm of the sum in (3.7) is bounded by a universal constant plus the maximum of $\log u, \log \log(1/\beta), 1$. Equations (3.5)-(3.7) imply

$$\log t_C \leq A_1(1 + \log C)\Lambda \quad (3.8)$$

for a numerical A_1 .

Because $d \geq 1$ and $\Lambda \geq 1$, (2.5) and (3.8) give

$$a_o(t_C) \leq A_2(1 + \log C)d\Lambda, \quad \log Q_o(t_C) \leq A_3(1 + \log C)\Lambda. \quad (3.9)$$

To see the second inequality without suppressing a parameter, use

$$Q_o(t_C) \leq 2e t_C d a_o(t_C) / \alpha$$

(the product on the right exceeds e) and take logarithms, invoking (3.6), (3.8), and the first inequality in (3.9).

The unrounded expression in (2.6) exceeds one. Consequently

$$m_o(t_C) \leq 2C_o \frac{d^2}{\alpha} a_o(t_C) \log Q_o(t_C), \quad (3.10)$$

and taking logarithms proves the remaining first-line term of (3.3). Furthermore

$$\begin{aligned} \log L_o(t_C) &= d \log(2^d n_o(t_C) d) + d^2 \log 2 \\ &= 2d^2 \log 2 + d \log t_C + d \log m_o(t_C) + d \log d \\ &\leq A_4(1 + \log C)d^2 \Lambda, \end{aligned} \quad (3.11)$$

which is stronger than the second line of (3.3).

It remains to expose the nonlogarithmic δ charge. From $\varepsilon_{SS,o} = \varepsilon/8$ and $\delta_{SS,o} = \delta/2$,

$$B_o(t_C) \leq 1 + \frac{80}{\varepsilon} [\log L_o(t_C) + \log(2/\delta)]. \quad (3.12)$$

Since $t_C L_o(t_C) \geq 2$,

$$\tau_{SS,o}(t_C) \leq \frac{8}{\varepsilon} [\log(8/\beta) + \log t_C + \log L_o(t_C)]. \quad (3.13)$$

Finally $g_\delta > 1$, so $\sqrt{g_\delta} + g_\delta \leq 2g_\delta \leq 2\ell$, while

$$\log(4(d+1)/\beta) \leq 4\Lambda.$$

Thus

$$\tau_{AT,o} \leq \frac{32c_{AT}\ell\Lambda}{\varepsilon}. \quad (3.14)$$

Each summand in (3.12)-(3.14), including the ceiling remainder, is at most a universal multiple of $d^2\ell\Lambda^2/\varepsilon$. Combining (3.8)-(3.14) and enlarging one universal A_o proves (3.3)-(3.4). No feasibility assumption was used to define t_C or any quantity evaluated at it. $\square$

$\square$

Proposition A.54 (Feasible least old teacher and exact margin). *Under Lemmas A.29 and A.30, let*

$$C_{\text{teach,o}} := (16A_o)^2, \quad \bar{k}_o := \left\lceil C_{\text{teach,o}} \frac{d^2 \log(64/(\delta\beta))\Lambda^2}{\varepsilon} \right\rceil. \quad (4.1)$$

Then

$$\frac{\bar{k}_o}{2} - \tau_{\text{AT,o}} \geq B_o(\bar{k}_o) + \tau_{\text{SS,o}}(\bar{k}_o) + 2. \quad (4.2)$$

Consequently the least feasible integer

$$k_o := \min \left\{ t \in \mathbb{Z} : t \geq 2, \quad \frac{t}{2} - \tau_{\text{AT,o}} \geq B_o(t) + \tau_{\text{SS,o}}(t) + 2 \right\} \quad (4.3)$$

exists and

$$2 \leq k_o \leq \bar{k}_o \leq 2C_{\text{teach,o}} \frac{d^2 \log(64/(\delta\beta))\Lambda^2}{\varepsilon}. \quad (4.4)$$

Define, only after this minimization,

$$\begin{aligned} a_o &:= a_o(k_o), & Q_o &:= Q_o(k_o), & m_o &:= m_o(k_o), \\ n_o &:= k_o m_o = N_o = N_{\text{old}}, & p_{o,r} &:= 2^r n_o d, & L_o &:= L_o(k_o), \\ B_o &:= B_o(k_o), & \tau_{\text{SS,o}} &:= \tau_{\text{SS,o}}(k_o). \end{aligned} \quad (4.5)$$

This exact tuple satisfies

$$\frac{k_o}{2} - \tau_{\text{AT,o}} \geq B_o + \tau_{\text{SS,o}} + 2, \quad p_{o,0} = n_o d \geq \max\{n_o, d + 1\}. \quad (4.6)$$

Proof. For $x \geq 1$, $\log x \leq \sqrt{x}$. Hence, with $C = C_{\text{teach,o}}$,

$$A_o(1 + \log C) \leq A_o + 16A_o^2 \leq \frac{C}{4}. \quad (4.7)$$

Lemma A.30 therefore bounds the three defects at $\bar{k}_o$ by

$$\frac{C_{\text{teach,o}} R_{\text{T,o}}}{4}. \quad (4.8)$$

On the other hand, $\bar{k}_o/2 \geq C_{\text{teach,o}} R_{\text{T,o}}/2$. Since $R_{\text{T,o}} > \log 256 > 5$, the remaining quarter of the teacher scale exceeds 2, proving (4.2). Thus the set in (4.3) is a nonempty subset of the integers bounded below by two, so it has a least element and (4.4) follows from $\lceil C_{\text{teach,o}} R_{\text{T,o}} \rceil \leq 2C_{\text{teach,o}} R_{\text{T,o}}$. Equation (4.6)'s first part is (4.3) evaluated at the least element. Finally $n_o \geq k_o \geq 2$; hence $n_o d \geq n_o$ and $n_o d \geq 2d \geq d + 1$. All ceilings remain in (4.5). $\square$

$\square$

Proposition A.55 (All-path totalization of the old quotient procedure). *Under Assumptions 1, 2, and 4, Lyu (Lyu, 2025, Claim 4.1, Definitions 4.2–4.3, Lemmas 4.1 and 4.3, and Corollary 4.1), and Proposition A.54, the positive-dimensional old quotient procedure can be fixed before data so that it outputs an element of H_C on every input in $Z_Q^{N_o}$ and every internal path. It uses only the old tuple (4.5), preserves every exact valid path, and sends every empty, invalid, failure-symbol, no-success, exhausted, or residual path to the fixed $\bar{c}_0$.*

Proof. Fix once and for all the old tuple, the default $\bar{c}_0$, a data-independent uniform partition rule into k_o indexed blocks of size m_o , and deterministic choices for identical local states. For block i and stage $r = 0, \dots, d$, form the exact restriction

$$H_{o,i}^r := \{h \in \bar{C} : e_{\bar{s}_{o,i}}(h) \leq \rho^{r+1} \gamma_o\}. \quad (5.1)$$

For every nonempty restriction, Lyu (Lyu, 2025, Claim 4.1 and Definition 4.3) supplies a valid optimal $(p_{o,r}, d)$ -decomposition. Define $\mathcal{L}_{o,i}^r$ to be its exact set of $(p_{o,r}, d)$ -essential SOA functions, in one fixed no-repetition order. By (2.10) it is a finite H_C -list of size at most L_o . Empty restrictions receive the empty list. If a purported list object is undefined, nonfinite, outside H_C , or over the cap, record an internal failure token and replace only that local list by the empty list. These checks do not alter a valid list.

At each stage define the total score

$$q_{o,r} := \begin{cases} \max_{h \in \cup_i \mathcal{L}_{o,i}^r} |\{i : h \in \mathcal{L}_{o,i}^r\}|, & \cup_i \mathcal{L}_{o,i}^r \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases} \quad (5.2)$$

Feed the $d+1$ scores in order to one stopped AboveThreshold process at threshold $k_o/2$. At its first legal Above stage, invoke the exact Sparse Sample law on the k_o lists using $(\varepsilon_{ss,o}, B_o)$. Return its actual union item if one is drawn. Return $\bar{c}_0$ if AboveThreshold never selects a legal stage, if Sparse Sample draws $\perp$, or if any transcript coordinate is invalid or exhausted. Because the failure-symbol weight is always positive and the effective union is finite, the valid Sparse Sample normalizer is finite and nonzero. Thus every path terminates in H_C , while exact restrictions, decompositions, lists, mechanism laws, and successful outputs are unchanged on valid paths. No event, realizability condition, privacy conclusion, or utility conclusion enters this definition. $\square$

Proposition A.56 (Old quotient kernel, raw pullback, and exact risk). *Under Assumption 2, Propositions A.6, A.8, and A.12, Lemma A.6, and Proposition A.55, the totalized pointwise old law defines a Markov kernel*

$$K_o = K_C^{\text{old-Lyu}} : Z_Q^{N_o} \rightsquigarrow H_C. \quad (6.1)$$

Its raw pullback

$$A_{o,N_o}(s, E) := K_o(T_{N_o}(s), E) \quad (6.2)$$

is a Markov kernel $Z_X^{N_o} \rightsquigarrow H_C$. For every $D, c, \bar{h}$, the released decoded risk is exactly (D1).

Proof. Proposition A.55 assigns a probability law on $(H_C, \mathcal{H}_C)$ to every atom of the countable-discrete input $Z_Q^{N_o}$. Lemma A.6 promotes this arbitrary pointwise family to the Markov kernel (6.1); in particular, data-dependent finite lists and fallback choices do not require a measurable global selector. Proposition A.6 makes T_{N_o} measurable, so Proposition A.12 proves (6.2) is a kernel. The output codomain is the full H_C , and Proposition A.8 applies to every such output, including improper SOA functions and $\bar{c}_0$. $\square$

Lemma A.31 (Coarse d -trace count for the old master sample). *Under Assumption 1 and Lemmas A.4 and A.5, suppose $d \geq 1$ and fix any labeled quotient tuple $\bar{s} = ((q_j, y_j))_{j=1}^{n_o}$. Then its old error-trace family*

$$\mathcal{E}_{\bar{C}}(\bar{s}) := \{(\mathbf{1}\{h(q_j) \neq y_j\})_{j=1}^{n_o} : h \in \bar{C}\} \quad (7.1)$$

is finite and satisfies

$$|\mathcal{E}_{\bar{C}}(\bar{s})| \leq \left(\frac{en_o}{d}\right)^d. \quad (7.2)$$

Proof. quotient preservation gives $\text{VC}(\bar{C}) = v \leq d$. On the indexed tuple, xor with the fixed label vector $(y_j)_j$ is a bijection between prediction traces and the error traces (7.1). Repeated quotient points can only identify prediction traces, never create new ones. Proposition A.54 gives $n_o \geq d$, so Sauer–Shelah (Sauer, 1972) applied with the coarser upper bound d yields

$$|\mathcal{E}_{\bar{C}}(\bar{s})| \leq \sum_{j=0}^d \binom{n_o}{j} \leq (en_o/d)^d.$$

No value of v , no finite cardinality of C , and no event from another arm is used. $\square$

$\square$

Lemma A.32 (Both fixed-trace without-replacement tails). *Under Assumption 4 and Proposition A.52 and Lemma A.29, condition on a complete indexed old master sample and fix one binary error trace of mean $\mu \in [0, 1]$. If $\hat{\mu}_i$ is its mean on one uniformly partitioned block of size m_o , then:*

$$\Pr[|\hat{\mu}_i - \mu| > \xi_d \mu \mid \bar{S}_o] \leq 2 \exp\left(-\frac{m_o \mu}{75d^2}\right) \leq 2 \exp\left(-\frac{m_o \alpha}{3600d^2}\right) \quad (8.1)$$

whenever $\mu > \gamma_o/3$, and

$$\Pr[\hat{\mu}_i \geq \gamma_o/2 \mid \bar{S}_o] \leq \exp[-m_o D(\gamma_o/2 \parallel \mu)] \leq \exp(-m_o \gamma_o/30) \quad (8.2)$$

whenever $0 \leq \mu \leq \gamma_o/3$. The second statement includes the boundary $\mu = \gamma_o/3$; at $\mu = 0$ its failure event is empty.

Proof. For (8.1), apply Lyu (Lyu, 2025) Proposition 2, equation (C1), with $\xi_d = 1/(5d)$. Its exponent is $\xi_d^2 m_o \mu/3 = m_o \mu/(75d^2)$. Since $\mu > \gamma_o/3 = \alpha/48$, the second inequality follows.

For (8.2), write the fixed population as $x_1, \dots, x_{n_o} \in \{0, 1\}$. We first prove (C6) directly. For $N \geq 1$, let $e_m(w)$ be the degree- m elementary symmetric polynomial of a nonnegative vector $w \in [0, \infty)^N$, with $e_0 = 1$. We establish

$$e_m(w_1, \dots, w_N) \leq \binom{N}{m} \left(\frac{1}{N} \sum_{j=1}^N w_j \right)^m, \quad 0 \leq m \leq N, \quad (C6)$$

where a zeroth power is one. The cases $m = 0$ and $m = 1$ are equalities in (C6), as is every case when $N = 1$. Suppose now that $N \geq 2$ and $2 \leq m \leq N$. Fix two coordinates a, b and all remaining coordinates z . Expanding by whether a monomial contains neither, one, or both of the selected coordinates, with $e_j(z) := 0$ outside $0 \leq j \leq N - 2$, gives

$$e_m(a, b, z) = A_{\text{sm}} + (a + b)B_{\text{sm}} + abC_{\text{sm}}, \quad A_{\text{sm}} = e_m(z), \quad B_{\text{sm}} = e_{m-1}(z), \quad C_{\text{sm}} = e_{m-2}(z) \geq 0. \quad (8.3a)$$

Replacing a, b by their common average preserves $a + b$, while

$$\left(\frac{a + b}{2} \right)^2 - ab = \frac{(a - b)^2}{4} \geq 0. \quad (8.3b)$$

Thus this averaging operation cannot decrease e_m .

Starting from $w^{(0)} = w$, repeatedly average a maximum and a minimum coordinate, and let $\bar{w} = N^{-1} \sum_j w_j$. The sum and hence $\bar{w}$ stay fixed. If

$$V_s := \sum_{j=1}^N (w_j^{(s)} - \bar{w})^2,$$

then one averaging step satisfies

$$V_s - V_{s+1} = \frac{1}{2} \left(\max_j w_j^{(s)} - \min_j w_j^{(s)} \right)^2. \quad (8.3c)$$

The nonnegative sequence $(V_s)_s$ decreases and converges, so the successive differences tend to zero. Hence the displayed maximum–minimum gap tends to zero. Since $\bar{w}$ always lies between the minimum and the maximum, every coordinate of $w^{(s)}$ tends to $\bar{w}$. Polynomial continuity and the monotonicity just proved now give

$$e_m(w) = e_m(w^{(0)}) \leq \lim_{s \rightarrow \infty} e_m(w^{(s)}) = \binom{N}{m} \bar{w}^m, \quad (8.3d)$$

which proves (C6), including $m = N$; the already separated $m = 0, 1$ cases cover the other endpoints. Here $N = n_o = k_o m_o$, $m = m_o$, $m_o \geq 1$, and $k_o \geq 2$, so $1 \leq m < N$.

For every $\lambda > 0$, apply the proved inequality (C6) to $w_j = e^{\lambda x_j}$. Since $N^{-1} \sum_j w_j = 1 - \mu + \mu e^\lambda$, it gives

$$\begin{aligned} \mathbb{E} \exp\{\lambda m_o \hat{\mu}_i\} &= \frac{e_{m_o}(e^{\lambda x_1}, \dots, e^{\lambda x_{n_o}})}{\binom{n_o}{m_o}} \\ &\leq (1 - \mu + \mu e^\lambda)^{m_o}. \end{aligned} \quad (8.3)$$

Markov's inequality and minimization over $\lambda > 0$ therefore yield, for $q > \mu$,

$$\Pr(\hat{\mu}_i \geq q) \leq \inf_{\lambda > 0} \exp\{m_o[\log(1 - \mu + \mu e^\lambda) - \lambda q]\} = e^{-m_o D(q||\mu)}. \quad (8.4)$$

For $\mu = 0$ the event is impossible. Otherwise, for fixed $q = \gamma_o/2$, the derivative of $D(q||\mu)$ in $\mu < q$ is $(\mu - q)/[\mu(1 - \mu)] < 0$. Hence the minimum over $\mu \leq \gamma_o/3$ occurs at $\gamma_o/3$. Using $\log(1 - u) \geq -u/(1 - u)$ for $0 \leq u < 1$,

$$\begin{aligned} D(\gamma_o/2||\gamma_o/3) &= \frac{\gamma_o}{2} \log \frac{3}{2} + \left(1 - \frac{\gamma_o}{2}\right) \log \frac{1 - \gamma_o/2}{1 - \gamma_o/3} \\ &\geq \frac{\gamma_o}{2} \log \frac{3}{2} - \frac{\gamma_o}{6} \geq \frac{\gamma_o}{30}, \end{aligned} \quad (8.5)$$

where the last line uses $\log(3/2) \geq 2/5$. Substitution in (8.4) proves (8.2). This local one-sided proof is why Proposition 2 is not misapplied at or near zero. $\square$

$\square$

Proposition A.57 (Generated old trace event and confidence charge). *Under Assumptions 1 and 4, Proposition A.54, and Lemmas A.31 and A.32, define $E_{\text{good},o}$ to be the event that, simultaneously for every trace in $\mathcal{E}_{\bar{C}}(\bar{S}_o)$ and every block i ,*

$$\hat{\mu}_i \in \begin{cases} [(1 - \xi_d)\mu, (1 + \xi_d)\mu], & \mu > \gamma_o/3, \\ [0, \gamma_o/2], & 0 \leq \mu \leq \gamma_o/3. \end{cases} \quad (9.1)$$

Then

$$\Pr(E_{\text{good},o}^c | \bar{S}_o) \leq 4k_o \left(\frac{en_o}{d} \right)^d \exp\left(-\frac{m_o \alpha}{3600d^2}\right), \quad (9.2)$$

and the exact old calibration gives

$$\Pr(E_{\text{good},o}^c) \leq \beta_{\text{tr},o} = \beta/4. \quad (9.3)$$

Proof. The two cases in (9.1) are disjoint and exhaustive, with equality at $\gamma_o/3$ assigned to the low case. Conditional on the master sample, each block is marginally a uniform subset; no conditional independence among different blocks is asserted. Lemma A.32, followed by the finite union over the k_o blocks and the traces counted in Lemma A.31, gives (9.2). The factor 4 absorbs the factor 2 in the high branch and the stronger low-branch exponent: since $d \geq 1$,

$$\frac{\gamma_o}{30} = \frac{\alpha}{480} \geq \frac{\alpha}{3600d^2}. \quad (9.4)$$

It remains to discharge the fixed point and the ceiling. The unrounded quantity in (2.6) exceeds one, so (3.10), now at k_o , gives

$$\frac{en_o}{d} \leq \frac{2eC_o k_o d a_o \log Q_o}{\alpha} \leq 2C_o Q_o \log Q_o. \quad (9.5)$$

Because $Q_o \geq e$,

$$\log(en_o/d) \leq C_5 \log Q_o, \quad C_5 := 2 + \log(2C_o). \quad (9.6)$$

Also $m_o \alpha / d^2 \geq C_o a_o \log Q_o$, while

$$\begin{aligned} \log(4k_o) + \log(1/\beta_{\text{tr},o}) &= \log(16k_o/\beta) \\ &\leq 2a_o \log Q_o, \\ d \log(en_o/d) &\leq C_5 a_o \log Q_o. \end{aligned} \quad (9.7)$$

The fixed choice $C_o = 2^{20}$ satisfies

$$\frac{C_o}{3600} > C_5 + 2. \quad (9.8)$$

Combining (9.6)-(9.8) yields

$$\frac{m_o \alpha}{3600d^2} - \log(4k_o) - d \log(en_o/d) \geq \log(1/\beta_{\text{tr},o}). \quad (9.9)$$

Insert (9.9) into (9.2), then integrate over the master sample. This proves (9.3) without using a VC-arm fixed point or event. $\square$

Lemma A.33 (Exact old source endpoint, half-scale map, and inclusion). *Under Propositions A.52 and A.57, define the source-indexed restrictions and scales, including the endpoint omitted from the source's pre-algorithm display, by*

$$H_{o,i,\text{src}}^s := \{h \in \bar{C} : e_{\bar{S}_{o,i}}(h) \leq \rho^s \gamma_o\}, \quad p_{o,s,\text{src}} := 2^s n_o d, \quad 1 \leq s \leq d+1. \quad (10.1)$$

Then the current construction (5.1) obeys the exact map

$$H_{o,i}^r = H_{o,i,\text{src}}^{r+1}, \quad p_{o,r} = \frac{1}{2} p_{o,r+1,\text{src}}, \quad 0 \leq r \leq d. \quad (10.2)$$

On $E_{\text{good},o}$, for every $0 \leq r < d$, every i_ , and every i ,*

$$H_{o,i_*}^{r+1} \subseteq H_{o,i}^r. \quad (10.3)$$

The two constants needed for this inclusion are

$$\rho^d \geq \frac{1}{2}, \quad \frac{1 + \xi_d}{1 - \xi_d} \rho \leq 1. \quad (10.4)$$

Proof. Equation (10.2) follows by substituting $s = r + 1$ into (10.1) and comparing with (5.1) and (4.5). The uniform factor $1/2$ preserves the exact doubling relation $p_{o,r+1} = 2p_{o,r}$ used by (C2).

Bernoulli's inequality gives

$$\rho^d = (1 - 1/(2d))^d \geq 1 - d/(2d) = 1/2.$$

For the second part of (10.4), direct multiplication reduces the desired inequality to

$$\frac{2}{5d} \leq \frac{1}{2d} \left(1 + \frac{1}{5d} \right),$$

which holds for $d \geq 1$.

Fix $h \in H_{o,i_*}^{r+1}$, and write $\mu = e_{\bar{S}_o}(h)$. If $\mu \leq \gamma_o/3$, (9.1) gives

$$e_{\bar{S}_{o,i}}(h) \leq \gamma_o/2 \leq \rho^d \gamma_o \leq \rho^{r+1} \gamma_o,$$

where $r+1 \leq d$ and $0 < \rho < 1$. If $\mu > \gamma_o/3$, then (9.1), membership in the left side, and (10.4) give

$$\begin{aligned} e_{\bar{S}_{o,i}}(h) &\leq (1 + \xi_d) \mu \\ &\leq \frac{1 + \xi_d}{1 - \xi_d} e_{\bar{S}_{o,i_*}}(h) \\ &\leq \frac{1 + \xi_d}{1 - \xi_d} \rho^{r+2} \gamma_o \leq \rho^{r+1} \gamma_o. \end{aligned}$$

Thus $h \in H_{o,i}^r$, proving (10.3). Both concentration branches, rather than an unstated relative estimate at zero, are needed. $\square$

$\square$

Lemma A.34 (Exact old essential lists and irreducible leaf witnesses). *Under Assumption 1, Proposition A.54, Lemma A.33, and Lyu (Lyu, 2025, Claim 4.1, Definition 4.3, Lemmas 4.1 and 4.3, and Corollary 4.1), every nonempty $H_{o,i}^r$ has the exact finite set $\mathcal{L}_{o,i}^r$ of $(p_{o,r}, d)$ -essential hypotheses, with*

$$|\mathcal{L}_{o,i}^r| \leq p_{o,r}^d 2^{d^2} \leq L_o. \quad (11.1)$$

If $f \in \mathcal{L}_{o,i}^r$ and $\mathcal{G}$ is a maximum-dimensional leaf of dimension t witnessing $f = \text{SOA}_{\mathcal{G}}$, then

$$\mathcal{G} \subseteq H_{o,i}^r, \quad \mathcal{G} \text{ is } p_{o,r} 2^{d-t} \text{-irreducible}, \quad (11.2)$$

and

$$p_{o,r} 2^{d-t} \geq p_{o,0} = n_o d \geq \max\{n_o, d + 1\}. \quad (11.3)$$

Empty restrictions retain the fixed empty-list convention and are not passed to a nonempty-class source assertion.

Proof. Every restriction is a subclass of $\bar{C}$ and therefore has Littlestone dimension at most d . Lyu (Lyu, 2025, Claim 4.1 and Definition 4.3) gives a valid optimal decomposition for each nonempty restriction, and Lyu (Lyu, 2025, Lemma 4.1 and Corollary 4.1) gives (11.1). By the definition of essentiality, f occurs as the SOA of a maximum-dimensional leaf in every optimal decomposition; validity of that leaf gives (11.2). Since $0 \leq t \leq d$, $r \geq 0$, and $p_{o,r} = 2^r p_{o,0}$, its irreducibility scale is at least $p_{o,0}$, and Proposition A.54 gives (11.3).

If a class is q -irreducible, it is q' -irreducible for $1 \leq q' \leq q$: extend any q' -tuple to length q by repeating a fixed quotient point, apply q -irreducibility, and use monotonicity of Littlestone dimension under restriction. Thus every leaf in (11.2) is in particular both n_o - and $(d+1)$ -irreducible. $\square$

$\square$

Proposition A.58 (Finite old DDim descent and common-score stage). *Under Assumptions 1 and 3, Proposition A.52, on $E_{\text{good},o}$, and under Lemmas A.33 and A.34, define*

$$q_{o,r} := \max_{h \in H_C} |\{i : h \in \mathcal{L}_{o,i}^r\}|, \quad M_{o,r} := \max_i \text{DDim}_{p_{o,r},d}(H_{o,i}^r). \quad (12.1)$$

Then, for $0 \leq r < d$,

$$q_{o,r} < k_o \implies M_{o,r+1} \leq M_{o,r} - 1, \quad (12.2)$$

and

$$\max_{0 \leq r \leq d} q_{o,r} = k_o. \quad (12.3)$$

Proof. Realizability puts $\bar{c}$ in every restriction (5.1), because its block empirical error is zero. Thus all classes used in the descent are nonempty.

Suppose $q_{o,r} < k_o$, and choose i_* attaining $M_{o,r+1}$. Lemma A.33 gives

$$H_{o,i_*}^{r+1} \subseteq H_{o,i_*}^r, \quad p_{o,r+1} = 2p_{o,r}$$

for every i . Applying the decomposition bound (C2) yields

$$M_{o,r+1} \leq \text{DDim}_{p_{o,r},d}(H_{o,i_*}^r) \leq M_{o,r}. \quad (12.4)$$

If equality held in (12.4), fix any optimal next-stage decomposition and one of its maximum-dimensional leaves, and write f for that leaf's SOA. The equality clause of (C2), applied separately to every i and to an arbitrary optimal current decomposition, says that f occurs at a maximum-dimensional leaf of every such current decomposition. Hence f is $(p_{o,r}, d)$ -essential to every $H_{o,i}^r$, so $f \in \mathcal{L}_{o,i}^r$ for all i . This gives $q_{o,r} = k_o$, a contradiction. DDim is integer-valued, proving the one-unit drop (12.2).

If $M_{o,r} = 0$, every current restriction has DDim zero. Lyu (Lyu, 2025, Corollary 4.1, item 4) says its essential list is the entire finite restriction. The common target $\bar{c}$ therefore lies in every list, and $q_{o,r} = k_o$. Finally $0 \leq M_{o,0} \leq d$. If every $q_{o,r}$ were smaller than k_o , applying (12.2) for $r = 0, \dots, d-1$ would give $M_{o,d} = 0$, which forces $q_{o,d} = k_o$, again a contradiction. This proves (12.3) by a finite potential argument with no imported success event. $\square$

Lemma A.35 (Old AboveThreshold detects a sufficient stage). *Under Assumption 4, Propositions A.54 and A.58, and conditional on $E_{\text{good},o}$, let $E_{\text{AT},o}$ be the event that all $d+1$ Laplace noises in the stopped old AboveThreshold transcript have magnitude at most $\tau_{\text{AT},o}$. Then*

$$\Pr(E_{\text{AT},o}^c) \leq \beta_{\text{AT},o}, \quad (13.1)$$

and on $E_{\text{AT},o}$ the process selects a legal stage $r_{o,*} \in \{0, \dots, d\}$ such that

$$q_{o,r_{o,*}} \geq \frac{k_o}{2} - \tau_{\text{AT},o} \geq B_o + \tau_{\text{SS},o} + 2. \quad (13.2)$$

Every earlier reported Below additionally satisfies

$$q_{o,r} < \frac{k_o}{2} + \tau_{\text{AT},o} < k_o. \quad (13.3)$$

Proof. For $Z \sim \text{Lap}(1/\eta_o)$, $\Pr(|Z| > u) = e^{-\eta_o u}$. The definition (2.9) and a union bound over at most $d+1$ noises prove (13.1); independence of those noises is more than is needed for this union bound.

On $E_{\text{AT},o}$, a Below report means $q_{o,r} + Z_r < k_o/2$, giving the first inequality in (13.3). The margin (4.6) implies $\tau_{\text{AT},o} < k_o/2$, which gives the second. Proposition A.58 supplies a stage with score exactly k_o . At that stage, even noise $-\tau_{\text{AT},o}$ leaves the noisy score above $k_o/2$, so the stopped transcript reports Above no later than that stage. At the selected stage, the Above inequality gives

$$q_{o,r_{o,*}} \geq k_o/2 - \tau_{\text{AT},o},$$

and (4.6) gives (13.2). Thus no no-success or illegal-stage fallback is reached on the stated event. $\square$

Proposition A.59 (Actual old Sparse Sample output). *Under Lemmas A.34 and A.35, conditional on $E_{\text{good},o} \cap E_{\text{AT},o}$, let $E_{\text{SS},o}$ be the event that Sparse Sample at $r_{o,*}$ returns an actual member of $\bigcup_i \mathcal{L}_{o,i}^{r_{o,*}}$, rather than $\perp$. Then*

$$\Pr(E_{\text{SS},o}^c \mid \text{the selected old lists}) \leq \frac{\beta_{\text{SS},o}}{k_o L_o + 1} \leq \beta_{\text{SS},o}, \quad (14.1)$$

and, on $E_{\text{SS},o}$,

$$\bar{H}_o \in \mathcal{L}_{o,i}^{r_{o,*}} \quad \text{for at least one } i. \quad (14.2)$$

Proof. The effective union has size at most $k_o L_o$. By (13.2), an item h_* has score $q_{o,r_{o,*}}$. In the exact Sparse Sample law, the failure symbol has weight $e^{\varepsilon_{\text{SS},o} B_o}$, while h_* has weight $e^{\varepsilon_{\text{SS},o} q_{o,r_{o,*}}}$. Discarding every other positive denominator term,

$$\begin{aligned} \Pr(\perp \mid \text{selected lists}) &\leq \exp[-\varepsilon_{\text{SS},o}(q_{o,r_{o,*}} - B_o)] \\ &\leq \exp(-\varepsilon_{\text{SS},o} \tau_{\text{SS},o}) \\ &= \frac{\beta_{\text{SS},o}}{k_o L_o + 1}. \end{aligned} \quad (14.3)$$

Here the unused $+2$ in (13.2) is harmless positive slack. Every nonfailure point in the exact effective domain is an actual union item, giving (14.2). This proves utility directly from the displayed weights and does not rely on any ambiguity about the failure symbol. $\square$

Lemma A.36 (Old list locality and query sensitivity on all inputs). *Under Proposition A.55, fix any old partition and any two quotient datasets differing in at most one record. At every stage, the two totalized list tuples differ in at most one block coordinate; all lists at every other coordinate are identical. Consequently every maximum-occurrence query (5.2) has global sensitivity at most one, and, conditional on every transcript-selected legal stage, the two Sparse Sample inputs have one-list replacement adjacency. These statements hold without realizability or $E_{\text{good},o}$.*

Proof. If the quotient datasets differ, the changed record lies in one fixed partition block i_0 . Every restriction, validity check, fixed decomposition choice, essential list ordering, sanitizer, and failure token for a block is a deterministic function of that block's local labeled state and public parameters. Thus all local states and lists for $i \neq i_0$ are identical at all stages; the i_0 -list may be replaced arbitrarily, including by or from the empty list.

Extend each list-occurrence count to all $h \in H_C$, assigning zero when an item is absent from every list. Replacement of one list changes every such count by at most one. If s, s' are the two count functions, then

$$\max_h s(h) \leq \max_h s'(h) + 1 \quad \text{and} \quad \max_h s'(h) \leq \max_h s(h) + 1,$$

including the all-empty convention. Hence the maximum query has sensitivity one. An adaptively selected stage merely chooses one of these already local tuples; for each fixed first-transcript value, its pair of second-mechanism inputs still differs in at most list i_0 . No event or stopping-stage equality across neighboring executions is assumed. $\square$

Proposition A.60 (All-input quotient and raw replacement privacy). *Under Assumptions 2 and 4, and under the conclusions of Lemma A.36 and Propositions A.6, A.52, and A.56, the released old quotient kernel is $(\varepsilon/2, \delta)$ -DP. Its raw pullback A_{o, N_o} is therefore (ε, δ) -DP for every pair of raw replace-one labeled inputs.*

Proof. For the stopped AboveThreshold process, (C5), (2.4), and $\log(1/\delta_{AT,o}) = \log(2/\delta) \leq g_\delta$ give

$$c_{AT}\eta_o[\sqrt{\log(1/\delta_{AT,o})} + \log(1/\delta_{AT,o})] \leq \varepsilon/4. \quad (16.1)$$

Lemma A.36 discharges sensitivity one, so this first transcript is $(\varepsilon/4, \delta/2)$ -DP. At every transcript-selected stage, the list cap (11.1), threshold (2.8), and one-list adjacency from Lemma A.36 discharge every premise of (C4). Thus Sparse Sample is

$$(2\varepsilon_{SS,o}, \delta_{SS,o}) = (\varepsilon/4, \delta/2)\text{-DP}. \quad (16.2)$$

The finite-transcript composition and postprocessing interface proved in Proposition A.52 therefore gives

$$(\varepsilon/4, \delta/2) + (\varepsilon/4, \delta/2) = (\varepsilon/2, \delta) \preceq (\varepsilon, \delta). \quad (16.3)$$

Mixing over the same data-independent partition preserves (16.3).

This argument applies to the total transcript on every labeled input. In particular, empty restrictions, sanitized invalid lists, different stopping stages, $\perp$, no-success, exhaustion, and fallback are already covered by deterministic postprocessing of the private transcript; neither realizability nor a utility event was used.

Finally, Proposition A.6 maps raw neighbors to equal or quotient replace-one inputs. Applying (16.3) to $K_o(T_{N_o}(\cdot), E)$ proves the claimed raw (ε, δ) -DP inequality for every $E \in \mathcal{H}_C$. $\square$

Proposition A.61 (Exact old SOA identity, fixed family, and empirical certificate). *Under Assumptions 1 and 3 and Proposition A.52, on*

$$\text{Core}_o := E_{\text{good},o} \cap E_{AT,o} \cap E_{SS,o}, \quad (17.1)$$

and under Lemma A.34 and Proposition A.59, the actual old output has a maximum-dimensional leaf witness $\mathcal{G} \subseteq \bar{C}$ satisfying

$$\bar{H}_o = \text{SOA}_{\mathcal{G}}, \quad \mathcal{G} \text{ is both } n_o\text{- and } (d+1)\text{-irreducible}, \quad (17.2)$$

$$\bar{H}_o \in \widehat{C}_{d+1} \subseteq H_C, \quad \text{LD}(\widehat{C}_{d+1}) \leq d, \quad (17.3)$$

and

$$e_{\bar{S}_o}(\bar{H}_o) \leq 2\gamma_o = \alpha/8. \quad (17.4)$$

Proof. By Proposition A.59, the actual output belongs to some selected list. Lemma A.34 and the definition of essentiality produce a maximum-dimensional leaf $\mathcal{G}$ with the exact identity and irreducibility in (17.2). The fixed-family result (C3) then gives (17.3). This is an improper family in general; no claim that $\bar{H}_o \in \bar{C}$ is made.

It remains to prove the empirical certificate. Every $g \in \mathcal{G} \subseteq H_{o,i}^{r_o,*}$ has producer block error at most $\rho^{r_o,*+1}\gamma_o \leq \gamma_o$. Write $\mu = e_{\bar{S}_o}(g)$. If $\mu \leq \gamma_o/3$, then already $\mu < 2\gamma_o$. If $\mu > \gamma_o/3$, the lower relative clause of $E_{\text{good},o}$ gives

$$\mu \leq \frac{e_{\bar{S}_o,i}(g)}{1 - \xi_d} \leq \frac{\gamma_o}{1 - \xi_d} \leq \frac{5}{4}\gamma_o < 2\gamma_o. \quad (17.5)$$

Thus every $g \in \mathcal{G}$ has master error at most $2\gamma_o$.

Suppose instead that $e_{\bar{S}_o}(\text{SOA}_{\mathcal{G}}) > 2\gamma_o$. Any $g \in \mathcal{G}$ agreeing with this SOA on all indexed master points would have exactly the same prediction vector and hence the same master error, contradicting (17.5). Therefore

$$\mathcal{G}|_{(q_1, \text{SOA}_{\mathcal{G}}(q_1)), \dots, (q_{n_o}, \text{SOA}_{\mathcal{G}}(q_{n_o}))} = \emptyset. \quad (17.6)$$

But n_o -irreducibility in (17.2) says this restriction preserves $\text{LD}(\mathcal{G})$, whereas the empty class has strictly smaller dimension. This contradiction proves (17.4). Equation (17.6) is the valid contradiction pattern in Lyu (Lyu, 2025) Theorem 3's proof; the malformed displayed output and unsupported probability-one sentence are not used. $\square$

Proposition A.62 (Old occurrence-mark kernel and exact projection). *Under Assumption 2, Lemmas A.6 and A.7, and Propositions A.55, A.56, A.59, A.52, A.13, and A.14, the old internal-state/output law admits a Markov-kernel lift*

$$\tilde{K}_o : Z_Q^{N_o} \rightsquigarrow H_C \times \{0, 1, \dots, k_o\}. \quad (18.1)$$

For every quotient input $\bar{s}$ and $E \in \mathcal{H}_C$,

$$\sum_{i=0}^{k_o} \tilde{K}_o(\bar{s}, E \times \{i\}) = K_o(\bar{s}, E). \quad (18.2)$$

On actual paths, positive marks are uniform over all distinct producer blocks whose all-stage list union contains the output; every nonactual path has mark 0.

Proof. Let $R_{o,\bar{s}}(d\omega, dh)$ be the joint law of the complete old transcript state and terminal H_C -output. For each block define

$$\mathcal{G}_{o,i}(\omega) := \bigcup_{r=0}^d \mathcal{L}_{o,i}^r(\omega). \quad (18.3)$$

On the measurable actual-output status event, set

$$I_o(\omega, h) := \{i \in [k_o] : h \in \mathcal{G}_{o,i}(\omega)\}. \quad (18.4)$$

The exact Sparse Sample support in Proposition A.59 makes this set nonempty on every actual path. Define

$$w_i(\omega, h) := \begin{cases} \mathbf{1}\{i \in I_o(\omega, h)\} / |I_o(\omega, h)|, & \text{on an actual path,} \\ 0, & \text{on a nonactual path,} \end{cases} \quad i \geq 1, \quad (18.5)$$

and let $w_0 = 0$ on actual paths and $w_0 = 1$ on nonactual paths. Finite-list membership, actual status, the finite sum $|I_o|$, and every w_i are measurable by the countable-input coding in Lemma A.6

and the finite-occurrence construction of Lemma A.7 and Proposition A.13. Repeated occurrences within a list or across stages do not duplicate a block coordinate in (18.4).

For $i = 0, \dots, k_o$, define

$$\tilde{K}_o(\bar{s}, E \times \{i\}) := \int \mathbf{1}_E(h) w_i(\omega, h) R_{o,\bar{s}}(d\omega, dh), \quad (18.6)$$

and extend by finite additivity over mark sections. The weights are nonnegative and sum pointwise to one, so (18.6) is a probability law. The quotient input is countable discrete, and the output product is standard Borel with a finite mark, so Lemma A.6 gives the kernel (18.1). Summing (18.6) over marks and using $\sum_i w_i = 1$ is the exact projection argument of Proposition A.14 and proves (18.2). Thus the mark is analysis-only and changes neither the released marginal nor the privacy proved in Proposition A.60. $\square$

$\square$

Lemma A.37 (Producer-local old reconstruction and pathwise inclusion). *Under Assumption 3, Propositions A.61 and A.62, fix the data-independent old partition and a positive mark i . Define*

$$\mathcal{G}_{o,i}(\bar{S}_{o,i}) := \bigcup_{r=0}^d \mathcal{L}_{o,i}^r, \quad |\mathcal{G}_{o,i}| \leq (d+1)L_o. \quad (19.1)$$

This finite list is a measurable function only of the m_o records in producer block i . On $\text{Core}_o \cap \{J_o = i\}$,

$$\bar{H}_o \in \mathcal{G}_{o,i}, \quad \widehat{\text{err}}_{o,-i}(\bar{H}_o) \leq \frac{k_o \alpha}{8(k_o - 1)} \leq \frac{\alpha}{4}. \quad (19.2)$$

Consequently the following inclusion holds pathwise, before conditioning on either the mark or block data:

$$\begin{aligned} & \{\text{Core}_o, J_o = i, \text{err}_{\bar{D}}(\bar{H}_o, \bar{c}) > \alpha\} \\ & \subseteq \bigcup_{\substack{h \in \mathcal{G}_{o,i}(\bar{S}_{o,i}): \\ \text{err}_{\bar{D}}(h, \bar{c}) > \alpha}} \{\widehat{\text{err}}_{o,-i}(h) \leq \alpha/4\}. \end{aligned} \quad (19.3)$$

Proof. Every restriction, decomposition choice, and essential list for coordinate i is block-local by the construction in Proposition A.55; concatenating the $d+1$ measurable finite lists proves the locality and measurability in (19.1). The size bound is the sum of the $d+1$ caps (11.1); duplicate functions only reduce it.

On an actual path with $J_o = i$, the mark definition (18.4) puts $\bar{H}_o$ in this block's union. On the core, Proposition A.61 bounds the number of mistakes on the whole master sample by $n_o \alpha / 8$. The complement has $n_o - m_o = (k_o - 1)m_o$ records, so its number of mistakes is no larger than the full-sample number and

$$\widehat{\text{err}}_{o,-i}(\bar{H}_o) \leq \frac{n_o}{n_o - m_o} \frac{\alpha}{8} = \frac{k_o \alpha}{8(k_o - 1)} \leq \frac{\alpha}{4},$$

where $k_o \geq 2$. If the left event of (19.3) occurs, its realized $\bar{H}_o$ is therefore one of the finitely many candidates indexing the right union and satisfies both defining conditions. This proves the inclusion without conditioning on an adaptive output. $\square$

$\square$

Lemma A.38 (Fixed-candidate complement lower tail). *Under Assumption 3, condition on the fixed partition, a producer block $\bar{S}_{o,i}$, and any fixed $h \in \mathcal{G}_{o,i}(\bar{S}_{o,i})$. If $p = \text{err}_{\bar{D}}(h, \bar{c}) > \alpha$, then the independent complement of $M = (k_o - 1)m_o$ records satisfies*

$$\Pr[\widehat{\text{err}}_{o,-i}(h) \leq \alpha/4 \mid \bar{S}_{o,i}] \leq \exp(-9\alpha M/32). \quad (20.1)$$

Proof. After conditioning on the producer block, h is fixed because of the locality in Lemma A.37. The complement records remain iid from $P_{\bar{D}, \bar{e}}$ and independent of that block. Their mistake indicators are therefore independent Bernoulli variables of mean p . Since $p > \alpha$,

$$\{\widehat{\text{err}}_{o,-i}(h) \leq \alpha/4\} \subseteq \{\widehat{\text{err}}_{o,-i}(h) \leq p/4\}.$$

We now derive (C7). Put $S = \sum_{j=1}^M X_j$, where the X_j 's are these mistake indicators. Here $M = (k_o - 1)m_o \geq 1$ and $p > \alpha > 0$. For completeness, if $M = 0$ or $p = 0$, the event in (C7) has probability one and its right side equals one. Otherwise fix $0 < \theta < 1$, set $q = (1 - \theta)p$, and take $\lambda = -\log(1 - \theta) > 0$. Exponential Markov applied to $e^{-\lambda S}$ gives

$$\begin{aligned} \Pr(S \leq Mq) &= \Pr(e^{-\lambda S} \geq e^{-\lambda Mq}) \\ &\leq e^{\lambda Mq} \mathbb{E}e^{-\lambda S} \\ &= \exp(M[\lambda q + \log(1 - p\theta)]) \\ &\leq \exp(M[\lambda(1 - \theta)p - p\theta]) \\ &= \exp(-pM[\theta + (1 - \theta)\log(1 - \theta)]), \end{aligned} \tag{20.2}$$

where the penultimate line uses $\log(1 - u) \leq -u$ for $0 \leq u < 1$. To finish the scalar estimate, define

$$g(\theta) := \theta + (1 - \theta)\log(1 - \theta) - \frac{\theta^2}{2}.$$

Then $g(0) = 0$ and

$$g'(\theta) = -\log(1 - \theta) - \theta \geq 0, \tag{20.3}$$

because $u(\theta) := -\log(1 - \theta) - \theta$ satisfies $u(0) = 0$ and $u'(\theta) = \theta/(1 - \theta) \geq 0$. Hence $g(\theta) \geq 0$, and (20.2) proves the locally derived inequality

$$\Pr[S \leq M(1 - \theta)p] \leq \exp(-\theta^2 pM/2). \tag{C7}$$

This does not invoke an external lower-tail theorem. Taking $\theta = 3/4$, then using $p > \alpha$, yields

$$\exp[-(3/4)^2 pM/2] = \exp(-9pM/32) \leq \exp(-9\alpha M/32),$$

which is (20.1). $\square$

$\square$

Proposition A.63 (Independent old-arm PAC ledger). *Under Assumptions 1, 2, 3, and 4, Propositions A.57, A.59, A.62, and A.61, and Lemmas A.35, A.37, and A.38, the released old raw learner satisfies*

$$\sup_D \sup_{c \in C} \Pr[\text{err}_D(\text{Dec}_C(\bar{H}_o), c) > \alpha] \leq \beta. \tag{21.1}$$

Proof. We first discharge the finite multiplicity with the same old block constant already used for the trace event. From (9.5),

$$n_o \leq (2C_o/e)dQ_o \log Q_o. \tag{21.2}$$

Hence, with $C_p := 5 + \log(2C_o)$,

$$\begin{aligned} \log p_{o,d} &= d \log 2 + \log n_o + \log d \\ &\leq C_p d \log Q_o, \end{aligned} \tag{21.3}$$

where $Q_o \geq e$, $\log d \leq d$, and $\log \log Q_o \leq \log Q_o$ were used explicitly. Therefore

$$\begin{aligned} \log \frac{k_o(d+1)L_o}{\beta_{\text{gen},o}} &= \log \frac{4k_o(d+1)}{\beta} + d \log p_{o,d} + d^2 \log 2 \\ &\leq C_6 d a_o \log Q_o, \quad C_6 := C_p + 3. \end{aligned} \quad (21.4)$$

Indeed $a_o \geq d$ and $a_o \geq \log(4k_o/\beta)$, while $\log(d+1) \leq d$.

The unrounded lower bound in (2.6) gives

$$\frac{9}{32} \alpha (k_o - 1) m_o \geq \frac{9C_o}{32} (k_o - 1) d^2 a_o \log Q_o. \quad (21.5)$$

For $C_o = 2^{20}$, $k_o \geq 2$, and $d \geq 1$, the right side of (21.5) is at least the right side of (21.4). Consequently

$$k_o(d+1)L_o \exp[-9\alpha(k_o-1)m_o/32] \leq \beta_{\text{gen},o}. \quad (21.6)$$

Now work under the marked lift. Sum the pathwise inclusion (19.3) over $i = 1, \dots, k_o$. Only after this finite inclusion is in place, condition on each producer block and apply Lemma A.38 to each of its at most $(d+1)L_o$ candidates. Equation (21.6) gives

$$\Pr[\text{Core}_o, \text{err}_{\bar{D}}(\bar{H}_o, \bar{c}) > \alpha] \leq \beta_{\text{gen},o}. \quad (21.7)$$

No uncountable supremum over H_C has been taken.

The other failures are charged independently:

$$\Pr(E_{\text{good},o}^c) \leq \beta_{\text{tr},o}, \quad \Pr(E_{\text{AT},o}^c) \leq \beta_{\text{AT},o}, \quad \Pr(E_{\text{good},o} \cap E_{\text{AT},o} \cap E_{\text{SS},o}^c) \leq \beta_{\text{SS},o}. \quad (21.8)$$

The last inequality follows by integrating the conditional bound (14.1) over the selected lists. Equations (21.7)-(21.8) and the exact confidence split (2.3) yield

$$\Pr[\text{err}_{\bar{D}}(\bar{H}_o, \bar{c}) > \alpha] \leq \beta_{\text{tr},o} + \beta_{\text{AT},o} + \beta_{\text{SS},o} + \beta_{\text{gen},o} = \beta. \quad (21.9)$$

Projection (18.2) removes the analysis-only mark without changing the released law. Proposition A.7 identifies the quotient sampling law, and Proposition A.8 turns (21.9) into (21.1), uniformly over arbitrary D and c . $\square$

Proposition A.64 (Ceiling-aware explicit R_{old} rate). *Under Assumptions 1 and 4, and Proposition A.54, there is a universal $K_O \geq 1$ such that, for $d \geq 1$,*

$$N_{\text{old}} \leq K_O \Lambda^6 R_{\text{old}}(d, \alpha, \beta, \varepsilon, \delta). \quad (22.1)$$

No positive power of d is hidden in $K_O \Lambda^6$.

Proof. The candidate functions in (2.5)-(2.9) are nondecreasing in t . Thus $k_o \leq \bar{k}_o$, the envelope derivation (3.8)-(3.10), and (4.4) give universal constants C_7, C_8, C_9, C_{10} such that

$$\begin{aligned} k_o &\leq C_7 \frac{d^2 \log(64/(\delta\beta)) \Lambda^2}{\varepsilon}, \quad a_o \leq C_8 d \Lambda^2, \\ \log Q_o &\leq C_9 \Lambda^2, \quad m_o \leq C_{10} \frac{d^3}{\alpha} \Lambda^4. \end{aligned} \quad (22.2)$$

The last inequality retains the ceiling: its unrounded term exceeds one, so $\lceil x \rceil \leq 2x$, exactly as in (3.10). Multiplication of the actual counts in (4.5) gives

$$N_{\text{old}} = k_o m_o \leq C_{11} \Lambda^6 \frac{d^5 \log(64/(\delta\beta))}{\varepsilon \alpha}. \quad (22.3)$$

Since $\delta\beta < 1/4$,

$$\log(64/(\delta\beta)) = \log 64 + \log(1/(\delta\beta)) \leq 4 \log(1/(\delta\beta)). \quad (22.4)$$

Substitution into (22.3), followed by the nonnegative second summand in the setting definition

$$R_{\text{old}} = \frac{d^5 \log(1/(\delta\beta))}{\varepsilon\alpha} + \frac{d + \log(1/\beta)}{\alpha},$$

proves (22.1) with $K_O = 4C_{11}$. All appearances of d outside R_{old} in this derivation are inside logarithms already bounded by Λ^6 ; the exposed polynomial is exactly d^5 . $\square$

Proposition A.65 (Old-arm boundaries and fixed versus scheduled δ). *Under Propositions A.53, A.60, A.63, and A.64, the complete old arm has the following boundary properties.*

1. At $d = 0$, it uses $N_{\text{old}} = 0$, deterministic mark 0, zero risk, and $(0, 0)$ -privacy. 2. At $d = 1$, all positive-branch scales, stages, restrictions, and inequalities are legal; in particular $\rho = 1/2$, $\rho^d = 1/2$, and $p_{o,0} \geq 2 = d + 1$. 3. The construction applies unchanged to finite and infinite C , and to $v = d$; it never introduces $|C|$, and it makes no VC-sensitive improvement claim. 4. Empty, invalid, failure, and fallback paths remain measurable and private on all inputs, while the old core excludes them for utility. Actual outputs may be improper elements of H_C . 5. Every integer ceiling and all four confidence shares are retained. 6. For every fixed allowed $0 < \delta < 1$, the kernel, privacy, PAC, and rate conclusions remain valid. Separately, along an asymptotic sequence,

$$\delta K_O \Lambda^6 R_{\text{old}} \longrightarrow 0 \implies N_{\text{old}} \delta \longrightarrow 0. \quad (23.1)$$

The setting's VC-arm schedule is not silently asserted to imply (23.1).

Proof. Item 1 is Proposition A.53; the mark-0 convention is the unique inactive occurrence convention. For item 2, substitute $d = 1$ in (2.2), (4.6), and (10.4); no denominator or stage disappears. Items 3 and 4 follow because the only class-size control is the finite trace restriction (7.2) and finite source list cap (11.1), both valid for infinite classes, while Propositions A.55 and A.60 cover every totalized path. At $v = d$, the old dictionary is literally unchanged and (22.1) is the d^5 source baseline. Proposition A.61 explicitly permits improper SOA outputs.

Item 5 follows from (2.6), (2.8), (4.1), (4.5), and the exact identity $4(\beta/4) = \beta$. None of Propositions A.56, A.60, A.63, or A.64 assumes a limiting relation involving δ , proving fixed- δ validity. Finally, multiplying (22.1) by δ proves (23.1). When $v \ll d$, R_{old} can be polynomially larger than R_{VC} , so the distinct VC-arm schedule in the setting cannot in general replace the explicit old-arm premise in (23.1). Proposition A.53 closes the exact $d = 0$ branch. On $d \geq 1$, Lemma A.29, Lemma A.30, and Proposition A.54 define the independent old tuple, prove a finite witness before minimization, retain all ceilings, and export the source margin. Proposition A.55 then defines every valid and fallback path, and Proposition A.56 turns that pointwise law into the required quotient and raw Markov kernels with exact decoded risk.

The old utility proof is independent of the VC-sensitive arm: Lemma A.31 uses only the coarse exponent d ; Lemma A.32 proves both fixed-trace branches; and Proposition A.57 closes the old fixed point and charges its own trace share. Lemma A.33 supplies the exact source endpoint and half-scale map. Lemma A.34 and Proposition A.58 discharge all decomposition, essentiality, irreducibility, and finite-potential obligations. Lemma A.35 and Proposition A.59 then produce an actual old list output with their own two confidence charges.

Privacy is event-free: Lemma A.36 proves sensitivity one and one-list adjacency on arbitrary inputs, and Proposition A.60 composes the exact old allocations and transfers them to raw neighbors. Utility continues through Proposition A.61, which proves the actual SOA identity, fixed-family

membership, and empirical error. Proposition A.62 constructs the old marked lift and exact projection. Lemmas A.37 and A.38 reduce the adaptive population-error event to a finite producer-local union, and Proposition A.63 absorbs its multiplicity and sums exactly the four old confidence shares.

Finally, Proposition A.64 eliminates every old auxiliary and proves (22.1), with the explicit choice $q_{\text{old}} = 6$. The boundary calculations at the start of this proof verify the null, first-stage, finite/infinite-class, $v = d$, improper-output, fallback, ceiling, confidence, and δ -schedule boundaries. Combining these results gives the old-arm interface

$$(K_C^{\text{old-Lyu}}, A_{\text{o}, N_{\text{o}}}, \tilde{K}_{\text{o}}, N_{\text{old}}) \text{ is measurable, raw } (\varepsilon, \delta)\text{-DP, } (\alpha, \beta)\text{-PAC, and } N_{\text{old}} \leq K_{\text{O}} \Lambda^6 R_{\text{old}}.$$

□

A.16 Finite Arm and Baseline Comparison

Proposition A.66 (Shared exact zero-sample learner). *Under Assumptions 1 and 2, Lemma A.5, Propositions A.4, A.8, and A.51, and Proposition A.53, if $d = 0$, then $v = 0$, C and $\bar{C}$ are singletons, and the finite, old, and VC arms all use the same $N = 0$ Dirac quotient law at the unique $\bar{c}_0 \in \bar{C}$. Its raw pullback along T_0 is the same Dirac kernel, is $(0, 0)$ -DP, and has decoded risk zero for every D, c . Consequently $m_C(\alpha, \beta; \varepsilon, \delta) = 0$.*

Proof. Lemma A.5 proves that a nonempty Littlestone-dimension-zero class cannot contain two distinct functions: two functions differing at one point would shatter a depth-one tree. Its stated conclusion also gives $v = 0$. Proposition A.4 gives the corresponding singleton conclusion for $\bar{C}$. Thus there is only one possible quotient concept $\bar{c}_0$.

Define the finite-arm law on the one-point empty-input space by

$$K_{C,0}^{\text{fin}}(\emptyset, E) = \mathbf{1}_E(\bar{c}_0). \quad (1.1)$$

This is a Markov kernel, and its raw pullback along the unique T_0 is identical. Since the kernel is data independent, it is $(0, 0)$ -DP. Propositions A.4 and A.8 show that its decoded output is the unique raw target and hence has risk zero for every D . Propositions A.51 and A.53 define their null branches by this same unique quotient output. Therefore the three laws are literally identical, not merely equivalent in risk. The definition of m_C then gives $m_C = 0$. No positive-dimensional denominator, ceiling, or limiting argument is used. □

□

Proposition A.67 (Measurable finite quotient exponential-weight kernel). *Under Assumptions 2 and 4, Propositions A.4, A.6, and A.11, and A.12, and Lemma A.6, if $M := |C| < \infty$ and $N \in \mathbb{N}_0$, then empirical-mistake exponential weights on the fixed finite class $\bar{C}$ define an everywhere-total Markov kernel $K_{C,N}^{\text{fin}} : Z_Q^N \rightsquigarrow H_C$. Its raw pullback*

$$A_N^{\text{fin}}(s, E) := K_{C,N}^{\text{fin}}(T_N(s), E) \quad (2.1)$$

is a Markov kernel $Z_X^N \rightsquigarrow H_C$. Both laws release in the setting's output space and use Dec_C only after release. The law is defined on every labeled input, including nonrealizable inputs.

Proof. Proposition A.4 is a bijection, so $|\bar{C}| = M \geq 1$. For a quotient sample $\bar{s} = ((q_r, y_r))_{r=1}^N$ and $\bar{h} \in \bar{C}$, define the empirical mistake count

$$L_{\bar{s}}(\bar{h}) := \sum_{r=1}^N \mathbf{1}\{\bar{h}(q_r) \neq y_r\}. \quad (2.2)$$

This definition uses arbitrary labels and has $L_\emptyset = 0$. Put

$$w_{\bar{s}}(\bar{h}) := \exp\left(-\frac{\varepsilon}{2}L_{\bar{s}}(\bar{h})\right), \quad Z_{\bar{s}} := \sum_{\bar{g} \in \bar{C}} w_{\bar{s}}(\bar{g}), \quad (2.3)$$

and, for $E \in \mathcal{H}_C$,

$$K_{C,N}^{\text{fin}}(\bar{s}, E) := \frac{\sum_{\bar{h} \in \bar{C}} \mathbf{1}_E(\bar{h}) w_{\bar{s}}(\bar{h})}{Z_{\bar{s}}}. \quad (2.4)$$

Every weight is strictly positive, and the fixed sum has exactly M terms. Thus $0 < Z_{\bar{s}} < \infty$, so (2.4) is an everywhere-defined Borel probability measure supported on the fixed set $\bar{C} \subseteq H_C$. No empirical subclass or data-dependent range is used.

The domain Z_Q^N is countable discrete. Lemma A.6 makes every event-mass coordinate of the pointwise family (2.4) measurable on its full power-set sigma-field. Each support singleton is Borel because H_C is standard Borel. Thus (2.4) is precisely a separately supplied, everywhere-defined pointwise law to which Proposition A.11 applies; in particular, $K_{C,N}^{\text{fin}}$ is a quotient kernel produced by that proposition. Proposition A.6 makes T_N measurable, so the exact premise of Proposition A.12 is met and that proposition gives (2.1). The codomain remains exactly H_C ; decoding is the unchanged setting map. $\square$

Proposition A.68 (Pure all-input privacy of finite private ERM). *Under Assumptions 2 and 4, Proposition A.6, and Proposition A.67, if $M < \infty$, then for every $N \in \mathbb{N}_0$, the quotient kernel $K_{C,N}^{\text{fin}}$ and its raw pullback A_N^{fin} are pure ε -DP on all replace-one labeled inputs. Hence they are (ε, δ) -DP for every allowed $0 < \delta < 1$.*

Proof. Let $\bar{s} \sim \bar{s}'$ be quotient replace-one samples. For every $\bar{h} \in \bar{C}$, only one summand in (2.2) can change, whence

$$|L_{\bar{s}}(\bar{h}) - L_{\bar{s}'}(\bar{h})| \leq 1. \quad (3.1)$$

Thus the score $-L_{\bar{s}}(\bar{h})$ has replacement sensitivity one, and (2.3) gives the two pointwise inequalities

$$e^{-\varepsilon/2} w_{\bar{s}'}(\bar{h}) \leq w_{\bar{s}}(\bar{h}) \leq e^{\varepsilon/2} w_{\bar{s}'}(\bar{h}). \quad (3.2)$$

For an arbitrary $E \in \mathcal{H}_C$, denote the numerator in (2.4) by $W_{\bar{s}}(E)$. Summing the upper inequality in (3.2) over $\bar{C} \cap E$, and the lower inequality over all of $\bar{C}$, yields

$$W_{\bar{s}}(E) \leq e^{\varepsilon/2} W_{\bar{s}'}(E), \quad Z_{\bar{s}} \geq e^{-\varepsilon/2} Z_{\bar{s}'}. \quad (3.3)$$

Consequently

$$K_{C,N}^{\text{fin}}(\bar{s}, E) = \frac{W_{\bar{s}}(E)}{Z_{\bar{s}}} \leq e^{\varepsilon} \frac{W_{\bar{s}'}(E)}{Z_{\bar{s}'}} = e^{\varepsilon} K_{C,N}^{\text{fin}}(\bar{s}', E). \quad (3.4)$$

This is the complete exponential-mechanism calculation; no privacy theorem is cited. It uses neither realizability nor a utility event.

Proposition A.6 maps every raw neighbor pair to equality or a quotient neighbor pair. Applying (3.4) after that map proves the same pure inequality for A_N^{fin} , including arbitrary nonrealizable labels and same-cell replacements. Finally, a pure ε -DP inequality implies the setting's (ε, δ) -DP inequality by adding any $\delta > 0$ to its right side. $\square$

Lemma A.39 (Direct unconditional realizable risk tail). *Under Assumptions 2, 3, and 4, Propositions A.7 and A.8, and Proposition A.67, if $M < \infty$, then for every $N \in \mathbb{N}_0$, every probability measure D , and every $c \in C$,*

$$\Pr_{S \sim P_{D,c}^N, \bar{H} \sim A_N^{\text{fin}}(S, \cdot)} [\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq M \exp\left(-\frac{N\varepsilon\alpha}{4}\right). \quad (4.1)$$

The probability is unconditional over both iid sampling and mechanism randomness, and the bound is uniform in D, c .

Proof. Work first on the quotient experiment. Fix D, c , write $\bar{c} \in \bar{C}$ for the quotient target, and for $\bar{h} \in \bar{C}$ put

$$p_{\bar{h}} := \text{err}_{\bar{D}}(\bar{h}, \bar{c}). \quad (4.2)$$

Let $\bar{S} \sim P_{\bar{D}, \bar{c}}^N$. Realizability gives $L_{\bar{S}}(\bar{c}) = 0$ for every realized sample. Hence the target has weight one and

$$Z_{\bar{S}} \geq w_{\bar{S}}(\bar{c}) = 1. \quad (4.3)$$

For each fixed $\bar{h}$, (2.4) and (4.3) imply the pointwise conditional bound

$$K_{C,N}^{\text{fin}}(\bar{S}, \{\bar{h}\}) \leq \exp\left(-\frac{\varepsilon}{2} L_{\bar{S}}(\bar{h})\right). \quad (4.4)$$

The N mistake indicators of this fixed $\bar{h}$ are iid Bernoulli with mean $p_{\bar{h}}$. Therefore

$$\begin{aligned} \mathbb{E} \exp\left(-\frac{\varepsilon}{2} L_{\bar{S}}(\bar{h})\right) &= \left(1 - p_{\bar{h}} + p_{\bar{h}} e^{-\varepsilon/2}\right)^N \\ &= \left[1 - p_{\bar{h}}(1 - e^{-\varepsilon/2})\right]^N \\ &\leq \exp\left[-N p_{\bar{h}}(1 - e^{-\varepsilon/2})\right]. \end{aligned} \quad (4.5)$$

The last line uses $1 - u \leq e^{-u}$, which follows directly from $e^t \geq 1 + t$ at $t = -u$. To lower-bound the remaining coefficient, set $x = \varepsilon/2 \in (0, 1/2]$. Since $e^x \geq 1 + x$,

$$1 - e^{-x} \geq 1 - \frac{1}{1+x} = \frac{x}{1+x} \geq \frac{x}{2} = \frac{\varepsilon}{4}. \quad (4.6)$$

The bad subset $\mathcal{B} := \{\bar{h} \in \bar{C} : p_{\bar{h}} > \alpha\}$ is finite. Taking expectations in (4.4), using (4.5)-(4.6), and summing only over this fixed finite set gives

$$\begin{aligned} \Pr[\text{err}_{\bar{D}}(\bar{H}, \bar{c}) > \alpha] &= \mathbb{E}_{\bar{S}} \sum_{\bar{h} \in \mathcal{B}} K_{C,N}^{\text{fin}}(\bar{S}, \{\bar{h}\}) \\ &\leq \sum_{\bar{h} \in \mathcal{B}} e^{-N p_{\bar{h}}(1 - e^{-\varepsilon/2})} \\ &\leq |\mathcal{B}| e^{-N \varepsilon \alpha / 4} \leq M e^{-N \varepsilon \alpha / 4}. \end{aligned} \quad (4.7)$$

This calculation does not condition on a generalization event and does not introduce a separate empirical-to-population sampling term.

Proposition A.7 identifies the raw sample's quotient image with the quotient iid law, and Proposition A.8 identifies the measurable bad-risk events pointwise. Thus (4.7) is exactly the raw probability in (4.1). Nothing in the bound depends on D or c , so both suprema are valid. $\square$

$\square$

Proposition A.69 (Ceiling-aware complete finite-class arm). *Under Assumptions 1, 2, 3, and 4, and Propositions A.66, A.67, and A.68, and Lemma A.39, if $M = |C| < \infty$, then the finite arm is a measurable quotient-first learner whose raw pullback is pure ε -DP, hence (ε, δ) -DP, and whose uniform realizable decoded-risk failure probability is at most β . It uses $N_{\text{fin}} = 0$ when $d = 0$. When $d \geq 1$, it uses*

$$N_{\text{fin}} := \left\lceil \frac{4(\log M + \log(1/\beta))}{\varepsilon\alpha} \right\rceil \quad (5.1)$$

and satisfies the exact normalized bound

$$N_{\text{fin}} \leq 8R_{\text{fin}}(M, \alpha, \beta, \varepsilon). \quad (5.2)$$

Thus one may take $K_{\text{fin}} = 8$ and $q_{\text{fin}} = 0$, with no hidden positive power or sampling term.

If $|C| = \infty$, define the finite arm's certified cost to be $B_{\text{fin}} := +\infty$ and construct no finite-arm law or finite surrogate.

Proof. The $d = 0$ case is exactly Proposition A.66. Here $M = 1$, $\log M = 0$, and $\log^+ M = 1$, but the shared exact learner uses no samples; in particular, $0 \leq 8R_{\text{fin}}$.

Suppose $d \geq 1$ and $M < \infty$. Then C is not a singleton, so $M \geq 2$, although the following normalization needs only $M \geq 1$. Write $b := \log(1/\beta)$. Definition (5.1) gives

$$\frac{N_{\text{fin}}\varepsilon\alpha}{4} \geq \log M + b. \quad (5.3)$$

Lemma A.39 and (5.3) therefore give

$$\sup_D \sup_{c \in C} \Pr[\text{err}_D(\text{Dec}_C(\bar{H}), c) > \alpha] \leq Me^{-\log M - b} = \beta. \quad (5.4)$$

Propositions A.67 and A.68 supply the complete kernel and all-input privacy interfaces at this actual integer sample size.

It remains to pay the ceiling and normalize to the setting's exact rate. Put

$$x := \frac{4(\log M + b)}{\varepsilon\alpha}. \quad (5.5)$$

Because $b > \log 4 > 1$, $0 < \varepsilon \leq 1$, and $0 < \alpha < 1/4$, one has $x > 16 > 1$. Hence

$$N_{\text{fin}} = \lceil x \rceil \leq x + 1 \leq 2x = \frac{8(\log M + b)}{\varepsilon\alpha}. \quad (5.6)$$

For every finite nonempty class, $\log M \leq \log^+ M = \max\{1, \log M\}$. Therefore

$$\begin{aligned} N_{\text{fin}} &\leq \frac{8(\log^+ M + b)}{\varepsilon\alpha} \\ &\leq 8 \left[\frac{\log^+ M + b}{\varepsilon\alpha} + \frac{b}{\alpha} \right] = 8R_{\text{fin}}. \end{aligned} \quad (5.7)$$

The second summand in the setting's R_{fin} is used only as nonnegative slack in the last inequality. The direct tail (4.7) did not create a $\log M/\alpha$ or any other sampling term. All constants in (5.1)-(5.7) are numerical, and no factor Λ is needed.

For infinite C , the setting already defines $\log^+ |C| = +\infty$. Assigning cost $+\infty$ disables precisely this arm and does not define a data-dependent finite subset of C or $\bar{C}$. $\square$

$\square$

Proposition A.70 (Minimum of three complete common-interface learners). *Under Assumptions 1, 2, 3, and 4, Proposition A.8, Propositions A.66 and A.69, Proposition A.51, and Propositions A.56, A.60, A.63, and A.64, there is a universal*

$$K_* := \max\{8, K_O, K_V\} \geq 1 \quad (6.1)$$

such that, for every fixed allowed $0 < \delta < 1$,

$$m_C(\alpha, \beta; \varepsilon, \delta) \leq K_* \Lambda^6 \min\{R_{\text{fin}}, R_{\text{old}}, R_{\text{VC}}\}. \quad (6.2)$$

The minimum selects one complete learner before data are observed; it does not mix learner laws. For infinite C , the extended-real value $R_{\text{fin}} = +\infty$ removes only the finite arm.

Proof. The $d = 0$ case is already exact: by Proposition A.66, $m_C = 0$, so (6.2) follows from the nonnegativity of the three displayed rates. Assume henceforth $d \geq 1$.

The completed arm certificates are as follows. When $M < \infty$, the finite arm uses $K_{C, N_{\text{fin}}}^{\text{fin}}$ and its raw pullback, outputs in H_C followed by Dec_C , is pure ε -DP (hence (ε, δ) -DP), has raw decoded-risk failure at most β , and has certified cost $B_{\text{fin}} := 8R_{\text{fin}}$. The old arm uses K_O and its raw pullback, with the same output, privacy, and PAC interfaces, and has cost $B_O := K_O \Lambda^6 R_{\text{old}}$. The VC arm has the same interfaces and cost $B_V := K_V \Lambda^4 R_{\text{VC}}$.

Every row is a complete Markov-kernel learner on all raw labeled inputs. The PAC statement in each row is unconditional, is uniform over the same D, c , uses binary population risk, and has confidence $1 - \beta$. Proposition A.8, recorded as (D1), is the common zero-residual quotient/raw risk bridge. The laws need not be equal on the positive-dimensional branch; their mathematical interfaces are equal.

If $M = \infty$, set $B_{\text{fin}} = +\infty$ and omit that row from selection. Both Lyu (Lyu, 2025) bounds are finite under the fixed parameter regime, so at least one arm remains. If $M < \infty$, all three rows are available. Choose an index j_* minimizing the certified real thresholds B_{fin}, B_O, B_V , with a fixed data-independent tie rule. This choice is a function only of $C, d, v, \alpha, \beta, \varepsilon, \delta$, on which the definition of m_C permits the learner to depend. Run the single complete learner in row j_* , using its own integer sample size N_{j_*} . No random arm choice, sample splitting across arms, post-selection, or mixture occurs.

Each row's actual sample size is bounded by its own threshold, so

$$m_C \leq N_{j_*} \leq B_{j_*} = \min\{B_{\text{fin}}, B_O, B_V\}. \quad (6.3)$$

The setting definition gives $\Lambda \geq 1$. With (6.1),

$$\begin{aligned} B_{\text{fin}} &\leq K_* \Lambda^6 R_{\text{fin}}, \\ B_O &\leq K_* \Lambda^6 R_{\text{old}}, \\ B_V &\leq K_* \Lambda^6 R_{\text{VC}}. \end{aligned} \quad (6.4)$$

Taking the minimum in (6.4) and applying (6.3) proves (6.2), including the extended-real infinite-class convention. None of these finite-parameter inequalities uses an asymptotic condition on δ . $\square$

$\square$

Proposition A.71 (Boundary-preserving fixed-parameter frontier). *Under Assumptions 1, 2, 3, and 4, Propositions A.66, A.69, and A.70, Proposition A.51, and Propositions A.64 and A.65, the following exact boundary and specialization statements hold.*

1. If $d = 0$, then $v = 0$, $|C| = 1$, and the selected law is the shared zero-sample law of Proposition A.66; in particular, $m_C = 0$. No positive-dimensional rate is evaluated. 2. If $d \geq 1$, then $1 \leq v \leq d$. At $v = 1$, the VC arm's exposed structural factor is d^4 . At $v = d$,

$$vd^4 = d d^4 = d^5 \quad (7.1)$$

exactly, with no hidden positive power changing the equality. More generally $v = \Theta(d)$ gives the d^5 polynomial scale. 3. If C is finite, all three complete arms are available and the finite arm exposes $\log^+ |C|$. If C is infinite, then $R_{\text{fin}} = +\infty$, no finite surrogate is constructed, and the old and VC arms remain unchanged and available. 4. For every fixed $0 < \delta < 1$, the kernel, privacy, PAC, and full finite-parameter minimum (6.2) are valid. No statement $N\delta \rightarrow 0$ follows from fixed positive δ . 5. On $d \geq 1$, let $N_* := N_{j_*}$ be the actual sample size selected in the proof of Proposition A.70. Since its threshold is minimal,

$$0 \leq N_*\delta \leq \delta \min\{B_{\text{fin}}, B_o, B_V\} \leq \delta K_V \Lambda^4 R_{VC}. \quad (7.2)$$

Hence the setting's explicit schedule $\delta K_V \Lambda^4 R_{VC} \rightarrow 0$ implies $N_*\delta \rightarrow 0$, whichever complete arm wins the deterministic comparison. If the standalone old arm's own sample size is being studied rather than the minimizing selected learner, Proposition A.65 supplies only

$$\delta K_O \Lambda^6 R_{\text{old}} \rightarrow 0 \implies N_o\delta = N_{\text{old}}\delta \rightarrow 0; \quad (7.3)$$

the VC schedule is not silently substituted for (7.3). The finite arm is pure DP and needs no δ -schedule for privacy; no numerical $N_{\text{fin}}\delta$ limit is claimed unless a corresponding scalar domination is supplied. 6. Fix $\alpha, \beta, \varepsilon$ and restrict the asymptotic comparison to the declared scheduled-small- δ convention. On $d \geq 1$, $\log(1/\beta)$ is fixed and $v \geq 1$, so the three displayed rate expressions have polynomial class-complexity profiles

$$R_{\text{fin}} : \log^+ |C|, \quad R_{\text{old}} : d^5, \quad R_{VC} : vd^4, \quad (7.4)$$

with only the setting-permitted displayed logarithmic factors suppressed. Thus (6.2) is exactly the formalized shorthand

$$m_C(\alpha, \beta; \varepsilon, \delta) = \tilde{O}(\min\{\log^+ |C|, d^5, vd^4\}). \quad (7.5)$$

Here the first entry is $+\infty$ for infinite C . 7. The comparison between the two Lyu (Lyu, 2025) certificates is an exposed structural upper-bound comparison:

$$\frac{vd^4}{d^5} = \frac{v}{d}. \quad (7.6)$$

Consequently $v = o(d)$ gives $vd^4 = o(d^5)$, while $v = \Theta(d)$ returns to the d^5 scale and $v = d$ gives the exact identity (7.1). This does not assert a lower bound on m_C , and arm-specific displayed logarithms remain governed by the setting's tilde convention. 8. All conclusions remain conditional on the primitive finite-or-countable measurable evaluation quotient. They do not cover uncountably many evaluation types, characterize private sample complexity, or prove a universal $\text{poly}(v, \log d)$ or $\text{poly}(v, \log^* d)$ bound.

Proof. Item 1 is Proposition A.66. Conversely, a singleton binary class shatters no one-point set or depth-one tree, so the singleton and $d = 0$ branches coincide. If $d \geq 1$, two concepts differ at some point, and that point is VC-shattered; hence $v \geq 1$. If a set $\{x_1, \dots, x_r\}$ is VC-shattered, label every node at tree depth $j < r$ by x_{j+1} . Every root-to-leaf binary labeling is realized by a concept, so this is a Littlestone-shattered tree of depth r . Therefore $v \leq d$. Direct substitution

now proves Items 2 and 7. Proposition A.69 and the extended-real convention in Proposition A.70 prove Item 3. The final sentence of that minimum proposition, together with pure privacy from Proposition A.69, the fixed-parameter statement of Proposition A.51, and the fixed-parameter statement of Proposition A.65, proves Item 4.

For Item 5, (6.3) makes N_* no larger than every available certified threshold, in particular B_V ; multiplying by δ gives (7.2). The squeeze proves the selected-learner limit only under the exact setting schedule. Equation (7.3) is the separate old-arm statement, not a consequence newly inferred here.

For Items 6 and 7, with $\alpha, \beta, \varepsilon$ fixed, the coefficients and additive confidence quantities are fixed. Since $v \geq 1$, the first term of R_{VC} has exposed polynomial factor vd^4 ; Proposition A.51 proves that the lower-order displayed term introduces no larger positive power. Proposition A.64 exposes exactly d^5 , and Proposition A.69 exposes exactly $\log^+ |C|$. Equation (6.2) then gives (7.5), with the universal $K_*\Lambda^6$ hiding only permitted logarithms. The quotient and open-problem scope in Item 8 is precisely the unchanged formalized setting. Proposition A.66 first isolates the exact common $d = 0$ law. On the finite positive-dimensional branch, Proposition A.67 constructs the total finite-range quotient law and raw pullback on the same H_C, T_N , and decoder interface as the other arms. Proposition A.68 proves pure all-input privacy by the two sensitivity-one weight comparisons (3.2)-(3.4). Lemma A.39 uses the realizable target's unit weight and the exact Bernoulli transform to prove the unconditional uniform risk tail (4.1), without a separate sampling event. Proposition A.69 chooses the actual integer sample size, pays the ceiling, handles $M = 1$ and $\log^+ M$, and proves $N_{\text{fin}} \leq 8R_{\text{fin}}$; it assigns cost $+\infty$ and constructs no law when C is infinite.

Proposition A.51 supplies the complete VC-arm tuple. Propositions A.56, A.60, A.63, and A.64 supply the independently complete old-arm tuple. Consequently, the deterministic threshold comparison in Proposition A.70 selects one entire certified learner with bound

$$K_*\Lambda^6 \min\{R_{\text{fin}}, R_{\text{old}}, R_{VC}\}.$$

No probability law is averaged or spliced across arms.

Finally, Items 1–8 and their preceding derivations prove each null, dimension, cardinality, fixed- δ , scheduled- δ , and scaling specialization. It separates the selected learner's setting schedule from the standalone old arm's condition and states the exact conditional frontier without upgrading it to a characterization or unrestricted solution. Together, these results complete the common-interface comparison. □

A.17 Proof of the Main Theorem

Proof of the main theorem. Proposition A.4 gives quotient factorization; Lemmas A.4 and A.5 give VC- and Littlestone-dimension preservation; and Propositions A.6, A.7, and A.8 give raw-neighbor transport, iid pushforward, and the exact risk identity. The countable-domain kernel and raw pullback are provided by Propositions A.11 and A.12. For the VC arm, Proposition A.51 supplies the measurable kernel, all-input privacy, PAC guarantee, and $K_V\Lambda^4 R_{VC}$ sample bound. For the comparison arm, Propositions A.56, A.60, A.63, A.64, and A.65 supply the old-Lyu kernel, privacy, PAC, rate, and boundary statements. The finite arm is supplied by Propositions A.67, A.68, A.69, and A.66. Proposition A.70 selects one complete arm before data are observed, and Proposition A.71 proves the null, dimension, cardinality, and scheduled- δ reductions. All three arms use the same output space, decoder, privacy quantifier, and risk identity, so the deterministic minimum is a single learner. This proves Theorem 3.1. □